HL7 FHIR R4 — Chapter 5: Conformance Module (sections 5 – 5.9.11)

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5 Conformance Module

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5.0 Conformance Module

The Conformance Module represents metadata about the datatypes, resources and API features of the FHIR specification and can be used to create derived specifications.

CapabilityStatement StructureDefinition OperationDefinition SearchParameter | CompartmentDefinition ImplementationGuide ElementDefinition (datatype) | Full profiling details Detailed conformance rules

5.0.1 Introduction

The core FHIR specification describes a set of resources, frameworks and APIs that are used in many different contexts in healthcare. However, there is wide variability between jurisdictions and across the healthcare ecosystem around practices, requirements, regulations, education and what actions are feasible and/or beneficial.

For this reason, the FHIR specification is a "platform specification" - it creates a common platform or foundation on which a variety of different solutions are implemented. As a consequence, this specification usually requires further adaptation to particular contexts of use.

Typically, these adaptations specify:

• Rules about which resource elements are or are not used, and what additional elements are added that are not part of the base specification

• Rules about which of FHIR's RESTful API, messaging and document features are used, and how

• Rules about which terminologies are used in particular elements

• Descriptions of how the Resource elements and API features map to local requirements and/or implementations

Note that because of the nature of the healthcare ecosystem, there may be multiple overlapping sets of adaptations - by healthcare domain, by country, by institution, and/or by vendor/implementation.

FHIR provides a set of resources that can be used to represent and share the adaptations listed above in a computable fashion. These resources are collectively called the conformance resources. Although these conformance resources can be used in isolation they are typically used in the context of an Implementation Guide or a Capability Statement:

• Implementation Guides are documents published by a domain, institution or vendor that describe how FHIR is adapted to support a certain use case (or set of use cases). An implementation guide combines a set of conformance resources and supporting narrative into a document for use by implementers.

• A capability statements use the conformance resources to document how a client or server has implemented FHIR, i.e. which aspects of the specification and API are implemented and how.

5.0.2 Contents

The content of an Implementation Guide is described using the ImplementationGuide resource, while the capability statement is represented by the CapabilityStatement resource. These two resources make use of the complete set of conformance resources to fully capture the set of adaptations they represent. Note that the CapabilityStatement resource is one of the conformance resources, the first just describing the capabilities of a system, while the latter is the set of all conformance resources, including:

• The StructureDefinition defines how a particular structure (Resource, Extension or Data Type) is used to: Describe how existing elements in resources and/or data types are used and identify existing elements that are not used. Exact rules on how to do this can be found in the section on conformance rules Define extensions that can be used in resources or data types. Reference the Value Sets that specify the content of coded elements, see the terminology module for more details.

• The MessageDefinition describes messages that can be sent and received, including the driving event, contents to be exchanged and responsibilities on receipt.

• The OperationDefinition describes additional operations in addition to those in the base specification.

• The SearchParameter describes additional search capabilities in addition to those in the base specification

• The CompartmentDefinition describes a logical grouping for resources, used in access control or search

Conformance resources may be used independently, not just within the context of an ImplementationGuide resource or capability statement. See the section Common use cases for examples of such uses.

The conformance resources and their relationships are shown below:

Resources shown with a dotted box are described in other sections of the specification: ValueSet, ConceptMap and StructureMap are from the section on terminology, TestScript is part of the section on Implementer Support.

5.0.3 Security and Privacy

The conformance resources do not represent patient-related data, and as such are less susceptible to data breaching. Some caution is required however:

• StructureDefinitions may contain invariants formulated as structured expressions that are evaluated by external engines (i.e. xpath), which -if improperly sandboxed- could provide low-level access to the system

• A server may be implemented in such a way that it will automatically support additional search parameters or compartments definitions uploaded to that system. This may result in searches that use large amounts of processing power, expose sensitive patient data, or both.

• If a server acts as a repository of conformance resources, it should tightly control modifications to its data: updates to conformance resources are usually managed by a review process to ensure backwards compatibility and stability. Worst case, changes to these resources may result in dependent systems running the risk of being misconfigured.

5.0.4 Common use Cases

Conformance resources are commonly used as part of an Implementation Guide or CapabilityStatement resource. There are many ways to use the resources independently however, including:

• An instance of a resource may reference one or more StructureDefinitions to claim conformance to the rules laid out in those StructureDefinitions

• A server may serve CapabilityStatement resources, effectively functioning as a discovery endpoint for services within an organization

• A code-generation tool may use StructureDefinitions and OperationDefinitions to generate code that represents the structures as classes and operations as remotely callable functions to provide an easier programming model to a software developer.

• A server may act as a repository of NamingSystem resources, so vendors and implementers can quickly look up the URLs or oids for a given terminology or identifier system.

5.0.5 Developmental Roadmap

A subset of the conformance resources has been tested and used in production tooling and are now normative. These include StructureDefinition and ValueSet. Others, like CapabilityStatement, have been used widely, but not across all elements. As a consequence, these resource have a considerable number of elements marked for "trial use", while other parts are normative and will no longer change in a substantive way.

Other resources are still under development:

• ImplementationGuide: used in the HL7 production tooling but has not received much use outside of these tools yet.

• CompartmentDefinition: was new in STU3, and as such has not undergone much production use

These resources have been mainly used in the tools used to build the FHIR publication, early-adopter implementation guides and the FHIR Foundation conformance resource registry. In future, we expect to see more widespread use of these resources in validation tooling, code-generators and more extensive model-based guide authoring tools.

5.1 Profiling FHIR

URL: https://hl7.org/fhir/R4/profiling.html

5.1.0 Profiling FHIR

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

The base FHIR specification (this specification) describes a set of base resources, frameworks and APIs that are used in many different contexts in healthcare. However, there is wide variability between jurisdictions and across the healthcare ecosystem around practices, requirements, regulations, education and what actions are feasible and/or beneficial.

For this reason, the FHIR specification is a "platform specification" - it creates a common platform or foundation on which a variety of different solutions are implemented. As a consequence, this specification usually requires further adaptation to particular contexts of use. Typically, these adaptations specify:

• Rules about which resource elements are or are not used, and what additional elements are added that are not part of the base specification

• Rules about which API features are used, and how

• Rules about which terminologies are used in particular elements

• Descriptions of how the Resource elements and API features map to local requirements and/or implementations

Note that because of the nature of the healthcare ecosystem, there may be multiple overlapping sets of adaptations - by healthcare domain, by country, by institution, and/or by vendor/implementation.

5.1.0.1 Glossary

FHIR defines a cascade of artifacts for this purpose:

Artifact | Description | US Core example
Implementation Guide (IG) | A coherent and bounded set of adaptations that are published as a single unit. Validation occurs within the context of the Implementation Guide | US Core IG
Package | A group of related adaptations that are published as a group within an Implementation Guide | US Core Capability Statements
Conformance Resource | A single resource in a package that makes rules about how an implementation works. These are described below | DAF Problem Value Set
Profile | A set of constraints on a resource represented as a structure definition with kind = constraint | DAF Medication Request

The verb 'profile', or 'profiling', is used to describe the process of creating a profile.

5.1.0.2 Conformance Resources

Typically, Implementation Guides both restrict and extend APIs, resources and terminologies. FHIR provides a set of resources that can be used to represent and share the decisions that have been made, and allows implementers to build useful services from them. These resources are known as the conformance resources. These conformance resources allow implementers to:

• Indicate that some API calls are not used for a particular situation, and provide additional details about how API calls are used (Capability Statement)

• Add additional operations or search parameters not in the base specification (using the OperationDefinition resource or the SearchParameter Resource)

• Define how a particular structure (Resource, Extension or Data Type) is used (StructureDefinition Resource): Describe how existing elements in resources are used Identify existing elements that are not used Define extensions that can be used in resources or data types

• Mix custom and standard terminologies and choose which codes from these to use for a particular coded element(Value Set and StructureDefinition Resources)

• Map between local and standard terminologies or content models (Concept Map Resource)

• Register system namespaces for identifiers and terminologies (NamingSystem Resource)

These resources need to be used as discussed below, and also following the basic concepts for extension that are described in "Extensibility". For implementer convenience, the specification itself publishes its base definitions using these same resources.

5.1.0.3 Two uses of Profiles

The CapabilityStatement resource describes two different uses for profiles on resources: Resource Profiles and Supported Profiles. Resource Profiles are specified using the CapabilityStatement.rest.resource.profile element and Supported Profiles are specified using the CapabilityStatement.rest.resource.supportedProfile element.

5.1.0.3.1 CapabilityStatement.rest.resource.profile

These profiles describe the general features that are supported by the system for each kind of resource. Typically, this is the superset of all the different use-cases implemented by the system. This is a resource-level perspective of a system's functionality.

5.1.0.3.2 CapabilityStatement.rest.resource.supportedProfile

These profiles describe the information handled/produced by the system on a per use case basis. Some examples of the uses for these kind of profiles:

• A Laboratory service producing a set of different reports - general chemistry, blood count, etc. Typical labs would support several hundred different reports

• A care manager which handles a set of different types of care plans and associated clinical resources

• A medications formulary that handles several different levels of sophistication in its medication representations

These profiles represent different use cases leading to handling resources of the type indicated by the CapabilityStatement.rest.resource.type differently. For instance:

• A decision support service that provides analysis on several different sets of data conforming to a particular pattern - tests x,y and z with particular codes and units

For a producer system and a consumer system to exchange data successfully based on one of these supported profiles, it is not enough to know that the systems happen to have profiles that overlap for the use case of interest; the consumer must be able to filter the total set of resources made available by the producer system and deal only with the ones relevant to the use case.

As an example, consider a laboratory system generating thousands of reports a day. 1% of those reports are a particular endocrine report that a decision support system knows how to process. Both systems declare that they support the particular endocrine report profile, but how does the decision support system actually find the endocrine reports that it knows how to process?

One possible option is for the decision support system to receive every single report coming from the lab system, check whether it conforms to the profile or not, and then decide whether to process it. Checking whether a resource conforms to a particular profile or not is a straight forward operation (one option is to use the provided tools for this), but this is a very inefficient way - the decision support system has to receive and process 100 times as many resources as it uses. To help a consumer find the correct set of reports for a use-case, a producer of resources also SHALL, for any profile declared in CapabilityStatement.rest.resource.supportedProfile:

• Mark resources with profile assertions documenting the profile(s) they conform to (this enables indexing by the profile)

• (if a server) support searching by the _profile parameter for the declared profiles

Beyond these requirements, a producer of resources SHOULD ensure that any resource instance that would reasonably be expected to conform to the declared profiles SHOULD be published in this form.

Trial-Use Note: There are many uninvestigated issues associated with this use of profiles. HL7 is actively seeking feedback from users who experiment in this area, and users should be prepared for changes to features and obligations in this area in the future.

Feedback is welcome here .

5.1.0.4 Extending and Restricting the API

A CapabilityStatement resource lists the REST interactions (read, update, search, etc.) that a server provides or that a client uses, along with some supporting information for each. It can also be used to define a set of desired behaviors (e.g. as part of a specification or a Request for Proposal). The only interaction that servers are required to support is the capabilities interaction itself - to retrieve the server's CapabilityStatement. Beyond that, servers and clients support and use whichever API calls are relevant to their use case.

In addition to the operations that FHIR provides, servers may provide additional operations that are not part of the FHIR specification. Implementers can safely do this by appending a custom operation name prefixed with '$' to an existing FHIR URL, as the Operations framework does. The Conformance resource supports defining what OperationDefinitions make use of particular names on an end-point. If services are defined that are not declared using OperationDefinition, it may be appropriate to use longer names, reducing the chance of collision (and confusion) with services declared by other interfaces. The base specification will never define operation names with a "." (period) in them, so implementers are recommended to use some appropriate prefix in their names (such as "ihe.someService") to reduce the likelihood of name conflicts.

Implementations are encouraged, but not required, to define operations using the standard FHIR operations framework - that is, to declare the operations using the OperationDefinition resource, but some operations may involve formats that can't be described that way.

Implementations are also able to extend the FHIR API using additional content types. For instance, it might be useful to read or update the appointment resources using a vCard based format. vCard defines its own mime type, and these additional mime types can safely be used in addition to those defined in this specification.

5.1.0.5 Extending and Restricting Resources

Extending and restricting resources (collectively known as 'profiling a resource') is done with a "StructureDefinition" resource, which is a statement of rules about how the elements in a resource are used, and where extensions are used in a resource.

5.1.0.6 Changing Cardinality

One key function of profiles is to change the cardinality of an element. A profile can restrict the cardinality of an element within the limits of the base structure it is constraining. This table summarizes what types of restrictions are allowed:

derived (across) base (down)	0..0 (Not used)	0..1 (optional)	0..n (optional, many)	1..1 (required)	1..n (at least 1)
0..1	yes	yes	no	yes	no
0..*	yes	yes	yes	yes	yes
1..1	no	no	no	yes	no
1..*	no	no	no	yes	yes

When a profile is constraining another profile where there are more cardinality options (e.g. low is not just 0 or 1, and high is not just 1 or *), the same principles still apply: the constraining profile can only allow what the base profile allows.

Note that though a profile can constrain an element from x..* to x..1, this doesn't make any difference to the representation in the JSON format - the element will still be represented in an array. As an example, take Patient.name which has a cardinality of 0..*. In an unprofiled Patient, this will be represented as:

```
{
  "resourceType" : "Patient",
  "name" : [{
     "text" : "Peter James"
  }]
}
```

Even if a profile is created on the resource that narrows the cardinality to 1..1, applications will still process the resource without knowledge of the profile. For this reason the representation will still be the same.

5.1.0.7 Limitations of Use

What StructureDefinitions can do when they are constraining existing resources and datatypes is limited in some respects:

• Profiles cannot break the rules established in the base specification (e.g. cardinality as described above)

• Profiles cannot specify default values or meanings for elements defined in the base specification (note that data types and resources do not define default values at all, but default values may be defined for logical models

• Profiles cannot give more specific names to elements defined in the base specification, or add new elements

• It must be safe to process a resource without knowing the profile

The consequence of this is that if a profile mandates extended behavior that cannot be ignored, it must also mandate the use of a modifier extension. Another way of saying this is that knowledge must be explicit in the instance, not implicit in the profile.

As an example, if a profile wished to describe that a Procedure resource was being negated (e.g. asserting that it never happened), it could not simply say in the profile itself that this is what the resource means; instead, the profile must say that the resource must have an extension that represents this knowledge.

There is a facility to mark resources to indicate that they can only be safely understood by a process that is aware of and understands a set of published rules. For more information, see Restricted Understanding of Resources.

5.1.0.8 Using StructureDefinitions

A "constraint" StructureDefinition specifies a set of restrictions on the content of a FHIR resource or data type, or an additional set of constraints on an existing profile. A given structure definition is identified by its canonical URL, which SHOULD be the URL at which it is published. The following kinds of statements can be made about how an element is used, using a series of Element Definitions:

• Restricting the cardinality of the element; e.g. the base might allow 0..*, and a particular application might support 1..2

• Ruling out use of an element by setting its maximum cardinality to 0

• Restricting the contents of an element to a single fixed value

• Making additional constraints on the content of nested elements within the resource (expressed as XPath statements)

• Restricting the types for an element that allows multiple types

• Requiring a typed element or the target of a resource reference to conform to another structure profile (declared in the same profile, or elsewhere)

• Specifying a binding to a different terminology value set (see below)

• Providing refined definitions, comments/usage notes and examples for the elements defined in a Resource to reflect the usage of the element within the context of the Profile

• Providing more specific or additional mappings (e.g. to HL7 v2 or HL7 v3) for the resource when used in a particular context

• Declaring that one or more elements in the structure must be 'supported' (see below)

Any changed definitions SHALL be restrictions that are consistent with the rules defined in the resource in the FHIR Specification from which the profile is derived. Note that some of these restrictions can be enforced by tooling (and are by the FHIR tooling), but others (e.g. alignment of changes to descriptive text) cannot be automatically enforced.

Note that structure definitions cannot 'remove' mappings and constraints that are defined in the base structure, but for purposes of clarity, they can refrain from repeating them.

A structure definition contains a linear list of element definitions. The inherent nested structure of the elements is derived from the path value of each element. For instance, a sequence of the element paths like this:

• Root

• Root.childA

• Root.childA.grandchild1

• Root.childB

defines the following structure:

```
<Root>
  <childA>
    <grandChild1/>
  </childA>
  <childB/>
</Root>
```

or its JSON equivalent. The structure is coherent - children are never implied, and the path statements are always in order. The element list is a linear list rather than being explicitly nested because element definitions are frequently re-used in multiple places within a single definition, and this re-use is easier with a flat structure.

5.1.0.9 Differential vs Snapshot

StructureDefinitions may contain a differential statement, a snapshot statement or both.

Differential statements describe only the differences that they make relative to the structure definition they constrain (which is most often the base FHIR resource or data type). For example, a profile may make a single element mandatory (cardinality 1..1). In the example of a differential structure, it will contain a single element with the path of the element being made mandatory, and a cardinality statement. Nothing else is stated - all the rest of the structural information is implied (note that this means that a differential profile can be sparse and only mention the elements that are changed, without having to list the full structure. This rule includes the root element - it is not needed in a sparse differential).

Note that a differential can choose not to constrain elements. Doing so means that the profile will be more flexible in terms of compatibility with other profiles, but will require more work to support from implementing systems. Alternatively, a profile can constrain all optional elements to be not present (max cardinality = 0) - this closes the content, which makes implementation easier, but also reduces its usefulness.

In order to properly understand a differential structure, it must be applied to the structure definition on which it is based. In order to save tools from needing to support this operation (which is computationally intensive - and impossible if the base structure is not available), a StructureDefinition can also carry a "snapshot" - a fully calculated form of the structure that is not dependent on any other structure. The FHIR project provides tools for the common platforms that can populate a snapshot from a differential (note that the tools generate complete verbose snapshots; they do not support suppressing mappings or constraints).

StructureDefinitions can contain both a differential and a snapshot view. In fact, this is the most useful form - the differential form serves the authoring process, while the snapshot serves the implementation tooling. StructureDefinition resources used in operational systems should always have the snapshot view populated.

5.1.0.10 Slicing

One common feature of constraining StructureDefinitions is to take an element that may occur more than once (e.g. in a list), and then split the list into a series of sub-lists, each with different restrictions on the elements in the sub-list with associated additional meaning. In FHIR, this operation is known as "Slicing" a list. It is common to "slice" a list into sub-lists with each containing just one element, effectively putting constraints on each element in the list.

Here is an example to illustrate the process:

In this example, the base structure definition for the resource Observation defines the "component" element which contains a nested code and a value for observations that have multiple values. A classic example of this kind of observation is a blood pressure measurement - it contains 2 values, one for systolic, and one for diastolic (example).

This diagram shows the conceptual process of 'slicing' the component list into systolic and diastolic slices (note that to avoid clutter, the "name" attribute of Observation is shown as just a code not a full CodeableConcept).

The structure definition for Blood Pressure splits the component list into two sub-lists of one element each: a systolic element, and a diastolic element. Each of these elements has a fixed value for the code element (a fixed LOINC code for the name), and both have a value of type Quantity. This process is known as "slicing" and the Systolic and Diastolic elements are called "slices".

Note that when the resource is exchanged, the serialization format that is exchanged is not altered by the constraining definition. This means that the item profile names defined in the structure definition ("systolic", etc. in this example) are never exchanged. A resource instance looks like this:

```
<Observation>
  ...
  <component>
    <code {LOINC="8480-6"}/>
    <value ...>
  </component>
  <component>
    <code {LOINC="8462-4"}/>
    <value ...>
  </component>
</Observation>
```

In order to determine that the first related item corresponds to "Systolic" in the structure definition, so that it can then determine to which additional constraints for a sub-list the item conforms, the system checks the values of the elements. In this case, the "code" element in the target resource can be used to determine which slice that target refers to. This element is called the "discriminator".

5.1.0.11 Discriminator

In the general case, systems processing resources using a structure definition that slices a list can determine the slice corresponding to an item in the list by checking whether the item's content meets the rules specified for the slice. This would require a processor to be able to check all the rules applied in the slice and to do so speculatively in a depth-first fashion. Both of these requirements are inappropriately difficult for an operational system, and particularly for generated code (e.g. software that is automatically produced based on the StructureDefinition). Thus, to provide a better way to distinguish slices, a sliced element can designate a field or set of fields that act as a "discriminator" used to tell the slices apart.

When a discriminator is provided, the composite of the values of the elements designated in the discriminator is unique and distinct for each possible slice and applications can easily determine which slice an item in a list is. The intention is that this can be done in generated code, e.g. using a switch/case statement.

When a constraining structure designates one or more discriminators, it SHALL ensure that the possible values for each slice are different and non-overlapping, so that the slices can easily be distinguished.

Each discriminator is a pair of values: a type that indicates how the field is processed when evaluating the discriminator, and a FHIRPath expression that identifies the element in which the discriminator is found. There are five different processing types for discriminators:

value | The slices have different values in the nominated element.

exists | The slices are differentiated by the presence or absence of the nominated element.

pattern | The slices have different values in the nominated element, as determined by testing them against the applicable ElementDefinition.pattern[x].

type | The slices are differentiated by type of the nominated element.

profile | The slices are differentiated by conformance of the nominated element to a specified profile. Note that if the path specifies .resolve() then the profile is the target profile on the reference. In this case, validation by the possible profiles is required to differentiate the slices.

The FHIRPath statement that allows for the selection of the element on which the discriminator is based is a restricted FHIRPath statement that is allowed to include:

- Element selections (e.g. FHIRPath statements without "()" such as component.value)
- The function extension(url) to allow selection of a particular extension
- The function resolve() to allow slicing across resource boundaries
- The function ofType() to allow choosing a type in a polymorphic element

See the full details about the restricted FHIRPath statement.

Further notes about the use of the different discriminator types:

value | This is the most commonly used discriminator type: to decide based on the value of an element. Elements used like this are mostly primitive types- code, uri. Typical example: slice on the value of Patient.telecom.system, for values phone, email etc.

pattern | This is mostly used with elements of type CodeableConcept where the elements are distinguished by the presence of a particular code but other codes are expected to be present, and are irrelevant for the slice matching process. Typical example: slice on the value of Observation.code, for values LOINC codes 1234-5, 4235-8 etc

exists | This is not used commonly - it only has 2 values, so not much discrimination power. It's mainly used as an adjunct slicing criteria along with other discriminators. Elements used like this are mostly complex backbone elements. Typical example: slice on the pattern of Observation.code and the presence of Observation.component.

type | Used to match slices based on the type of the item. While it can be used with polymorphic elements such as Observation.value[x], mostly it is used with Resource types on references, to apply different profiles based on the different resource type. Typical example: slice on the type of List.item.resolve() for the types Patient, RelatedPerson.

profile | Used to match slices based on the whether the item conforms to the specified profile. This provides the most power, since the full range of profiling capabilities are available, but it is also the hardest to implement, and requires the most processing (>1000-fold compared to the others). Implementers should use this only where absolutely required. Typical example: slice on the type of Composition.section.entry() for the profiles Current-Clinical-Condition, Past-Medical-Event, etc

Each slice must use the element definition for the element(s) in the discriminator(s) to ensure that the slices are clearly differentiated by assigning an appropriate value domain, depending on the discriminator type. If the type is value, or pattern, then the element definition must use either:

- ElementDefinition.fixed[x], or
- ElementDefinition.pattern[x], or

• if the element has a terminology binding, a required binding with a Value Set that enumerates the list of possible codes in the value set ("extensional definition")

It is the composite (combined) values of the discriminators that are unique, not each discriminator alone. For example, a slice on a list of items that are references to other resources could designate fields from different resources, where each resource only has one of the designated elements, as long as they are distinct across slices.

A structure definition is not required to designate any discriminator at all for a slice, but those that don't identify discriminators are describing content that is very difficult to process, and so this is discouraged.

Within a structure definition, a slice is defined using multiple element entries that share a path but have distinct names. These entries together form a "slice group" that is:

• Initiated by a "slicing entry" That is, the first element in a slice group must contain a slicing property that defines the discriminator for all members of the group. It also contains the unconstrained definition of the element that is sliced, potentially including children of the unconstrained element, if there are any

• Mutually exclusive. This means that each element in a slice group SHALL describe a distinct set of values for the group's discriminators. Because of this constraint, an element in a resource instance will never match more than one element in a given slice group. If no discriminators are named, it SHOULD still be possible to differentiate the slices based on their properties, though it may be substantially harder to do so.

• Serialized as a group. The entries in a slice group must be adjacent in a serialized structure definition, or, if there are any intervening elements, those elements must be "compatible with" the group. Concretely, this means that any intervening elements must have a path that starts with the slice group's path. For example, an element with a path of Observation.name.extension would be compatible with (and thus, would not "break up") a slice group whose path was Observation.name

Some examples of discriminators:

Context | Discriminator Type | Discriminator Path | Interpretation
List.entry | value | item.resolve().name | Entries are differentiated by the name element on the target resource - probably an observation, which could be determined by other information in the profile
List.entry | type | item.resolve() | Entries are differentiated by the type of the target element that the reference points to
List.entry | profile | item.resolve() | Entries are differentiated by a profile tag on the target of the reference, as specified by a structure definition in the profile
List.entry | value | item.extension('http://acme.org/extensions/test').value | Entries are differentiated by the value of the code element in the extension with the designated URL
List.entry.extension | value | url | Extensions are differentiated by the value of their url property (usually how extensions are sliced)
List.entry | type, value | item.resolve(), item.resolve().value | Extensions are differentiated by the combination of the type of the referenced resource, and, if it has one, the code element of that resource. This would be appropriate for where a List might be composed of a Condition, and set of observations, each differentiated by its name - the condition has no name, so that is evaluated as a null in the discriminator set
Observation.value[x] | type | $this | Different constraints (e.g. "must support", usage notes, vocabulary bindings, etc.) are asserted for different supported types for the multi-typed element Observation.value[x]

Note that discriminator types of type and profile can also be used where a repeating element contains a resource directly (e.g. DomainResource.contained, Bundle.entry, Parameters.parameter.resource).

The examples of slicing and discriminators show exactly how this and other typical uses of slicing are represented in profiles.

Note that extensions are always sliced by the url element, though they may be resliced on additional elements where required.

5.1.0.12 Slice Cardinality

When an element of a fixed cardinality m..n is sliced, the following rules apply:

• Each slice cannot have a greater cardinality than the maximum number of slices allowed

• The sum of the maximum cardinalities can be larger than n

• The sum of the minimum cardinalities must be less or equal to n

• Each individual slice can have a minimum cardinality of 0 (less than m - the only situation where this is allowed), but the total number of elements in the instance must still be greater or equal to m

5.1.0.13 Default Slice

There is a special slice, called the default slice. This allows a profile to describe a set of specific slices, and then make a set of rules that apply to all of the remaining content that is not in one of the defined slices. Some rules about the default slice:

• It is identified because the name of the slice is @default. The sliceName '@default' is reserved and cannot be used in any other context

• Default slices are only allowed when the slicing rule = closed

• Default slices must not fix the value of the discriminator elements

• Default slices can be re-sliced in dependent profiles

One use of a default slice would be the case where the profile slices an identifier element to require a set of known identifiers, where the type element is prohibited (since they are known identifiers) but requires type on all other identifiers if any are present. In this case, the default slice makes no rules about the identifier.system (which is the slicing discriminator), but fixes the cardinality of type to 1..1 in the @default slice.

5.1.0.14 Re-profiling and Re-slicing

Profiles can be based on other profiles and can apply further constraints to those already specified. This is a useful technique, but implementers should be wary of over-use - humans have trouble understanding the implications of deep stacks of constraining profiles.

When a profile constrains another profile, it can make additional constraints, including extending the discriminator, adding new slices (if the slices are not already closed), and slicing inside the existing slices.

The rules for constraining ElementDefinition.slicing are as follows:

• ElementDefinition.slicing.rule can be constrained from open to closed

• ElementDefinition.slicing.ordered can be constrained from false to true

• If a discriminator for an element is declared in a parent profile, child profiles referencing that element: SHALL include all the same discriminators MAY add additional discriminators

It's sometimes necessary to slice data that has already been sliced in the base profile - that is, create new slices within the existing slices. This is called "Re-slicing". The rules for re-slicing are as follows:

When you slice, you define a name for each new slice. The name has to be unique across the set of slices in the profile. So if profile A defines an element X with cardinality 0..*, and profile B is derived from profile A, then profile B can either:

• make a constraint on X with no ElementDefinition.sliceName - in which case the profile is adding constraints to all slices of X; or

• make a constraint on X with an ElementDefinition.sliceName - in which case the profile is describing a specific slice on X, and the constraints only apply to that slice; or

• it can do both

Then, profile C derives from profile B. Profile C can do the following:

• make a constraint on X with no ElementDefinition.sliceName - in which case the profile is constraining all slices of X; or

• make a constraint on X with a different ElementDefinition.sliceName from that used in profile B - in which case the profile is describing a specific new slice on X, and the constraints only apply to that slice; or

• make a constraint on X with the same ElementDefinition.sliceName as that used in profile B - in which case the profile is making new constraints on the slice defined in profile B; or

• some combination of the above options

Note that it is possible for Profile C to make rules that are incompatible with profile B, in which case there is no set of instances that can be valid against profile C

In addition to the above, there are times when Profile C will need to further slice a slice defined in B. In this case, there's a need to reference both the ElementDefinition.sliceName of the original slice from Profile B as well as to define an ElementDefinition.sliceName for the slice defined within Profile C. This is done by separating the names using "/". For example, if Profile B defines the slice "example", and profile C defines the slice "example/example1", then this is deemed to be "example1" slice of the example slice. This process can continue indefinitely by separating each layer of slicing names with the "/" character. This pattern applies to @default too: @default/@default.

5.1.0.15 Extension Definitions

An extension definition defines the URL that identifies the extension and is used to refer to the extension definition when it is used in a resource.

The extension definition also defines the context where the extension can be used (usually a particular path or a data type) and then defines the extension element using the same details used to profile the structural elements that are part of resources. This means that a single extension can be defined once and used on different resources and/or datatypes, e.g. one would only have to define an extension for "hair color" once, and then specify that it can be used on both Patient and Practitioner.

For further discussion of defining and using extensions, along with some examples, see Extensibility.

5.1.0.15.1 Using Extensions in Profiles

Once defined, an extension can be used in an instance of a resource without any Profile declaring that it can, should or must be, but Profiles can be used to describe how an extension is used.

To prescribe the use of an extension in an instance, the extension list on the resource needs to be sliced. This is shown in the extensibility examples

Note that the minimum cardinality of an extension SHALL be a valid restriction on the minimum cardinality in the definition of the extension. If the minimum cardinality of the extension is 1 when it is defined, it can only

be mandatory when it is added to a profile. This is not recommended - the minimum cardinality of an extension should usually be 0.

5.1.0.16 Binding Definitions

Coded elements have bindings that link from the element to a definition of the set of possible codes that the element may contain. The binding identifies the definition of the set of possible codes and controls how tightly the set of the possible codes is interpreted.

The set of possible codes is either a formal reference to a ValueSet resource, which may be version specific, or a general reference to some web content that defines a set of codes. The second is most appropriate where a set of values is defined by some external standard (such as mime types). Alternatively, where the binding is incomplete (e.g. under development) just a text description of the possible codes can be provided.

Bindings have a property that defines the degree of flexibility associated with the use of the codes in the value set. See Binding Strength for further information.

5.1.0.17 Mixing Custom and Standard Terminologies

CodeSystem resources can be used to carry definitions of local codes (Example) and ValueSets can mix a combination of local codes and standard codes (e.g. LOINC, SNOMED), or just to choose a particular set of standard codes (examples: LOINC, SNOMED, RxNorm). Profiles can bind to these value sets instead of the ones defined in the base specification, following these rules:

Binding Strength in base specification | Customization Rules in Profiles
required | The value set can only contain codes contained in the value set specified by the FHIR specification
extensible | The value set can contain codes not found in the base value set. These additional codes SHOULD NOT have the same meaning as existing codes in the base value set
preferred or example | The value set can contain whatever is appropriate for local use

Note that local codes are not as interoperable as standard published code systems (e.g. LOINC, SNOMED CT), so it is preferable to use standard code systems.

5.1.0.18 Changing Binding Strength in Profiles

A profile can change the terminology binding of an element - both strength and value set - within the limits of the base structure it is constraining. This table summarizes the changes that can be made to the binding strength:

derived (across) base (down) | required | extensible | preferred | example
required | yes | no | no | no
extensible | yes | yes | no | no
preferred | yes | yes | yes | no
example | yes | yes | yes | yes

Note that a constraining profile may leave the binding strength the same and change the value set instead. Whatever the constraining profile does, it cannot make codes valid that are invalid in the base profile.

5.1.0.19 Must Support

One of the properties that can be declared on profiles but not on resource or data type definitions is 'mustSupport'. This is a boolean property. If true, it means that systems claiming to conform to a given profile must "support" the element. This is distinct from cardinality. It is possible to have an element with a minimum cardinality of "0", but still expect systems to support the element.

The meaning of "support" is not defined by the base FHIR specification, but can be set to true in a profile. When a profile does this, it SHALL also make clear exactly what kind of "support" is required. Examples include:

- The system must be able to store and retrieve the element
- The system must display the element to the user and/or allow the user to capture the element via the UI
- The element must appear in an output report
- The element must be taken into account when performing decision support, calculations or other processing
- etc.

The specific meaning of "Must Support" for the purposes of a profile SHALL be described in the ElementDefinition.definition, the general StructureDefinition.description or in other documentation for the implementation guide that includes the profile.

If creating a profile based on another profile, Must Support can be changed from false to true, but cannot be changed from true to false. Note that an element that has the property IsModifier is not necessarily a "key" element (e.g. one of the important elements to make use of the resource), nor is it automatically mustSupport - however both of these things are more likely to be true for IsModifier elements than for other elements.

5.1.0.20 Search Criteria

Implementations can define search criteria in addition to those defined in the specification itself. Search criteria fall into one of four categories:

• Enabling search on core elements that don't have standard search criteria defined (e.g. searching Observation by normal range)

• Enabling search on elements that already have standard search criteria defined, but with custom matching rules; e.g. a sounds-like search on Practitioner name

• Enabling search on an extension

• Enabling search that doesn't correspond to a single element but rather a combination of elements or computation on an element; e.g. searching for patients by age

Additional Search Parameters can be defined using the SearchParameter resource.

5.1.0.21 Presenting Profiles

When this specification describes a profile, the profile is presented in 5 different forms:

Text Summary | This presents a short summary human readable summary of the profile - a combination of the author's summary, and some automatically generated summary content
Differential Table | This is a view of the differential statement (see above). For context, additional information not in the differential is also shown partially transparent
Snapshot Table | This is a view of the snapshot produced by the profile (see above). The information is a comprehensive view of what the profile means
XML Template | An example of what the profile looks like in XML format
JSON Template | An example of what the profile looks like in JSON format

5.1.0.22 Supporting Multiple Profiles

Applications may be required to support more than one profile at a time. A typical example might be an EHR application that is required to support a general purpose data sharing profile (such as DAF), and also must support specific profiles for decision support using the same interface.

The impact of supporting two sets of profiles depends on whether resources are being created or consumed. When an application is creating content, it must create content that conforms to both sets of profiles - that is, the intersection of the profiles. When an application is consuming information, then it must be able to consume content that conforms to either set of profiles - that is, the union of the profiles.

Since applications generally consume and produce resources at the same time, conforming to more than one profile might not be possible, unless the profiles are designed to make statements at different levels - and the case above is one such case, where one profile is focused on data access, provenance, and availability, the other profile is focused on clinical content.

Accordingly, profiles can relate to each other in four different ways. Each profile can be thought of in terms of the set of instances that conform to the profile:

• Non-overlapping: there no instances that conform to profiles A & B (technically, the intersection of profiles A & B is an empty set)

• Partly overlapping: some instances conform to both A & B, but others only conform to A or B

• One set contained in the other: all resources that conform to A conform to B, but only some of the ones that conform to B conform to A (or vice versa)

• Identical sets: the set of resources that conform to A is the same as the set of resource that conform to B and the set of resources that don't conform to A is the same as the set of resources that don't conform to B

Profiles can be compared to determine their compatibility. One such comparison can be found (Todo: bring this into the build) between DAF and QICore . Note that this comparison is generated by tooling under ongoing development, and is purely draft content to demonstrate the idea of profile comparison.

5.1.1 Slicing and Discriminator Examples

URL: https://hl7.org/fhir/R4/profiling-examples.html

5.1.1 Slicing and Discriminator Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

5.1.1.1 Slicing Patient Telecom element

One common use of slicing is to describe different constraints on different kinds of patient contact details. In this example, Patient.telecom is defined as: ContactPoint [0..*] where the ContactPoint has system, value and use.

Consider the case where the profile should say:

• There must be a single home phone number

• The patient may have a work phone number

• The patient may have an email address

• No other types of contact are allowed

An example of a patient resource that meets these rules:

```
<Patient>
... snip ...
  <telecom>
    <system value="phone" />
    <use value="home" />
    <value value="5551234567" />
  </telecom>
  <telecom>
    <system value="email" />
    <value value="someone@acme.org" />
  </telecom>
... snip ...
</Patient>
```

To do this, the profile that implements these rules needs to do the following:

• On the base Patient.telecom element: define that slicing is discriminated by system and use, and that the slices are closed ("No other types of contact are allowed"). Order is left unfixed

• Then define 3 slices: home phone: fixed values for system and use, and slice cardinality 1..1 work phone: fixed values for system and use, and slice cardinality 0..1 email: fixed value for system, cardinality 0..0 for use, and slice cardinality 0..1

In a StructureDefinition, this will look like:

```
  <!-- setting up the slicing -->
  <element>
    <path value="Patient.telecom"/>
    <slicing>
      <discriminator>
        <type value="value"/>
        <path value="system"/>
      </discriminator/>
      <discriminator>
        <type value="value"/>
        <path value="use"/>
      </discriminator/>
      <rules value="closed"/>
    </slicing>
    <!-- net cardinality rules -->
    <min value="1"/>
    <max value="3"/>
  </element>

  <!-- first slice: home phone -->
  <element>
    <path value="Patient.telecom"/>
    <name value="HomePhone"/> <!-- mandatory - gives the slice a name -->
    <min value="1"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.system"/>
    <min value="1"/>
    <fixedCode value="phone"/>
  </element>
  <element>
    <path value="Patient.telecom.value"/>
    <min value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.use"/>
    <min value="1"/>
    <fixedCode value="home"/>
  </element>

  <!-- second slice: work phone -->
  <element>
    <path value="Patient.telecom"/>
    <name value="WorkPhone"/> <!-- mandatory - gives the slice a name -->
    <min value="0"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.system"/>
    <min value="1"/>
    <fixedCode value="phone"/>
  </element>
  <element>
    <path value="Patient.telecom.value"/>
    <min value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.use"/>
    <min value="1"/>
    <fixedCode value="work"/>
  </element>

  <!-- third slice: email -->
  <element>
```

```
    <path value="Patient.telecom"/>
    <name value="Email"/> <!-- mandatory - gives the slice a name -->
    <min value="0"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.system"/>
    <min value="1"/>
    <fixedCode value="email"/>
  </element>
  <element>
    <path value="Patient.telecom.value"/>
    <min value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.use"/>
    <max value="0"/>
  </element>
```

Note: much of the definition detail has been left out, and only the parts relevant to the pattern are shown. Also, providing a fixed value makes the minimum cardinality irrelevant, but it is shown here for completeness.

This table illustrates the relationship between the instance and the ElementDefinitions:

| Path | Name | Min | Max | Fixed
<Patient> | Patient
| Patient.telecom | | 1 | 3
<telecom> | Patient.telecom | HomePhone | 1 | 1
<system value="phone" /> | Patient.telecom.system | | 1 | 1 | phone
<value value="5551234567" /> | Patient.telecom.value | | 1 | 1
<use value="home" /> | Patient.telecom.use | | 1 | 1 | home
</telecom>
| Patient.telecom | WorkPhone | 0 | 1
<telecom> | Patient.telecom | Email | 0 | 1
<system value="email" /> | Patient.telecom.system | | 1 | 1 | email
<value value="someone@acme.org" /> | Patient.telecom.value | | 1 | 1
</telecom> | Patient.telecom.use

5.1.1.2 Fixed Order Slicing

A variant to this that looks simpler but turns out to be deceptively complicated is to fix the order and make all slices mandatory. In this case the profile says:

- The cardinality of telecom is 3..3
- The slicing is closed, and ordered, no discriminator
- The first slice is home phone number
- The second slice is work phone number
- The third slice is email address

An example of a patient resource that meets these rules:

```
{
  "resourceType" : "Patient",
  "telecom" : [
    {
      "system" : "phone",
      "value" : "5551234567",
      "use" : "home"
    },
    {
      "system" : "phone",
      "value" : "5551234567",
      "use" : "work"
    },
    {
      "system" : "email",
      "value" : "someone@acme.org"
    }
  ]
}
```

In a StructureDefinition, this will look like:

```
  <!-- setting up the slicing -->
  <element>
    <path value="Patient.telecom"/>
    <slicing>
      <!-- invariant: there SHALL be discriminator or description -->
      <description value="No discriminator needed since offsets are fixed"/>
      <rules value="closed"/>
      <ordered value="true"/>
    </slicing>
    <!-- cardinality rules -->
    <min value="3"/>
    <max value="3"/>
```

```
  </element>

  <!-- first slice: home phone -->
  <element>
    <path value="Patient.telecom"/>
    <name value="HomePhone"/> <!-- mandatory - gives the slice a name -->
    <min value="1"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.system"/>
    <min value="1"/>
    <fixedCode value="phone"/>
  </element>
  <element>
    <path value="Patient.telecom.value"/>
    <min value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.use"/>
    <min value="1"/>
    <fixedCode value="home"/>
  </element>

  <!-- second slice: work phone -->
  <element>
    <path value="Patient.telecom"/>
    <name value="WorkPhone"/> <!-- mandatory - gives the slice a name -->
    <min value="0"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.system"/>
    <min value="1"/>
    <fixedCode value="phone"/>
  </element>
  <element>
    <path value="Patient.telecom.value"/>
    <min value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.use"/>
    <min value="1"/>
    <fixedCode value="work"/>
  </element>

  <!-- third slice: email -->
  <element>
    <path value="Patient.telecom"/>
    <name value="Email"/> <!-- mandatory - gives the slice a name -->
    <min value="0"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.system"/>
    <min value="1"/>
    <fixedCode value="email"/>
  </element>
  <element>
    <path value="Patient.telecom.value"/>
    <min value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.use"/>
    <max value="0"/>
  </element>
```

A simple content model like this is attractive because it's both easier to specify and to work with. There's no need to think about the values of system and use - implementers can just grab first telecom value etc.

The problem with this approach is that such simple requirements are rarely valid in production healthcare systems, and even if they are valid during implementation, they rarely stay that way. Implementing multiple applications based on simple offsets will make the overall eco-system fragile against change, either internally, or as the scope of the eco-system grows.

Since the discriminator could be system+use in this case, it's best for future compatibility to specify it anyway.

5.1.1.3 Blood Pressure Example

Another use of Slicing is for Blood Pressure Measurements, where the profile says:

- There must be two components
- The first has LOINC code 8480-6, and a quantity
- The second has LOINC code 8462-4, and a quantity
- Other components are allowed (posture, etc., but not profiled in the base blood pressure profile)

An example of an observation resource that conforms such constrained StructureDefinitions looks like this:

```
<Observation>
  ...
  <component>
    <code>
      <coding>
        <system value="http://loinc.org" />
        <code value="8480-6" />
        <display value="Systolic blood pressure" />
      </coding>
    </code>
    <valueQuantity ... />
  </component>
  <component>
    <code>
      <coding>
        <system value="http://loinc.org" />
        <code value="8462-4" />
        <display value="Diastolic blood pressure" />
      </coding>
    </code>
    <valueQuantity .../>
  </component>
</Patient>
```

To do this, the profile that implements these rules needs to do the following:

• On the base Observation.component element: define that slicing is discriminated by code. Order is left unfixed, and rules are left open.

• Then define 2 slices: systolic: fixed values for code, cardinality 1..1, value is a Quantity diastolic: fixed values for code, cardinality 1..1, value is a Quantity

In a StructureDefinition, this will look like:

```
  <!-- setting up the slicing -->
  <element>
    <path value="Observation.component"/>
    <slicing>
      <discriminator>
        <type value="value"/>
        <path value="code"/>
      </discriminator/>
    </slicing>
    <!-- net cardinality rules -->
    <min value="2"/>
    <max value="*"/>
  </element>

  <!-- first slice: systolic -->
  <element>
    <path value="Observation.component"/>
    <name value="systolic"/> <!-- mandatory - gives the slice a name -->
    <min value="1"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Observation.component.code"/>
    <min value="1"/>
    <fixedCodeableConcept>
      <coding>
        <system value="http://loinc.org" />
        <code value="8480-6" />
        <display value="Systolic blood pressure" />
      </coding>
    </fixedCodeableConcept>
  </element>
  <element>
    <path value="Observation.component.valueQuantity"/>
    <min value="1"/>
  </element>

  <!-- second slice: diastolic -->
  <element>
    <path value="Observation.component"/>
    <name value="diastolic"/> <!-- mandatory - gives the slice a name -->
    <min value="1"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Observation.component.code"/>
    <min value="1"/>
    <fixedCodeableConcept>
      <coding>
        <system value="http://loinc.org" />
        <code value="8462-4" />
        <display value="Diastolic blood pressure" />
      </coding>
    </fixedCodeableConcept>
```

```
  </element>
  <element>
    <path value="Observation.component.valueQuantity"/>
    <min value="1"/>
  </element>
```

Note: much of the definition detail has been left out, and only the parts relevant to the pattern are shown. E.g. A real blood pressure profile would fix unit, an overall Observation code etc.

5.1.1.4 Extensions

For another example, consider slicing extensions. The base extension on every element is defined as a list (0..*) of extensions, and each extension has a url that identifies it, and a value. Consider an example where a profile defines that for a particular element (named Patient), there are two extensions, with URLs http://acme.com/a and http://acme.com/b. In addition, the profile allows other extensions to be used.

Technically, the profile achieves this by "slicing" the extension list, into two slices, and saying that the slicing is "open" - that there can be other slices introduced. Here are the relevant parts of the Profile on patient:

```
<StructureDefinition xmlns="http://hl7.org/fhir">
  <!-- snip -->
    <baseType value="Patient" />
    <baseDefinition value="http://hl7.org/fhir/StructureDefinition/Patient" />
    <derivation value="constraint" />
    <snapshot>
      <element>
        <path value="Patient"/>
        <!-- snip -->
      </element>
      <element>
        <path value="Patient.extension"/>
        <!-- this first element defines the slicing, and carries the base definition forward -->
        <slicing>
          <!-- Extensions are always discriminated by URL -->
          <discriminator>
            <type value="value"/>
            <path value="url"/>
          </discriminator/>
          <ordered value="false"/>     <!-- we don't care what order they appear in -->
          <rules value="open"/>        <!-- other extensions can be used -->
        </slicing>
        <!-- -- snip definition -->
      </element>
      <!-- first extension -->
      <element>
        <path value="Patient.extension"/>
        <!-- snip most of definition -->
        <type>
         <code value="Extension"/>
         <!-- the profile for an extension is a reference to the extension definition itself -
           this implies a profile, and happens to fix the @url value to the desired URL -->
         <profile value="http://acme.com/a"/>
        </type>
      </element>
      <!-- second extension -->
      <element>
        <path value="Patient.extension"/>
        <!-- snip most of definition -->
        <type>
         <code value="Extension"/>
         <!-- the profile for an extension is a reference to the extension definition itself -
           this implies a profile, and happens to fix the @url value to the desired URL -->
         <profile value="http://acme.com/b"/>
        </type>
      </element>
      <!-- snip rest of profile -->
    </snapshot>
</StructureDefinition>
```

Here is a patient example that conforms to this profile:

```
<Patient xmlns="http://hl7.org/fhir">
  <!-- two extensions, the order doesn't matter -->
  <extension url="http://acme.com/b">
    <!-- this has the right url, and so matches the second slice -->
    <!-- snip whatever value extension would have -->
  </extension>
  <extension url="http://acme.com/a">
    <!-- this has the right url, and so matches the first slice -->
    <!-- snip whatever value extension would have -->
  </extension>
  <!-- the rest of patient -->
</Patient>
```

5.1.1.5 Diagnostic Report & Observation

In this example, a profile on a diagnostic report says that it must have four observations, each with a different LOINC code (e.g. a classic lab panel). In this case (taken from the Example Lipid Profile), the structure that applies to DiagnosticReport will say that there are four slices on DiagnosticReport.result, each conforming to a

different structure, which are also contained in the same profile. Each of those structures will constrain the LOINC code in the observation.

```
<!-- first structure, the DiagnosticReport -->
<StructureDefinition xmlns="http://hl7.org/fhir">
  <!-- snip -->
  <url value="http://acme.org/fhir/StructureDefinition/lipid-report"/>
  <name value="LipidProfile"/>
  <baseType value="DiagnosticReport"/>
  <baseDefinition value="http://hl7.org/fhir/StructureDefinition/DiagnosticReport"/>
  <derivation value="constraint" />

  <!-- snip -->
  <snapshot>
    <!-- snip elements -->
    <element>
      <!-- first definition for result -->
      <path value="DiagnosticReport.result"/>
      <slicing>
        <!-- this is sliced by the code value of the target of the reference -->
        <discriminator>
          <type value="value"/>
          <path value="resolve().code"/>
        </discriminator/>
        <!-- have to be in the specified order -->
        <ordered value="true"/>
        <!-- this profile says, no other observations allowed -->
        <rules value="closed"/>
      </slicing>
      <!-- snip definition -->
    </element>
    <!-- first slice: Cholesterol -->
    <element>
      <path value="DiagnosticReport.result"/>
      <name value="Cholesterol"/>
      <!-- snip definition parts -->
      <type>
        <code value="Reference"/>
        <!-- this target must conform to the "Cholesterol" structure -->
        <targetProfile value="http://acme.org/fhir/StructureDefinition/Cholesterol"/>
      </type>
    </element>
    <!-- next 3 slices all the same, but different names for profile -->
    <element>
      <path value="DiagnosticReport.result"/>
      <name value="Triglyceride"/>
      <!-- snip definition parts -->
      <type>
        <code value="Reference"/>
        <!-- this target must conform to the "Triglyceride" structure -->
        <targetProfile value="http://acme.org/fhir/StructureDefinition/Triglyceride"/>
      </type>
    </element>
    <element>
      <path value="DiagnosticReport.result"/>
      <name value="LDLCholesterol"/>
      <!-- snip definition parts -->
      <type>
        <code value="Reference"/>
        <!-- this target must conform to the "LDLCholesterol" structure -->
        <targetProfile value="http://acme.org/fhir/StructureDefinition/LDLCholesterol"/>
      </type>
    </element>
    <element>
      <path value="DiagnosticReport.result"/>
      <name value="HDLCholesterol"/>
      <!-- snip definition parts -->
      <type>
        <code value="Reference"/>
        <!-- this target must conform to the "HDLCholesterol" structure -->
        <targetProfile value="http://acme.org/fhir/StructureDefinition/HDLCholesterol"/>
      </type>
    </element>
    <!-- snip elements -->
  </snapshot>
</StructureDefinition>

<!-- now, the second structure, for the Cholesterol profile -->
<StructureDefinition>
  <url value="http://acme.org/fhir/StructureDefinition/Cholesterol"/>
  <name value="Cholesterol"/>
  <baseType value="Observation"/>
  <baseDefinition value="http://hl7.org/fhir/StructureDefinition/Observation"/>
  <derivation value="constraint" />
  <snapshot>
    <!-- snip elements -->
    <element>
      <!-- this the element definition for name. Because of the
        slicing / discriminator rules in the LipidReport profile
```

```
        that references it, it is required to fix the value of
        the name element -->
      <path value="Observation.code"/>
      <definition>
         <!-- there are actually 3 ways to fix a CodeableConcept
         to a single fixed value. Here, we used the simplest one -->
         <valueCodeableConcept>
           <!-- just fix the value to the right code -->
           <coding>
             <system value="http://loinc.org"/>
             <code value="35200-5"/>
             <display value="Cholesterol"/>
           </coding>
         </valueCodeableConcept>
      </definition>
    </element>
    <!-- snip elements -->
  </snapshot>
</StructureDefinition>

<!-- Triglyceride profile -->
<StructureDefinition>
  <url value="http://acme.org/fhir/StructureDefinition/Triglyceride"/>
  <name value="Triglyceride"/>
  <baseType value="Observation"/>
  <baseDefinition value="http://hl7.org/fhir/StructureDefinition/Observation"/>
  <derivation value="constraint" />
  <snapshot>
    <!-- snip elements -->
    <element>
      <!-- this the element definition for name. Because of the
        slicing / discriminator rules in the LipidReport profile
        that references it, it is required to fix the value of
        the name element -->
      <path value="Observation.code"/>
      <definition>
         <!-- there's actually 3 ways to fix a CodeableConcept
         to a single fixed value. Here, we used the simplest one -->
         <valueCodeableConcept>
           <!-- just fix the value to the right code -->
           <coding>
             <system value="http://loinc.org"/>
             <code value="35217-9"/>
             <display value="Triglyceride"/>
           </coding>
         </valueCodeableConcept>
      </definition>
    </element>
    <!-- snip elements -->
  </snapshot>
</StructureDefinition>

<!-- LDLCholesterol profile -->
<StructureDefinition>
  <url value="http://acme.org/fhir/StructureDefinition/LDLCholesterol"/>
  <name value="LDLCholesterol"/>
  <baseType value="Observation"/>
  <baseDefinition value="http://hl7.org/fhir/StructureDefinition/Observation"/>
  <derivation value="constraint" />
  <snapshot>
    <!-- snip elements -->
    <element>
      <!-- this the element definition for name. Because of the
        slicing / discriminator rules in the LipidReport profile
        that references it, it is required to fix the value of
        the name element -->
      <path value="Observation.code"/>
      <definition>
         <!-- because of the way that LDL cholesterol measurements works
           (well, in this context- it varies), there's 2 different LOINC
           codes for either measured or calculated. So here, we bind to
           a value set -->
       <binding>
        <conformance value="required"/> <!-- must be required if this is a discriminator -->
        <!-- snip the actual value set reference, but it refers to a value
             set with two LOINC codes, one for each kind of LDL, which in
             this case are LOINC codes 18262-6 and 13457-7 -->
       </binding>
     </definition>
    </element>
    <!-- snip elements -->
  </snapshot>
</StructureDefinition>

<!-- HDLCholesterol profile -->
<StructureDefinition>
  <url value="http://acme.org/fhir/StructureDefinition/HDLCholesterol"/>
  <name value="HDLCholesterol"/>
  <baseType value="Observation"/>
```

```
  <baseDefinition value="http://hl7.org/fhir/StructureDefinition/Observation"/>
  <derivation value="constraint" />
  <snapshot>
    <!-- snip elements -->
    <element>
      <!-- this the element definition for name. Because of the
        slicing / discriminator rules in the LipidReport profile
        that references it, it is required to fix the value of
        the name element -->
      <path value="Observation.code"/>
      <definition>
         <!-- there's actually 3 ways to fix a CodeableConcept
        to a single fixed value. Here, we used the simplest one -->
        <valueCodeableConcept>
          <!-- just fix the value to the right code -->
          <coding>
            <system value="http://loinc.org"/>
            <code value="2085-9"/>
            <display value="LDL Cholesterol"/>
          </coding>
        </valueCodeableConcept>
      </definition>
    </element>
    <!-- snip elements -->
  </snapshot>
</StructureDefinition>
```

Here is an instance that meets the rules for this profile:

```
<!-- first, the diagnostic report -->
<DiagnosticReport xmlns="http://hl7.org/fhir">
  <!-- snip -->
  <!-- here's the set of results. We don't know what
    slices they are or anything until we go off, find
    the references, and look in them -->
  <result>
    <reference value="Observation/cholesterol"/>
  </result>
  <result>
    <reference value="Observation/triglyceride"/>
  </result>
  <result>
    <reference value="Observation/ldlcholesterol"/>
  </result>
  <result>
    <reference value="Observation/hdlcholesterol"/>
  </result>
  <!-- snip -->
</DiagnosticReport>

<!-- Observation, id = cholesterol -->
<Observation xmlns="http://hl7.org/fhir">
  <!-- the observation starts with the name, as specified
    by the profile for the first slice  -->
  <code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="35200-5"/>
      <display value="Cholesterol"/>
    </coding>
  </code>
  <!-- snip -->
</Observation>

<!-- Observation, id = triglyceride -->
<!-- this code matches the second slice. good -->
<Observation xmlns="http://hl7.org/fhir">
  <code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="35217-9"/>
      <display value="Triglyceride"/>
    </coding>
  </code>
  <!-- snip -->
</Observation>

<!-- Observation, id = hdlcholesterol -->
<!-- this code matches the fourth slice. good -->
<Observation xmlns="http://hl7.org/fhir">
  <code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="2085-9"/>
      <display value="HDL Cholesterol"/>
    </coding>
  </code>
  <!-- snip -->
</Observation>
```

```
<!-- Observation, id = ldlcholesterol -->
<!-- this code matches the third slice. good -->
<Observation id="ldlcholesterol">
  <code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="13457-7"/>
      <display value="LDL Chol. (Calc)"/>
    </coding>
  </code>
  <!-- snip -->
</Observation>
```

Note that this version is not valid, because the slices are not in the correct order:

```
<!-- first, the diagnostic report -->
<DiagnosticReport xmlns="http://hl7.org/fhir">
  <!-- snip -->
  <!-- here's the set of results. We don't know what
    slices they are or anything until we go off, find
    the references, and look in them -->
  <result>
    <reference value="Observation/cholesterol"/>
  </result>
  <result>
    <reference value="Observation/triglyceride"/>
  </result>
    <!-- Oops the last two are out of order -->
  <result>
    <reference value="Observation/hdlcholesterol"/>
  </result>
  <result>
    <reference value="Observation/ldlcholesterol"/>
  </result>
  <!-- snip -->
</DiagnosticReport>
```

5.1.1.6 Composition Sections

Most uses of Composition involve conformance to a profile that specifies which sections will exist, and what their contents will be. This is yet another example of slicing. A typical document content profile might specify a section structure something like this:

• Reason for visit Narrative, LOINC Code 29299-5

• Medications, LOINC Code 46057-6 Prescribed Medications, LOINC Code 66149-6 OTC medications, 66150-4 (optional)

• Vital Signs, LOINC Code 8716-3

Real profiles will contain lots of detail about the sections, but these are omitted here in the interests of clarity.

An example of a Composition that meets these rules:

```
<Composition>
... snip ...
  <section>
    <code>
      <coding>
        <system value="http://loinc.org" />
        <use value="29299-5" />
        <value value="Reason for visit Narrative" />
      </coding>
    </code>
    ... snip ...
  </section>
  <section>
    <code>
      <coding>
        <system value="http://loinc.org" />
        <use value="46057-6" />
        <value value="Medications section" />
      </coding>
    </code>
    ... snip ...
    <section>
      <code>
        <coding>
          <system value="http://loinc.org" />
          <use value="66149-6" />
          <value value="Prescribed medications" />
        </coding>
      </code>
      ... snip ...
    </section>
    <section>
      <code>
        <coding>
          <system value="http://loinc.org" />
          <use value="66150-4" />
          <value value="O medications" />
```

```
      </coding>
    </code>
    ... snip ...
  </section>
</section>
<section>
  <code>
    <coding>
      <system value="http://loinc.org" />
      <use value="8716-3" />
      <value value="Vital signs" />
    </coding>
  </code>
  ... snip ...
</section>
</Composition>
```

To do this, the profile that implements these rules needs to do the following:

• On the base Composition.section element: define that slicing is discriminated by code, and that the slices are closed and ordered

• Then define 3 slices on Composition.section: reason-for-visit: fixed code, cardinality 1..1 medication: fixed code, cardinality 1..1 vital-signs: fixed code, cardinality 1..1

• Then, in the medication slice, slice the Composition.section.section: define that slicing is discriminated by code, and that the slices are closed and ordered

• Then define 2 slices on the Composition.section.section in medication: prescribed: fixed code, cardinality 1..1 otc: fixed code, cardinality 0..1

In a StructureDefinition, this will look like:

```
<!-- setting up the slicing on Composition.section -->
<element>
  <path value="Composition.section"/>
  <slicing>
    <discriminator>
      <type value="value"/>
      <path value="code"/>
    </discriminator/>
    <ordered value="true"/>
    <rules value="closed"/>
  </slicing>
  <!-- net cardinality rules -->
  <min value="3"/>
  <max value="3"/>
</element>

<!-- first slice: reason for visit -->
<element>
  <path value="Composition.section"/>
  <name value="reason-for-visit"/>
  <min value="1"/>
  <max value="1"/>
</element>
<element>
  <path value="Composition.section.code"/>
  <min value="1"/>
  <fixedCodeableConcept>
    <coding>
      <system value="http://loinc.org" />
      <use value="29299-5" />
      <value value="Reason for visit Narrative" />
    </coding>
  </fixedCodeableConcept>
</element>

<!-- second slice: medications -->
<element>
  <path value="Composition.section"/>
  <name value="medications"/>
  <min value="1"/>
  <max value="1"/>
</element>
<element>
  <path value="Composition.section.code"/>
  <min value="1"/>
  <fixedCodeableConcept>
    <coding>
      <system value="http://loinc.org" />
      <use value="46057-6" />
      <value value="Medications section" />
    </coding>
  </fixedCodeableConcept>
</element>

<!-- setting up the inner slicing on medication Composition.section.section -->
<element>
  <path value="Composition.section.section"/>
```

```
    <slicing>
      <discriminator>
        <type value="value"/>
        <path value="code"/>
      </discriminator/>
      <ordered value="true"/>
      <rules value="closed"/>
    </slicing>
    <!-- net cardinality rules -->
    <min value="1"/>
    <max value="2"/>
  </element>

  <!-- first inner slice: prescribed medications -->
  <element>
    <path value="Composition.section.section"/>
    <name value="prescribed"/>
    <min value="1"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Composition.section.section.code"/>
    <min value="1"/>
    <fixedCodeableConcept>
      <coding>
        <system value="http://loinc.org" />
        <use value="66149-6" />
        <value value="Prescribed medications" />
      </coding>
    </fixedCodeableConcept>
  </element>

  <!-- second inner slice: over the counter medications -->
  <element>
    <path value="Composition.section.section"/>
    <name value="otc"/>
    <min value="0"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Composition.section.section.code"/>
    <min value="1"/>
    <fixedCodeableConcept>
      <coding>
        <system value="http://loinc.org" />
        <use value="66150-4" />
        <value value="Over the counter medications" />
      </coding>
    </fixedCodeableConcept>
  </element>

  <!-- third slice: Vital Signs -->
  <element>
    <path value="Composition.section"/>
    <name value="vital-signs"/>
    <min value="1"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Composition.section.code"/>
    <min value="1"/>
    <fixedCodeableConcept>
      <coding>
        <system value="http://loinc.org" />
        <use value="8716-3" />
        <value value="Vital signs" />
      </coding>
    </fixedCodeableConcept>
  </element>
```

5.1.1.7 Re-slicing

For some use cases, it is necessary to further re-slice in a derivative profile. This is when Profile A derives from Profile B, Profile A slices an element in Slice A and B, and Profile B further slices Slice A into A1 and A2.

As an example, assume that institution implementation guide specifies that a medication list is represented by a List resource, where the items into the list are split up into 3 slices, based on resource type:

- MedicationRequest
- MedicationAdministration
- MedicationStatement

These slices must come in order.

Now, a particular application creates an additional profile that specifies the following rules in addition to the base rules of the institution:

• the medicationRequests will be sliced into active and non-active, where all the active requests will come first, and then the inactive ones

• only active MedicationAdministrations will be encountered

• Medication statements are not supported

An example of a List resource that meets these rules:

```
<List>
... snip ...
  <entry>
    <item>
      <reference value="MedicationRequest/ex-active-1"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="MedicationRequest/ex-active-2"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="MedicationRequest/ex-inactive-1"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="MedicationAdministration/ex-any-1"/>
    </item>
  </entry>

  ... snip ...
</List>
```

The StructureDefinition differential for the first profile sets up the slicing, and then defines 3 slices:

```
  <!-- setting up the slicing -->
  <element>
    <path value="List.entry"/>
    <slicing>
      <discriminator>
        <type value="profile"/>
        <path value="entry.item.resolve()"/>
      </discriminator>
      <ordered value="true"/>
      <rules value="closed"/>
    </slicing>
  </element>

  <!-- first slice: medrequest -->
  <element>
    <path value="List.entry"/>
    <name value="medrequest"/> <!-- mandatory - gives the slice a name -->
  </element>
  <element>
    <path value="List.entry.item"/>
    <type>
      <code value="Reference"/>
      <targetProfile value="http://example.org/StructureDefinition/medrequest"/>
    </type>
  </element>

  <!-- second slice: medadmin -->
  <element>
    <path value="List.entry"/>
    <name value="medadmin"/> <!-- mandatory - gives the slice a name -->
  </element>
  <element>
    <path value="List.entry.item"/>
    <type>
      <code value="Reference"/>
      <targetProfile value="http://example.org/StructureDefinition/medadmin"/>
    </type>
  </element>

  <!-- third slice: medstmt -->
  <element>
    <path value="List.entry"/>
    <name value="medstmt"/> <!-- mandatory - gives the slice a name -->
  </element>
  <element>
    <path value="List.entry.item"/>
    <type>
      <code value="Reference"/>
      <targetProfile value="http://example.org/StructureDefinition/medstmt"/>
    </type>
  </element>
```

The derived StructureDefinition differential re-slices the first slice into 2 slices, constrains the second slice, and prohibits the 3rd:

```
<!-- setting up the slicing - this profile doesn't change the slicing
  (though it could add new discriminators if it needed) -->
<element>
  <path value="List.entry"/>
  <slicing>
    <discriminator>
      <type value="profile"/>
      <path value="entry.item.resolve()"/>
    </discriminator>
    <ordered value="true"/>
    <rules value="closed"/>
  </slicing>
</element>

<!-- first slice: medrequest/active -->
<element>
  <path value="List.entry"/>
  <!-- name/sub-name: splitting an existing slice into new slices  -->
  <name value="medrequest/active"/>
</element>
<element>
  <path value="List.entry.item"/>
  <type>
    <code value="Reference"/>
    <targetProfile value="http://example.org/StructureDefinition/medrequest-active"/>
  </type>
</element>

<!-- second slice: medrequest/inactive -->
<element>
  <path value="List.entry"/>
  <name value="medrequest/inactive"/>
</element>
<element>
  <path value="List.entry.item"/>
  <type>
    <code value="Reference"/>
    <targetProfile value="http://example.org/StructureDefinition/medrequest-active"/>
  </type>
</element>

<!-- third slice: medadmin (rule on the second slice in the parent profile) -->
<element>
  <path value="List.entry"/>
  <!-- use the same name - constraining the existing slice -->
  <name value="medadmin"/>
</element>
<element>
  <path value="List.entry.item"/>
  <type>
    <code value="Reference"/>
    <targetProfile value="http://example.org/StructureDefinition/medadmin-active"/>
  </type>
</element>

<!-- fourth slice: medstmt -->
<element>
  <path value="List.entry"/>
  <!-- use the same name - constraining the existing slice -->
  <name value="medstmt"/>
  <max value="0"/>
</element>
```

5.2 Resource CapabilityStatement - Content

URL: https://hl7.org/fhir/R4/capabilitystatement.html
FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Conformance Package for further details.

A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a particular version of FHIR that may be used as a statement of actual server functionality or a statement of required or desired server implementation.

5.2.1 Scope and Usage

The capability statement is a key part of the overall conformance framework in FHIR. It is used as a statement of the features of actual software, or of a set of rules for an application to provide. This statement connects to all the detailed statements of functionality, such as StructureDefinitions and ValueSets. This composite statement of application capability may be used for system compatibility testing, code generation, or as the basis for a conformance assessment. For further information about Conformance testing, see Conformance Rules and Profiling FHIR.

Specifically, capability statements are used in one of three ways:

Instance | implementation must be present and software may be present
Capability | implementation must be absent, software must be present
Requirements | implementation and software must be absent

5.2.1.1 Instance: Describe an actual implementation

In this scenario, the capability statement describes the capabilities of a deployed and configured solution available at a particular access point or set of access points. The statement describes exactly how to interface with that deployed solution and thus provides for a degree of self-configuration of software solutions.

This is the type of statement that FHIR restful solutions are expected to make available on invocation of the capabilities operation. It is also the type of statement that forms a basis for the testing, certification or commissioning of specific software installations.

5.2.1.2 Capability: Describe software solution capabilities

In this scenario, the capability statement describes generic capabilities of a software application or component solution. The solution might be available for purchase or other acquisition and might be deployed and configured at any number of independent sites. Because it is not dependent on any particular implementation, the profile cannot provide specific details such as endpoint addresses. It may also need to document various configurations in which the application can be set up or describe the degree of customizability associated with the solution.

This type of statement may be used as a marketing tool by software and system developers to formally describe their capabilities. It can also be used as the basis for conformance testing of software solutions independent of a particular installation.

5.2.1.3 Requirements: Describe a desired solution

In this scenario, the capability statement describes the capabilities of a desired system. It might be used as part of an architectural design process to document needed system capabilities, or might be used as part of an RFP process to formally document the requirements of a requested solution and to document the criteria by which proposals will be evaluated.

These three types of profiles can be used together. A requirements statement can be compared against the solution statements proffered by respondents to an RFP. A solution statement for a software package forms the starting point for the implementation statement associated with a particular installation of that software package.

CapabilityStatements of type "requirement" describe what capabilities are potentially relevant; additional documentation or extensions (see capabilitystatement-expectation) within the CapabilityStatement are expected to make more explicit statements of degree of expectation associated with each capability.

5.2.2 Background and Context

Capability Statements provide for a degree of automatic configuration and adaptation. However, capturing absolutely every variation that could impact the interoperability of two systems, let alone keeping that detailed information up-to-date as systems evolve through maintenance and upgrades, is rarely practical. Therefore, capability statements should be seen as an interim step. They provide a degree of automation. However, they also provide a great deal of human-readable content that can minimize the need for direct communication between the operators of the systems being configured to interoperate.

5.2.2.1 Supporting Multiple Versions

Applications may implement multiple versions. If they do, then a CapabilityStatement describes the system's support for a particular version of FHIR, and the system will have multiple statements, one for each version it supports. For further information, see Managing Multiple Versions, and the $versions operation.

5.2.2.2 Mixed Normative Content

While the core of the CapabilityStatement resource is Normative, many of the flags that indicate exactly how the system operates are marked as trial-use. Roughly, the portions of the resource that correspond to OpenAPI document elements are normative.

Applications looking for normative stability should only use the normative parts of the resource, and not populate or ignore the portions labelled trial-use. To assist with this, clients can ask for the server to return a 'Normative content only' CapabilityStatement using the mode parameter on /metadata.

Community discussion regarding more capable, efficient and computable representations of an applications capabilities may lead to change to the trial-use parts of this resource or the creation of new resources and/or functionality in future versions of this specification.

This resource is referenced by itself and TestScript

5.2.3 Resource Content

- Structure
- UML

- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
CapabilityStatement | IN | | DomainResource | A statement of system capabilities + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: A Capability Statement SHALL have at least one of REST, messaging or document element. + Rule: A Capability Statement SHALL have at least one of description, software, or implementation element. + Rule: Messaging end-point is required (and is only permitted) when a statement is for an implementation. + Rule: The set of documents must be unique by the combination of profile and mode. + Rule: If kind = instance, implementation must be present and software may be present + Rule: If kind = capability, implementation must be absent, software must be present + Rule: If kind = requirements, implementation and software must be absent Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this capability statement, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the capability statement
name | ΣI | 0..1 | string | Name for this capability statement (computer friendly)
title | Σ | 0..1 | string | Name for this capability statement (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 1..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | I | 0..1 | markdown | Natural language description of the capability statement
useContext | ΣTU | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for capability statement (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this capability statement is defined
copyright | | 0..1 | markdown | Use and/or publishing restrictions
kind | ΣI | 1..1 | code | instance | capability | requirements CapabilityStatementKind (Required)
instantiates | Σ | 0..* | canonical(CapabilityStatement) | Canonical URL of another capability statement this implements
imports | ΣTU | 0..* | canonical(CapabilityStatement) | Canonical URL of another capability statement this adds to
software | ΣI | 0..1 | BackboneElement | Software that is covered by this capability statement
name | Σ | 1..1 | string | A name the software is known by
version | Σ | 0..1 | string | Version covered by this statement
releaseDate | Σ | 0..1 | dateTime | Date this version was released
implementation | ΣI | 0..1 | BackboneElement | If this describes a specific instance
description | Σ | 1..1 | string | Describes this specific instance
url | Σ | 0..1 | url | Base URL for the installation
custodian | ΣTU | 0..1 | Reference(Organization) | Organization that manages the data
fhirVersion | Σ | 1..1 | code | FHIR Version the system supports FHIRVersion (Required)
format | Σ | 1..* | code | formats supported (xml | json | ttl | mime type) MimeType (Required)
patchFormat | Σ | 0..* | code | Patch formats supported MimeType (Required)
implementationGuide | Σ | 0..* | canonical(ImplementationGuide) | Implementation guides supported
rest | ΣI | 0..* | BackboneElement | If the endpoint is a RESTful one + Rule: A given resource can only be described once per RESTful mode.
mode | Σ | 1..1 | code | client | server RestfulCapabilityMode (Required)
documentation | | 0..1 | markdown | General description of implementation
security | ΣTU | 0..1 | BackboneElement | Information about security of implementation
cors | Σ | 0..1 | boolean | Adds CORS Headers (http://enable-cors.org/)
service | Σ | 0..* | CodeableConcept | OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates RestfulSecurityService (Extensible)
description | | 0..1 | markdown | General description of how security works
resource | ΣI | 0..* | BackboneElement | Resource served on the REST interface + Rule: Search parameter names must be unique in the context of a resource.
type | Σ | 1..1 | code | A resource type that is supported ResourceType (Required)
profile | Σ | 0..1 | canonical(StructureDefinition) | Base System profile for all uses of resource
supportedProfile | ΣTU | 0..* | canonical(StructureDefinition) | Profiles for use cases supported
documentation | | 0..1 | markdown | Additional information about the use of the resource type
interaction | | 0..* | BackboneElement | What operations are supported?
code | | 1..1 | code | read | vread | update | patch | delete | history-instance | history-type | create | search-type TypeRestfulInteraction (Required)
documentation | | 0..1 | markdown | Anything special about operation behavior
versioning | TU | 0..1 | code | no-version | versioned | versioned-update ResourceVersionPolicy (Required)
readHistory | TU | 0..1 | boolean | Whether vRead can return past versions
updateCreate | TU | 0..1 | boolean | If update can commit to a new identity
conditionalCreate | TU | 0..1 | boolean | If allows/uses conditional create
conditionalRead | TU | 0..1 | code | not-supported | modified-since | not-match | full-support ConditionalReadStatus (Required)
conditionalUpdate | TU | 0..1 | boolean | If allows/uses conditional update
conditionalDelete | TU | 0..1 | code | not-supported | single | multiple - how conditional delete is supported ConditionalDeleteStatus (Required)

referencePolicy | TU | 0..* | code | literal | logical | resolves | enforced | local ReferenceHandlingPolicy (Required)
searchInclude | TU | 0..* | string | _include values supported by the server
searchRevInclude | TU | 0..* | string | _revinclude values supported by the server
searchParam | | 0..* | BackboneElement | Search parameters supported by implementation
name | | 1..1 | string | Name of search parameter
definition | | 0..1 | canonical(SearchParameter) | Source of definition for parameter
type | | 1..1 | code | number | date | string | token | reference | composite | quantity | uri | special SearchParamType (Required)
documentation | | 0..1 | markdown | Server-specific usage
operation | Σ | 0..* | BackboneElement | Definition of a resource operation
name | Σ | 1..1 | string | Name by which the operation/query is invoked
definition | Σ | 1..1 | canonical(OperationDefinition) | The defined operation/query
documentation | | 0..1 | markdown | Specific details about operation behavior
interaction | | 0..* | BackboneElement | What operations are supported?
code | | 1..1 | code | transaction | batch | search-system | history-system SystemRestfulInteraction (Required)
documentation | | 0..1 | markdown | Anything special about operation behavior
searchParam | | 0..* | see searchParam | Search parameters for searching all resources
operation | Σ | 0..* | see operation | Definition of a system level operation
compartment | | 0..* | canonical(CompartmentDefinition) | Compartments served/used by system
messaging | ΣITU | 0..* | BackboneElement | If messaging is supported
endpoint | | 0..* | BackboneElement | Where messages should be sent
protocol | | 1..1 | Coding | http | ftp | mllp + MessageTransport (Extensible)
address | | 1..1 | url | Network address or identifier of the end-point
reliableCache | | 0..1 | unsignedInt | Reliable Message Cache Length (min)
documentation | | 0..1 | markdown | Messaging interface behavior details
supportedMessage | Σ | 0..* | BackboneElement | Messages supported by this system
mode | Σ | 1..1 | code | sender | receiver EventCapabilityMode (Required)
definition | Σ | 1..1 | canonical(MessageDefinition) | Message supported by this system
document | ΣITU | 0..* | BackboneElement | Document definition
mode | Σ | 1..1 | code | producer | consumer DocumentMode (Required)
documentation | | 0..1 | markdown | Description of document support
profile | Σ | 1..1 | canonical(StructureDefinition) | Constraint on the resources used in the document
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CapabilityStatement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this capability statement, represented as a URI (globally
    unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the capability statement -->
 <name value="[string]"/><!--  0..1 Name for this capability statement (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this capability statement (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 1..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  0..1 Natural language description of the capability statement -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for capability statement (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this capability statement is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <kind value="[code]"/><!--  1..1 instance | capability | requirements -->
 <instantiates><!-- 0..* canonical(CapabilityStatement) Canonical URL of another capability statement this implements
    --></instantiates>
 <imports><!-- 0..* canonical(CapabilityStatement) Canonical URL of another capability statement this adds to
    --></imports>
 <software>  <!--  0..1 Software that is covered by this capability statement -->
  <name value="[string]"/><!-- 1..1 A name the software is known by -->
  <version value="[string]"/><!-- 0..1 Version covered by this statement -->
  <releaseDate value="[dateTime]"/><!-- 0..1 Date this version was released -->
 </software>
 <implementation>  <!--  0..1 If this describes a specific instance -->
  <description value="[string]"/><!-- 1..1 Describes this specific instance -->
  <url value="[url]"/><!-- 0..1 Base URL for the installation -->
  <custodian><!-- 0..1 Reference(Organization) Organization that manages the data --></custodian>
 </implementation>
 <fhirVersion value="[code]"/><!-- 1..1 FHIR Version the system supports -->
 <format value="[code]"/><!-- 1..* formats supported (xml | json | ttl | mime type) -->
 <patchFormat value="[code]"/><!-- 0..* Patch formats supported -->
 <implementationGuide><!-- 0..* canonical(ImplementationGuide) Implementation guides supported --></implementationGuide>
 <rest>  <!--  0..* If the endpoint is a RESTful one -->
  <mode value="[code]"/><!-- 1..1 client | server -->
  <documentation value="[markdown]"/><!-- 0..1 General description of implementation -->
  <security>  <!-- 0..1 Information about security of implementation -->
   <cors value="[boolean]"/><!-- 0..1 Adds CORS Headers (http://enable-cors.org/) -->
   <service><!-- 0..* CodeableConcept OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates --></service>
   <description value="[markdown]"/><!-- 0..1 General description of how security works -->
  </security>
```

```
  <resource>  <!-- 0..* Resource served on the REST interface -->
   <type value="[code]"/><!-- 1..1 A resource type that is supported -->
   <profile><!-- 0..1 canonical(StructureDefinition) Base System profile for all uses of resource --></profile>
   <supportedProfile><!-- 0..* canonical(StructureDefinition) Profiles for use cases supported --></supportedProfile>
   <documentation value="[markdown]"/><!-- 0..1 Additional information about the use of the resource type -->
   <interaction>  <!-- 0..* What operations are supported? -->
    <code value="[code]"/><!-- 1..1 read | vread | update | patch | delete | history-instance | history-type | create |
    search-type -->
    <documentation value="[markdown]"/><!-- 0..1 Anything special about operation behavior -->
   </interaction>
   <versioning value="[code]"/><!-- 0..1 no-version | versioned | versioned-update -->
   <readHistory value="[boolean]"/><!-- 0..1 Whether vRead can return past versions -->
   <updateCreate value="[boolean]"/><!-- 0..1 If update can commit to a new identity -->
   <conditionalCreate value="[boolean]"/><!-- 0..1 If allows/uses conditional create -->
   <conditionalRead value="[code]"/><!-- 0..1 not-supported | modified-since | not-match | full-support -->
   <conditionalUpdate value="[boolean]"/><!-- 0..1 If allows/uses conditional update -->
   <conditionalDelete value="[code]"/><!-- 0..1 not-supported | single | multiple - how conditional delete is supported
    -->
   <referencePolicy value="[code]"/><!-- 0..* literal | logical | resolves | enforced | local -->
   <searchInclude value="[string]"/><!-- 0..* _include values supported by the server -->
   <searchRevInclude value="[string]"/><!-- 0..* _revinclude values supported by the server -->
   <searchParam>  <!-- 0..* Search parameters supported by implementation -->
    <name value="[string]"/><!-- 1..1 Name of search parameter -->
    <definition><!-- 0..1 canonical(SearchParameter) Source of definition for parameter --></definition>
    <type value="[code]"/><!-- 1..1 number | date | string | token | reference | composite | quantity | uri | special
    -->
    <documentation value="[markdown]"/><!-- 0..1 Server-specific usage -->
   </searchParam>
   <operation>  <!-- 0..* Definition of a resource operation -->
    <name value="[string]"/><!-- 1..1 Name by which the operation/query is invoked -->
    <definition><!-- 1..1 canonical(OperationDefinition) The defined operation/query --></definition>
    <documentation value="[markdown]"/><!-- 0..1 Specific details about operation behavior -->
   </operation>
  </resource>
  <interaction>  <!-- 0..* What operations are supported? -->
   <code value="[code]"/><!-- 1..1 transaction | batch | search-system | history-system -->
   <documentation value="[markdown]"/><!-- 0..1 Anything special about operation behavior -->
  </interaction>
  <searchParam><!-- 0..* Content as for CapabilityStatement.rest.resource.searchParam Search parameters for searching
    all resources --></searchParam>
  <operation><!-- 0..* Content as for CapabilityStatement.rest.resource.operation Definition of a system level operation
     --></operation>
  <compartment><!-- 0..* canonical(CompartmentDefinition) Compartments served/used by system --></compartment>
 </rest>
 <messaging>  <!--  0..* If messaging is supported -->
  <endpoint>  <!-- 0..* Where messages should be sent -->
   <protocol><!-- 1..1 Coding http | ftp | mllp + --></protocol>
   <address value="[url]"/><!-- 1..1 Network address or identifier of the end-point -->
  </endpoint>
  <reliableCache value="[unsignedInt]"/><!-- 0..1 Reliable Message Cache Length (min) -->
  <documentation value="[markdown]"/><!-- 0..1 Messaging interface behavior details -->
  <supportedMessage>  <!-- 0..* Messages supported by this system -->
   <mode value="[code]"/><!-- 1..1 sender | receiver -->
   <definition><!-- 1..1 canonical(MessageDefinition) Message supported by this system --></definition>
  </supportedMessage>
 </messaging>
 <document>  <!--  0..* Document definition -->
  <mode value="[code]"/><!-- 1..1 producer | consumer -->
  <documentation value="[markdown]"/><!-- 0..1 Description of document support -->
  <profile><!-- 1..1 canonical(StructureDefinition) Constraint on the resources used in the document --></profile>
 </document>
</CapabilityStatement>
```

JSON Template

```
{
  "resourceType" : "CapabilityStatement",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this capability statement, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the capability statement
  "name" : "<string>", // C? Name for this capability statement (computer friendly)
  "title" : "<string>", // Name for this capability statement (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // R!  Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // C? Natural language description of the capability statement
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for capability statement (if applicable)
  "purpose" : "<markdown>", // Why this capability statement is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "kind" : "<code>", // C? R!  instance | capability | requirements
  "instantiates" : [{ canonical(CapabilityStatement) }], // Canonical URL of another capability statement this
    implements
  "imports" : [{ canonical(CapabilityStatement) }], // Canonical URL of another capability statement this adds to
  "software" : { // C? Software that is covered by this capability statement
    "name" : "<string>", // R!  A name the software is known by
    "version" : "<string>", // Version covered by this statement
```

```
    "releaseDate" : "<dateTime>" // Date this version was released
  },
  "implementation" : { // C? If this describes a specific instance
    "description" : "<string>", // R!  Describes this specific instance
    "url" : "<url>", // Base URL for the installation
    "custodian" : { Reference(Organization) } // Organization that manages the data
  },
  "fhirVersion" : "<code>", // R!  FHIR Version the system supports
  "format" : ["<code>"], // R!  formats supported (xml | json | ttl | mime type)
  "patchFormat" : ["<code>"], // Patch formats supported
  "implementationGuide" : [{ canonical(ImplementationGuide) }], // Implementation guides supported
  "rest" : [{ // C? If the endpoint is a RESTful one
    "mode" : "<code>", // R!  client | server
    "documentation" : "<markdown>", // General description of implementation
    "security" : { // Information about security of implementation
      "cors" : <boolean>, // Adds CORS Headers (http://enable-cors.org/)
      "service" : [{ CodeableConcept }], // OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates
      "description" : "<markdown>" // General description of how security works
    },
    "resource" : [{ // Resource served on the REST interface
      "type" : "<code>", // R!  A resource type that is supported
      "profile" : { canonical(StructureDefinition) }, // Base System profile for all uses of resource
      "supportedProfile" : [{ canonical(StructureDefinition) }], // Profiles for use cases supported
      "documentation" : "<markdown>", // Additional information about the use of the resource type
      "interaction" : [{ // What operations are supported?
        "code" : "<code>", // R!  read | vread | update | patch | delete | history-instance | history-type | create |
    search-type
        "documentation" : "<markdown>" // Anything special about operation behavior
      }],
      "versioning" : "<code>", // no-version | versioned | versioned-update
      "readHistory" : <boolean>, // Whether vRead can return past versions
      "updateCreate" : <boolean>, // If update can commit to a new identity
      "conditionalCreate" : <boolean>, // If allows/uses conditional create
      "conditionalRead" : "<code>", // not-supported | modified-since | not-match | full-support
      "conditionalUpdate" : <boolean>, // If allows/uses conditional update
      "conditionalDelete" : "<code>", // not-supported | single | multiple - how conditional delete is supported
      "referencePolicy" : ["<code>"], // literal | logical | resolves | enforced | local
      "searchInclude" : ["<string>"], // _include values supported by the server
      "searchRevInclude" : ["<string>"], // _revinclude values supported by the server
      "searchParam" : [{ // Search parameters supported by implementation
        "name" : "<string>", // R!  Name of search parameter
        "definition" : { canonical(SearchParameter) }, // Source of definition for parameter
        "type" : "<code>", // R!  number | date | string | token | reference | composite | quantity | uri | special
        "documentation" : "<markdown>" // Server-specific usage
      }],
      "operation" : [{ // Definition of a resource operation
        "name" : "<string>", // R!  Name by which the operation/query is invoked
        "definition" : { canonical(OperationDefinition) }, // R!  The defined operation/query
        "documentation" : "<markdown>" // Specific details about operation behavior
      }]
    }],
    "interaction" : [{ // What operations are supported?
      "code" : "<code>", // R!  transaction | batch | search-system | history-system
      "documentation" : "<markdown>" // Anything special about operation behavior
    }],
    "searchParam" : [{ Content as for CapabilityStatement.rest.resource.searchParam }], // Search parameters for
    searching all resources
    "operation" : [{ Content as for CapabilityStatement.rest.resource.operation }], // Definition of a system level
    operation
    "compartment" : [{ canonical(CompartmentDefinition) }] // Compartments served/used by system
  }],
  "messaging" : [{ // C? If messaging is supported
    "endpoint" : [{ // Where messages should be sent
      "protocol" : { Coding }, // R!  http | ftp | mllp +
      "address" : "<url>" // R!  Network address or identifier of the end-point
    }],
    "reliableCache" : "<unsignedInt>", // Reliable Message Cache Length (min)
    "documentation" : "<markdown>", // Messaging interface behavior details
    "supportedMessage" : [{ // Messages supported by this system
      "mode" : "<code>", // R!  sender | receiver
      "definition" : { canonical(MessageDefinition) } // R!  Message supported by this system
    }]
  }],
  "document" : [{ // C? Document definition
    "mode" : "<code>", // R!  producer | consumer
    "documentation" : "<markdown>", // Description of document support
    "profile" : { canonical(StructureDefinition) } // R!  Constraint on the resources used in the document
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CapabilityStatement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
```

```
fhir:CapabilityStatement.url [ uri ]; # 0..1 Canonical identifier for this capability statement, represented as a URI
  (globally unique)
fhir:CapabilityStatement.version [ string ]; # 0..1 Business version of the capability statement
fhir:CapabilityStatement.name [ string ]; # 0..1 Name for this capability statement (computer friendly)
fhir:CapabilityStatement.title [ string ]; # 0..1 Name for this capability statement (human friendly)
fhir:CapabilityStatement.status [ code ]; # 1..1 draft | active | retired | unknown
fhir:CapabilityStatement.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
fhir:CapabilityStatement.date [ dateTime ]; # 1..1 Date last changed
fhir:CapabilityStatement.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
fhir:CapabilityStatement.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
fhir:CapabilityStatement.description [ markdown ]; # 0..1 Natural language description of the capability statement
fhir:CapabilityStatement.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
fhir:CapabilityStatement.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for capability statement
   (if applicable)
fhir:CapabilityStatement.purpose [ markdown ]; # 0..1 Why this capability statement is defined
fhir:CapabilityStatement.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
fhir:CapabilityStatement.kind [ code ]; # 1..1 instance | capability | requirements
fhir:CapabilityStatement.instantiates [ canonical(CapabilityStatement) ], ... ; # 0..* Canonical URL of another
  capability statement this implements
fhir:CapabilityStatement.imports [ canonical(CapabilityStatement) ], ... ; # 0..* Canonical URL of another capability
  statement this adds to
fhir:CapabilityStatement.software [ # 0..1 Software that is covered by this capability statement
  fhir:CapabilityStatement.software.name [ string ]; # 1..1 A name the software is known by
  fhir:CapabilityStatement.software.version [ string ]; # 0..1 Version covered by this statement
  fhir:CapabilityStatement.software.releaseDate [ dateTime ]; # 0..1 Date this version was released
];
fhir:CapabilityStatement.implementation [ # 0..1 If this describes a specific instance
  fhir:CapabilityStatement.implementation.description [ string ]; # 1..1 Describes this specific instance
  fhir:CapabilityStatement.implementation.url [ url ]; # 0..1 Base URL for the installation
  fhir:CapabilityStatement.implementation.custodian [ Reference(Organization) ]; # 0..1 Organization that manages the
  data
];
fhir:CapabilityStatement.fhirVersion [ code ]; # 1..1 FHIR Version the system supports
fhir:CapabilityStatement.format [ code ], ... ; # 1..* formats supported (xml | json | ttl | mime type)
fhir:CapabilityStatement.patchFormat [ code ], ... ; # 0..* Patch formats supported
fhir:CapabilityStatement.implementationGuide [ canonical(ImplementationGuide) ], ... ; # 0..* Implementation guides
  supported
fhir:CapabilityStatement.rest [ # 0..* If the endpoint is a RESTful one
  fhir:CapabilityStatement.rest.mode [ code ]; # 1..1 client | server
  fhir:CapabilityStatement.rest.documentation [ markdown ]; # 0..1 General description of implementation
  fhir:CapabilityStatement.rest.security [ # 0..1 Information about security of implementation
    fhir:CapabilityStatement.rest.security.cors [ boolean ]; # 0..1 Adds CORS Headers (http://enable-cors.org/)
    fhir:CapabilityStatement.rest.security.service [ CodeableConcept ], ... ; # 0..* OAuth | SMART-on-FHIR | NTLM |
  Basic | Kerberos | Certificates
    fhir:CapabilityStatement.rest.security.description [ markdown ]; # 0..1 General description of how security works
  ];
  fhir:CapabilityStatement.rest.resource [ # 0..* Resource served on the REST interface
    fhir:CapabilityStatement.rest.resource.type [ code ]; # 1..1 A resource type that is supported
    fhir:CapabilityStatement.rest.resource.profile [ canonical(StructureDefinition) ]; # 0..1 Base System profile for
  all uses of resource
    fhir:CapabilityStatement.rest.resource.supportedProfile [ canonical(StructureDefinition) ], ... ; # 0..* Profiles
  for use cases supported
    fhir:CapabilityStatement.rest.resource.documentation [ markdown ]; # 0..1 Additional information about the use of
  the resource type
    fhir:CapabilityStatement.rest.resource.interaction [ # 0..* What operations are supported?
      fhir:CapabilityStatement.rest.resource.interaction.code [ code ]; # 1..1 read | vread | update | patch | delete
  | history-instance | history-type | create | search-type
      fhir:CapabilityStatement.rest.resource.interaction.documentation [ markdown ]; # 0..1 Anything special about
  operation behavior
    ], ...;
    fhir:CapabilityStatement.rest.resource.versioning [ code ]; # 0..1 no-version | versioned | versioned-update
    fhir:CapabilityStatement.rest.resource.readHistory [ boolean ]; # 0..1 Whether vRead can return past versions
    fhir:CapabilityStatement.rest.resource.updateCreate [ boolean ]; # 0..1 If update can commit to a new identity
    fhir:CapabilityStatement.rest.resource.conditionalCreate [ boolean ]; # 0..1 If allows/uses conditional create
    fhir:CapabilityStatement.rest.resource.conditionalRead [ code ]; # 0..1 not-supported | modified-since | not-match
   | full-support
    fhir:CapabilityStatement.rest.resource.conditionalUpdate [ boolean ]; # 0..1 If allows/uses conditional update
    fhir:CapabilityStatement.rest.resource.conditionalDelete [ code ]; # 0..1 not-supported | single | multiple - how
  conditional delete is supported
    fhir:CapabilityStatement.rest.resource.referencePolicy [ code ], ... ; # 0..* literal | logical | resolves |
  enforced | local
    fhir:CapabilityStatement.rest.resource.searchInclude [ string ], ... ; # 0..* _include values supported by the
  server
    fhir:CapabilityStatement.rest.resource.searchRevInclude [ string ], ... ; # 0..* _revinclude values supported by
  the server
    fhir:CapabilityStatement.rest.resource.searchParam [ # 0..* Search parameters supported by implementation
      fhir:CapabilityStatement.rest.resource.searchParam.name [ string ]; # 1..1 Name of search parameter
      fhir:CapabilityStatement.rest.resource.searchParam.definition [ canonical(SearchParameter) ]; # 0..1 Source of
  definition for parameter
      fhir:CapabilityStatement.rest.resource.searchParam.type [ code ]; # 1..1 number | date | string | token |
  reference | composite | quantity | uri | special
      fhir:CapabilityStatement.rest.resource.searchParam.documentation [ markdown ]; # 0..1 Server-specific usage
    ], ...;
    fhir:CapabilityStatement.rest.resource.operation [ # 0..* Definition of a resource operation
      fhir:CapabilityStatement.rest.resource.operation.name [ string ]; # 1..1 Name by which the operation/query is
  invoked
      fhir:CapabilityStatement.rest.resource.operation.definition [ canonical(OperationDefinition) ]; # 1..1 The
  defined operation/query
      fhir:CapabilityStatement.rest.resource.operation.documentation [ markdown ]; # 0..1 Specific details about
```

```
    operation behavior
      ], ...;
    ], ...;
    fhir:CapabilityStatement.rest.interaction [ # 0..* What operations are supported?
      fhir:CapabilityStatement.rest.interaction.code [ code ]; # 1..1 transaction | batch | search-system | history-
    system
      fhir:CapabilityStatement.rest.interaction.documentation [ markdown ]; # 0..1 Anything special about operation
    behavior
    ], ...;
    fhir:CapabilityStatement.rest.searchParam [ See CapabilityStatement.rest.resource.searchParam ], ... ; # 0..* Search
     parameters for searching all resources
    fhir:CapabilityStatement.rest.operation [ See CapabilityStatement.rest.resource.operation ], ... ; # 0..* Definition
     of a system level operation
    fhir:CapabilityStatement.rest.compartment [ canonical(CompartmentDefinition) ], ... ; # 0..* Compartments
    served/used by system
  ], ...;
  fhir:CapabilityStatement.messaging [ # 0..* If messaging is supported
    fhir:CapabilityStatement.messaging.endpoint [ # 0..* Where messages should be sent
      fhir:CapabilityStatement.messaging.endpoint.protocol [ Coding ]; # 1..1 http | ftp | mllp +
      fhir:CapabilityStatement.messaging.endpoint.address [ url ]; # 1..1 Network address or identifier of the end-point
    ], ...;
    fhir:CapabilityStatement.messaging.reliableCache [ unsignedInt ]; # 0..1 Reliable Message Cache Length (min)
    fhir:CapabilityStatement.messaging.documentation [ markdown ]; # 0..1 Messaging interface behavior details

    fhir:CapabilityStatement.messaging.supportedMessage [ # 0..* Messages supported by this system
      fhir:CapabilityStatement.messaging.supportedMessage.mode [ code ]; # 1..1 sender | receiver
      fhir:CapabilityStatement.messaging.supportedMessage.definition [ canonical(MessageDefinition) ]; # 1..1 Message
    supported by this system
    ], ...;
  ], ...;
  fhir:CapabilityStatement.document [ # 0..* Document definition
    fhir:CapabilityStatement.document.mode [ code ]; # 1..1 producer | consumer
    fhir:CapabilityStatement.document.documentation [ markdown ]; # 0..1 Description of document support
    fhir:CapabilityStatement.document.profile [ canonical(StructureDefinition) ]; # 1..1 Constraint on the resources
    used in the document
  ], ...;
]
```

Changes since R3

CapabilityStatement

CapabilityStatement | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *

CapabilityStatement.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

CapabilityStatement.experimental | No longer marked as Modifier

CapabilityStatement.kind | Change value set from http://hl7.org/fhir/ValueSet/capability-statement-kind to http://hl7.org/fhir/ValueSet/capability-statement-kind|4.0.1

CapabilityStatement.instantiates | Type changed from uri to canonical(CapabilityStatement)

CapabilityStatement.imports | Added Element

CapabilityStatement.implementation.url | Type changed from uri to url

CapabilityStatement.implementation.custodian | Added Element

CapabilityStatement.fhirVersion | Type changed from id to code Add Binding http://hl7.org/fhir/ValueSet/FHIR-version|4.0.1 (required)

CapabilityStatement.format | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1

CapabilityStatement.patchFormat | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1

CapabilityStatement.implementationGuide | Type changed from uri to canonical(ImplementationGuide)

CapabilityStatement.rest.mode | Change value set from http://hl7.org/fhir/ValueSet/restful-capability-mode to http://hl7.org/fhir/ValueSet/restful-capability-mode|4.0.1

CapabilityStatement.rest.documentation | Type changed from string to markdown

CapabilityStatement.rest.security.service | Change code system for extensibly bound codes from "http://hl7.org/fhir/restful-security-service" to "http://terminology.hl7.org/CodeSystem/restful-security-service"

CapabilityStatement.rest.security.description | Type changed from string to markdown

CapabilityStatement.rest.resource.type | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1

CapabilityStatement.rest.resource.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)

CapabilityStatement.rest.resource.supportedProfile | Moved from CapabilityStatement.profile to supportedProfile Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)

CapabilityStatement.rest.resource.interaction | Min Cardinality changed from 1 to 0

CapabilityStatement.rest.resource.interaction.code | Change value set from http://hl7.org/fhir/ValueSet/type-restful-interaction to http://hl7.org/fhir/ValueSet/type-restful-interaction|4.0.1

CapabilityStatement.rest.resource.interaction.documentation | Type changed from string to markdown

CapabilityStatement.rest.resource.versioning | Change value set from http://hl7.org/fhir/ValueSet/versioning-policy to http://hl7.org/fhir/ValueSet/versioning-policy|4.0.1

CapabilityStatement.rest.resource.conditionalRead | Change value set from http://hl7.org/fhir/ValueSet/conditional-read-status to http://hl7.org/fhir/ValueSet/conditional-read-status|4.0.1

CapabilityStatement.rest.resource.conditionalDelete | Change value set from http://hl7.org/fhir/ValueSet/conditional-delete-status to http://hl7.org/fhir/ValueSet/conditional-delete-status|4.0.1

CapabilityStatement.rest.resource.referencePolicy | Change value set from http://hl7.org/fhir/ValueSet/reference-handling-policy to http://hl7.org/fhir/ValueSet/reference-handling-policy|4.0.1

CapabilityStatement.rest.resource.searchParam.definition | Type changed from uri to canonical(SearchParameter)

CapabilityStatement.rest.resource.searchParam.type | Change value set from http://hl7.org/fhir/ValueSet/search-param-type to http://hl7.org/fhir/ValueSet/search-param-type|4.0.1
CapabilityStatement.rest.resource.searchParam.documentation | Type changed from string to markdown
CapabilityStatement.rest.resource.operation | Added Element
CapabilityStatement.rest.resource.operation.name | Added Mandatory Element
CapabilityStatement.rest.resource.operation.definition | Added Mandatory Element
CapabilityStatement.rest.resource.operation.documentation | Added Element
CapabilityStatement.rest.interaction.code | Change value set from http://hl7.org/fhir/ValueSet/system-restful-interaction to http://hl7.org/fhir/ValueSet/system-restful-interaction|4.0.1
CapabilityStatement.rest.interaction.documentation | Type changed from string to markdown
CapabilityStatement.rest.operation | Remove Type BackboneElement
CapabilityStatement.rest.compartment | Type changed from uri to canonical(CompartmentDefinition)
CapabilityStatement.messaging.endpoint.protocol | Change code system for extensibly bound codes from "http://hl7.org/fhir/message-transport" to "http://terminology.hl7.org/CodeSystem/message-transport"
CapabilityStatement.messaging.endpoint.address | Type changed from uri to url
CapabilityStatement.messaging.documentation | Type changed from string to markdown
CapabilityStatement.messaging.supportedMessage.mode | Change value set from http://hl7.org/fhir/ValueSet/event-capability-mode to http://hl7.org/fhir/ValueSet/event-capability-mode|4.0.1
CapabilityStatement.messaging.supportedMessage.definition | Type changed from Reference(MessageDefinition) to canonical(MessageDefinition)
CapabilityStatement.document.mode | Change value set from http://hl7.org/fhir/ValueSet/document-mode to http://hl7.org/fhir/ValueSet/document-mode|4.0.1
CapabilityStatement.document.documentation | Type changed from string to markdown
CapabilityStatement.document.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
CapabilityStatement.acceptUnknown | deleted
CapabilityStatement.rest.security.certificate | deleted
CapabilityStatement.rest.operation.name | deleted
CapabilityStatement.rest.operation.definition | deleted
CapabilityStatement.messaging.event | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 9 tests that all execute ok. 1 fail round-trip testing and 9 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
CapabilityStatement | IN | | DomainResource | A statement of system capabilities + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: A Capability Statement SHALL have at least one of REST, messaging or document element. + Rule: A Capability Statement SHALL have at least one of description, software, or implementation element. + Rule: Messaging end-point is required (and is only permitted) when a statement is for an implementation. + Rule: The set of documents must be unique by the combination of profile and mode. + Rule: If kind = instance, implementation must be present and software may be present + Rule: If kind = capability, implementation must be absent, software must be present + Rule: If kind = requirements, implementation and software must be absent Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this capability statement, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the capability statement
name | ΣI | 0..1 | string | Name for this capability statement (computer friendly)
title | Σ | 0..1 | string | Name for this capability statement (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 1..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | I | 0..1 | markdown | Natural language description of the capability statement
useContext | ΣTU | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for capability statement (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this capability statement is defined
copyright | | 0..1 | markdown | Use and/or publishing restrictions
kind | ΣI | 1..1 | code | instance | capability | requirements CapabilityStatementKind (Required)
instantiates | Σ | 0..* | canonical(CapabilityStatement) | Canonical URL of another capability statement this implements
imports | ΣTU | 0..* | canonical(CapabilityStatement) | Canonical URL of another capability statement this adds to
software | ΣI | 0..1 | BackboneElement | Software that is covered by this capability statement
name | Σ | 1..1 | string | A name the software is known by
version | Σ | 0..1 | string | Version covered by this statement
releaseDate | Σ | 0..1 | dateTime | Date this version was released
implementation | ΣI | 0..1 | BackboneElement | If this describes a specific instance
description | Σ | 1..1 | string | Describes this specific instance
url | Σ | 0..1 | url | Base URL for the installation
custodian | ΣTU | 0..1 | Reference(Organization) | Organization that manages the data
fhirVersion | Σ | 1..1 | code | FHIR Version the system supports FHIRVersion (Required)
format | Σ | 1..* | code | formats supported (xml | json | ttl | mime type) MimeType (Required)
patchFormat | Σ | 0..* | code | Patch formats supported MimeType (Required)

implementationGuide | Σ | 0..* | canonical(ImplementationGuide) | Implementation guides supported
rest | ΣI | 0..* | BackboneElement | If the endpoint is a RESTful one + Rule: A given resource can only be described once per RESTful mode.
mode | Σ | 1..1 | code | client | server RestfulCapabilityMode (Required)
documentation | | 0..1 | markdown | General description of implementation
security | ΣTU | 0..1 | BackboneElement | Information about security of implementation
cors | Σ | 0..1 | boolean | Adds CORS Headers (http://enable-cors.org/)
service | Σ | 0..* | CodeableConcept | OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates RestfulSecurityService (Extensible)
description | | 0..1 | markdown | General description of how security works
resource | ΣI | 0..* | BackboneElement | Resource served on the REST interface + Rule: Search parameter names must be unique in the context of a resource.
type | Σ | 1..1 | code | A resource type that is supported ResourceType (Required)
profile | Σ | 0..1 | canonical(StructureDefinition) | Base System profile for all uses of resource
supportedProfile | ΣTU | 0..* | canonical(StructureDefinition) | Profiles for use cases supported
documentation | | 0..1 | markdown | Additional information about the use of the resource type
interaction | | 0..* | BackboneElement | What operations are supported?
code | | 1..1 | code | read | vread | update | patch | delete | history-instance | history-type | create | search-type TypeRestfulInteraction (Required)
documentation | | 0..1 | markdown | Anything special about operation behavior
versioning | TU | 0..1 | code | no-version | versioned | versioned-update ResourceVersionPolicy (Required)
readHistory | TU | 0..1 | boolean | Whether vRead can return past versions
updateCreate | TU | 0..1 | boolean | If update can commit to a new identity
conditionalCreate | TU | 0..1 | boolean | If allows/uses conditional create
conditionalRead | TU | 0..1 | code | not-supported | modified-since | not-match | full-support ConditionalReadStatus (Required)
conditionalUpdate | TU | 0..1 | boolean | If allows/uses conditional update
conditionalDelete | TU | 0..1 | code | not-supported | single | multiple - how conditional delete is supported ConditionalDeleteStatus (Required)
referencePolicy | TU | 0..* | code | literal | logical | resolves | enforced | local ReferenceHandlingPolicy (Required)
searchInclude | TU | 0..* | string | _include values supported by the server
searchRevInclude | TU | 0..* | string | _revinclude values supported by the server
searchParam | | 0..* | BackboneElement | Search parameters supported by implementation
name | | 1..1 | string | Name of search parameter
definition | | 0..1 | canonical(SearchParameter) | Source of definition for parameter
type | | 1..1 | code | number | date | string | token | reference | composite | quantity | uri | special SearchParamType (Required)
documentation | | 0..1 | markdown | Server-specific usage
operation | Σ | 0..* | BackboneElement | Definition of a resource operation
name | Σ | 1..1 | string | Name by which the operation/query is invoked
definition | Σ | 1..1 | canonical(OperationDefinition) | The defined operation/query
documentation | | 0..1 | markdown | Specific details about operation behavior
interaction | | 0..* | BackboneElement | What operations are supported?
code | | 1..1 | code | transaction | batch | search-system | history-system SystemRestfulInteraction (Required)
documentation | | 0..1 | markdown | Anything special about operation behavior
searchParam | | 0..* | see searchParam | Search parameters for searching all resources
operation | Σ | 0..* | see operation | Definition of a system level operation
compartment | | 0..* | canonical(CompartmentDefinition) | Compartments served/used by system
messaging | ΣITU | 0..* | BackboneElement | If messaging is supported
endpoint | | 0..* | BackboneElement | Where messages should be sent
protocol | | 1..1 | Coding | http | ftp | mllp + MessageTransport (Extensible)
address | | 1..1 | url | Network address or identifier of the end-point
reliableCache | | 0..1 | unsignedInt | Reliable Message Cache Length (min)
documentation | | 0..1 | markdown | Messaging interface behavior details
supportedMessage | Σ | 0..* | BackboneElement | Messages supported by this system
mode | Σ | 1..1 | code | sender | receiver EventCapabilityMode (Required)
definition | Σ | 1..1 | canonical(MessageDefinition) | Message supported by this system
document | ΣITU | 0..* | BackboneElement | Document definition
mode | Σ | 1..1 | code | producer | consumer DocumentMode (Required)
documentation | | 0..1 | markdown | Description of document support
profile | Σ | 1..1 | canonical(StructureDefinition) | Constraint on the resources used in the document
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CapabilityStatement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this capability statement, represented as a URI (globally
    unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the capability statement -->
 <name value="[string]"/><!--  0..1 Name for this capability statement (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this capability statement (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 1..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
```

```
<contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
<description value="[markdown]"/><!--  0..1 Natural language description of the capability statement -->
<useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
<jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for capability statement (if applicable)
   --></jurisdiction>
<purpose value="[markdown]"/><!-- 0..1 Why this capability statement is defined -->
<copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
<kind value="[code]"/><!--  1..1 instance | capability | requirements -->
<instantiates><!-- 0..* canonical(CapabilityStatement) Canonical URL of another capability statement this implements
   --></instantiates>
<imports><!-- 0..* canonical(CapabilityStatement) Canonical URL of another capability statement this adds to
   --></imports>
<software>  <!--  0..1 Software that is covered by this capability statement -->
 <name value="[string]"/><!-- 1..1 A name the software is known by -->
 <version value="[string]"/><!-- 0..1 Version covered by this statement -->
 <releaseDate value="[dateTime]"/><!-- 0..1 Date this version was released -->
</software>
<implementation>  <!--  0..1 If this describes a specific instance -->
 <description value="[string]"/><!-- 1..1 Describes this specific instance -->
 <url value="[url]"/><!-- 0..1 Base URL for the installation -->
 <custodian><!-- 0..1 Reference(Organization) Organization that manages the data --></custodian>
</implementation>
<fhirVersion value="[code]"/><!-- 1..1 FHIR Version the system supports -->
<format value="[code]"/><!-- 1..* formats supported (xml | json | ttl | mime type) -->
<patchFormat value="[code]"/><!-- 0..* Patch formats supported -->
<implementationGuide><!-- 0..* canonical(ImplementationGuide) Implementation guides supported --></implementationGuide>
<rest>  <!--  0..* If the endpoint is a RESTful one -->
 <mode value="[code]"/><!-- 1..1 client | server -->
 <documentation value="[markdown]"/><!-- 0..1 General description of implementation -->
 <security>  <!-- 0..1 Information about security of implementation -->
  <cors value="[boolean]"/><!-- 0..1 Adds CORS Headers (http://enable-cors.org/) -->
  <service><!-- 0..* CodeableConcept OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates --></service>
  <description value="[markdown]"/><!-- 0..1 General description of how security works -->
 </security>
 <resource>  <!-- 0..* Resource served on the REST interface -->
  <type value="[code]"/><!-- 1..1 A resource type that is supported -->
  <profile><!-- 0..1 canonical(StructureDefinition) Base System profile for all uses of resource --></profile>
  <supportedProfile><!-- 0..* canonical(StructureDefinition) Profiles for use cases supported --></supportedProfile>
  <documentation value="[markdown]"/><!-- 0..1 Additional information about the use of the resource type -->
  <interaction>  <!-- 0..* What operations are supported? -->
   <code value="[code]"/><!-- 1..1 read | vread | update | patch | delete | history-instance | history-type | create |
   search-type -->
   <documentation value="[markdown]"/><!-- 0..1 Anything special about operation behavior -->
  </interaction>
  <versioning value="[code]"/><!-- 0..1 no-version | versioned | versioned-update -->
  <readHistory value="[boolean]"/><!-- 0..1 Whether vRead can return past versions -->
  <updateCreate value="[boolean]"/><!-- 0..1 If update can commit to a new identity -->
  <conditionalCreate value="[boolean]"/><!-- 0..1 If allows/uses conditional create -->
  <conditionalRead value="[code]"/><!-- 0..1 not-supported | modified-since | not-match | full-support -->
  <conditionalUpdate value="[boolean]"/><!-- 0..1 If allows/uses conditional update -->
  <conditionalDelete value="[code]"/><!-- 0..1 not-supported | single | multiple - how conditional delete is supported
   -->
  <referencePolicy value="[code]"/><!-- 0..* literal | logical | resolves | enforced | local -->
  <searchInclude value="[string]"/><!-- 0..* _include values supported by the server -->
  <searchRevInclude value="[string]"/><!-- 0..* _revinclude values supported by the server -->
  <searchParam>  <!-- 0..* Search parameters supported by implementation -->
   <name value="[string]"/><!-- 1..1 Name of search parameter -->
   <definition><!-- 0..1 canonical(SearchParameter) Source of definition for parameter --></definition>
   <type value="[code]"/><!-- 1..1 number | date | string | token | reference | composite | quantity | uri | special
   -->
   <documentation value="[markdown]"/><!-- 0..1 Server-specific usage -->
  </searchParam>
  <operation>  <!-- 0..* Definition of a resource operation -->
   <name value="[string]"/><!-- 1..1 Name by which the operation/query is invoked -->
   <definition><!-- 1..1 canonical(OperationDefinition) The defined operation/query --></definition>
   <documentation value="[markdown]"/><!-- 0..1 Specific details about operation behavior -->
  </operation>
 </resource>
 <interaction>  <!-- 0..* What operations are supported? -->
  <code value="[code]"/><!-- 1..1 transaction | batch | search-system | history-system -->
  <documentation value="[markdown]"/><!-- 0..1 Anything special about operation behavior -->
 </interaction>
 <searchParam><!-- 0..* Content as for CapabilityStatement.rest.resource.searchParam Search parameters for searching
   all resources --></searchParam>
 <operation><!-- 0..* Content as for CapabilityStatement.rest.resource.operation Definition of a system level operation
    --></operation>
 <compartment><!-- 0..* canonical(CompartmentDefinition) Compartments served/used by system --></compartment>
</rest>
<messaging>  <!--  0..* If messaging is supported -->
 <endpoint>  <!-- 0..* Where messages should be sent -->
  <protocol><!-- 1..1 Coding http | ftp | mllp + --></protocol>
  <address value="[url]"/><!-- 1..1 Network address or identifier of the end-point -->
 </endpoint>
 <reliableCache value="[unsignedInt]"/><!-- 0..1 Reliable Message Cache Length (min) -->
 <documentation value="[markdown]"/><!-- 0..1 Messaging interface behavior details -->
 <supportedMessage>  <!-- 0..* Messages supported by this system -->
  <mode value="[code]"/><!-- 1..1 sender | receiver -->
  <definition><!-- 1..1 canonical(MessageDefinition) Message supported by this system --></definition>
 </supportedMessage>
```

```
 </messaging>
 <document>  <!--  0..* Document definition -->
  <mode value="[code]"/><!-- 1..1 producer | consumer -->
  <documentation value="[markdown]"/><!-- 0..1 Description of document support -->
  <profile><!-- 1..1 canonical(StructureDefinition) Constraint on the resources used in the document --></profile>
 </document>
</CapabilityStatement>
```

JSON Template

```
{
  "resourceType" : "CapabilityStatement",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this capability statement, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the capability statement
  "name" : "<string>", // C? Name for this capability statement (computer friendly)
  "title" : "<string>", // Name for this capability statement (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // R!  Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // C? Natural language description of the capability statement
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for capability statement (if applicable)
  "purpose" : "<markdown>", // Why this capability statement is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "kind" : "<code>", // C? R!  instance | capability | requirements
  "instantiates" : [{ canonical(CapabilityStatement) }], // Canonical URL of another capability statement this
    implements
  "imports" : [{ canonical(CapabilityStatement) }], // Canonical URL of another capability statement this adds to
  "software" : { // C? Software that is covered by this capability statement
    "name" : "<string>", // R!  A name the software is known by
    "version" : "<string>", // Version covered by this statement
    "releaseDate" : "<dateTime>" // Date this version was released
  },
  "implementation" : { // C? If this describes a specific instance
    "description" : "<string>", // R!  Describes this specific instance
    "url" : "<url>", // Base URL for the installation
    "custodian" : { Reference(Organization) } // Organization that manages the data
  },
  "fhirVersion" : "<code>", // R!  FHIR Version the system supports
  "format" : ["<code>"], // R!  formats supported (xml | json | ttl | mime type)
  "patchFormat" : ["<code>"], // Patch formats supported
  "implementationGuide" : [{ canonical(ImplementationGuide) }], // Implementation guides supported
  "rest" : [{ // C? If the endpoint is a RESTful one
    "mode" : "<code>", // R!  client | server
    "documentation" : "<markdown>", // General description of implementation
    "security" : { // Information about security of implementation
      "cors" : <boolean>, // Adds CORS Headers (http://enable-cors.org/)
      "service" : [{ CodeableConcept }], // OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates
      "description" : "<markdown>" // General description of how security works
    },
    "resource" : [{ // Resource served on the REST interface
      "type" : "<code>", // R!  A resource type that is supported
      "profile" : { canonical(StructureDefinition) }, // Base System profile for all uses of resource
      "supportedProfile" : [{ canonical(StructureDefinition) }], // Profiles for use cases supported
      "documentation" : "<markdown>", // Additional information about the use of the resource type
      "interaction" : [{ // What operations are supported?
        "code" : "<code>", // R!  read | vread | update | patch | delete | history-instance | history-type | create |
    search-type
        "documentation" : "<markdown>" // Anything special about operation behavior
      }],
      "versioning" : "<code>", // no-version | versioned | versioned-update
      "readHistory" : <boolean>, // Whether vRead can return past versions
      "updateCreate" : <boolean>, // If update can commit to a new identity
      "conditionalCreate" : <boolean>, // If allows/uses conditional create
      "conditionalRead" : "<code>", // not-supported | modified-since | not-match | full-support
      "conditionalUpdate" : <boolean>, // If allows/uses conditional update
      "conditionalDelete" : "<code>", // not-supported | single | multiple - how conditional delete is supported
      "referencePolicy" : ["<code>"], // literal | logical | resolves | enforced | local
      "searchInclude" : ["<string>"], // _include values supported by the server
      "searchRevInclude" : ["<string>"], // _revinclude values supported by the server
      "searchParam" : [{ // Search parameters supported by implementation
        "name" : "<string>", // R!  Name of search parameter
        "definition" : { canonical(SearchParameter) }, // Source of definition for parameter
        "type" : "<code>", // R!  number | date | string | token | reference | composite | quantity | uri | special
        "documentation" : "<markdown>" // Server-specific usage
      }],
      "operation" : [{ // Definition of a resource operation
        "name" : "<string>", // R!  Name by which the operation/query is invoked
        "definition" : { canonical(OperationDefinition) }, // R!  The defined operation/query
        "documentation" : "<markdown>" // Specific details about operation behavior
      }]
    }],
    "interaction" : [{ // What operations are supported?
      "code" : "<code>", // R!  transaction | batch | search-system | history-system
      "documentation" : "<markdown>" // Anything special about operation behavior
    }],
```

```
    "searchParam" : [{ Content as for CapabilityStatement.rest.resource.searchParam }], // Search parameters for
    searching all resources
    "operation" : [{ Content as for CapabilityStatement.rest.resource.operation }], // Definition of a system level
    operation
    "compartment" : [{ canonical(CompartmentDefinition) }] // Compartments served/used by system
  }],
  "messaging" : [{ // C? If messaging is supported
    "endpoint" : [{ // Where messages should be sent
      "protocol" : { Coding }, // R!  http | ftp | mllp +
      "address" : "<url>" // R!  Network address or identifier of the end-point
    }],
    "reliableCache" : "<unsignedInt>", // Reliable Message Cache Length (min)
    "documentation" : "<markdown>", // Messaging interface behavior details
    "supportedMessage" : [{ // Messages supported by this system
      "mode" : "<code>", // R!  sender | receiver
      "definition" : { canonical(MessageDefinition) } // R!  Message supported by this system
    }]
  }],
  "document" : [{ // C? Document definition
    "mode" : "<code>", // R!  producer | consumer
    "documentation" : "<markdown>", // Description of document support
    "profile" : { canonical(StructureDefinition) } // R!  Constraint on the resources used in the document
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CapabilityStatement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CapabilityStatement.url [ uri ]; # 0..1 Canonical identifier for this capability statement, represented as a URI
    (globally unique)
  fhir:CapabilityStatement.version [ string ]; # 0..1 Business version of the capability statement
  fhir:CapabilityStatement.name [ string ]; # 0..1 Name for this capability statement (computer friendly)
  fhir:CapabilityStatement.title [ string ]; # 0..1 Name for this capability statement (human friendly)
  fhir:CapabilityStatement.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:CapabilityStatement.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:CapabilityStatement.date [ dateTime ]; # 1..1 Date last changed
  fhir:CapabilityStatement.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:CapabilityStatement.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:CapabilityStatement.description [ markdown ]; # 0..1 Natural language description of the capability statement
  fhir:CapabilityStatement.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:CapabilityStatement.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for capability statement
     (if applicable)
  fhir:CapabilityStatement.purpose [ markdown ]; # 0..1 Why this capability statement is defined
  fhir:CapabilityStatement.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:CapabilityStatement.kind [ code ]; # 1..1 instance | capability | requirements
  fhir:CapabilityStatement.instantiates [ canonical(CapabilityStatement) ], ... ; # 0..* Canonical URL of another
    capability statement this implements
  fhir:CapabilityStatement.imports [ canonical(CapabilityStatement) ], ... ; # 0..* Canonical URL of another capability
    statement this adds to
  fhir:CapabilityStatement.software [ # 0..1 Software that is covered by this capability statement
    fhir:CapabilityStatement.software.name [ string ]; # 1..1 A name the software is known by
    fhir:CapabilityStatement.software.version [ string ]; # 0..1 Version covered by this statement
    fhir:CapabilityStatement.software.releaseDate [ dateTime ]; # 0..1 Date this version was released
  ];
  fhir:CapabilityStatement.implementation [ # 0..1 If this describes a specific instance
    fhir:CapabilityStatement.implementation.description [ string ]; # 1..1 Describes this specific instance
    fhir:CapabilityStatement.implementation.url [ url ]; # 0..1 Base URL for the installation
    fhir:CapabilityStatement.implementation.custodian [ Reference(Organization) ]; # 0..1 Organization that manages the
    data
  ];
  fhir:CapabilityStatement.fhirVersion [ code ]; # 1..1 FHIR Version the system supports
  fhir:CapabilityStatement.format [ code ], ... ; # 1..* formats supported (xml | json | ttl | mime type)
  fhir:CapabilityStatement.patchFormat [ code ], ... ; # 0..* Patch formats supported
  fhir:CapabilityStatement.implementationGuide [ canonical(ImplementationGuide) ], ... ; # 0..* Implementation guides
    supported
  fhir:CapabilityStatement.rest [ # 0..* If the endpoint is a RESTful one
    fhir:CapabilityStatement.rest.mode [ code ]; # 1..1 client | server
    fhir:CapabilityStatement.rest.documentation [ markdown ]; # 0..1 General description of implementation
    fhir:CapabilityStatement.rest.security [ # 0..1 Information about security of implementation
      fhir:CapabilityStatement.rest.security.cors [ boolean ]; # 0..1 Adds CORS Headers (http://enable-cors.org/)
      fhir:CapabilityStatement.rest.security.service [ CodeableConcept ], ... ; # 0..* OAuth | SMART-on-FHIR | NTLM |
    Basic | Kerberos | Certificates
      fhir:CapabilityStatement.rest.security.description [ markdown ]; # 0..1 General description of how security works
    ];
    fhir:CapabilityStatement.rest.resource [ # 0..* Resource served on the REST interface
      fhir:CapabilityStatement.rest.resource.type [ code ]; # 1..1 A resource type that is supported
      fhir:CapabilityStatement.rest.resource.profile [ canonical(StructureDefinition) ]; # 0..1 Base System profile for
    all uses of resource
      fhir:CapabilityStatement.rest.resource.supportedProfile [ canonical(StructureDefinition) ], ... ; # 0..* Profiles
    for use cases supported
      fhir:CapabilityStatement.rest.resource.documentation [ markdown ]; # 0..1 Additional information about the use of
    the resource type
      fhir:CapabilityStatement.rest.resource.interaction [ # 0..* What operations are supported?
```

```
      fhir:CapabilityStatement.rest.resource.interaction.code [ code ]; # 1..1 read | vread | update | patch | delete
    | history-instance | history-type | create | search-type
      fhir:CapabilityStatement.rest.resource.interaction.documentation [ markdown ]; # 0..1 Anything special about
    operation behavior
    ], ...;
    fhir:CapabilityStatement.rest.resource.versioning [ code ]; # 0..1 no-version | versioned | versioned-update
    fhir:CapabilityStatement.rest.resource.readHistory [ boolean ]; # 0..1 Whether vRead can return past versions
    fhir:CapabilityStatement.rest.resource.updateCreate [ boolean ]; # 0..1 If update can commit to a new identity
    fhir:CapabilityStatement.rest.resource.conditionalCreate [ boolean ]; # 0..1 If allows/uses conditional create
    fhir:CapabilityStatement.rest.resource.conditionalRead [ code ]; # 0..1 not-supported | modified-since | not-match
     | full-support
    fhir:CapabilityStatement.rest.resource.conditionalUpdate [ boolean ]; # 0..1 If allows/uses conditional update
    fhir:CapabilityStatement.rest.resource.conditionalDelete [ code ]; # 0..1 not-supported | single | multiple - how
    conditional delete is supported
    fhir:CapabilityStatement.rest.resource.referencePolicy [ code ], ... ; # 0..* literal | logical | resolves |
    enforced | local
    fhir:CapabilityStatement.rest.resource.searchInclude [ string ], ... ; # 0..* _include values supported by the
    server
    fhir:CapabilityStatement.rest.resource.searchRevInclude [ string ], ... ; # 0..* _revinclude values supported by
    the server
    fhir:CapabilityStatement.rest.resource.searchParam [ # 0..* Search parameters supported by implementation
      fhir:CapabilityStatement.rest.resource.searchParam.name [ string ]; # 1..1 Name of search parameter
      fhir:CapabilityStatement.rest.resource.searchParam.definition [ canonical(SearchParameter) ]; # 0..1 Source of
    definition for parameter
      fhir:CapabilityStatement.rest.resource.searchParam.type [ code ]; # 1..1 number | date | string | token |
    reference | composite | quantity | uri | special
      fhir:CapabilityStatement.rest.resource.searchParam.documentation [ markdown ]; # 0..1 Server-specific usage
    ], ...;
    fhir:CapabilityStatement.rest.resource.operation [ # 0..* Definition of a resource operation
      fhir:CapabilityStatement.rest.resource.operation.name [ string ]; # 1..1 Name by which the operation/query is
    invoked
      fhir:CapabilityStatement.rest.resource.operation.definition [ canonical(OperationDefinition) ]; # 1..1 The
    defined operation/query
      fhir:CapabilityStatement.rest.resource.operation.documentation [ markdown ]; # 0..1 Specific details about
    operation behavior
    ], ...;
  ], ...;
  fhir:CapabilityStatement.rest.interaction [ # 0..* What operations are supported?
    fhir:CapabilityStatement.rest.interaction.code [ code ]; # 1..1 transaction | batch | search-system | history-
    system
    fhir:CapabilityStatement.rest.interaction.documentation [ markdown ]; # 0..1 Anything special about operation
    behavior
  ], ...;
  fhir:CapabilityStatement.rest.searchParam [ See CapabilityStatement.rest.resource.searchParam ], ... ; # 0..* Search
   parameters for searching all resources
  fhir:CapabilityStatement.rest.operation [ See CapabilityStatement.rest.resource.operation ], ... ; # 0..* Definition
   of a system level operation
  fhir:CapabilityStatement.rest.compartment [ canonical(CompartmentDefinition) ], ... ; # 0..* Compartments
  served/used by system
], ...;
fhir:CapabilityStatement.messaging [ # 0..* If messaging is supported
  fhir:CapabilityStatement.messaging.endpoint [ # 0..* Where messages should be sent
    fhir:CapabilityStatement.messaging.endpoint.protocol [ Coding ]; # 1..1 http | ftp | mllp +
    fhir:CapabilityStatement.messaging.endpoint.address [ url ]; # 1..1 Network address or identifier of the end-point
  ], ...;
  fhir:CapabilityStatement.messaging.reliableCache [ unsignedInt ]; # 0..1 Reliable Message Cache Length (min)
  fhir:CapabilityStatement.messaging.documentation [ markdown ]; # 0..1 Messaging interface behavior details
  fhir:CapabilityStatement.messaging.supportedMessage [ # 0..* Messages supported by this system
    fhir:CapabilityStatement.messaging.supportedMessage.mode [ code ]; # 1..1 sender | receiver
    fhir:CapabilityStatement.messaging.supportedMessage.definition [ canonical(MessageDefinition) ]; # 1..1 Message
  supported by this system
  ], ...;
], ...;
fhir:CapabilityStatement.document [ # 0..* Document definition
  fhir:CapabilityStatement.document.mode [ code ]; # 1..1 producer | consumer
  fhir:CapabilityStatement.document.documentation [ markdown ]; # 0..1 Description of document support
  fhir:CapabilityStatement.document.profile [ canonical(StructureDefinition) ]; # 1..1 Constraint on the resources
  used in the document
], ...;
]
```

Changes since Release 3

CapabilityStatement

CapabilityStatement | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *

CapabilityStatement.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

CapabilityStatement.experimental | No longer marked as Modifier

CapabilityStatement.kind | Change value set from http://hl7.org/fhir/ValueSet/capability-statement-kind to http://hl7.org/fhir/ValueSet/capability-statement-kind|4.0.1

CapabilityStatement.instantiates | Type changed from uri to canonical(CapabilityStatement)

CapabilityStatement.imports | Added Element

CapabilityStatement.implementation.url | Type changed from uri to url

CapabilityStatement.implementation.custodian | Added Element

CapabilityStatement.fhirVersion | Type changed from id to code Add Binding http://hl7.org/fhir/ValueSet/FHIR-version|4.0.1 (required)

CapabilityStatement.format | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1
CapabilityStatement.patchFormat | Change value set from http://hl7.org/fhir/ValueSet/mimetypes to http://hl7.org/fhir/ValueSet/mimetypes|4.0.1
CapabilityStatement.implementationGuide | Type changed from uri to canonical(ImplementationGuide)
CapabilityStatement.rest.mode | Change value set from http://hl7.org/fhir/ValueSet/restful-capability-mode to http://hl7.org/fhir/ValueSet/restful-capability-mode|4.0.1
CapabilityStatement.rest.documentation | Type changed from string to markdown
CapabilityStatement.rest.security.service | Change code system for extensibly bound codes from "http://hl7.org/fhir/restful-security-service" to "http://terminology.hl7.org/CodeSystem/restful-security-service"
CapabilityStatement.rest.security.description | Type changed from string to markdown
CapabilityStatement.rest.resource.type | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
CapabilityStatement.rest.resource.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
CapabilityStatement.rest.resource.supportedProfile | Moved from CapabilityStatement.profile to supportedProfile Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
CapabilityStatement.rest.resource.interaction | Min Cardinality changed from 1 to 0
CapabilityStatement.rest.resource.interaction.code | Change value set from http://hl7.org/fhir/ValueSet/type-restful-interaction to http://hl7.org/fhir/ValueSet/type-restful-interaction|4.0.1
CapabilityStatement.rest.resource.interaction.documentation | Type changed from string to markdown
CapabilityStatement.rest.resource.versioning | Change value set from http://hl7.org/fhir/ValueSet/versioning-policy to http://hl7.org/fhir/ValueSet/versioning-policy|4.0.1
CapabilityStatement.rest.resource.conditionalRead | Change value set from http://hl7.org/fhir/ValueSet/conditional-read-status to http://hl7.org/fhir/ValueSet/conditional-read-status|4.0.1
CapabilityStatement.rest.resource.conditionalDelete | Change value set from http://hl7.org/fhir/ValueSet/conditional-delete-status to http://hl7.org/fhir/ValueSet/conditional-delete-status|4.0.1
CapabilityStatement.rest.resource.referencePolicy | Change value set from http://hl7.org/fhir/ValueSet/reference-handling-policy to http://hl7.org/fhir/ValueSet/reference-handling-policy|4.0.1
CapabilityStatement.rest.resource.searchParam.definition | Type changed from uri to canonical(SearchParameter)
CapabilityStatement.rest.resource.searchParam.type | Change value set from http://hl7.org/fhir/ValueSet/search-param-type to http://hl7.org/fhir/ValueSet/search-param-type|4.0.1
CapabilityStatement.rest.resource.searchParam.documentation | Type changed from string to markdown
CapabilityStatement.rest.resource.operation | Added Element
CapabilityStatement.rest.resource.operation.name | Added Mandatory Element
CapabilityStatement.rest.resource.operation.definition | Added Mandatory Element
CapabilityStatement.rest.resource.operation.documentation | Added Element
CapabilityStatement.rest.interaction.code | Change value set from http://hl7.org/fhir/ValueSet/system-restful-interaction to http://hl7.org/fhir/ValueSet/system-restful-interaction|4.0.1
CapabilityStatement.rest.interaction.documentation | Type changed from string to markdown
CapabilityStatement.rest.operation | Remove Type BackboneElement
CapabilityStatement.rest.compartment | Type changed from uri to canonical(CompartmentDefinition)
CapabilityStatement.messaging.endpoint.protocol | Change code system for extensibly bound codes from "http://hl7.org/fhir/message-transport" to "http://terminology.hl7.org/CodeSystem/message-transport"
CapabilityStatement.messaging.endpoint.address | Type changed from uri to url
CapabilityStatement.messaging.documentation | Type changed from string to markdown
CapabilityStatement.messaging.supportedMessage.mode | Change value set from http://hl7.org/fhir/ValueSet/event-capability-mode to http://hl7.org/fhir/ValueSet/event-capability-mode|4.0.1
CapabilityStatement.messaging.supportedMessage.definition | Type changed from Reference(MessageDefinition) to canonical(MessageDefinition)
CapabilityStatement.document.mode | Change value set from http://hl7.org/fhir/ValueSet/document-mode to http://hl7.org/fhir/ValueSet/document-mode|4.0.1
CapabilityStatement.document.documentation | Type changed from string to markdown
CapabilityStatement.document.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
CapabilityStatement.acceptUnknown | deleted
CapabilityStatement.rest.security.certificate | deleted
CapabilityStatement.rest.operation.name | deleted
CapabilityStatement.rest.operation.definition | deleted
CapabilityStatement.messaging.event | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 9 tests that all execute ok. 1 fail round-trip testing and 9 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

5.2.3.1 Terminology Bindings

Path | Definition | Type | Reference
CapabilityStatement.status | The lifecycle status of an artifact. | Required | PublicationStatus
CapabilityStatement.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
CapabilityStatement.kind | How a capability statement is intended to be used. | Required | CapabilityStatementKind
CapabilityStatement.fhirVersion | All published FHIR Versions. | Required | FHIRVersion

CapabilityStatement.format CapabilityStatement.patchFormat | The mime type of an attachment. Any valid mime type is allowed. | Required | Mime Types
CapabilityStatement.rest.mode | The mode of a RESTful capability statement. | Required | RestfulCapabilityMode
CapabilityStatement.rest.security.service | Types of security services used with FHIR. | Extensible | RestfulSecurityService
CapabilityStatement.rest.resource.type | One of the resource types defined as part of this version of FHIR. | Required | Resource Types
CapabilityStatement.rest.resource.interaction.code | Operations supported by REST at the type or instance level. | Required | TypeRestfulInteraction
CapabilityStatement.rest.resource.versioning | How the system supports versioning for a resource. | Required | ResourceVersionPolicy
CapabilityStatement.rest.resource.conditionalRead | A code that indicates how the server supports conditional read. | Required | ConditionalReadStatus
CapabilityStatement.rest.resource.conditionalDelete | A code that indicates how the server supports conditional delete. | Required | ConditionalDeleteStatus
CapabilityStatement.rest.resource.referencePolicy | A set of flags that defines how references are supported. | Required | ReferenceHandlingPolicy
CapabilityStatement.rest.resource.searchParam.type | Data types allowed to be used for search parameters. | Required | SearchParamType
CapabilityStatement.rest.interaction.code | Operations supported by REST at the system level. | Required | SystemRestfulInteraction
CapabilityStatement.messaging.endpoint.protocol | The protocol used for message transport. | Extensible | MessageTransport
CapabilityStatement.messaging.supportedMessage.mode | The mode of a message capability statement. | Required | EventCapabilityMode
CapabilityStatement.document.mode | Whether the application produces or consumes documents. | Required | DocumentMode

5.2.3.2 Constraints

id | Level | Location | Description | Expression
cpb-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
cpb-1 | Rule | (base) | A Capability Statement SHALL have at least one of REST, messaging or document element. | rest.exists() or messaging.exists() or document.exists()
cpb-2 | Rule | (base) | A Capability Statement SHALL have at least one of description, software, or implementation element. | (description.count() + software.count() + implementation.count()) > 0
cpb-3 | Rule | (base) | Messaging end-point is required (and is only permitted) when a statement is for an implementation. | messaging.endpoint.empty() or kind = 'instance'
cpb-7 | Rule | (base) | The set of documents must be unique by the combination of profile and mode. | document.select(profile&mode).isDistinct()
cpb-9 | Rule | CapabilityStatement.rest | A given resource can only be described once per RESTful mode. | resource.select(type).isDistinct()
cpb-12 | Rule | CapabilityStatement.rest.resource | Search parameter names must be unique in the context of a resource. | searchParam.select(name).isDistinct()
cpb-14 | Rule | (base) | If kind = instance, implementation must be present and software may be present | (kind != 'instance') or implementation.exists()
cpb-15 | Rule | (base) | If kind = capability, implementation must be absent, software must be present | (kind != 'capability') or (implementation.exists().not() and software.exists())
cpb-16 | Rule | (base) | If kind = requirements, implementation and software must be absent | (kind!='requirements') or (implementation.exists().not() and software.exists().not())

5.2.4 Notes:

• The CapabilityStatement resource provides for an application to describe its use of the RESTful paradigm messaging events, or FHIR documents. Usually, an application would only describe one, but more than one may be described

• RESTful CapabilityStatement rules: RESTful servers are required to provide this resource on demand. Servers SHALL specify what resource types and operations are supported, and SHOULD also specify profiles for each resource type. The CapabilityStatement returned on demand may represent the specific capabilities granted to a specific user if retrieved with that specific user's credentials, if one is in context. Servers that require authentication SHOULD still return a CapabilityStatemnt before authentication/authorization is performed RESTful clients SHOULD publish a capability statement The search parameters that a server supports (or a client makes use of) are specified in the resource profile that the capability statement references Resource Types or operations that are not listed are not supported

• Messaging CapabilityStatement rules: The interpretation of request and response depends on the mode. If the mode is sender, then request specifies what the application sends, and response specifies what it accepts. If the mode is "receiver", then this is reversed If a request or response is not specified for an event, then no rules are made for it Events that are not listed are not supported The MessageDefinition resource is newly proposed and is still considered 'draft'. The supportedMessage element can be used in place of the event and the work group believes it may meet implementer needs better, however because the new mechanism has not yet been reviewed by ballot, the older 'event' mechanism has been retained. Implementers may use one or the other to define their capabilities. Feedback is welcome.

• Document CapabilityStatement rules: Document profiles should directly constrain the Document.information.class & type elements so that there is no ambiguity concerning which profile any given

document conforms to.

• Other service-based use of resources: Due to the variability of these services, the CapabilityStatement resource does not attempt to describe service-based use of resources. The various service specifications will need to describe this usage in their own way.

5.2.4.1 Supporting Profiles

A CapabilityStatement declares two different kinds of profiles for the functionality it describes. For a discussion of the use of these two types of resources, see two uses for profiles.

5.2.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
context TU | token | A use context assigned to the capability statement | (CapabilityStatement.useContext.value as CodeableConcept)
context-quantity TU | quantity | A quantity- or range-valued use context assigned to the capability statement | (CapabilityStatement.useContext.value as Quantity) | (CapabilityStatement.useContext.value as Range)
context-type TU | token | A type of use context assigned to the capability statement | CapabilityStatement.useContext.code
context-type-quantity TU | composite | A use context type and quantity- or range-based value assigned to the capability statement | On CapabilityStatement.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value TU | composite | A use context type and value assigned to the capability statement | On CapabilityStatement.useContext: context-type: code context: value.as(CodeableConcept)
date TU | date | The capability statement publication date | CapabilityStatement.date
description TU | string | The description of the capability statement | CapabilityStatement.description
fhirversion TU | token | The version of FHIR | CapabilityStatement.version
format TU | token | formats supported (xml | json | ttl | mime type) | CapabilityStatement.format
guide TU | reference | Implementation guides supported | CapabilityStatement.implementationGuide (ImplementationGuide)
jurisdiction TU | token | Intended jurisdiction for the capability statement | CapabilityStatement.jurisdiction
mode TU | token | Mode - restful (server/client) or messaging (sender/receiver) | CapabilityStatement.rest.mode
name TU | string | Computationally friendly name of the capability statement | CapabilityStatement.name
publisher TU | string | Name of the publisher of the capability statement | CapabilityStatement.publisher
resource TU | token | Name of a resource mentioned in a capability statement | CapabilityStatement.rest.resource.type
resource-profile TU | reference | A profile id invoked in a capability statement | CapabilityStatement.rest.resource.profile (StructureDefinition)
security-service TU | token | OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates | CapabilityStatement.rest.security.service
software TU | string | Part of the name of a software application | CapabilityStatement.software.name
status TU | token | The current status of the capability statement | CapabilityStatement.status
supported-profile TU | reference | Profiles for use cases supported | CapabilityStatement.rest.resource.supportedProfile (StructureDefinition)
title TU | string | The human-friendly name of the capability statement | CapabilityStatement.title
url TU | uri | The uri that identifies the capability statement | CapabilityStatement.url
version TU | token | The business version of the capability statement | CapabilityStatement.version

5.2.6 Resource CapabilityStatement - Examples

URL: https://hl7.org/fhir/R4/capabilitystatement-examples.html

5.2.6 Resource CapabilityStatement - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
General Condition Example | example | XML | JSON | Turtle
PHR Example | phr | XML | JSON | Turtle
Complete Capability Statement | capabilitystatement-base | XML | JSON | Turtle
EmptyCapabilityStatement | capabilitystatement-base2 | XML | JSON | Turtle
Terminology Server Base Capability Statement | terminology-server | XML | JSON | Turtle
Knowledge Repository Base Capability Statement | knowledge-repository | XML | JSON | Turtle
Measure Processor Base Capability Statement | measure-processor | XML | JSON | Turtle
Showing new message definition structure | messagedefinition | XML | JSON | Turtle
System which records information about EHR events | null | XML | JSON | Turtle | from EHRS Functional model - Record Lifecycle Events IG

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.2.7 Resource CapabilityStatement - Detailed Descriptions

URL: https://hl7.org/fhir/R4/capabilitystatement-definitions.html

5.2.7 Resource CapabilityStatement - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the CapabilityStatement resource.

CapabilityStatement

Element Id | CapabilityStatement

Definition | A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a particular version of FHIR that may be used as a statement of actual server functionality or a statement of required or desired server implementation.

Cardinality | 0..*

Type | DomainResource

Comments | Applications may implement multiple versions (see Managing Multiple Versions, and the $versions operation). If they do, then a CapabilityStatement describes the system's support for a particular version of FHIR, and the server will have multiple statements, one for each version.

Invariants | Defined on this element cpb-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}') cpb-1 Rule A Capability Statement SHALL have at least one of REST, messaging or document element. rest.exists() or messaging.exists() or document.exists() cpb-2 Rule A Capability Statement SHALL have at least one of description, software, or implementation element. (description.count() + software.count() + implementation.count()) > 0 cpb-3 Rule Messaging end-point is required (and is only permitted) when a statement is for an implementation. messaging.endpoint.empty() or kind = 'instance' cpb-7 Rule The set of documents must be unique by the combination of profile and mode. document.select(profile&mode).isDistinct() cpb-14 Rule If kind = instance, implementation must be present and software may be present (kind != 'instance') or implementation.exists() cpb-15 Rule If kind = capability, implementation must be absent, software must be present (kind != 'capability') or (implementation.exists().not() and software.exists()) cpb-16 Rule If kind = requirements, implementation and software must be absent (kind!='requirements') or (implementation.exists().not() and software.exists().not())

Defined on this element

cpb-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

cpb-1 | Rule | A Capability Statement SHALL have at least one of REST, messaging or document element. | rest.exists() or messaging.exists() or document.exists()

cpb-2 | Rule | A Capability Statement SHALL have at least one of description, software, or implementation element. | (description.count() + software.count() + implementation.count()) > 0

cpb-3 | Rule | Messaging end-point is required (and is only permitted) when a statement is for an implementation. | messaging.endpoint.empty() or kind = 'instance'

cpb-7 | Rule | The set of documents must be unique by the combination of profile and mode. | document.select(profile&mode).isDistinct()

cpb-14 | Rule | If kind = instance, implementation must be present and software may be present | (kind != 'instance') or implementation.exists()

cpb-15 | Rule | If kind = capability, implementation must be absent, software must be present | (kind != 'capability') or (implementation.exists().not() and software.exists())

cpb-16 | Rule | If kind = requirements, implementation and software must be absent | (kind!='requirements') or (implementation.exists().not() and software.exists().not())

CapabilityStatement.url

Element Id | CapabilityStatement.url

Definition | An absolute URI that is used to identify this capability statement when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this capability statement is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the capability statement is stored on different servers.

Cardinality | 0..1

Type | uri

Requirements | Allows the capability statement to be referenced by a single globally unique identifier.

Summary | true

Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.

CapabilityStatement.version

Element Id | CapabilityStatement.version

Definition | The identifier that is used to identify this version of the capability statement when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the capability statement author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.

Note | This is a business versionId, not a resource version id (see discussion)

Cardinality | 0..1

Type | string

Summary | true

Comments | There may be different capability statement instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the capability statement with the format [url]|[version].

CapabilityStatement.name

Element Id | CapabilityStatement.name

Definition | A natural language name identifying the capability statement. This name should be usable as an identifier for the module by machine processing applications such as code generation.

Cardinality | 0..1

Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
CapabilityStatement.title
Element Id | CapabilityStatement.title
Definition | A short, descriptive, user-friendly title for the capability statement.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
CapabilityStatement.status
Element Id | CapabilityStatement.status
Definition | The status of this capability statement. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of capability statements that are appropriate for use versus not.This is not intended for use with actual capability statements, but where capability statements are used to describe possible or desired systems.
CapabilityStatement.experimental
Element Id | CapabilityStatement.experimental
Definition | A Boolean value to indicate that this capability statement is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level capability statement.
Summary | true
Comments | Allows filtering of capability statements that are appropriate for use versus not.
CapabilityStatement.date
Element Id | CapabilityStatement.date
Definition | The date (and optionally time) when the capability statement was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the capability statement changes.
Cardinality | 1..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the capability statement. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
CapabilityStatement.publisher
Element Id | CapabilityStatement.publisher
Definition | The name of the organization or individual that published the capability statement.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the capability statement. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the capability statement is the organization or individual primarily responsible for the maintenance and upkeep of the capability statement. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the capability statement. This item SHOULD be populated unless the information is available from context.
CapabilityStatement.contact
Element Id | CapabilityStatement.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
CapabilityStatement.description
Element Id | CapabilityStatement.description
Definition | A free text natural language description of the capability statement from a consumer's perspective. Typically, this is used when the capability statement describes a desired rather than an actual solution, for example as a formal expression of requirements as part of an RFP.

Cardinality | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the capability statement was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the capability statement as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the capability statement is presumed to be the predominant language in the place the capability statement was created).This does not need to be populated if the description is adequately implied by the software or implementation details.
Invariants | Affect this element cpb-2 Rule A Capability Statement SHALL have at least one of description, software, or implementation element. (description.count() + software.count() + implementation.count()) > 0
Affect this element
cpb-2 | Rule | A Capability Statement SHALL have at least one of description, software, or implementation element. | (description.count() + software.count() + implementation.count()) > 0
CapabilityStatement.useContext
Element Id | CapabilityStatement.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate capability statement instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
CapabilityStatement.jurisdiction
Element Id | CapabilityStatement.jurisdiction
Definition | A legal or geographic region in which the capability statement is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the capability statement to be used in jurisdictions other than those for which it was originally designed or intended.
CapabilityStatement.purpose
Element Id | CapabilityStatement.purpose
Definition | Explanation of why this capability statement is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the capability statement. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this capability statement.
CapabilityStatement.copyright
Element Id | CapabilityStatement.copyright
Definition | A copyright statement relating to the capability statement and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the capability statement.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the capability statement and/or its content.
Alternate Names | License; Restrictions
CapabilityStatement.kind
Element Id | CapabilityStatement.kind
Definition | The way that this statement is intended to be used, to describe an actual running instance of software, a particular product (kind, not instance of software) or a class of implementation (e.g. a desired purchase).
Cardinality | 1..1
Terminology Binding | CapabilityStatementKind (Required)
Type | code
Requirements | Allow searching the 3 modes.
Summary | true
Invariants | Affect this element cpb-15 Rule If kind = capability, implementation must be absent, software must be present (kind != 'capability') or (implementation.exists().not() and software.exists())
Affect this element
cpb-15 | Rule | If kind = capability, implementation must be absent, software must be present | (kind != 'capability') or (implementation.exists().not() and software.exists())
CapabilityStatement.instantiates
Element Id | CapabilityStatement.instantiates
Definition | Reference to a canonical URL of another CapabilityStatement that this software implements. This capability statement is a published API description that corresponds to a business service. The server may actually implement a subset of the capability statement it claims to implement, so the capability statement must specify the full capability details.
Cardinality | 0..*
Type | canonical(CapabilityStatement)
Summary | true
Comments | HL7 defines the following Services: Terminology Service. Many Implementation Guides define additional services.
CapabilityStatement.imports
Element Id | CapabilityStatement.imports

Definition | Reference to a canonical URL of another CapabilityStatement that this software adds to. The capability statement automatically includes everything in the other statement, and it is not duplicated, though the server may repeat the same resources, interactions and operations to add additional details to them.
Cardinality | 0..*
Type | canonical(CapabilityStatement)
Summary | true
Comments | the contents of any directly or indirectly imported CapabilityStatements SHALL NOT overlap, i.e. they cannot refer to the same rest/resource, operations/name, searchparam/name, interaction/code, messaging/endpoint, document/mode pair. A capability statement that imports another CapabilityStatement automatically instantiates it too (though this is often not a very useful statement for the kinds of CapabilityStatements that are suitable for importing).
CapabilityStatement.software
Element Id | CapabilityStatement.software
Definition | Software that is covered by this capability statement. It is used when the capability statement describes the capabilities of a particular software version, independent of an installation.
Cardinality | 0..1
Summary | true
Invariants | Affect this element cpb-2 Rule A Capability Statement SHALL have at least one of description, software, or implementation element. (description.count() + software.count() + implementation.count()) > 0
Affect this element
cpb-2 | Rule | A Capability Statement SHALL have at least one of description, software, or implementation element. | (description.count() + software.count() + implementation.count()) > 0
CapabilityStatement.software.name
Element Id | CapabilityStatement.software.name
Definition | Name the software is known by.
Cardinality | 1..1
Type | string
Summary | true
CapabilityStatement.software.version
Element Id | CapabilityStatement.software.version
Definition | The version identifier for the software covered by this statement.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | If possible, a version should be specified, as statements are likely to be different for different versions of software.
CapabilityStatement.software.releaseDate
Element Id | CapabilityStatement.software.releaseDate
Definition | Date this version of the software was released.
Cardinality | 0..1
Type | dateTime
Summary | true
CapabilityStatement.implementation
Element Id | CapabilityStatement.implementation
Definition | Identifies a specific implementation instance that is described by the capability statement - i.e. a particular installation, rather than the capabilities of a software program.
Cardinality | 0..1
Summary | true
Invariants | Affect this element cpb-2 Rule A Capability Statement SHALL have at least one of description, software, or implementation element. (description.count() + software.count() + implementation.count()) > 0 cpb-15 Rule If kind = capability, implementation must be absent, software must be present (kind != 'capability') or (implementation.exists().not() and software.exists())
Affect this element
cpb-2 | Rule | A Capability Statement SHALL have at least one of description, software, or implementation element. | (description.count() + software.count() + implementation.count()) > 0
cpb-15 | Rule | If kind = capability, implementation must be absent, software must be present | (kind != 'capability') or (implementation.exists().not() and software.exists())
CapabilityStatement.implementation.description
Element Id | CapabilityStatement.implementation.description
Definition | Information about the specific installation that this capability statement relates to.
Cardinality | 1..1
Type | string
Summary | true
CapabilityStatement.implementation.url
Element Id | CapabilityStatement.implementation.url
Definition | An absolute base URL for the implementation. This forms the base for REST interfaces as well as the mailbox and document interfaces.
Cardinality | 0..1
Type | url
Summary | true
CapabilityStatement.implementation.custodian
Element Id | CapabilityStatement.implementation.custodian
Definition | The organization responsible for the management of the instance and oversight of the data on the server at the specified URL.
Cardinality | 0..1

Type | Reference(Organization)
Summary | true
CapabilityStatement.fhirVersion
Element Id | CapabilityStatement.fhirVersion
Definition | The version of the FHIR specification that this CapabilityStatement describes (which SHALL be the same as the FHIR version of the CapabilityStatement itself). There is no default value.
Cardinality | 1..1
Terminology Binding | FHIRVersion (Required)
Type | code
Summary | true
Comments | Servers may implement multiple versions (see Managing Multiple Versions, and the $versions operation). If they do, and the CapabilityStatement is requested from the server, then this fhirVersion will be either the version requested, or the server's default version.
CapabilityStatement.format
Element Id | CapabilityStatement.format
Definition | A list of the formats supported by this implementation using their content types.
Cardinality | 1..*
Terminology Binding | MimeType (Required)
Type | code
Summary | true
Comments | "xml", "json" and "ttl" are allowed, which describe the simple encodings described in the specification (and imply appropriate bundle support). Otherwise, mime types are legal here.
CapabilityStatement.patchFormat
Element Id | CapabilityStatement.patchFormat
Definition | A list of the patch formats supported by this implementation using their content types.
Cardinality | 0..*
Terminology Binding | MimeType (Required)
Type | code
Summary | true
Comments | At present, the patch mime types application/json-patch+json and application/xml-patch+xml are legal. Generally, if a server supports PATCH, it would be expected to support the patch formats and match the formats it supports, but this is not always possible or necessary.
CapabilityStatement.implementationGuide
Element Id | CapabilityStatement.implementationGuide
Definition | A list of implementation guides that the server does (or should) support in their entirety.
Cardinality | 0..*
Type | canonical(ImplementationGuide)
Summary | true
CapabilityStatement.rest
Element Id | CapabilityStatement.rest
Definition | A definition of the restful capabilities of the solution, if any.
Cardinality | 0..*
Summary | true
Comments | Multiple repetitions allow definition of both client and/or server behaviors or possibly behaviors under different configuration settings (for software or requirements statements).
Invariants | Defined on this element cpb-9 Rule A given resource can only be described once per RESTful mode. resource.select(type).isDistinct() Affect this element cpb-1 Rule A Capability Statement SHALL have at least one of REST, messaging or document element. rest.exists() or messaging.exists() or document.exists()
Defined on this element
cpb-9 | Rule | A given resource can only be described once per RESTful mode. | resource.select(type).isDistinct()
Affect this element
cpb-1 | Rule | A Capability Statement SHALL have at least one of REST, messaging or document element. | rest.exists() or messaging.exists() or document.exists()
CapabilityStatement.rest.mode
Element Id | CapabilityStatement.rest.mode
Definition | Identifies whether this portion of the statement is describing the ability to initiate or receive restful operations.
Cardinality | 1..1
Terminology Binding | RestfulCapabilityMode (Required)
Type | code
Summary | true
CapabilityStatement.rest.documentation
Element Id | CapabilityStatement.rest.documentation
Definition | Information about the system's restful capabilities that apply across all applications, such as security.
Cardinality | 0..1
Type | markdown
CapabilityStatement.rest.security
Element Id | CapabilityStatement.rest.security
Definition | Information about security implementation from an interface perspective - what a client needs to know.
Cardinality | 0..1
Summary | true
CapabilityStatement.rest.security.cors
Element Id | CapabilityStatement.rest.security.cors
Definition | Server adds CORS headers when responding to requests - this enables Javascript applications to use the server.

Cardinality | 0..1
Type | boolean
Summary | true
Comments | The easiest CORS headers to add are Access-Control-Allow-Origin: * & Access-Control-Request-Method: GET, POST, PUT, DELETE. All servers SHOULD support CORS.
CapabilityStatement.rest.security.service
Element Id | CapabilityStatement.rest.security.service
Definition | Types of security services that are supported/required by the system.
Cardinality | 0..*
Terminology Binding | RestfulSecurityService (Extensible)
Type | CodeableConcept
Summary | true
CapabilityStatement.rest.security.description
Element Id | CapabilityStatement.rest.security.description
Definition | General description of how security works.
Cardinality | 0..1
Type | markdown
CapabilityStatement.rest.resource
Element Id | CapabilityStatement.rest.resource
Definition | A specification of the restful capabilities of the solution for a specific resource type.
Cardinality | 0..*
Summary | true
Comments | Max of one repetition per resource type.
Invariants | Defined on this element cpb-12 Rule Search parameter names must be unique in the context of a resource. searchParam.select(name).isDistinct()
Defined on this element
cpb-12 | Rule | Search parameter names must be unique in the context of a resource. | searchParam.select(name).isDistinct()
CapabilityStatement.rest.resource.type
Element Id | CapabilityStatement.rest.resource.type
Definition | A type of resource exposed via the restful interface.
Cardinality | 1..1
Terminology Binding | Any defined Resource Type name
Type | code
Summary | true
CapabilityStatement.rest.resource.profile
Element Id | CapabilityStatement.rest.resource.profile
Definition | A specification of the profile that describes the solution's overall support for the resource, including any constraints on cardinality, bindings, lengths or other limitations. See further discussion in Using Profiles.
Cardinality | 0..1
Type | canonical(StructureDefinition)
Summary | true
Comments | The profile applies to all resources of this type - i.e. it is the superset of what is supported by the system.
CapabilityStatement.rest.resource.supportedProfile
Element Id | CapabilityStatement.rest.resource.supportedProfile
Definition | A list of profiles that represent different use cases supported by the system. For a server, "supported by the system" means the system hosts/produces a set of resources that are conformant to a particular profile, and allows clients that use its services to search using this profile and to find appropriate data. For a client, it means the system will search by this profile and process data according to the guidance implicit in the profile. See further discussion in Using Profiles.
Cardinality | 0..*
Type | canonical(StructureDefinition)
Summary | true
Comments | Supported profiles are different than the profile that applies to a particular resource in .rest.resource.profile. The resource profile is a general statement of what features of the resource are supported overall by the system - the sum total of the facilities it supports. A supported profile is a deeper statement about the functionality of the data and services provided by the server (or used by the client). A typical case is a laboratory system that produces a set of different reports - this is the list of types of data that it publishes. A key aspect of declaring profiles here is the question of how the client converts knowledge that the server publishes this data into working with the data; the client can inspect individual resources to determine whether they conform to a particular profile, but how does it find the ones that do? It does so by searching using the _profile parameter, so any resources listed here must be valid values for the _profile resource (using the identifier in the target profile).
CapabilityStatement.rest.resource.documentation
Element Id | CapabilityStatement.rest.resource.documentation
Definition | Additional information about the resource type used by the system.
Cardinality | 0..1
Type | markdown
CapabilityStatement.rest.resource.interaction
Element Id | CapabilityStatement.rest.resource.interaction
Definition | Identifies a restful operation supported by the solution.
Cardinality | 0..*
Comments | In general, a Resource will only appear in a CapabilityStatement if the server actually has some capabilities - e.g. there is at least one interaction supported. However interactions can be omitted to support summarization (_summary = true).
CapabilityStatement.rest.resource.interaction.code
Element Id | CapabilityStatement.rest.resource.interaction.code
Definition | Coded identifier of the operation, supported by the system resource.

Cardinality | 1..1
Terminology Binding | TypeRestfulInteraction (Required)
Type | code
CapabilityStatement.rest.resource.interaction.documentation
Element Id | CapabilityStatement.rest.resource.interaction.documentation
Definition | Guidance specific to the implementation of this operation, such as 'delete is a logical delete' or 'updates are only allowed with version id' or 'creates permitted from pre-authorized certificates only'.
Cardinality | 0..1
Type | markdown
Requirements | REST allows a degree of variability in the implementation of RESTful solutions that is useful for exchange partners to be aware of.
CapabilityStatement.rest.resource.versioning
Element Id | CapabilityStatement.rest.resource.versioning
Definition | This field is set to no-version to specify that the system does not support (server) or use (client) versioning for this resource type. If this has some other value, the server must at least correctly track and populate the versionId meta-property on resources. If the value is 'versioned-update', then the server supports all the versioning features, including using e-tags for version integrity in the API.
Cardinality | 0..1
Terminology Binding | ResourceVersionPolicy (Required)
Type | code
Comments | If a server supports versionIds correctly, it SHOULD support vread too, but is not required to do so.
CapabilityStatement.rest.resource.readHistory
Element Id | CapabilityStatement.rest.resource.readHistory
Definition | A flag for whether the server is able to return past versions as part of the vRead operation.
Cardinality | 0..1
Type | boolean
Comments | It is useful to support the vRead operation for current operations, even if past versions aren't available.
CapabilityStatement.rest.resource.updateCreate
Element Id | CapabilityStatement.rest.resource.updateCreate
Definition | A flag to indicate that the server allows or needs to allow the client to create new identities on the server (that is, the client PUTs to a location where there is no existing resource). Allowing this operation means that the server allows the client to create new identities on the server.
Cardinality | 0..1
Type | boolean
Comments | Allowing the clients to create new identities on the server means that the system administrator needs to have confidence that the clients do not create clashing identities between them. Obviously, if there is only one client, this won't happen. While creating identities on the client means that the clients need to be managed, it's much more convenient for many scenarios if such management can be put in place.
CapabilityStatement.rest.resource.conditionalCreate
Element Id | CapabilityStatement.rest.resource.conditionalCreate
Definition | A flag that indicates that the server supports conditional create.
Cardinality | 0..1
Type | boolean
Comments | Conditional Create is mainly appropriate for interface engine scripts converting from other formats, such as v2.
CapabilityStatement.rest.resource.conditionalRead
Element Id | CapabilityStatement.rest.resource.conditionalRead
Definition | A code that indicates how the server supports conditional read.
Cardinality | 0..1
Terminology Binding | ConditionalReadStatus (Required)
Type | code
Comments | Conditional Read is mainly appropriate for interface engine scripts converting from other formats, such as v2.
CapabilityStatement.rest.resource.conditionalUpdate
Element Id | CapabilityStatement.rest.resource.conditionalUpdate
Definition | A flag that indicates that the server supports conditional update.
Cardinality | 0..1
Type | boolean
Comments | Conditional Update is mainly appropriate for interface engine scripts converting from other formats, such as v2.
CapabilityStatement.rest.resource.conditionalDelete
Element Id | CapabilityStatement.rest.resource.conditionalDelete
Definition | A code that indicates how the server supports conditional delete.
Cardinality | 0..1
Terminology Binding | ConditionalDeleteStatus (Required)
Type | code
Comments | Conditional Delete is mainly appropriate for interface engine scripts converting from other formats, such as v2.
CapabilityStatement.rest.resource.referencePolicy
Element Id | CapabilityStatement.rest.resource.referencePolicy
Definition | A set of flags that defines how references are supported.
Cardinality | 0..*
Terminology Binding | ReferenceHandlingPolicy (Required)
Type | code
CapabilityStatement.rest.resource.searchInclude
Element Id | CapabilityStatement.rest.resource.searchInclude
Definition | A list of _include values supported by the server.

Cardinality | 0..*
Type | string
Comments | If this list is empty, the server does not support includes.
CapabilityStatement.rest.resource.searchRevInclude
Element Id | CapabilityStatement.rest.resource.searchRevInclude
Definition | A list of _revinclude (reverse include) values supported by the server.
Cardinality | 0..*
Type | string
Comments | If this list is empty, the server does not support reverse includes.
CapabilityStatement.rest.resource.searchParam
Element Id | CapabilityStatement.rest.resource.searchParam
Definition | Search parameters for implementations to support and/or make use of - either references to ones defined in the specification, or additional ones defined for/by the implementation.
Cardinality | 0..*
Comments | The search parameters should include the control search parameters such as _sort, _count, etc. that also apply to this resource (though many will be listed at CapabilityStatement.rest.searchParam). The behavior of some search parameters may be further described by other code or extension elements, or narrative within the capability statement or linked SearchParameter definitions.
CapabilityStatement.rest.resource.searchParam.name
Element Id | CapabilityStatement.rest.resource.searchParam.name
Definition | The name of the search parameter used in the interface.
Cardinality | 1..1
Type | string
Comments | Parameter names cannot overlap with standard parameter names, and standard parameters cannot be redefined.
CapabilityStatement.rest.resource.searchParam.definition
Element Id | CapabilityStatement.rest.resource.searchParam.definition
Definition | An absolute URI that is a formal reference to where this parameter was first defined, so that a client can be confident of the meaning of the search parameter (a reference to SearchParameter.url). This element SHALL be populated if the search parameter refers to a SearchParameter defined by the FHIR core specification or externally defined IGs.
Cardinality | 0..1
Type | canonical(SearchParameter)
Comments | This SHOULD be present, and matches refers to a SearchParameter by its canonical URL. If systems wish to document their support for modifiers, comparators, target resource types, and chained parameters, they should do using a search parameter resource. This element SHALL be populated if the search parameter refers to a SearchParameter defined by the FHIR core specification or externally defined IGs.
CapabilityStatement.rest.resource.searchParam.type
Element Id | CapabilityStatement.rest.resource.searchParam.type
Definition | The type of value a search parameter refers to, and how the content is interpreted.
Cardinality | 1..1
Terminology Binding | SearchParamType (Required)
Type | code
Comments | While this can be looked up from the definition, it is included here as a convenience for systems that autogenerate a query interface based on the server capability statement. It SHALL be the same as the type in the search parameter definition.
CapabilityStatement.rest.resource.searchParam.documentation
Element Id | CapabilityStatement.rest.resource.searchParam.documentation
Definition | This allows documentation of any distinct behaviors about how the search parameter is used. For example, text matching algorithms.
Cardinality | 0..1
Type | markdown
CapabilityStatement.rest.resource.operation
Element Id | CapabilityStatement.rest.resource.operation
Definition | Definition of an operation or a named query together with its parameters and their meaning and type. Consult the definition of the operation for details about how to invoke the operation, and the parameters.
Cardinality | 0..*
Summary | true
Comments | Operations linked from CapabilityStatement.rest.resource.operation must have OperationDefinition.type = true or OperationDefinition.instance = true. If an operation that is listed in multiple CapabilityStatement.rest.resource.operation (e.g. for different resource types), then clients should understand that the operation is only supported on the specified resource types, and that may be a subset of those listed in OperationDefinition.resource.
CapabilityStatement.rest.resource.operation.name
Element Id | CapabilityStatement.rest.resource.operation.name
Definition | The name of the operation or query. For an operation, this is the name prefixed with $ and used in the URL. For a query, this is the name used in the _query parameter when the query is called.
Cardinality | 1..1
Type | string
Summary | true
Comments | The name here SHOULD be the same as the name in the definition, unless there is a name clash and the name cannot be used. The name does not include the "$" portion that is always included in the URL.
CapabilityStatement.rest.resource.operation.definition
Element Id | CapabilityStatement.rest.resource.operation.definition
Definition | Where the formal definition can be found. If a server references the base definition of an Operation (i.e. from the specification itself such as http://hl7.org/fhir/OperationDefinition/ValueSet-expand), that means it supports the full capabilities of the operation - e.g. both GET and POST invocation. If it only supports a subset, it must define its own custom OperationDefinition with a 'base' of the original OperationDefinition. The custom definition would describe the specific subset of functionality

supported.
Cardinality | 1..1
Type | canonical(OperationDefinition)
Summary | true
Comments | This can be used to build an HTML form to invoke the operation, for instance.
CapabilityStatement.rest.resource.operation.documentation
Element Id | CapabilityStatement.rest.resource.operation.documentation
Definition | Documentation that describes anything special about the operation behavior, possibly detailing different behavior for system, type and instance-level invocation of the operation.
Cardinality | 0..1
Type | markdown
CapabilityStatement.rest.interaction
Element Id | CapabilityStatement.rest.interaction
Definition | A specification of restful operations supported by the system.
Cardinality | 0..*
CapabilityStatement.rest.interaction.code
Element Id | CapabilityStatement.rest.interaction.code
Definition | A coded identifier of the operation, supported by the system.
Cardinality | 1..1
Terminology Binding | SystemRestfulInteraction (Required)
Type | code
CapabilityStatement.rest.interaction.documentation
Element Id | CapabilityStatement.rest.interaction.documentation
Definition | Guidance specific to the implementation of this operation, such as limitations on the kind of transactions allowed, or information about system wide search is implemented.
Cardinality | 0..1
Type | markdown
CapabilityStatement.rest.searchParam
Element Id | CapabilityStatement.rest.searchParam
Definition | Search parameters that are supported for searching all resources for implementations to support and/or make use of - either references to ones defined in the specification, or additional ones defined for/by the implementation.
Cardinality | 0..*
Type | See CapabilityStatement.rest.resource.searchParam
Comments | Typically, the only search parameters supported for all searches are those that apply to all resources - tags, profiles, text search etc. These search parameters should include the control search parameters such as _sort, _count, etc. that also apply to this resource (though many will be listed at CapabilityStatement.rest.searchParam). The behavior of some search parameters may be further described by other code or extension elements, or narrative within the capability statement or linked SearchParameter definitions.
CapabilityStatement.rest.operation
Element Id | CapabilityStatement.rest.operation
Definition | Definition of an operation or a named query together with its parameters and their meaning and type.
Cardinality | 0..*
Type | See CapabilityStatement.rest.resource.operation
Summary | true
Comments | CapabilityStatement.rest.operation is for operations invoked at the system level, or for operations that are supported across multiple resource types. Operations linked from CapabilityStatement.rest.operation must have OperationDefinition.system = true, or more than one Operation.resource.
CapabilityStatement.rest.compartment
Element Id | CapabilityStatement.rest.compartment
Definition | An absolute URI which is a reference to the definition of a compartment that the system supports. The reference is to a CompartmentDefinition resource by its canonical URL .
Cardinality | 0..*
Type | canonical(CompartmentDefinition)
Comments | At present, the only defined compartments are at CompartmentDefinition.
CapabilityStatement.messaging
Element Id | CapabilityStatement.messaging
Definition | A description of the messaging capabilities of the solution.
Cardinality | 0..*
Summary | true
Comments | Multiple repetitions allow the documentation of multiple endpoints per solution.
Invariants | Affect this element cpb-1 Rule A Capability Statement SHALL have at least one of REST, messaging or document element. rest.exists() or messaging.exists() or document.exists()
Affect this element
cpb-1 | Rule | A Capability Statement SHALL have at least one of REST, messaging or document element. | rest.exists() or messaging.exists() or document.exists()
CapabilityStatement.messaging.endpoint
Element Id | CapabilityStatement.messaging.endpoint
Definition | An endpoint (network accessible address) to which messages and/or replies are to be sent.
Cardinality | 0..*
Alternate Names | 3
CapabilityStatement.messaging.endpoint.protocol
Element Id | CapabilityStatement.messaging.endpoint.protocol
Definition | A list of the messaging transport protocol(s) identifiers, supported by this endpoint.

Cardinality | 1..1
Terminology Binding | MessageTransport (Extensible)
Type | Coding
CapabilityStatement.messaging.endpoint.address
Element Id | CapabilityStatement.messaging.endpoint.address
Definition | The network address of the endpoint. For solutions that do not use network addresses for routing, it can be just an identifier.
Cardinality | 1..1
Type | url
CapabilityStatement.messaging.reliableCache
Element Id | CapabilityStatement.messaging.reliableCache
Definition | Length if the receiver's reliable messaging cache in minutes (if a receiver) or how long the cache length on the receiver should be (if a sender).
Cardinality | 0..1
Type | unsignedInt
Comments | If this value is missing then the application does not implement (receiver) or depend on (sender) reliable messaging.
CapabilityStatement.messaging.documentation
Element Id | CapabilityStatement.messaging.documentation
Definition | Documentation about the system's messaging capabilities for this endpoint not otherwise documented by the capability statement. For example, the process for becoming an authorized messaging exchange partner.
Cardinality | 0..1
Type | markdown
CapabilityStatement.messaging.supportedMessage
Element Id | CapabilityStatement.messaging.supportedMessage
Definition | References to message definitions for messages this system can send or receive.
Cardinality | 0..*
Summary | true
Comments | This is a proposed alternative to the messaging.event structure.
CapabilityStatement.messaging.supportedMessage.mode
Element Id | CapabilityStatement.messaging.supportedMessage.mode
Definition | The mode of this event declaration - whether application is sender or receiver.
Cardinality | 1..1
Terminology Binding | EventCapabilityMode (Required)
Type | code
Summary | true
CapabilityStatement.messaging.supportedMessage.definition
Element Id | CapabilityStatement.messaging.supportedMessage.definition
Definition | Points to a message definition that identifies the messaging event, message structure, allowed responses, etc.
Cardinality | 1..1
Type | canonical(MessageDefinition)
Summary | true
CapabilityStatement.document
Element Id | CapabilityStatement.document
Definition | A document definition.
Cardinality | 0..*
Summary | true
Invariants | Affect this element cpb-1 Rule A Capability Statement SHALL have at least one of REST, messaging or document element. rest.exists() or messaging.exists() or document.exists()
Affect this element
cpb-1 | Rule | A Capability Statement SHALL have at least one of REST, messaging or document element. | rest.exists() or messaging.exists() or document.exists()
CapabilityStatement.document.mode
Element Id | CapabilityStatement.document.mode
Definition | Mode of this document declaration - whether an application is a producer or consumer.
Cardinality | 1..1
Terminology Binding | DocumentMode (Required)
Type | code
Summary | true
CapabilityStatement.document.documentation
Element Id | CapabilityStatement.document.documentation
Definition | A description of how the application supports or uses the specified document profile. For example, when documents are created, what action is taken with consumed documents, etc.
Cardinality | 0..1
Type | markdown
CapabilityStatement.document.profile
Element Id | CapabilityStatement.document.profile
Definition | A profile on the document Bundle that constrains which resources are present, and their contents.
Cardinality | 1..1
Type | canonical(StructureDefinition)
Summary | true
Comments | The profile is actually on the Bundle.

5.2.8 Resource CapabilityStatement - Mappings

URL: https://hl7.org/fhir/R4/capabilitystatement-mappings.html

5.2.8 Resource CapabilityStatement - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the capabilitystatement resource (see Mappings to Other Standards for further information & status).

5.2.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

CapabilityStatement
url | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

5.2.8.2 Workflow Pattern (http://hl7.org/fhir/workflow)

CapabilityStatement
url | Definition.url
version | Definition.version
title | Definition.title
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright

5.2.8.3 RIM Mapping (http://hl7.org/v3)

CapabilityStatement | n/a

5.2.9 Resource CapabilityStatement - Extensions & Profiles

URL: https://hl7.org/fhir/R4/capabilitystatement-profiles.html

5.2.9 Resource CapabilityStatement - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the CapabilityStatement resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

5.2.9.1 Profiles

No Profiles defined for this resource

5.2.9.2 Extensions

capabilities | CapabilityStatement.rest.security | capabilities | for Smart App Launch Capability Statements
capabilitystatement-expectation | CapabilityStatement.rest.resource.interaction, CapabilityStatement.rest.resource.searchParam, CapabilityStatement.rest.searchParam, CapabilityStatement.rest.operation, CapabilityStatement.document, CapabilityStatement.rest.interaction, CapabilityStatement.rest.resource.searchInclude, CapabilityStatement.rest.resource.searchRevInclude | expectation | for Common CapabilityStatement extensions
capabilitystatement-prohibited | CapabilityStatement.rest.resource.interaction, CapabilityStatement.rest.resource.searchParam, CapabilityStatement.rest.searchParam, CapabilityStatement.rest.operation, CapabilityStatement.document, CapabilityStatement.rest.interaction | prohibited | for Common CapabilityStatement extensions
capabilitystatement-search-parameter-combination | CapabilityStatement.rest.resource | search-parameter-combination | for Common CapabilityStatement extensions

capabilitystatement-supported-system | CapabilityStatement | supported-system | for Common CapabilityStatement extensions
capabilitystatement-websocket | CapabilityStatement.rest | websocket | for Common CapabilityStatement extensions
oauth-uris | CapabilityStatement.rest.security | oauth-uris | for Smart App Launch Capability Statements
replaces | CapabilityStatement | replaces | for General Extensions for use by FHIR Implementers
resource-approvalDate | CapabilityStatement | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | CapabilityStatement | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | CapabilityStatement | lastReviewDate | for Resource HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

5.2.9.3 Search Extensions

No Search Extensions defined for this resource

5.2.10 Resource CapabilityStatement - Operations

URL: https://hl7.org/fhir/R4/capabilitystatement-operations.html

5.2.10 Resource CapabilityStatement - Operations

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This resource has 4 operations operations associated with it:

$subset | Fetch a subset of the CapabilityStatement resource | Trial Use
$implements | Test if a server implements a client's required operations | Trial Use
$conforms | Test if a server implements a client's required operations | Trial Use
$versions | Discover what versions a server supports | Trial Use

For more information about operations, including how they are invoked, see Operations.

5.2.10.1 Operation $subset on CapabilityStatement

URL: https://hl7.org/fhir/R4/capabilitystatement-operation-subset.html

5.2.10.1 Operation $subset on CapabilityStatement

FHIR Infrastructure Work Group | Maturity Level: 5 | Trial Use | Compartments: N/A

This operation asks the server to return a subset of the CapabilityStatement resource - just the REST parts that relate to a set of nominated resources - the resources that the client is interested in.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/CapabilityStatement-subset
```

Formal Definition (as a OperationDefinition).

URL: [base]/CapabilityStatement/$subset

URL: [base]/CapabilityStatement/[id]/$subset

This is an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
server | 0..1 | uri | | | The canonical URL - use this if the subset is not invoked on an instance (or on the /metadata end-point)
resource | 1..* | code | | | A resource that the client would like to include in the return
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
return | 1..1 | CapabilityStatement | | | The subsetted CapabilityStatement resource that is returned. This should be tagged with the SUBSETTED code Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

5.2.10.1.1 Examples

Request: Request a subset of Patient capabilities from a known CapabilityStatement

```
POST [base]/CapabilityStatement/example/$subset
[other headers]
```

```
<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="resource"/>
    <valueCode value="Patient"/>
  </parameter>
</Parameters>
```

Response:

```
HTTP/1.1 200 OK
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<CapabilityStatement xmlns="http://hl7.org/fhir">
  <id value="example"/>
  <meta>
    <tag>
      <system value="http://hl7.org/fhir/v3/ObservationValue"/>
      <code value="SUBSETTED"/> <display value="subsetted"/>
    </tag>
  </meta>
  <rest>
    <resource>
      <type value="Patient"/>
      <interaction>
        <code value="read"/>
      </interaction>
      <interaction>
        <code value="vread"/>
        <documentation value="Only supported for patient records since 12-Dec 2012"/>
      </interaction>
      <interaction>
        <code value="update"/>
      </interaction>
      <interaction>
        <code value="history-instance"/>
      </interaction>
      <interaction>
        <code value="create"/>
      </interaction>
      <interaction>
        <code value="history-type"/>
      </interaction>
      <versioning value="versioned-update"/>
      <readHistory value="true"/>
      <updateCreate value="false"/>
      <conditionalCreate value="true"/>
      <conditionalUpdate value="false"/>
      <conditionalDelete value="not-supported"/>
      <searchInclude value="Organization"/>
      <searchRevInclude value="Person"/>
      <searchParam>
        <name value="identifier"/>
        <definition value="http://hl7.org/fhir/SearchParameter/Patient-identifier"/>
        <type value="token"/>
        <documentation value="Only supports search by institution MRN"/>
      </searchParam>
      <searchParam>
        <name value="careprovider"/>
        <definition value="http://hl7.org/fhir/SearchParameter/Patient-careprovider"/>
        <type value="reference"/>
     </searchParam>
    </resource>
  </rest>
</CapabilityStatement>
```

For more information about operations, including how they are invoked, see Operations.

5.2.10.2 Operation $implements on CapabilityStatement

URL: https://hl7.org/fhir/R4/capabilitystatement-operation-implements.html

5.2.10.2 Operation $implements on CapabilityStatement

FHIR Infrastructure Work Group | Maturity Level: 5 | Trial Use | Compartments: N/A

This operation asks the server to check that it implements all the resources, interactions, search parameters, and operations that the client provides in its capability statement. The client provides its capability statement inline, or by referring the server to the canonical URL of its capability statement.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/CapabilityStatement-implements
```

Formal Definition (as a OperationDefinition).

URL: [base]/CapabilityStatement/$implements

URL: [base]/CapabilityStatement/[id]/$implements

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

server | 0..1 | canonical | | | A canonical reference to the server capability statement - use this if the implements is not invoked on an instance (or on the /metadata end-point)

client | 0..1 | canonical | | | A canonical reference to the client capability statement - use this if the implements is not invoked on an instance (or on the /metadata end-point)

resource | 0..1 | CapabilityStatement | | | The client capability statement, provided inline

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | OperationOutcome | | | Outcome of the CapabilityStatement test Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The operation does not perform a full conformance check; in particular it does not check that the profiles align. It merely checks that the behaviors the client wishes to use are provided Technically, this operation is implemented as follows:

• The server's capability statement must have an entry for each resource in the client's capability statement

• The server's resource support must have matching flags for updateCreate, conditionalCreate, conditionalRead, conditionalUpdate, conditionalDelete, searchInclude, searchRevInclude

• The server's capability statement must have a matching interaction for each interaction in the client capability statement (whether or not it is on a resource)

• The server's capability statement must have a search parameter with matching name and definition for any search parameters in the client capability statement

• The server must have an operation definition with a matching reference for any operations in the client capability statement

If the capability statements match by these rules, then the return value is a 200 OK with an operation outcome that contains no issues with severity >= error. If the capability statement doesn't match, the return value is a 4xx error, with an OperationOutcome with at least one issue with severity >= error

5.2.10.2.1 Examples

Request: Request a CapabilityStatement implements comparison between a client and server systems

```
POST [base]/CapabilityStatement/$implements
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="server"/>
    <valueUri value="http://server.example.org/fhir/metadata"/>
  </parameter>
  <parameter>
    <name value="client"/>
    <valueUri value="http://client.example.org/fhir/metadata"/>
  </parameter>
</Parameters>
```

Response:

```
HTTP/1.1 200 OK
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<OperationOutcome xmlns="http://hl7.org/fhir">
  <id value="implementsok"/>
  <text>
    <status value="additional"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>Server http://server.example.org/fhir/metadata implements client http://client.example.org/fhir/metadata
    capabilities.</p>
    </div>
  </text>
  <issue>
    <severity value="information"/>
    <code value="informational"/>
    <details>
      <text value="Server http://server.example.org/fhir/metadata implements client
    http://client.example.org/fhir/metadata capabilities."/>
    </details>
 </issue>
</OperationOutcome>
```

For more information about operations, including how they are invoked, see Operations.

5.2.10.3 Operation $conforms on CapabilityStatement

URL: https://hl7.org/fhir/R4/capabilitystatement-operation-conforms.html

5.2.10.3 Operation $conforms on CapabilityStatement

FHIR Infrastructure Work Group | Maturity Level: 5 | Trial Use | Compartments: N/A

This operation asks the server to check that it implements all the resources, interactions, search parameters, and operations that the client provides in its capability statement. The client provides both capability statements by reference, and must ensure that all the referenced resources are available to the conformance server.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/CapabilityStatement-conforms
```

Formal Definition (as a OperationDefinition).

URL: [base]/CapabilityStatement/$conforms

This is an idempotent operation

In Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
left	0..1	canonical			A canonical reference to the left-hand system's capability statement
right	0..1	canonical			A canonical reference to the right-hand system's capability statement
mode	0..1	code			What kind of comparison to perform - server to server, or client to server (use the codes 'server/server' or 'client/server')

Out Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
issues	1..1	OperationOutcome			Outcome of the CapabilityStatement test
union	0..1	CapabilityStatement			The intersection of the functionality described by the CapabilityStatement resources
intersection	0..1	CapabilityStatement			The union of the functionality described by the CapabilityStatement resources

The operation performs a full comparison of the functionality described by the two capability statements, including the profiles and value sets they reference, and also including concept maps and structure maps.

The full execution of this operation is still a matter of research, but it is intended to support comparison of systems to see if they will interoperate

If the capability statements can be successfully compared, then the return value is a 200 OK with an OperationOutcome along with intersection and union capability statements. The operation outcome can contain errors relating to differences between the capability statements. If the capability statements cannot be compared, because dependencies cannot be located, the return value is a 4xx error, with an OperationOutcome with at least one issue with severity >= error

5.2.10.3.1 Examples

Request: Request a CapabilityStatement conforms comparison between two server systems

```
POST [base]/CapabilityStatement/$conforms
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="left"/>
    <valueUri value="http://server1.example.org/fhir/metadata"/>
  </parameter>
  <parameter>
    <name value="right"/>
    <valueUri value="http://server2.example.org/fhir/metadata"/>
  </parameter>
  <parameter>
    <name value="mode"/>
    <valueCode value="server/server"/>
  </parameter>
</Parameters>
```

Response:

```
HTTP/1.1 200 OK
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<OperationOutcome xmlns="http://hl7.org/fhir">
  <id value="conformsnotok"/>
  <text>
    <status value="additional"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>Request timeout waiting for server http://server1.example.org/fhir/metadata CapabilitiesStatement.</p>
    </div>
  </text>
  <issue>
    <severity value="error"/>
    <code value="timeout"/>
    <diagnostics value="Request timeout waiting for server http://server1.example.org/fhir/metadata
    CapabilitiesStatement."/>
 </issue>
</OperationOutcome>
```

For more information about operations, including how they are invoked, see Operations.

5.2.10.4 Operation $versions on CapabilityStatement

URL: https://hl7.org/fhir/R4/capabilitystatement-operation-versions.html

5.2.10.4 Operation $versions on CapabilityStatement

FHIR Infrastructure Work Group | Maturity Level: 5 | Trial Use | Compartments: N/A

Using the FHIR Version Mime Type Parameter, a server can support multiple versions on the same end-point. The only way for client to find out what versions a server supports in this fashion is the $versions operation. The client invokes the operation with no parameters. and the server returns the list of supported versions, along with the default version it will use if no fhirVersion parameter is present.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/CapabilityStatement-versions
```

Formal Definition (as a OperationDefinition).

URL: [base]/$versions

This is an idempotent operation

Out Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
version	1..*	code			A version supported by the server. Use the major.minor version like 3.0
default	1..1	code			The default version for the server. Use the major.minor version like 3.0

5.2.10.4.1 Examples

Request: Request a list of versions supported by the server

```
GET [base]/$versions
Accept: application/fhir+xml
[other headers]
```

Response:

```
HTTP/1.1 200 OK
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="version"/>
    <valueString value="3.0"/>
  </parameter>
  <parameter>
    <name value="version"/>
    <valueString value="4.0"/>
  </parameter>
  <parameter>
    <name value="default"/>
    <valueString value="4.0"/>
  </parameter>
</Parameters>

or in JSON:

{
  "resourceType": "Parameters",
  "parameter": [{
    "name": "version",
    "valueString": "3.0"
  }, {
    "name": "version",
    "valueString": "4.0"
  }, {
    "name": "default",
    "valueString": "4.0"
  }]
}
```

For the convenience of clients, servers that implement this operation SHALL also support the relevant XML or JSON format:

Request: Request a list of versions supported by the server

```
GET [base]/$versions
Accept: application/json
[other headers]

or

GET [base]/$versions
Accept: application/xml
[other headers]
```

Response:

```
HTTP/1.1 200 OK
[other headers]

{
  "versions": ["3.0", "4.0"],
  "default" : "4.0"
```

```
}

on in XML:

<versions>
  <version>3.0</version>
  <version>4.0</version>
  <default>4.0</default>
</versions>
```

For more information about operations, including how they are invoked, see Operations.

5.2.11 Resource CapabilityStatement - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/capabilitystatement-version-maps.html

No validation errors - all conversions are clean

5.2.11 Resource CapabilityStatement - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for CapabilityStatement.

Functional status for this map: 9 tests that all execute ok. 1 fail round-trip testing and 9 r3 resources are invalid (0 errors). (see documentation)

5.2.11.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/CapabilityStatement3to4" = "R3 to R4 Conversions for CapabilityStatement"

uses "http://hl7.org/fhir/3.0/StructureDefinition/CapabilityStatement" alias CapabilityStatementR3 as source
uses "http://hl7.org/fhir/StructureDefinition/CapabilityStatement" alias CapabilityStatement as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group CapabilityStatement(source src : CapabilityStatementR3, target tgt : CapabilityStatement) extends DomainResource
    <<type+>> {
  src.url -> tgt.url;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.copyright -> tgt.copyright;
  src.kind -> tgt.kind;
  src.instantiates -> tgt.instantiates;
  src.software as vs0 -> tgt.software as vt0 then software(vs0, vt0);
  src.implementation as vs0 -> tgt.implementation as vt0 then implementation(vs0, vt0);
  src.fhirVersion -> tgt.fhirVersion;
  src.acceptUnknown as vs ->  tgt.extension as ext,  ext.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    CapabilityStatement.acceptUnknown',  ext.value = vs;
  src.format -> tgt.format;
  src.patchFormat -> tgt.patchFormat;
  src.implementationGuide -> tgt.implementationGuide;
  src.rest as vs0 -> tgt.rest as vt0 then rest(vs0, vt0);
  src.messaging as vs0 -> tgt.messaging as vt0 then messaging(vs0, vt0);
  src.document as vs0 -> tgt.document as vt0 then document(vs0, vt0);
}

group software(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.version -> tgt.version;
  src.releaseDate -> tgt.releaseDate;
}

group implementation(source src, target tgt) extends BackboneElement {
  src.description -> tgt.description;
  src.url -> tgt.url;
}

group rest(source src, target tgt) extends BackboneElement {
  src.mode -> tgt.mode;
  src.documentation -> tgt.documentation;
  src.security as vs0 -> tgt.security as vt0 then security(vs0, vt0);
  src.resource as vs0 -> tgt.resource as vt0 then resource(vs0, vt0);
  src.interaction as vs0 -> tgt.interaction as vt0 then interaction(vs0, vt0);
  src.searchParam as vs0 -> tgt.searchParam as vt0 then searchParam(vs0, vt0);
  src.operation as vs0 -> tgt.operation as vt0 then operation(vs0, vt0);
  src.compartment -> tgt.compartment;
}
```

```
group security(source src, target tgt) extends BackboneElement {
  src.cors -> tgt.cors;
  src.service -> tgt.service;
  src.description -> tgt.description;
}

group resource(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.profile -> tgt.profile;
  src.documentation -> tgt.documentation;
  src.interaction as vs0 -> tgt.interaction as vt0 then interaction(vs0, vt0);
  src.versioning -> tgt.versioning;
  src.readHistory -> tgt.readHistory;
  src.updateCreate -> tgt.updateCreate;
  src.conditionalCreate -> tgt.conditionalCreate;
  src.conditionalRead -> tgt.conditionalRead;
  src.conditionalUpdate -> tgt.conditionalUpdate;
  src.conditionalDelete -> tgt.conditionalDelete;
  src.referencePolicy -> tgt.referencePolicy;
  src.searchInclude -> tgt.searchInclude;
  src.searchRevInclude -> tgt.searchRevInclude;
  src.documentation -> tgt.documentation;
  src.searchParam as vs0 -> tgt.searchParam as vt0 then searchParam(vs0, vt0);
}

group interaction(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.documentation -> tgt.documentation;
}

group searchParam(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.definition -> tgt.definition;
  src.type -> tgt.type;
  src.documentation -> tgt.documentation;
}

group operation(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.definition -> tgt.definition;
}

group messaging(source src, target tgt) extends BackboneElement {
  src.endpoint as vs0 -> tgt.endpoint as vt0 then endpoint(vs0, vt0);
  src.reliableCache -> tgt.reliableCache;
  src.documentation -> tgt.documentation;
  src.supportedMessage as vs0 -> tgt.supportedMessage as vt0 then supportedMessage(vs0, vt0);
}

group endpoint(source src, target tgt) extends BackboneElement {
  src.protocol -> tgt.protocol;
  src.address -> tgt.address;
}

group supportedMessage(source src, target tgt) extends BackboneElement {
  src.mode -> tgt.mode;
  src.definition -> tgt.definition;
}

group document(source src, target tgt) extends BackboneElement {
  src.mode -> tgt.mode;
  src.documentation -> tgt.documentation;
  src.profile -> tgt.profile;
}
```

5.2.11.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/CapabilityStatement4to3" = "R4 to R3 Conversion for CapabilityStatement"

uses "http://hl7.org/fhir/StructureDefinition/CapabilityStatement" alias CapabilityStatement as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/CapabilityStatement" alias CapabilityStatementR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group CapabilityStatement(source src : CapabilityStatement, target tgt : CapabilityStatementR3) extends DomainResource
    <<type+>> {
  src.url -> tgt.url;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
```

```
  src.copyright -> tgt.copyright;
  src.kind -> tgt.kind;
  src.instantiates -> tgt.instantiates;
  src.software as vs0 -> tgt.software as vt0 then software(vs0, vt0);
  src.implementation as vs0 -> tgt.implementation as vt0 then implementation(vs0, vt0);
  src.fhirVersion -> tgt.fhirVersion;
  src.extension as ext where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    CapabilityStatement.acceptUnknown' then {
    ext.value : code as vs0 -> tgt.acceptUnknown = vs0 "acceptUnknown2";
  } "acceptUnknown";
  src.format -> tgt.format;
  src.patchFormat -> tgt.patchFormat;
  src.implementationGuide -> tgt.implementationGuide;
  src.rest as vs0 -> tgt.rest as vt0 then rest(vs0, vt0);
  src.messaging as vs0 -> tgt.messaging as vt0 then messaging(vs0, vt0);
  src.document as vs0 -> tgt.document as vt0 then document(vs0, vt0);
}

group software(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.version -> tgt.version;
  src.releaseDate -> tgt.releaseDate;
}

group implementation(source src, target tgt) extends BackboneElement {
  src.description -> tgt.description;
  src.url -> tgt.url;
}

group rest(source src, target tgt) extends BackboneElement {
  src.mode -> tgt.mode;
  src.documentation -> tgt.documentation;
  src.security as vs0 -> tgt.security as vt0 then security(vs0, vt0);
  src.resource as vs0 -> tgt.resource as vt0 then resource(vs0, vt0);
  src.interaction as vs0 -> tgt.interaction as vt0 then interaction(vs0, vt0);
  src.searchParam as vs0 -> tgt.searchParam as vt0 then searchParam(vs0, vt0);
  src.operation as vs0 -> tgt.operation as vt0 then operation(vs0, vt0);
  src.compartment -> tgt.compartment;
}

group security(source src, target tgt) extends BackboneElement {
  src.cors -> tgt.cors;
  src.service -> tgt.service;
  src.description -> tgt.description;
}

group resource(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.profile -> tgt.profile;
  src.documentation -> tgt.documentation;
  src.interaction as vs0 -> tgt.interaction as vt0 then interaction(vs0, vt0);
  src.versioning -> tgt.versioning;
  src.readHistory -> tgt.readHistory;
  src.updateCreate -> tgt.updateCreate;
  src.conditionalCreate -> tgt.conditionalCreate;
  src.conditionalRead -> tgt.conditionalRead;
  src.conditionalUpdate -> tgt.conditionalUpdate;
  src.conditionalDelete -> tgt.conditionalDelete;
  src.referencePolicy -> tgt.referencePolicy;
  src.searchInclude -> tgt.searchInclude;
  src.searchRevInclude -> tgt.searchRevInclude;
  src.documentation -> tgt.documentation;
  src.searchParam as vs0 -> tgt.searchParam as vt0 then searchParam(vs0, vt0);
}

group interaction(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.documentation -> tgt.documentation;
}

group searchParam(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.definition -> tgt.definition;
  src.type -> tgt.type;
  src.documentation -> tgt.documentation;
}

group operation(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.definition -> tgt.definition;
}

group messaging(source src, target tgt) extends BackboneElement {
  src.endpoint as vs0 -> tgt.endpoint as vt0 then endpoint(vs0, vt0);
  src.reliableCache -> tgt.reliableCache;
  src.documentation -> tgt.documentation;
  src.supportedMessage as vs0 -> tgt.supportedMessage as vt0 then supportedMessage(vs0, vt0);
}

group endpoint(source src, target tgt) extends BackboneElement {
```

```
  src.protocol -> tgt.protocol;
  src.address -> tgt.address;
}

group supportedMessage(source src, target tgt) extends BackboneElement {
  src.mode -> tgt.mode;
  src.definition -> tgt.definition;
}

group document(source src, target tgt) extends BackboneElement {
  src.mode -> tgt.mode;
  src.documentation -> tgt.documentation;
  src.profile -> tgt.profile;
}
```

5.2.11.3 R4 Validation Errors

5.2.12 Common CapabilityStatement extensions

URL: https://hl7.org/fhir/R4/capabilitystatement-common.html

Common extensions for use with the CapabilityStatement resource

5.2.12 Common CapabilityStatement extensions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

5.2.12.1 Content

Extensions:

capabilitystatement-expectation | expectation : Defines the level of expectation associated with a given system capability.

capabilitystatement-prohibited | prohibited : If set to true, indicates that support for the specified behavior would make a system non-conformant with the specification.

capabilitystatement-supported-system | supported-system : A code system that is supported by the system that is not defined in a value set resource.

capabilitystatement-search-parameter-combination | search-parameter-combination : This extension defines a possible search parameter combination, by listing a set of search parameters and indicating whether they are required or optional. If a search combination is specified, clients should expect that they must submit a search that meets one of the required combinations or the search will be unsuccessful. If multiple search parameter combinations are specified, a client may pick between them, and supply the minimal required parameters for any of the combinations.

capabilitystatement-websocket | websocket : Where the server provides its web socket end-point.

5.2.13 JSON Schema for CapabilityStatement

URL: https://hl7.org/fhir/R4/capabilitystatement.schema.json.html

5.2.13 JSON Schema for CapabilityStatement

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for CapabilityStatement, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/CapabilityStatement",
  "$ref": "#/definitions/CapabilityStatement",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "CapabilityStatement": {
      "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a
    particular version of FHIR that may be used as a statement of actual server functionality or a statement of required
     or desired server implementation.",
      "properties": {
        "resourceType": {
          "description": "This is a CapabilityStatement resource",
          "const": "CapabilityStatement"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
```

```
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this capability statement when it is referenced in a
specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which at which an authoritative instance of this capability statement is (or will
 be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the capability
statement is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the capability statement when it is
referenced in a specification, model, design or instance. This is an arbitrary value managed by the capability
statement author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if
a managed version is not available. There is also no expectation that versions can be placed in a lexicographical
sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the capability statement. This name should be usable as an
 identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the capability statement.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
```

```
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this capability statement. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this capability statement is authored for testing purposes
(or education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the capability statement was published. The date must
change when the business version changes and it must change if the status code changes. In addition, it should
change when the substantive content of the capability statement changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the capability statement.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the capability statement from a consumer\u0027s
perspective. Typically, this is used when the capability statement describes a desired rather than an actual
solution, for example as a formal expression of requirements as part of an RFP.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate capability statement
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the capability statement is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this capability statement is needed and why it has been designed as it
has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the capability statement and/or its contents. Copyright
statements are generally legal restrictions on the use and publishing of the capability statement.",
      "$ref": "#/definitions/markdown"
```

```
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "kind": {
      "description": "The way that this statement is intended to be used, to describe an actual running instance of
software, a particular product (kind, not instance of software) or a class of implementation (e.g. a desired
purchase).",
      "enum": [
        "instance",
        "capability",
        "requirements"
      ]
    },
    "_kind": {
      "description": "Extensions for kind",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "instantiates": {
      "description": "Reference to a canonical URL of another CapabilityStatement that this software implements.
This capability statement is a published API description that corresponds to a business service. The server may
actually implement a subset of the capability statement it claims to implement, so the capability statement must
specify the full capability details.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "imports": {
      "description": "Reference to a canonical URL of another CapabilityStatement that this software adds to. The
capability statement automatically includes everything in the other statement, and it is not duplicated, though the
server may repeat the same resources, interactions and operations to add additional details to them.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "software": {
      "description": "Software that is covered by this capability statement.  It is used when the capability
statement describes the capabilities of a particular software version, independent of an installation.",
      "$ref": "#/definitions/CapabilityStatement_Software"
    },
    "implementation": {
      "description": "Identifies a specific implementation instance that is described by the capability statement -
i.e. a particular installation, rather than the capabilities of a software program.",
      "$ref": "#/definitions/CapabilityStatement_Implementation"
    },
    "fhirVersion": {
      "description": "The version of the FHIR specification that this CapabilityStatement describes (which SHALL be
the same as the FHIR version of the CapabilityStatement itself). There is no default value.",
      "enum": [
        "0.01",
        "0.05",
        "0.06",
        "0.11",
        "0.0.80",
        "0.0.81",
        "0.0.82",
        "0.4.0",
        "0.5.0",
        "1.0.0",
        "1.0.1",
        "1.0.2",
        "1.1.0",
        "1.4.0",
        "1.6.0",
        "1.8.0",
        "3.0.0",
        "3.0.1",
        "3.3.0",
        "3.5.0",
        "4.0.0",
        "4.0.1"
      ]
    },
    "_fhirVersion": {
      "description": "Extensions for fhirVersion",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "format": {
      "description": "A list of the formats supported by this implementation using their content types.",
      "items": {
        "$ref": "#/definitions/code"
      },
      "type": "array"
    },
    "_format": {
      "description": "Extensions for format",
```

```
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "patchFormat": {
      "description": "A list of the patch formats supported by this implementation using their content types.",
      "items": {
        "$ref": "#/definitions/code"
      },
      "type": "array"
    },
    "_patchFormat": {
      "description": "Extensions for patchFormat",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "implementationGuide": {
      "description": "A list of implementation guides that the server does (or should) support in their entirety.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "rest": {
      "description": "A definition of the restful capabilities of the solution, if any.",
      "items": {
        "$ref": "#/definitions/CapabilityStatement_Rest"
      },
      "type": "array"
    },
    "messaging": {
      "description": "A description of the messaging capabilities of the solution.",
      "items": {
        "$ref": "#/definitions/CapabilityStatement_Messaging"
      },
      "type": "array"
    },
    "document": {
      "description": "A document definition.",
      "items": {
        "$ref": "#/definitions/CapabilityStatement_Document"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"CapabilityStatement_Software": {
  "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a
particular version of FHIR that may be used as a statement of actual server functionality or a statement of required
 or desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "Name the software is known by.",
```

```
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "The version identifier for the software covered by this statement.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "releaseDate": {
      "description": "Date this version of the software was released.",
      "$ref": "#/definitions/dateTime"
    },
    "_releaseDate": {
      "description": "Extensions for releaseDate",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"CapabilityStatement_Implementation": {
  "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a
particular version of FHIR that may be used as a statement of actual server functionality or a statement of required
 or desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "Information about the specific installation that this capability statement relates to.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "url": {
      "description": "An absolute base URL for the implementation.  This forms the base for REST interfaces as well
as the mailbox and document interfaces.",
      "$ref": "#/definitions/url"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "custodian": {
      "description": "The organization responsible for the management of the instance and oversight of the data on
the server at the specified URL.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false
},
"CapabilityStatement_Rest": {
  "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a
particular version of FHIR that may be used as a statement of actual server functionality or a statement of required
 or desired server implementation.",
  "properties": {
```

```
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "mode": {
      "description": "Identifies whether this portion of the statement is describing the ability to initiate or receive restful operations.",
      "enum": [
        "client",
        "server"
      ]
    },
    "_mode": {
      "description": "Extensions for mode",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "documentation": {
      "description": "Information about the system\u0027s restful capabilities that apply across all applications, such as security.",
      "$ref": "#/definitions/markdown"
    },
    "_documentation": {
      "description": "Extensions for documentation",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "security": {
      "description": "Information about security implementation from an interface perspective - what a client needs to know.",
      "$ref": "#/definitions/CapabilityStatement_Security"
    },
    "resource": {
      "description": "A specification of the restful capabilities of the solution for a specific resource type.",
      "items": {
        "$ref": "#/definitions/CapabilityStatement_Resource"
      },
      "type": "array"
    },
    "interaction": {
      "description": "A specification of restful operations supported by the system.",
      "items": {
        "$ref": "#/definitions/CapabilityStatement_Interaction1"
      },
      "type": "array"
    },
    "searchParam": {
      "description": "Search parameters that are supported for searching all resources for implementations to support and/or make use of - either references to ones defined in the specification, or additional ones defined for/by the implementation.",
      "items": {
        "$ref": "#/definitions/CapabilityStatement_SearchParam"
      },
      "type": "array"
    },
    "operation": {
      "description": "Definition of an operation or a named query together with its parameters and their meaning and type.",
      "items": {
        "$ref": "#/definitions/CapabilityStatement_Operation"
      },
      "type": "array"
    },
    "compartment": {
      "description": "An absolute URI which is a reference to the definition of a compartment that the system supports. The reference is to a CompartmentDefinition resource by its canonical URL .",
```

```
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"CapabilityStatement_Security": {
  "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a
particular version of FHIR that may be used as a statement of actual server functionality or a statement of required
 or desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "cors": {
      "description": "Server adds CORS headers when responding to requests - this enables Javascript applications to
 use the server.",
      "$ref": "#/definitions/boolean"
    },
    "_cors": {
      "description": "Extensions for cors",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "service": {
      "description": "Types of security services that are supported/required by the system.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "description": {
      "description": "General description of how security works.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"CapabilityStatement_Resource": {
  "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a
particular version of FHIR that may be used as a statement of actual server functionality or a statement of required
 or desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
```

```
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "A type of resource exposed via the restful interface.",
      "$ref": "#/definitions/code"
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "profile": {
      "description": "A specification of the profile that describes the solution\u0027s overall support for the
resource, including any constraints on cardinality, bindings, lengths or other limitations. See further discussion
in [Using Profiles](profiling.html#profile-uses).",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "supportedProfile": {
      "description": "A list of profiles that represent different use cases supported by the system. For a server,
\"supported by the system\" means the system hosts/produces a set of resources that are conformant to a particular
profile, and allows clients that use its services to search using this profile and to find appropriate data. For a
client, it means the system will search by this profile and process data according to the guidance implicit in the
profile. See further discussion in [Using Profiles](profiling.html#profile-uses).",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "documentation": {
      "description": "Additional information about the resource type used by the system.",
      "$ref": "#/definitions/markdown"
    },
    "_documentation": {
      "description": "Extensions for documentation",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "interaction": {
      "description": "Identifies a restful operation supported by the solution.",
      "items": {
        "$ref": "#/definitions/CapabilityStatement_Interaction"
      },
      "type": "array"
    },
    "versioning": {
      "description": "This field is set to no-version to specify that the system does not support (server) or use
(client) versioning for this resource type. If this has some other value, the server must at least correctly track
and populate the versionId meta-property on resources. If the value is \u0027versioned-update\u0027, then the server
 supports all the versioning features, including using e-tags for version integrity in the API.",
      "enum": [
        "no-version",
        "versioned",
        "versioned-update"
      ]
    },
    "_versioning": {
      "description": "Extensions for versioning",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "readHistory": {
      "description": "A flag for whether the server is able to return past versions as part of the vRead
operation.",
      "$ref": "#/definitions/boolean"
    },
    "_readHistory": {
      "description": "Extensions for readHistory",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "updateCreate": {
      "description": "A flag to indicate that the server allows or needs to allow the client to create new
identities on the server (that is, the client PUTs to a location where there is no existing resource). Allowing this
 operation means that the server allows the client to create new identities on the server.",
      "$ref": "#/definitions/boolean"
    },
    "_updateCreate": {
      "description": "Extensions for updateCreate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "conditionalCreate": {
```

```
  "description": "A flag that indicates that the server supports conditional create.",
  "$ref": "#/definitions/boolean"
},
"_conditionalCreate": {
  "description": "Extensions for conditionalCreate",
  "$ref": "Element.schema.json#/definitions/Element"
},
"conditionalRead": {
  "description": "A code that indicates how the server supports conditional read.",
  "enum": [
    "not-supported",
    "modified-since",
    "not-match",
    "full-support"
  ]
},
"_conditionalRead": {
  "description": "Extensions for conditionalRead",
  "$ref": "Element.schema.json#/definitions/Element"
},
"conditionalUpdate": {
  "description": "A flag that indicates that the server supports conditional update.",
  "$ref": "#/definitions/boolean"
},
"_conditionalUpdate": {
  "description": "Extensions for conditionalUpdate",
  "$ref": "Element.schema.json#/definitions/Element"
},
"conditionalDelete": {
  "description": "A code that indicates how the server supports conditional delete.",
  "enum": [
    "not-supported",
    "single",
    "multiple"
  ]
},
"_conditionalDelete": {
  "description": "Extensions for conditionalDelete",
  "$ref": "Element.schema.json#/definitions/Element"
},
"referencePolicy": {
  "description": "A set of flags that defines how references are supported.",
  "items": {
    "enum": [
      "literal",
      "logical",
      "resolves",
      "enforced",
      "local"
    ]
  },
  "type": "array"
},
"_referencePolicy": {
  "description": "Extensions for referencePolicy",
  "items": {
    "$ref": "Element.schema.json#/definitions/Element"
  },
  "type": "array"
},
"searchInclude": {
  "description": "A list of _include values supported by the server.",
  "items": {
    "$ref": "#/definitions/string"
  },
  "type": "array"
},
"_searchInclude": {
  "description": "Extensions for searchInclude",
  "items": {
    "$ref": "Element.schema.json#/definitions/Element"
  },
  "type": "array"
},
"searchRevInclude": {
  "description": "A list of _revinclude (reverse include) values supported by the server.",
  "items": {
    "$ref": "#/definitions/string"
  },
  "type": "array"
},
"_searchRevInclude": {
  "description": "Extensions for searchRevInclude",
  "items": {
    "$ref": "Element.schema.json#/definitions/Element"
  },
  "type": "array"
},
"searchParam": {
```

```
      "description": "Search parameters for implementations to support and/or make use of - either references to
ones defined in the specification, or additional ones defined for/by the implementation.",
      "items": {
        "$ref": "#/definitions/CapabilityStatement_SearchParam"
      },
      "type": "array"
    },
    "operation": {
      "description": "Definition of an operation or a named query together with its parameters and their meaning and
 type. Consult the definition of the operation for details about how to invoke the operation, and the parameters.",
      "items": {
        "$ref": "#/definitions/CapabilityStatement_Operation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"CapabilityStatement_Interaction": {
  "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a
particular version of FHIR that may be used as a statement of actual server functionality or a statement of required
 or desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "Coded identifier of the operation, supported by the system resource.",
      "enum": [
        "read",
        "vread",
        "update",
        "patch",
        "delete",
        "history-instance",
        "history-type",
        "create",
        "search-type"
      ]
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "documentation": {
      "description": "Guidance specific to the implementation of this operation, such as \u0027delete is a logical
delete\u0027 or \u0027updates are only allowed with version id\u0027 or \u0027creates permitted from pre-authorized
certificates only\u0027.",
      "$ref": "#/definitions/markdown"
    },
    "_documentation": {
      "description": "Extensions for documentation",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"CapabilityStatement_SearchParam": {
  "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a
particular version of FHIR that may be used as a statement of actual server functionality or a statement of required
 or desired server implementation.",
  "properties": {
    "id": {
```

```
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "The name of the search parameter used in the interface.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "definition": {
      "description": "An absolute URI that is a formal reference to where this parameter was first defined, so that a client can be confident of the meaning of the search parameter (a reference to [[[SearchParameter.url]]]). This element SHALL be populated if the search parameter refers to a SearchParameter defined by the FHIR core specification or externally defined IGs.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "type": {
      "description": "The type of value a search parameter refers to, and how the content is interpreted.",
      "enum": [
        "number",
        "date",
        "string",
        "token",
        "reference",
        "composite",
        "quantity",
        "uri",
        "special"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "documentation": {
      "description": "This allows documentation of any distinct behaviors about how the search parameter is used. For example, text matching algorithms.",
      "$ref": "#/definitions/markdown"
    },
    "_documentation": {
      "description": "Extensions for documentation",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"CapabilityStatement_Operation": {
  "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a particular version of FHIR that may be used as a statement of actual server functionality or a statement of required or desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
```

```
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "The name of the operation or query. For an operation, this is the name  prefixed with $ and
used in the URL. For a query, this is the name used in the _query parameter when the query is called.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "definition": {
      "description": "Where the formal definition can be found. If a server references the base definition of an
Operation (i.e. from the specification itself such as ```http://hl7.org/fhir/OperationDefinition/ValueSet-
expand```), that means it supports the full capabilities of the operation - e.g. both GET and POST invocation.  If
it only supports a subset, it must define its own custom [[[OperationDefinition]]] with a \u0027base\u0027 of the
original OperationDefinition.  The custom definition would describe the specific subset of functionality
supported.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "documentation": {
      "description": "Documentation that describes anything special about the operation behavior, possibly detailing
 different behavior for system, type and instance-level invocation of the operation.",
      "$ref": "#/definitions/markdown"
    },
    "_documentation": {
      "description": "Extensions for documentation",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false,
  "required": [
    "definition"
  ]
},
"CapabilityStatement_Interaction1": {
  "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a
particular version of FHIR that may be used as a statement of actual server functionality or a statement of required
 or desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "A coded identifier of the operation, supported by the system.",
      "enum": [
        "transaction",
```

```
        "batch",
        "search-system",
        "history-system"
      ]
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "documentation": {
      "description": "Guidance specific to the implementation of this operation, such as limitations on the kind of
transactions allowed, or information about system wide search is implemented.",
      "$ref": "#/definitions/markdown"
    },
    "_documentation": {
      "description": "Extensions for documentation",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"CapabilityStatement_Messaging": {
  "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a
particular version of FHIR that may be used as a statement of actual server functionality or a statement of required
 or desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "endpoint": {
      "description": "An endpoint (network accessible address) to which messages and/or replies are to be sent.",
      "items": {
        "$ref": "#/definitions/CapabilityStatement_Endpoint"
      },
      "type": "array"
    },
    "reliableCache": {
      "description": "Length if the receiver\u0027s reliable messaging cache in minutes (if a receiver) or how long
the cache length on the receiver should be (if a sender).",
      "$ref": "#/definitions/unsignedInt"
    },
    "_reliableCache": {
      "description": "Extensions for reliableCache",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "documentation": {
      "description": "Documentation about the system\u0027s messaging capabilities for this endpoint not otherwise
documented by the capability statement.  For example, the process for becoming an authorized messaging exchange
partner.",
      "$ref": "#/definitions/markdown"
    },
    "_documentation": {
      "description": "Extensions for documentation",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "supportedMessage": {
      "description": "References to message definitions for messages this system can send or receive.",
      "items": {
        "$ref": "#/definitions/CapabilityStatement_SupportedMessage"
      },
      "type": "array"
    }
  },
```

```
  "additionalProperties": false
},
"CapabilityStatement_Endpoint": {
  "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a
particular version of FHIR that may be used as a statement of actual server functionality or a statement of required
 or desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "protocol": {
      "description": "A list of the messaging transport protocol(s) identifiers, supported by this endpoint.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "address": {
      "description": "The network address of the endpoint. For solutions that do not use network addresses for
routing, it can be just an identifier.",
      "$ref": "#/definitions/url"
    },
    "_address": {
      "description": "Extensions for address",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false,
  "required": [
    "protocol"
  ]
},
"CapabilityStatement_SupportedMessage": {
  "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a
particular version of FHIR that may be used as a statement of actual server functionality or a statement of required
 or desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
```

```
      "mode": {
        "description": "The mode of this event declaration - whether application is sender or receiver.",
        "enum": [
          "sender",
          "receiver"
        ]
      },
      "_mode": {
        "description": "Extensions for mode",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "definition": {
        "description": "Points to a message definition that identifies the messaging event, message structure, allowed
 responses, etc.",
        "$ref": "canonical.schema.json#/definitions/canonical"
      }
    },
    "additionalProperties": false,
    "required": [
      "definition"
    ]
  },
  "CapabilityStatement_Document": {
    "description": "A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server for a
particular version of FHIR that may be used as a statement of actual server functionality or a statement of required
 or desired server implementation.",
    "properties": {
      "id": {
        "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
        "$ref": "string.schema.json#/definitions/string"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "mode": {
        "description": "Mode of this document declaration - whether an application is a producer or consumer.",
        "enum": [
          "producer",
          "consumer"
        ]
      },
      "_mode": {
        "description": "Extensions for mode",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "documentation": {
        "description": "A description of how the application supports or uses the specified document profile.  For
example, when documents are created, what action is taken with consumed documents, etc.",
        "$ref": "#/definitions/markdown"
      },
      "_documentation": {
        "description": "Extensions for documentation",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "profile": {
        "description": "A profile on the document Bundle that constrains which resources are present, and their
contents.",
        "$ref": "canonical.schema.json#/definitions/canonical"
      }
    },
    "additionalProperties": false,
    "required": [
      "profile"
    ]
  }
 }
}
```

5.2.14 Resource CapabilityStatement - Maturity Dependency View

URL: https://hl7.org/fhir/R4/capabilitystatement-dependencies.html

5.2.14 Resource CapabilityStatement - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| Some of the content on this page (marked clearly) has been approved as part of an ANSI standard. See the Conformance Package for further details.

Dependency Graph for CapabilityStatement FMM level 5

Name | Card. | Type | Dependency Analysis
CapabilityStatement | | DomainResource
url | 0..1 | uri
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
date | 1..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext TU | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
copyright | 0..1 | markdown
kind | 1..1 | code
instantiates | 0..* | canonical(CapabilityStatement)
imports TU | 0..* | canonical(CapabilityStatement)
software | 0..1 | BackboneElement
name | 1..1 | string
version | 0..1 | string
releaseDate | 0..1 | dateTime
implementation | 0..1 | BackboneElement
description | 1..1 | string
url | 0..1 | url
custodian TU | 0..1 | Reference(Organization) | Type Warning: (Organization = FMM3-Trial Use vs. Element = FMM5-Trial Use)
fhirVersion | 1..1 | code
format | 1..* | code
patchFormat | 0..* | code
implementationGuide | 0..* | canonical(ImplementationGuide) | Type Warning: (ImplementationGuide = FMM1-Trial Use vs. Element = FMM5-Normative)
rest | 0..* | BackboneElement
mode | 1..1 | code
documentation | 0..1 | markdown
security TU | 0..1 | BackboneElement
cors | 0..1 | boolean
service | 0..* | CodeableConcept
description | 0..1 | markdown
resource | 0..* | BackboneElement
type | 1..1 | code
profile | 0..1 | canonical(StructureDefinition)
supportedProfile TU | 0..* | canonical(StructureDefinition)
documentation | 0..1 | markdown
interaction | 0..* | BackboneElement
code | 1..1 | code
documentation | 0..1 | markdown
versioning TU | 0..1 | code
readHistory TU | 0..1 | boolean
updateCreate TU | 0..1 | boolean
conditionalCreate TU | 0..1 | boolean
conditionalRead TU | 0..1 | code
conditionalUpdate TU | 0..1 | boolean
conditionalDelete TU | 0..1 | code
referencePolicy TU | 0..* | code
searchInclude TU | 0..* | string
searchRevInclude TU | 0..* | string
searchParam | 0..* | BackboneElement
name | 1..1 | string

definition | 0..1 | canonical(SearchParameter) | Type Warning: (SearchParameter = FMM3-Trial Use vs. Element = FMM5-Normative)
type | 1..1 | code
documentation | 0..1 | markdown
operation | 0..* | BackboneElement
name | 1..1 | string
definition | 1..1 | canonical(OperationDefinition)
documentation | 0..1 | markdown
interaction | 0..* | BackboneElement
code | 1..1 | code
documentation | 0..1 | markdown
searchParam | 0..* | see searchParam
operation | 0..* | see operation
compartment | 0..* | canonical(CompartmentDefinition) | Type Warning: (CompartmentDefinition = FMM1-Trial Use vs. Element = FMM5-Normative)
messaging TU | 0..* | BackboneElement
endpoint | 0..* | BackboneElement
protocol | 1..1 | Coding
address | 1..1 | url
reliableCache | 0..1 | unsignedInt
documentation | 0..1 | markdown
supportedMessage | 0..* | BackboneElement
mode | 1..1 | code
definition | 1..1 | canonical(MessageDefinition) | Type Warning: (MessageDefinition = FMM1-Trial Use vs. Element = FMM5-Normative)
document TU | 0..* | BackboneElement
mode | 1..1 | code
documentation | 0..1 | markdown
profile | 1..1 | canonical(StructureDefinition)

5.3 Resource StructureDefinition - Content

URL: https://hl7.org/fhir/R4/structuredefinition.html
FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Conformance Package for further details.

A definition of a FHIR structure. This resource is used to describe the underlying resources, data types defined in FHIR, and also for describing extensions and constraints on resources and data types.

5.3.1 Scope and Usage

The StructureDefinition resource describes a structure - a set of data element definitions, and their associated rules of usage. These structure definitions are used to describe both the content defined in the FHIR specification itself - Resources, data types, the underlying infrastructural types, and also are used to describe how these structures are used in implementations. This allows the definitions of the structures to be shared and published through repositories of structure definitions, compared with each other, and used as the basis for code, report and UI generation.

Note that as part of the specification itself, a full set of structure definitions for all resources and data types is published.

5.3.2 Boundaries and Relationships

• StructureDefinitions are used by CapabilityStatement instances for specifying how resources are used

• StructureDefinitions use Value Sets to specify the content of coded elements

5.3.3 Background and Context

Implementers should be familiar with the background and concepts described in Profiling FHIR before working with this resource.

5.3.3.1 Metadata

The StructureDefinition resource has a set of metadata that is mostly shared with the Value Set, CapabilityStatement and other infrastructure resources. The metadata describes the structure and helps find the structure when registered in repositories.

url | The identifier that is used to identify this structure when it is referenced in a specification, model, design or an instance. This URL is where the structure can be accessed
identifier | Other identifiers that are used to identify this structure
version | The identifier that is used to identify this version of the structure when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the structure author manually and the value should be a timestamp. Note that there may be multiple resource versions of the structure that have this same identifier. The resource will have updates that create new versions for technical reasons, whereas the stated version number needs to be under the author's control

name | A Computer-ready name (e.g. a token) that identifies the structure - suitable for code generation. Note that this name (and other names relevant for code generation, including element & slice names, codes etc) may collide with reserved words in the relevant target language, and code generators will need to handle this
title | A free text natural language name identifying the structure
status | The status of the structure allows filtering of StructureDefinitions that are appropriate for use vs. not. See the Status Codes
experimental | This structure was authored for testing purposes (or education/evaluation/marketing), and is not intended for genuine usage
date | The date this version of the structure was published
publisher | Details of the individual or organization who accepts responsibility for publishing the structure. This helps establish the "authority/credibility" of the structure.
contact | Contact details to assist a user in finding and communicating with the publisher
description | A free text natural language description of the structure and its use
purpose | Why this structure was created - what the intent of it is
useContext & Jurisdiction | Computable details about purpose and scope of use
keyword | A set of terms from external terminologies that may be used to assist with indexing and searching of StructureDefinitionss
copyright | Use and/or publishing restrictions
fhirVersion | The version of the FHIR specification on which this structure is based. It is not necessary to specify the version, as most SructureDefinitions are valid across multiple versions, and the validity of a structure against a particular version of FHIR can easily be checked by tooling.
type | The type the structure describes.
contextType & context | For extensions, the types of contexts in which the extension can be used. For further details, see Defining Extensions

Notes:

• The name and title of the structure are not required to be globally unique, but the name amd title should have some scoping information (e.g. AcmeAllergies / Acme Inc. (USA), Allergy List)

• Multiple keywords may be assigned to the structure. These may either describe the structure, the focus of the structure or both. They are solely to help find the structure by searching for structured concepts

• The 3 status codes (draft, active, and retired) are the codes that are relevant to structure consumers. Authors may wish to use the authoring-status extension to track the life cycle of a structure as it is prepared

This resource is referenced by DataRequirement, ElementDefinition, Meta, ParameterDefinition, ActivityDefinition, CapabilityStatement, GraphDefinition, ImplementationGuide, MessageDefinition, OperationDefinition, itself and StructureMap

5.3.4 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
StructureDefinition | IN | | DomainResource | Structural Definition + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: Element paths must be unique unless the structure is a constraint + Rule: If the first element in a differential has no "." in the path and it's not a logical model, it has no type + Rule: If the structure is not abstract, then there SHALL be a baseDefinition + Rule: If the structure defines an extension then the structure must have context information + Rule: A structure must have either a differential, or a snapshot (or both) + Rule: In any snapshot or differential, no label, code or requirements on an element without a "." in the path (e.g. the first element) + Rule: If there's a type, its content must match the path name in the first element of a snapshot + Rule: All element definitions must have an id + Rule: The first element in a snapshot has no type unless model is a logical model. + Rule: All element definitions must have unique ids (snapshot) + Rule: All element definitions must have unique ids (diff) + Rule: Context Invariants can only be used for extensions + Rule: FHIR Specification models only use FHIR defined types + Rule: Default values can only be specified on specializations + Rule: FHIR Specification models never have default values + Rule: No slice name on root Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 1..1 | uri | Canonical identifier for this structure definition, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the structure definition
version | Σ | 0..1 | string | Business version of the structure definition
name | ΣI | 1..1 | string | Name for this structure definition (computer friendly)
title | Σ | 0..1 | string | Name for this structure definition (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the structure definition

useContext | ΣTU | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for structure definition (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this structure definition is defined
copyright | | 0..1 | markdown | Use and/or publishing restrictions
keyword | Σ | 0..* | Coding | Assist with indexing and finding Structure Definition Use Codes / Keywords (Extensible)
fhirVersion | Σ | 0..1 | code | FHIR Version this StructureDefinition targets FHIRVersion (Required)
mapping | I | 0..* | BackboneElement | External specification that the content is mapped to + Rule: Must have at least a name or a uri (or both)
identity | | 1..1 | id | Internal id when this mapping is used
uri | I | 0..1 | uri | Identifies what this mapping refers to
name | I | 0..1 | string | Names what this mapping refers to
comment | | 0..1 | string | Versions, Issues, Scope limitations etc.
kind | Σ | 1..1 | code | primitive-type | complex-type | resource | logical StructureDefinitionKind (Required)
abstract | Σ | 1..1 | boolean | Whether the structure is abstract
context | ΣI | 0..* | BackboneElement | If an extension, where it can be used in instances
type | Σ | 1..1 | code | fhirpath | element | extension ExtensionContextType (Required)
expression | Σ | 1..1 | string | Where the extension can be used in instances
contextInvariant | ΣI | 0..* | string | FHIRPath invariants - when the extension can be used
type | ΣI | 1..1 | uri | Type defined or constrained by this structure FHIRDefinedType (Extensible)
baseDefinition | ΣI | 0..1 | canonical(StructureDefinition) | Definition that this type is constrained/specialized from
derivation | Σ | 0..1 | code | specialization | constraint - How relates to base definition TypeDerivationRule (Required)
snapshot | I | 0..1 | BackboneElement | Snapshot view of the structure + Rule: Each element definition in a snapshot must have a formal definition and cardinalities + Rule: All snapshot elements must start with the StructureDefinition's specified type for non-logical models, or with the same type name for logical models + Rule: All snapshot elements must have a base definition
element | I | 1..* | ElementDefinition | Definition of elements in the resource (if no StructureDefinition) + Rule: provide either a binding reference or a description (or both)
differential | I | 0..1 | BackboneElement | Differential view of the structure + Rule: No slicing on the root element + Rule: In any differential, all the elements must start with the StructureDefinition's specified type for non-logical models, or with the same type name for logical models
element | | 1..* | ElementDefinition | Definition of elements in the resource (if no StructureDefinition)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<StructureDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Canonical identifier for this structure definition, represented as a URI (globally
    unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the structure definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the structure definition -->
 <name value="[string]"/><!--  1..1 Name for this structure definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this structure definition (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the structure definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for structure definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this structure definition is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <keyword><!-- 0..* Coding Assist with indexing and finding --></keyword>
 <fhirVersion value="[code]"/><!-- 0..1 FHIR Version this StructureDefinition targets -->
 <mapping>  <!-- 0..* External specification that the content is mapped to -->
  <identity value="[id]"/><!-- 1..1 Internal id when this mapping is used -->
  <uri value="[uri]"/><!--  0..1 Identifies what this mapping refers to -->
  <name value="[string]"/><!--  0..1 Names what this mapping refers to -->
  <comment value="[string]"/><!-- 0..1 Versions, Issues, Scope limitations etc. -->
 </mapping>
 <kind value="[code]"/><!-- 1..1 primitive-type | complex-type | resource | logical -->
 <abstract value="[boolean]"/><!-- 1..1 Whether the structure is abstract -->
 <context>  <!--  0..* If an extension, where it can be used in instances -->
  <type value="[code]"/><!-- 1..1 fhirpath | element | extension -->
  <expression value="[string]"/><!-- 1..1 Where the extension can be used in instances -->
 </context>
 <contextInvariant value="[string]"/><!--  0..* FHIRPath invariants - when the extension can be used -->
 <type value="[uri]"/><!--  1..1 Type defined or constrained by this structure -->
 <baseDefinition><!--  0..1 canonical(StructureDefinition) Definition that this type is constrained/specialized from
    --></baseDefinition>
 <derivation value="[code]"/><!-- 0..1 specialization | constraint - How relates to base definition -->
 <snapshot>  <!--  0..1 Snapshot view of the structure -->
  <element><!--  1..* ElementDefinition Definition of elements in the resource (if no StructureDefinition) --></element>
 </snapshot>
 <differential>  <!--  0..1 Differential view of the structure -->
  <element><!-- 1..* ElementDefinition Definition of elements in the resource (if no StructureDefinition) --></element>
 </differential>
</StructureDefinition>
```

JSON Template

```
{
  "resourceType" : "StructureDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Canonical identifier for this structure definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the structure definition
  "version" : "<string>", // Business version of the structure definition
  "name" : "<string>", // C? R!  Name for this structure definition (computer friendly)
  "title" : "<string>", // Name for this structure definition (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the structure definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for structure definition (if applicable)
  "purpose" : "<markdown>", // Why this structure definition is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "keyword" : [{ Coding }], // Assist with indexing and finding
  "fhirVersion" : "<code>", // FHIR Version this StructureDefinition targets
  "mapping" : [{ // External specification that the content is mapped to
    "identity" : "<id>", // R!  Internal id when this mapping is used
    "uri" : "<uri>", // C? Identifies what this mapping refers to
    "name" : "<string>", // C? Names what this mapping refers to
    "comment" : "<string>" // Versions, Issues, Scope limitations etc.
  }],
  "kind" : "<code>", // R!  primitive-type | complex-type | resource | logical
  "abstract" : <boolean>, // R!  Whether the structure is abstract
  "context" : [{ // C? If an extension, where it can be used in instances
    "type" : "<code>", // R!  fhirpath | element | extension
    "expression" : "<string>" // R!  Where the extension can be used in instances
  }],
  "contextInvariant" : ["<string>"], // C? FHIRPath invariants - when the extension can be used
  "type" : "<uri>", // C? R!  Type defined or constrained by this structure
  "baseDefinition" : { canonical(StructureDefinition) }, // C? Definition that this type is constrained/specialized from
  "derivation" : "<code>", // specialization | constraint - How relates to base definition
  "snapshot" : { // C? Snapshot view of the structure
    "element" : [{ ElementDefinition }] // C? R!  Definition of elements in the resource (if no StructureDefinition)
  },
  "differential" : { // C? Differential view of the structure
    "element" : [{ ElementDefinition }] // R!  Definition of elements in the resource (if no StructureDefinition)
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:StructureDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:StructureDefinition.url [ uri ]; # 1..1 Canonical identifier for this structure definition, represented as a URI
    (globally unique)
  fhir:StructureDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the structure definition
  fhir:StructureDefinition.version [ string ]; # 0..1 Business version of the structure definition
  fhir:StructureDefinition.name [ string ]; # 1..1 Name for this structure definition (computer friendly)
  fhir:StructureDefinition.title [ string ]; # 0..1 Name for this structure definition (human friendly)
  fhir:StructureDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:StructureDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:StructureDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:StructureDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:StructureDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:StructureDefinition.description [ markdown ]; # 0..1 Natural language description of the structure definition
  fhir:StructureDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:StructureDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for structure definition
     (if applicable)
  fhir:StructureDefinition.purpose [ markdown ]; # 0..1 Why this structure definition is defined
  fhir:StructureDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:StructureDefinition.keyword [ Coding ], ... ; # 0..* Assist with indexing and finding
  fhir:StructureDefinition.fhirVersion [ code ]; # 0..1 FHIR Version this StructureDefinition targets
  fhir:StructureDefinition.mapping [ # 0..* External specification that the content is mapped to
    fhir:StructureDefinition.mapping.identity [ id ]; # 1..1 Internal id when this mapping is used
    fhir:StructureDefinition.mapping.uri [ uri ]; # 0..1 Identifies what this mapping refers to
    fhir:StructureDefinition.mapping.name [ string ]; # 0..1 Names what this mapping refers to
    fhir:StructureDefinition.mapping.comment [ string ]; # 0..1 Versions, Issues, Scope limitations etc.
  ], ...;
  fhir:StructureDefinition.kind [ code ]; # 1..1 primitive-type | complex-type | resource | logical
  fhir:StructureDefinition.abstract [ boolean ]; # 1..1 Whether the structure is abstract
  fhir:StructureDefinition.context [ # 0..* If an extension, where it can be used in instances
    fhir:StructureDefinition.context.type [ code ]; # 1..1 fhirpath | element | extension
    fhir:StructureDefinition.context.expression [ string ]; # 1..1 Where the extension can be used in instances
  ], ...;
  fhir:StructureDefinition.contextInvariant [ string ], ... ; # 0..* FHIRPath invariants - when the extension can be
    used
  fhir:StructureDefinition.type [ uri ]; # 1..1 Type defined or constrained by this structure
  fhir:StructureDefinition.baseDefinition [ canonical(StructureDefinition) ]; # 0..1 Definition that this type is
```

```
    constrained/specialized from
  fhir:StructureDefinition.derivation [ code ]; # 0..1 specialization | constraint - How relates to base definition
  fhir:StructureDefinition.snapshot [ # 0..1 Snapshot view of the structure
    fhir:StructureDefinition.snapshot.element [ ElementDefinition ], ... ; # 1..* Definition of elements in the resource
     (if no StructureDefinition)
  ];
  fhir:StructureDefinition.differential [ # 0..1 Differential view of the structure
    fhir:StructureDefinition.differential.element [ ElementDefinition ], ... ; # 1..* Definition of elements in the
    resource (if no StructureDefinition)
  ];
]
```

Changes since R3

StructureDefinition

StructureDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

StructureDefinition.experimental | No longer marked as Modifier

StructureDefinition.keyword | Add Binding http://hl7.org/fhir/ValueSet/definition-use (extensible)

StructureDefinition.fhirVersion | Type changed from id to code Add Binding http://hl7.org/fhir/ValueSet/FHIR-version|4.0.1 (required)

StructureDefinition.kind | Change value set from http://hl7.org/fhir/ValueSet/structure-definition-kind to http://hl7.org/fhir/ValueSet/structure-definition-kind|4.0.1

StructureDefinition.context | Type changed from string to BackboneElement

StructureDefinition.context.type | Added Mandatory Element

StructureDefinition.context.expression | Added Mandatory Element

StructureDefinition.type | Type changed from code to uri

StructureDefinition.baseDefinition | Type changed from uri to canonical(StructureDefinition)

StructureDefinition.derivation | Change value set from http://hl7.org/fhir/ValueSet/type-derivation-rule to http://hl7.org/fhir/ValueSet/type-derivation-rule|4.0.1

StructureDefinition.contextType | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 206 tests that all execute ok. 4 fail round-trip testing and 206 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

StructureDefinition | IN | | DomainResource | Structural Definition + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: Element paths must be unique unless the structure is a constraint + Rule: If the first element in a differential has no "." in the path and it's not a logical model, it has no type + Rule: If the structure is not abstract, then there SHALL be a baseDefinition + Rule: If the structure defines an extension then the structure must have context information + Rule: A structure must have either a differential, or a snapshot (or both) + Rule: In any snapshot or differential, no label, code or requirements on an element without a "." in the path (e.g. the first element) + Rule: If there's a type, its content must match the path name in the first element of a snapshot + Rule: All element definitions must have an id + Rule: The first element in a snapshot has no type unless model is a logical model. + Rule: All element definitions must have unique ids (snapshot) + Rule: All element definitions must have unique ids (diff) + Rule: Context Invariants can only be used for extensions + Rule: FHIR Specification models only use FHIR defined types + Rule: Default values can only be specified on specializations + Rule: FHIR Specification models never have default values + Rule: No slice name on root Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | Σ | 1..1 | uri | Canonical identifier for this structure definition, represented as a URI (globally unique)

identifier | Σ | 0..* | Identifier | Additional identifier for the structure definition

version | Σ | 0..1 | string | Business version of the structure definition

name | ΣI | 1..1 | string | Name for this structure definition (computer friendly)

title | Σ | 0..1 | string | Name for this structure definition (human friendly)

status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | Σ | 0..1 | boolean | For testing purposes, not real usage

date | Σ | 0..1 | dateTime | Date last changed

publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)

contact | Σ | 0..* | ContactDetail | Contact details for the publisher

description | | 0..1 | markdown | Natural language description of the structure definition

useContext | ΣTU | 0..* | UsageContext | The context that the content is intended to support

jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for structure definition (if applicable) Jurisdiction (Extensible)

purpose | | 0..1 | markdown | Why this structure definition is defined

copyright | | 0..1 | markdown | Use and/or publishing restrictions

keyword | Σ | 0..* | Coding | Assist with indexing and finding Structure Definition Use Codes / Keywords (Extensible)

fhirVersion | Σ | 0..1 | code | FHIR Version this StructureDefinition targets FHIRVersion (Required)

mapping | I | 0..* | BackboneElement | External specification that the content is mapped to + Rule: Must have at least a name or a uri (or both)

identity | | 1..1 | id | Internal id when this mapping is used

uri | I | 0..1 | uri | Identifies what this mapping refers to

name | I | 0..1 | string | Names what this mapping refers to

comment | | 0..1 | string | Versions, Issues, Scope limitations etc.

kind | Σ | 1..1 | code | primitive-type | complex-type | resource | logical StructureDefinitionKind (Required)

abstract | Σ | 1..1 | boolean | Whether the structure is abstract

context | ΣI | 0..* | BackboneElement | If an extension, where it can be used in instances

type | Σ | 1..1 | code | fhirpath | element | extension ExtensionContextType (Required)
expression | Σ | 1..1 | string | Where the extension can be used in instances
contextInvariant | ΣI | 0..* | string | FHIRPath invariants - when the extension can be used
type | ΣI | 1..1 | uri | Type defined or constrained by this structure FHIRDefinedType (Extensible)
baseDefinition | ΣI | 0..1 | canonical(StructureDefinition) | Definition that this type is constrained/specialized from
derivation | Σ | 0..1 | code | specialization | constraint - How relates to base definition TypeDerivationRule (Required)
snapshot | I | 0..1 | BackboneElement | Snapshot view of the structure + Rule: Each element definition in a snapshot must have a formal definition and cardinalities + Rule: All snapshot elements must start with the StructureDefinition's specified type for non-logical models, or with the same type name for logical models + Rule: All snapshot elements must have a base definition
element | I | 1..* | ElementDefinition | Definition of elements in the resource (if no StructureDefinition) + Rule: provide either a binding reference or a description (or both)
differential | I | 0..1 | BackboneElement | Differential view of the structure + Rule: No slicing on the root element + Rule: In any differential, all the elements must start with the StructureDefinition's specified type for non-logical models, or with the same type name for logical models
element | | 1..* | ElementDefinition | Definition of elements in the resource (if no StructureDefinition)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<StructureDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Canonical identifier for this structure definition, represented as a URI (globally
    unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the structure definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the structure definition -->
 <name value="[string]"/><!--  1..1 Name for this structure definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this structure definition (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the structure definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for structure definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this structure definition is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <keyword><!-- 0..* Coding Assist with indexing and finding --></keyword>
 <fhirVersion value="[code]"/><!-- 0..1 FHIR Version this StructureDefinition targets -->
 <mapping>  <!-- 0..* External specification that the content is mapped to -->
  <identity value="[id]"/><!-- 1..1 Internal id when this mapping is used -->
  <uri value="[uri]"/><!--  0..1 Identifies what this mapping refers to -->
  <name value="[string]"/><!--  0..1 Names what this mapping refers to -->
  <comment value="[string]"/><!-- 0..1 Versions, Issues, Scope limitations etc. -->
 </mapping>
 <kind value="[code]"/><!-- 1..1 primitive-type | complex-type | resource | logical -->
 <abstract value="[boolean]"/><!-- 1..1 Whether the structure is abstract -->
 <context>  <!--  0..* If an extension, where it can be used in instances -->
  <type value="[code]"/><!-- 1..1 fhirpath | element | extension -->
  <expression value="[string]"/><!-- 1..1 Where the extension can be used in instances -->
 </context>
 <contextInvariant value="[string]"/><!--  0..* FHIRPath invariants - when the extension can be used -->
 <type value="[uri]"/><!--  1..1 Type defined or constrained by this structure -->
 <baseDefinition><!--  0..1 canonical(StructureDefinition) Definition that this type is constrained/specialized from
    --></baseDefinition>
 <derivation value="[code]"/><!-- 0..1 specialization | constraint - How relates to base definition -->
 <snapshot>  <!--  0..1 Snapshot view of the structure -->
  <element><!--  1..* ElementDefinition Definition of elements in the resource (if no StructureDefinition) --></element>
 </snapshot>
 <differential>  <!--  0..1 Differential view of the structure -->
  <element><!-- 1..* ElementDefinition Definition of elements in the resource (if no StructureDefinition) --></element>
 </differential>
</StructureDefinition>
```

JSON Template

```
{
  "resourceType" : "StructureDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Canonical identifier for this structure definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the structure definition
  "version" : "<string>", // Business version of the structure definition
  "name" : "<string>", // C? R!  Name for this structure definition (computer friendly)
  "title" : "<string>", // Name for this structure definition (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the structure definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for structure definition (if applicable)
  "purpose" : "<markdown>", // Why this structure definition is defined
```

```
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "keyword" : [{ Coding }], // Assist with indexing and finding
  "fhirVersion" : "<code>", // FHIR Version this StructureDefinition targets
  "mapping" : [{ // External specification that the content is mapped to
    "identity" : "<id>", // R!  Internal id when this mapping is used
    "uri" : "<uri>", // C? Identifies what this mapping refers to
    "name" : "<string>", // C? Names what this mapping refers to
    "comment" : "<string>" // Versions, Issues, Scope limitations etc.
  }],
  "kind" : "<code>", // R!  primitive-type | complex-type | resource | logical
  "abstract" : <boolean>, // R!  Whether the structure is abstract
  "context" : [{ // C? If an extension, where it can be used in instances
    "type" : "<code>", // R!  fhirpath | element | extension
    "expression" : "<string>" // R!  Where the extension can be used in instances
  }],
  "contextInvariant" : ["<string>"], // C? FHIRPath invariants - when the extension can be used
  "type" : "<uri>", // C? R!  Type defined or constrained by this structure
  "baseDefinition" : { canonical(StructureDefinition) }, // C? Definition that this type is constrained/specialized from
  "derivation" : "<code>", // specialization | constraint - How relates to base definition
  "snapshot" : { // C? Snapshot view of the structure
    "element" : [{ ElementDefinition }] // C? R!  Definition of elements in the resource (if no StructureDefinition)
  },
  "differential" : { // C? Differential view of the structure
    "element" : [{ ElementDefinition }] // R!  Definition of elements in the resource (if no StructureDefinition)
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:StructureDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:StructureDefinition.url [ uri ]; # 1..1 Canonical identifier for this structure definition, represented as a URI
    (globally unique)
  fhir:StructureDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the structure definition
  fhir:StructureDefinition.version [ string ]; # 0..1 Business version of the structure definition
  fhir:StructureDefinition.name [ string ]; # 1..1 Name for this structure definition (computer friendly)
  fhir:StructureDefinition.title [ string ]; # 0..1 Name for this structure definition (human friendly)
  fhir:StructureDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:StructureDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:StructureDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:StructureDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:StructureDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:StructureDefinition.description [ markdown ]; # 0..1 Natural language description of the structure definition
  fhir:StructureDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:StructureDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for structure definition
     (if applicable)
  fhir:StructureDefinition.purpose [ markdown ]; # 0..1 Why this structure definition is defined
  fhir:StructureDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:StructureDefinition.keyword [ Coding ], ... ; # 0..* Assist with indexing and finding
  fhir:StructureDefinition.fhirVersion [ code ]; # 0..1 FHIR Version this StructureDefinition targets
  fhir:StructureDefinition.mapping [ # 0..* External specification that the content is mapped to
    fhir:StructureDefinition.mapping.identity [ id ]; # 1..1 Internal id when this mapping is used
    fhir:StructureDefinition.mapping.uri [ uri ]; # 0..1 Identifies what this mapping refers to
    fhir:StructureDefinition.mapping.name [ string ]; # 0..1 Names what this mapping refers to
    fhir:StructureDefinition.mapping.comment [ string ]; # 0..1 Versions, Issues, Scope limitations etc.
  ], ...;
  fhir:StructureDefinition.kind [ code ]; # 1..1 primitive-type | complex-type | resource | logical
  fhir:StructureDefinition.abstract [ boolean ]; # 1..1 Whether the structure is abstract
  fhir:StructureDefinition.context [ # 0..* If an extension, where it can be used in instances
    fhir:StructureDefinition.context.type [ code ]; # 1..1 fhirpath | element | extension
    fhir:StructureDefinition.context.expression [ string ]; # 1..1 Where the extension can be used in instances
  ], ...;
  fhir:StructureDefinition.contextInvariant [ string ], ... ; # 0..* FHIRPath invariants - when the extension can be
    used
  fhir:StructureDefinition.type [ uri ]; # 1..1 Type defined or constrained by this structure
  fhir:StructureDefinition.baseDefinition [ canonical(StructureDefinition) ]; # 0..1 Definition that this type is
    constrained/specialized from
  fhir:StructureDefinition.derivation [ code ]; # 0..1 specialization | constraint - How relates to base definition
  fhir:StructureDefinition.snapshot [ # 0..1 Snapshot view of the structure
    fhir:StructureDefinition.snapshot.element [ ElementDefinition ], ... ; # 1..* Definition of elements in the resource
     (if no StructureDefinition)
  ];
  fhir:StructureDefinition.differential [ # 0..1 Differential view of the structure
    fhir:StructureDefinition.differential.element [ ElementDefinition ], ... ; # 1..* Definition of elements in the
    resource (if no StructureDefinition)
  ];
]
```

Changes since Release 3

StructureDefinition

StructureDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

StructureDefinition.experimental | No longer marked as Modifier

StructureDefinition.keyword | Add Binding http://hl7.org/fhir/ValueSet/definition-use (extensible)
StructureDefinition.fhirVersion | Type changed from id to code Add Binding http://hl7.org/fhir/ValueSet/FHIR-version|4.0.1 (required)
StructureDefinition.kind | Change value set from http://hl7.org/fhir/ValueSet/structure-definition-kind to http://hl7.org/fhir/ValueSet/structure-definition-kind|4.0.1
StructureDefinition.context | Type changed from string to BackboneElement
StructureDefinition.context.type | Added Mandatory Element
StructureDefinition.context.expression | Added Mandatory Element
StructureDefinition.type | Type changed from code to uri
StructureDefinition.baseDefinition | Type changed from uri to canonical(StructureDefinition)
StructureDefinition.derivation | Change value set from http://hl7.org/fhir/ValueSet/type-derivation-rule to http://hl7.org/fhir/ValueSet/type-derivation-rule|4.0.1
StructureDefinition.contextType | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 206 tests that all execute ok. 4 fail round-trip testing and 206 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

5.3.4.1 Terminology Bindings

Path | Definition | Type | Reference
StructureDefinition.status | The lifecycle status of an artifact. | Required | PublicationStatus
StructureDefinition.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
StructureDefinition.keyword | Codes for the meaning of the defined structure (SNOMED CT and LOINC codes, as an example). | Extensible | DefinitionUseCodes
StructureDefinition.fhirVersion | All published FHIR Versions. | Required | FHIRVersion
StructureDefinition.kind | Defines the type of structure that a definition is describing. | Required | StructureDefinitionKind
StructureDefinition.context.type | How an extension context is interpreted. | Required | ExtensionContextType
StructureDefinition.type | Either a resource or a data type, including logical model types. | Extensible | FHIRDefinedType
StructureDefinition.derivation | How a type relates to its baseDefinition. | Required | TypeDerivationRule

5.3.4.2 Constraints

id | Level | Location | Description | Expression
sdf-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
sdf-1 | Rule | (base) | Element paths must be unique unless the structure is a constraint | derivation = 'constraint' or snapshot.element.select(path).isDistinct()
sdf-15a | Rule | (base) | If the first element in a differential has no "." in the path and it's not a logical model, it has no type | (kind!='logical' and differential.element.first().path.contains('.').not()) implies differential.element.first().type.empty()
sdf-2 | Rule | StructureDefinition.mapping | Must have at least a name or a uri (or both) | name.exists() or uri.exists()
sdf-3 | Rule | StructureDefinition.snapshot | Each element definition in a snapshot must have a formal definition and cardinalities | element.all(definition.exists() and min.exists() and max.exists())
sdf-4 | Rule | (base) | If the structure is not abstract, then there SHALL be a baseDefinition | abstract = true or baseDefinition.exists()
sdf-5 | Rule | (base) | If the structure defines an extension then the structure must have context information | type != 'Extension' or derivation = 'specialization' or (context.exists())
sdf-6 | Rule | (base) | A structure must have either a differential, or a snapshot (or both) | snapshot.exists() or differential.exists()
sdf-8 | Rule | StructureDefinition.snapshot | All snapshot elements must start with the StructureDefinition's specified type for non-logical models, or with the same type name for logical models | (%resource.kind = 'logical' or element.first().path = %resource.type) and element.tail().all(path.startsWith(%resource.snapshot.element.first().path&'.'))
sdf-10 | Rule | StructureDefinition.snapshot.element | provide either a binding reference or a description (or both) | binding.empty() or binding.valueSet.exists() or binding.description.exists()
sdf-11 | Rule | (base) | If there's a type, its content must match the path name in the first element of a snapshot | kind != 'logical' implies snapshot.empty() or snapshot.element.first().path = type
sdf-14 | Rule | (base) | All element definitions must have an id | snapshot.element.all(id.exists()) and differential.element.all(id.exists())
sdf-15 | Rule | (base) | The first element in a snapshot has no type unless model is a logical model. | kind!='logical' implies snapshot.element.first().type.empty()
sdf-16 | Rule | (base) | All element definitions must have unique ids (snapshot) | snapshot.element.all(id.exists()) and snapshot.element.id.trace('ids').isDistinct()
sdf-17 | Rule | (base) | All element definitions must have unique ids (diff) | differential.element.all(id.exists()) and differential.element.id.trace('ids').isDistinct()
sdf-18 | Rule | (base) | Context Invariants can only be used for extensions | contextInvariant.exists() implies type = 'Extension'
sdf-19 | Rule | (base) | FHIR Specification models only use FHIR defined types | url.startsWith('http://hl7.org/fhir/StructureDefinition') implies (differential.element.type.code.all(matches('^[a-zA-Z0-9]+$') or matches('^http:\\/\\/hl7\\.org\\/fhirpath\\/System\\.[A-Z][A-Za-z]+$')) and snapshot.element.type.code.all(matches('^[a-zA-Z0-9\\.]+$') or matches('^http:\\/\\/hl7\\.org\\/fhirpath\\/System\\.[A-Z][A-Za-z]+$')))
sdf-20 | Rule | StructureDefinition.differential | No slicing on the root element | element.where(path.contains('.').not()).slicing.empty()

sdf-21 | Rule | (base) | Default values can only be specified on specializations | differential.element.defaultValue.exists() implies (derivation = 'specialization')
sdf-22 | Rule | (base) | FHIR Specification models never have default values | url.startsWith('http://hl7.org/fhir/StructureDefinition') implies (snapshot.element.defaultValue.empty() and differential.element.defaultValue.empty())
sdf-23 | Rule | (base) | No slice name on root | (snapshot | differential).element.all(path.contains('.').not() implies sliceName.empty())
sdf-8a | Rule | StructureDefinition.differential | In any differential, all the elements must start with the StructureDefinition's specified type for non-logical models, or with the same type name for logical models | (%resource.kind = 'logical' or element.first().path.startsWith(%resource.type)) and (element.tail().empty() or element.tail().all(path.startsWith(%resource.differential.element.first().path.replaceMatches('\\..*','')&'.')))
sdf-8b | Rule | StructureDefinition.snapshot | All snapshot elements must have a base definition | element.all(base.exists())
sdf-9 | Rule | (base) | In any snapshot or differential, no label, code or requirements on an element without a "." in the path (e.g. the first element) | children().element.where(path.contains('.').not()).label.empty() and children().element.where(path.contains('.').not()).code.empty() and children().element.where(path.contains('.').not()).requirements.empty()

5.3.5 Interpretation Notes:

• A structure is represented as a flat list of elements. The element.path provides the overall structure.

• Differentials in constraints need only specify elements that they are making rules about. Other elements can be inferred as defined in the base resource

• Elements specified in the differential (and all elements in the snapshot) must be ordered as such: Elements from the baseDefinition appear before new elements in a StructureDefinition with derivation 'specialization' Elements must be in the same order as the baseDefinition, and child elements appear in depth-first order. Unsliced descendants of sliced elements appear before slices

• element.short, element.formal, element.comments and element.mapping provide the specific definition for the field in a specific context.

• element.type is used to specify which types may be used for this element. If there is more than one type, the element offers a choice of types and must have a name that terminates in "[x]". (Note: when substituting [x] with a specific data type, always capitalize the first letter. Choice types are always camel-case. Ex: "effectiveDateTime" is correct, "effectivedateTime" is NOT correct.) When profiling a resource and only one type is specified, the name of the element is changed to include the type instead of '[x]'.

• When using XPath to define constraints about the relationship between the contents of the narrative and the contents of the data elements, the element against which the rule is expressed is the one against which the error or warning will be reported by a validator.

• When expression constraints are placed on elements that offer a choice of types, the expression has to select the type to which it applies using the FHIRPath is operator.

• For XPath constraints, the prefix "f" should be used for "http://hl7.org/fhir", and "h" for "http://www.w3.org/1999/xhtml". XPath constraints are written against the XML representation

• The condition element is used to assert that a constraint defined on another element affects the allowed cardinality of this element.

5.3.5.1 Different Uses for StructureDefinition

The base structure definition is used in a number of different ways to support the FHIR specification. The various uses of the StructureDefinition are controlled by the elements kind, type, base, and url, using this basic pattern:

```
{
  "url": the identity of this structure definition,
  "kind": (primitive-type | complex-type | resource | logical),
  "type": the type being constrained (if it's a constraint)
  "baseDefinition": the structure definition from which this is derived
}
```

This list shows a number of examples, with links to real examples for each:

Quantity

XML

JSON

```
{
  "resourceType": "StructureDefinition",
  "url": "http://hl7.org/fhir/StructureDefinition/Quantity",
  "name": "Quantity",
  "kind": "complex-type",
  "baseDefinition": "http://hl7.org/fhir/StructureDefinition/Element"
}
```

Money

XML

JSON

```
{
```

```
  "resourceType": "StructureDefinition",
  "url": "http://hl7.org/fhir/StructureDefinition/Money",
  "name": "Money",
  "kind": "complex-type",
  "type": "Quantity",
  "baseDefinition": "http://hl7.org/fhir/StructureDefinition/Quantity"
}
```


Patient

XML

JSON


```
{
  "resourceType": "StructureDefinition",
  "url": "http://hl7.org/fhir/StructureDefinition/Patient",
  "name": "Patient",
  "kind": "resource",
  "baseDefinition": "http://hl7.org/fhir/StructureDefinition/DomainResource"
}
```


Clinical Document Profile for Composition

XML

JSON


```
{
  "resourceType": "StructureDefinition",
  "url": "http://hl7.org/fhir/StructureDefinition/clinicaldocument",
  "name": "Clinical Document Profile for Composition",
  "kind": "resource",
  "type": "Composition",
  "baseDefinition": "http://hl7.org/fhir/StructureDefinition/Composition"
}
```


Extension

XML

JSON


```
{
  "resourceType": "StructureDefinition",
  "url": "http://hl7.org/fhir/StructureDefinition/Extension",
  "name": "Extension",
  "kind": "datatype",
  "baseDefinition": "http://hl7.org/fhir/StructureDefinition/Element"
}
```


Extension Data Absent Reason

XML

JSON


```
{
  "resourceType": "StructureDefinition",
  "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason",
  "name": "Data Absent Reason",
  "kind": "complex-type",
  "type": "Extension",
  "baseDefinition": "http://hl7.org/fhir/StructureDefinition/Extension"
}
```

```
{
  "resourceType": "StructureDefinition",
  "url": "http://example.org/fhir/StructureDefinition/race",
  "name": "Race codes used by institution (a subset of meaningful use codes)",
  "kind": "complex-type",
  "type": "Extension",
  "baseDefinition": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race"
}
```


On this list, structure definitions of type 1, 3, and 5 can only be defined by the FHIR specification itself. The other kinds of structure definitions are (or may be) created by the specification but can also be defined by other implementers.

5.3.5.2 Rules for Constrained Types

When the structure is a constraint (derivation = constraint), see Extending and Restricting Resources for the rules that apply.

5.3.5.3 Common Mapping Targets

Structures are able to map elements to concepts in other definition systems. Generally, these are used to map the elements to local implementation models, data paths, and concepts. However they are also used to map to other standards and published terminologies. These are the standard URIs used for common targets of the mapping:

Name | Details
Logical Mapping | Formal URL: http://hl7.org/fhir/logical
ClinicalTrials.gov Mapping | Formal URL: http://clinicaltrials.gov

Workflow Pattern | Formal URL: http://hl7.org/fhir/workflow
CDISC Define XML 2.0 metadata standard | Formal URL: http://www.cdisc.org/define-xml
Mapping to NCPDP SCRIPT 10.6 | Formal URL: http://ncpdp.org/SCRIPT10_6
FiveWs Pattern Mapping | Formal URL: http://hl7.org/fhir/fivews
BRIDG 5.1 Mapping | Formal URL: https://bridgmodel.nci.nih.gov
FHIR Composition | Formal URL: http://hl7.org/fhir/composition
IHE Structured Data Capture | Formal URL: http://siframework.org/ihe-sdc-profile
Canadian Dental Association eclaims standard | Formal URL: http://www.cda-adc.ca/en/services/cdanet/
SNOMED CT Concept Domain Binding | Formal URL: http://snomed.info/conceptdomain Concept domain bindings link a resource or an element to a set of SNOMED CT concepts that represent the intended semantics of the instances (whether or not SNOMED CT is used to encode that data element). This set of concepts is represented using a SNOMED CT expression constraint. Note that the 'Concept domain binding' may be a superset of the 'value set binding'. These bindings help to support: Quality checking FHIR resources by ensuring that (a) the intended semantics of the instances matches the valid range of the corresponding SNOMED CT attribute, and (b) the intended value set is appropriate for the intended semantics of the instances Semantic checking of data instances by helping to detect potential inconsistencies caused by overlap between the semantics incorporated in two concept domains
R3-ReferalRequest Mapping | Formal URL: http://hl7.org/fhir/rr
HL7 v2 Mapping | Formal URL: http://hl7.org/v2
RIM Mapping | Formal URL: http://hl7.org/v3
ISO 11179 | Formal URL: http://metadata-standards.org/11179/ These mappings are included to indicate where properties of the data elements defined by the FHIR specification relate to similar fields in the ISO 11179 specification.
Mapping to Quality Improvement Domain Analysis Model | Formal URL: http://hl7.org/qidam
DICOM Tag Mapping | Formal URL: http://nema.org/dicom
MDMI Referent Index Mapping | Formal URL: http://github.com/MDMI/ReferentIndexContent
V3 Pharmacy Dispense RMIM | Formal URL: http://www.hl7.org/v3/PORX_RM020070UV
XDS metadata equivalent | Formal URL: http://ihe.net/xds
Equivalent CAP Code | Formal URL: http://cap.org/ecc
CDA (R2) | Formal URL: http://hl7.org/v3/cda
Quality Improvement and Clinical Knowledge (QUICK) | Formal URL: http://siframework.org/cqf
FHIR DocumentReference | Formal URL: http://hl7.org/fhir/documentreference
Canadian Pharmacy Associaiton eclaims standard | Formal URL: http://www.pharmacists.ca/
UDI Mapping | Formal URL: http://fda.gov/UDI UDI is a unique numeric or alphanumeric code that consists of two parts: (1) a device identifier (DI), a mandatory, fixed portion of a UDI that identifies the labeler and the specific version or model of a device, and (2) a production identifier (PI), a conditional, variable portion of a UDI that identifies one or more of the following when included on the label of a device: (2a) the lot or batch number within which a device was manufactured; (2b) the serial number of a specific device; (2c) the expiration date of a specific device; (2d) the date a specific device was manufactured; (2e) the distinct identification code required by §1271.290(c) for a human cell, tissue, or cellular and tissue-based product (HCT/P) regulated as a device.
Quality Data Model | Formal URL: http://www.healthit.gov/quality-data-model
FHIR AuditEvent Mapping | Formal URL: http://hl7.org/fhir/auditevent
IHE Data Element Exchange (DEX) | Formal URL: http://ihe.net/data-element-exchange
SNOMED CT Attribute Binding | Formal URL: http://snomed.org/attributebinding Attribute bindings link coded data elements in FHIR resources to a corresponding attribute in the SNOMED CT concept model. These bindings help to support: clarifying the intended meaning of the data element Quality checking the alignment between FHIR resource design and any coresponding SNOMED CT concept model Composition and decomposition of data instances by indicating the SNOMED CT concept model attribute whose value may be used to decompose a precoordinated concept into this data element
W3C PROV | Formal URL: http://www.w3.org/ns/prov The provenance resource is based on known practices in the HL7 implementation space, particularly those found in the v2 EVN segment, the v3 ControlAct Wrapper, the CDA header, and IHE ATNA. The conceptual model underlying the design is the W3C provenance Specification . Though the content and format of the resource is designed to meet specific requirements for FHIR, all the parts of the resource are formally mapped to the PROV-O specification, and FHIR resources can be transformed to their W3C PROV equivalent.
Ontological RIM Mapping | Formal URL: http://hl7.org/orim
LOINC code for the element | Formal URL: http://loinc.org
vCard Mapping | Formal URL: http://w3.org/vcard
Open EHR Archetype Mapping | Formal URL: http://openehr.org
ServD | Formal URL: http://www.omg.org/spec/ServD/1.0/
FHIR Provenance Mapping | Formal URL: http://hl7.org/fhir/provenance
iCalendar | Formal URL: http://ietf.org/rfc/2445

5.3.5.4 Logical Models

StructureDefinitions are used to define the basic structures of FHIR: data types, resources, extensions, and profiles. The same definition structure can also be used to define any arbitrary structures that are a directed acyclic graph with typed nodes, where the primitive types are those defined by the FHIR specification.

This technique has many uses:

- Describing any arbitrary content model
- Describing existing HL7 content models (e.g. v2, CDA) using FHIR
- Describing common design patterns used in FHIR
- Defining a content model to support the mapping language

5.3.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
abstract TU | token | Whether the structure is abstract | StructureDefinition.abstract
base TU | reference | Definition that this type is constrained/specialized from | StructureDefinition.baseDefinition (StructureDefinition)
base-path TU | token | Path that identifies the base element | StructureDefinition.snapshot.element.base.path | StructureDefinition.differential.element.base.path
context TU | token | A use context assigned to the structure definition | (StructureDefinition.useContext.value as CodeableConcept)
context-quantity TU | quantity | A quantity- or range-valued use context assigned to the structure definition | (StructureDefinition.useContext.value as Quantity) | (StructureDefinition.useContext.value as Range)
context-type TU | token | A type of use context assigned to the structure definition | StructureDefinition.useContext.code
context-type-quantity TU | composite | A use context type and quantity- or range-based value assigned to the structure definition | On StructureDefinition.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value TU | composite | A use context type and value assigned to the structure definition | On StructureDefinition.useContext: context-type: code context: value.as(CodeableConcept)
date TU | date | The structure definition publication date | StructureDefinition.date
derivation TU | token | specialization | constraint - How relates to base definition | StructureDefinition.derivation
description TU | string | The description of the structure definition | StructureDefinition.description
experimental TU | token | For testing purposes, not real usage | StructureDefinition.experimental
ext-context TU | token | The system is the URL for the context-type: e.g. http://hl7.org/fhir/extension-context-type#element|CodeableConcept.text | StructureDefinition.context.type
identifier TU | token | External identifier for the structure definition | StructureDefinition.identifier
jurisdiction TU | token | Intended jurisdiction for the structure definition | StructureDefinition.jurisdiction
keyword TU | token | A code for the StructureDefinition | StructureDefinition.keyword
kind TU | token | primitive-type | complex-type | resource | logical | StructureDefinition.kind
name TU | string | Computationally friendly name of the structure definition | StructureDefinition.name
path TU | token | A path that is constrained in the StructureDefinition | StructureDefinition.snapshot.element.path | StructureDefinition.differential.element.path
publisher TU | string | Name of the publisher of the structure definition | StructureDefinition.publisher
status TU | token | The current status of the structure definition | StructureDefinition.status
title TU | string | The human-friendly name of the structure definition | StructureDefinition.title
type TU | uri | Type defined or constrained by this structure | StructureDefinition.type
url TU | uri | The uri that identifies the structure definition | StructureDefinition.url
valueset TU | reference | A vocabulary binding reference | StructureDefinition.snapshot.element.binding.valueSet (ValueSet)
version TU | token | The business version of the structure definition | StructureDefinition.version

5.3.7 Resource StructureDefinition - Examples

URL: https://hl7.org/fhir/R4/structuredefinition-examples.html

5.3.7 Resource StructureDefinition - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

- Base Types
- Resources
- Constraints
- Extensions
- Examples

Abstract Types
Element | XML | JSON
BackBoneElement | XML | JSON
Resource | XML | JSON
DomainResource | XML | JSON
Primitive Types
base64Binary | XML | JSON
boolean | XML | JSON
canonical | XML | JSON
code | XML | JSON
date | XML | JSON
dateTime | XML | JSON
decimal | XML | JSON
id | XML | JSON
instant | XML | JSON
integer | XML | JSON
markdown | XML | JSON
oid | XML | JSON
positiveInt | XML | JSON
string | XML | JSON

time | XML | JSON
unsignedInt | XML | JSON
uri | XML | JSON
url | XML | JSON
uuid | XML | JSON
Data Types
Address | XML | JSON
Age | XML | JSON
Annotation | XML | JSON
Attachment | XML | JSON
BackboneElement | XML | JSON
CodeableConcept | XML | JSON
Coding | XML | JSON
ContactDetail | XML | JSON
ContactPoint | XML | JSON
Contributor | XML | JSON
Count | XML | JSON
DataRequirement | XML | JSON
Distance | XML | JSON
Dosage | XML | JSON
Duration | XML | JSON
Element | XML | JSON
ElementDefinition | XML | JSON
Expression | XML | JSON
Extension | XML | JSON
HumanName | XML | JSON
Identifier | XML | JSON
MarketingStatus | XML | JSON
Meta | XML | JSON
Money | XML | JSON
Narrative | XML | JSON
ParameterDefinition | XML | JSON
Period | XML | JSON
Population | XML | JSON
ProdCharacteristic | XML | JSON
ProductShelfLife | XML | JSON
Quantity | XML | JSON
Range | XML | JSON
Ratio | XML | JSON
Reference | XML | JSON
RelatedArtifact | XML | JSON
SampledData | XML | JSON
Signature | XML | JSON
SubstanceAmount | XML | JSON
Timing | XML | JSON
TriggerDefinition | XML | JSON
UsageContext | XML | JSON

Resources
Account | XML | JSON
ActivityDefinition | XML | JSON
AdverseEvent | XML | JSON
AllergyIntolerance | XML | JSON
Appointment | XML | JSON
AppointmentResponse | XML | JSON
AuditEvent | XML | JSON
Basic | XML | JSON
Binary | XML | JSON
BiologicallyDerivedProduct | XML | JSON
BodyStructure | XML | JSON
Bundle | XML | JSON
CapabilityStatement | XML | JSON
CarePlan | XML | JSON
CareTeam | XML | JSON
CatalogEntry | XML | JSON
ChargeItem | XML | JSON
ChargeItemDefinition | XML | JSON
Claim | XML | JSON
ClaimResponse | XML | JSON
ClinicalImpression | XML | JSON
CodeSystem | XML | JSON
Communication | XML | JSON
CommunicationRequest | XML | JSON

CompartmentDefinition | XML | JSON
Composition | XML | JSON
ConceptMap | XML | JSON
Condition | XML | JSON
Consent | XML | JSON
Contract | XML | JSON
Coverage | XML | JSON
CoverageEligibilityRequest | XML | JSON
CoverageEligibilityResponse | XML | JSON
DetectedIssue | XML | JSON
Device | XML | JSON
DeviceDefinition | XML | JSON
DeviceMetric | XML | JSON
DeviceRequest | XML | JSON
DeviceUseStatement | XML | JSON
DiagnosticReport | XML | JSON
DocumentManifest | XML | JSON
DocumentReference | XML | JSON
EffectEvidenceSynthesis | XML | JSON
Encounter | XML | JSON
Endpoint | XML | JSON
EnrollmentRequest | XML | JSON
EnrollmentResponse | XML | JSON
EpisodeOfCare | XML | JSON
EventDefinition | XML | JSON
Evidence | XML | JSON
EvidenceVariable | XML | JSON
ExampleScenario | XML | JSON
ExplanationOfBenefit | XML | JSON
FamilyMemberHistory | XML | JSON
Flag | XML | JSON
Goal | XML | JSON
GraphDefinition | XML | JSON
Group | XML | JSON
GuidanceResponse | XML | JSON
HealthcareService | XML | JSON
ImagingStudy | XML | JSON
Immunization | XML | JSON
ImmunizationEvaluation | XML | JSON
ImmunizationRecommendation | XML | JSON
ImplementationGuide | XML | JSON
InsurancePlan | XML | JSON
Invoice | XML | JSON
Library | XML | JSON
Linkage | XML | JSON
List | XML | JSON
Location | XML | JSON
Measure | XML | JSON
MeasureReport | XML | JSON
Media | XML | JSON
Medication | XML | JSON
MedicationAdministration | XML | JSON
MedicationDispense | XML | JSON
MedicationKnowledge | XML | JSON
MedicationRequest | XML | JSON
MedicationStatement | XML | JSON
MedicinalProduct | XML | JSON
MedicinalProductAuthorization | XML | JSON
MedicinalProductContraindication | XML | JSON
MedicinalProductIndication | XML | JSON
MedicinalProductIngredient | XML | JSON
MedicinalProductInteraction | XML | JSON
MedicinalProductManufactured | XML | JSON
MedicinalProductPackaged | XML | JSON
MedicinalProductPharmaceutical | XML | JSON
MedicinalProductUndesirableEffect | XML | JSON
MessageDefinition | XML | JSON
MessageHeader | XML | JSON
MolecularSequence | XML | JSON
NamingSystem | XML | JSON
NutritionOrder | XML | JSON
Observation | XML | JSON

ObservationDefinition | XML | JSON
OperationDefinition | XML | JSON
OperationOutcome | XML | JSON
Organization | XML | JSON
OrganizationAffiliation | XML | JSON
Patient | XML | JSON
PaymentNotice | XML | JSON
PaymentReconciliation | XML | JSON
Person | XML | JSON
PlanDefinition | XML | JSON
Practitioner | XML | JSON
PractitionerRole | XML | JSON
Procedure | XML | JSON
Provenance | XML | JSON
Questionnaire | XML | JSON
QuestionnaireResponse | XML | JSON
RelatedPerson | XML | JSON
RequestGroup | XML | JSON
ResearchDefinition | XML | JSON
ResearchElementDefinition | XML | JSON
ResearchStudy | XML | JSON
ResearchSubject | XML | JSON
RiskAssessment | XML | JSON
RiskEvidenceSynthesis | XML | JSON
Schedule | XML | JSON
SearchParameter | XML | JSON
ServiceRequest | XML | JSON
Slot | XML | JSON
Specimen | XML | JSON
SpecimenDefinition | XML | JSON
StructureDefinition | XML | JSON
StructureMap | XML | JSON
Subscription | XML | JSON
Substance | XML | JSON
SubstanceNucleicAcid | XML | JSON
SubstancePolymer | XML | JSON
SubstanceProtein | XML | JSON
SubstanceReferenceInformation | XML | JSON
SubstanceSourceMaterial | XML | JSON
SubstanceSpecification | XML | JSON
SupplyDelivery | XML | JSON
SupplyRequest | XML | JSON
Task | XML | JSON
TerminologyCapabilities | XML | JSON
TestReport | XML | JSON
TestScript | XML | JSON
ValueSet | XML | JSON
VerificationResult | XML | JSON
VisionPrescription | XML | JSON

FHIR Specification Core
ActualGroup | XML | JSON
BMI | XML | JSON
BodyHeight | XML | JSON
BodyTemp | XML | JSON
BodyWeight | XML | JSON
BP | XML | JSON
Catalog | XML | JSON
Cholesterol | XML | JSON
ClinicalDocument | XML | JSON
DeviceMetricObservation | XML | JSON
DiagnosticReport-Genetics | XML | JSON
elementdefinition-de | XML | JSON
FamilyMemberHistory-Genetic | XML | JSON
GroupDefinition | XML | JSON
HDLCholesterol | XML | JSON
HeadCircum | XML | JSON
HeartRate | XML | JSON
HLAResult | XML | JSON
LDLCholesterol | XML | JSON
LipidProfile | XML | JSON
Observation-Genetics | XML | JSON
OxygenSat | XML | JSON

PICOElement | XML | JSON
provenance-relevant-history | XML | JSON
RespRate | XML | JSON
ServiceRequest-Genetics | XML | JSON
ShareableCodeSystem | XML | JSON
ShareableValueSet | XML | JSON
Synthesis | XML | JSON
Triglyceride | XML | JSON
VitalSigns | XML | JSON
VitalsPanel | XML | JSON
Clinical Reasoning
CDSHooksGuidanceResponse | XML | JSON
CDSHooksRequestGroup | XML | JSON
CDSHooksServicePlanDefinition | XML | JSON
ComputablePlanDefinition | XML | JSON
CQF-Questionnaire | XML | JSON
CQLLibrary | XML | JSON
ShareableActivityDefinition | XML | JSON
ShareableLibrary | XML | JSON
ShareableMeasure | XML | JSON
ShareablePlanDefinition | XML | JSON
EHRS Functional model - Record Lifecycle Events
ehrsrle-auditevent | XML | JSON
ehrsrle-provenance | XML | JSON

FHIR Specification Core
http://fhir-registry.smarthealthit.org/StructureDefinition/capabilities capabilities | XML | JSON
http://fhir-registry.smarthealthit.org/StructureDefinition/capabilities capabilities | XML | JSON
http://fhir-registry.smarthealthit.org/StructureDefinition/oauth-uris oauth-uris | XML | JSON
http://fhir-registry.smarthealthit.org/StructureDefinition/oauth-uris oauth-uris | XML | JSON
11179-objectClass objectClass | XML | JSON
11179-objectClass objectClass | XML | JSON
11179-objectClassProperty objectClassProperty | XML | JSON
11179-objectClassProperty objectClassProperty | XML | JSON
11179-permitted-value-conceptmap permitted-value-conceptmap | XML | JSON
11179-permitted-value-conceptmap permitted-value-conceptmap | XML | JSON
11179-permitted-value-valueset permitted-value-valueset | XML | JSON
11179-permitted-value-valueset permitted-value-valueset | XML | JSON
DiagnosticReport-geneticsAnalysis Analysis | XML | JSON
DiagnosticReport-geneticsAnalysis Analysis | XML | JSON
DiagnosticReport-geneticsAssessedCondition AssessedCondition | XML | JSON
DiagnosticReport-geneticsAssessedCondition AssessedCondition | XML | JSON
DiagnosticReport-geneticsFamilyMemberHistory FamilyMemberHistory | XML | JSON
DiagnosticReport-geneticsFamilyMemberHistory FamilyMemberHistory | XML | JSON
DiagnosticReport-geneticsReferences References | XML | JSON
DiagnosticReport-geneticsReferences References | XML | JSON
allergyintolerance-assertedDate assertedDate | XML | JSON
allergyintolerance-assertedDate assertedDate | XML | JSON
allergyintolerance-certainty certainty | XML | JSON
allergyintolerance-certainty certainty | XML | JSON
allergyintolerance-duration duration | XML | JSON
allergyintolerance-duration duration | XML | JSON
allergyintolerance-reasonRefuted reasonRefuted | XML | JSON
allergyintolerance-reasonRefuted reasonRefuted | XML | JSON
allergyintolerance-resolutionAge resolutionAge | XML | JSON
allergyintolerance-resolutionAge resolutionAge | XML | JSON
allergyintolerance-substanceExposureRisk substanceExposureRisk | XML | JSON
allergyintolerance-substanceExposureRisk substanceExposureRisk | XML | JSON
auditevent-Accession Accession | XML | JSON
auditevent-Accession Accession | XML | JSON
auditevent-Anonymized Anonymized | XML | JSON
auditevent-Anonymized Anonymized | XML | JSON
auditevent-Encrypted Encrypted | XML | JSON
auditevent-Encrypted Encrypted | XML | JSON
auditevent-Instance Instance | XML | JSON
auditevent-Instance Instance | XML | JSON
auditevent-MPPS MPPS | XML | JSON
auditevent-MPPS MPPS | XML | JSON
auditevent-NumberOfInstances NumberOfInstances | XML | JSON
auditevent-NumberOfInstances NumberOfInstances | XML | JSON
auditevent-ParticipantObjectContainsStudy ParticipantObjectContainsStudy | XML | JSON
auditevent-ParticipantObjectContainsStudy ParticipantObjectContainsStudy | XML | JSON
auditevent-SOPClass SOPClass | XML | JSON

auditevent-SOPClass SOPClass | XML | JSON
bodySite BodyStructure Reference | XML | JSON
bodySite BodyStructure Reference | XML | JSON
capabilitystatement-expectation expectation | XML | JSON
capabilitystatement-expectation expectation | XML | JSON
capabilitystatement-prohibited prohibited | XML | JSON
capabilitystatement-prohibited prohibited | XML | JSON
capabilitystatement-search-parameter-combination search-parameter-combination | XML | JSON
capabilitystatement-search-parameter-combination search-parameter-combination | XML | JSON
capabilitystatement-supported-system supported-system | XML | JSON
capabilitystatement-supported-system supported-system | XML | JSON
capabilitystatement-websocket websocket | XML | JSON
capabilitystatement-websocket websocket | XML | JSON
careplan-activity-title activity-title | XML | JSON
careplan-activity-title activity-title | XML | JSON
codesystem-alternate alternate | XML | JSON
codesystem-alternate alternate | XML | JSON
codesystem-author author | XML | JSON
codesystem-author author | XML | JSON
codesystem-concept-comments concept-comments | XML | JSON
codesystem-concept-comments concept-comments | XML | JSON
codesystem-conceptOrder conceptOrder | XML | JSON
codesystem-conceptOrder conceptOrder | XML | JSON
codesystem-effectiveDate effectiveDate | XML | JSON
codesystem-effectiveDate effectiveDate | XML | JSON
codesystem-expirationDate expirationDate | XML | JSON
codesystem-expirationDate expirationDate | XML | JSON
codesystem-history history | XML | JSON
codesystem-history history | XML | JSON
codesystem-keyWord keyWord | XML | JSON
codesystem-keyWord keyWord | XML | JSON
codesystem-label label | XML | JSON
codesystem-label label | XML | JSON
codesystem-map map | XML | JSON
codesystem-map map | XML | JSON
codesystem-otherName otherName | XML | JSON
codesystem-otherName otherName | XML | JSON
codesystem-replacedby replacedby | XML | JSON
codesystem-replacedby replacedby | XML | JSON
codesystem-sourceReference sourceReference | XML | JSON
codesystem-sourceReference sourceReference | XML | JSON
codesystem-trusted-expansion trusted-expansion | XML | JSON
codesystem-trusted-expansion trusted-expansion | XML | JSON
codesystem-usage usage | XML | JSON
codesystem-usage usage | XML | JSON
codesystem-warning warning | XML | JSON
codesystem-warning warning | XML | JSON
codesystem-workflowStatus workflowStatus | XML | JSON
codesystem-workflowStatus workflowStatus | XML | JSON
coding-sctdescid sctdescid | XML | JSON
coding-sctdescid sctdescid | XML | JSON
communication-media media | XML | JSON
communication-media media | XML | JSON
communicationrequest-initiatingLocation initiatingLocation | XML | JSON
communicationrequest-initiatingLocation initiatingLocation | XML | JSON
composition-clinicaldocument-otherConfidentiality otherConfidentiality | XML | JSON
composition-clinicaldocument-otherConfidentiality otherConfidentiality | XML | JSON
composition-clinicaldocument-versionNumber versionNumber | XML | JSON
composition-clinicaldocument-versionNumber versionNumber | XML | JSON
composition-section-subject section-subject | XML | JSON
composition-section-subject section-subject | XML | JSON
concept-bidirectional bidirectional | XML | JSON
concept-bidirectional bidirectional | XML | JSON
condition-assertedDate assertedDate | XML | JSON
condition-assertedDate assertedDate | XML | JSON
condition-dueTo dueTo | XML | JSON
condition-dueTo dueTo | XML | JSON
condition-occurredFollowing occurredFollowing | XML | JSON
condition-occurredFollowing occurredFollowing | XML | JSON
condition-outcome outcome | XML | JSON
condition-outcome outcome | XML | JSON
condition-related related | XML | JSON

condition-related related | XML | JSON
condition-ruledOut ruledOut | XML | JSON
condition-ruledOut ruledOut | XML | JSON
consent-NotificationEndpoint NotificationEndpoint | XML | JSON
consent-NotificationEndpoint NotificationEndpoint | XML | JSON
consent-Transcriber Transcriber | XML | JSON
consent-Transcriber Transcriber | XML | JSON
consent-Witness Witness | XML | JSON
consent-Witness Witness | XML | JSON
consent-location location | XML | JSON
consent-location location | XML | JSON
contactpoint-area area | XML | JSON
contactpoint-area area | XML | JSON
contactpoint-country country | XML | JSON
contactpoint-country country | XML | JSON
contactpoint-extension extension | XML | JSON
contactpoint-extension extension | XML | JSON
contactpoint-local local | XML | JSON
contactpoint-local local | XML | JSON
cqf-calculatedValue calculatedValue | XML | JSON
cqf-calculatedValue calculatedValue | XML | JSON
cqf-cdsHooksEndpoint cdsHooksEndpoint | XML | JSON
cqf-cdsHooksEndpoint cdsHooksEndpoint | XML | JSON
cqf-citation citation | XML | JSON
cqf-citation citation | XML | JSON
cqf-encounterClass encounterClass | XML | JSON
cqf-encounterClass encounterClass | XML | JSON
cqf-encounterType encounterType | XML | JSON
cqf-encounterType encounterType | XML | JSON
cqf-expression expression | XML | JSON
cqf-expression expression | XML | JSON
cqf-initialValue initialValue | XML | JSON
cqf-initialValue initialValue | XML | JSON
cqf-initiatingOrganization initiatingOrganization | XML | JSON
cqf-initiatingOrganization initiatingOrganization | XML | JSON
cqf-initiatingPerson initiatingPerson | XML | JSON
cqf-initiatingPerson initiatingPerson | XML | JSON
cqf-library library | XML | JSON
cqf-library library | XML | JSON
cqf-measureInfo measureInfo | XML | JSON
cqf-measureInfo measureInfo | XML | JSON
cqf-qualityOfEvidence qualityOfEvidence | XML | JSON
cqf-qualityOfEvidence qualityOfEvidence | XML | JSON
cqf-receivingOrganization receivingOrganization | XML | JSON
cqf-receivingOrganization receivingOrganization | XML | JSON
cqf-receivingPerson receivingPerson | XML | JSON
cqf-receivingPerson receivingPerson | XML | JSON
cqf-recipientLanguage recipientLanguage | XML | JSON
cqf-recipientLanguage recipientLanguage | XML | JSON
cqf-recipientType recipientType | XML | JSON
cqf-recipientType recipientType | XML | JSON
cqf-relativeDateTime relativeDateTime | XML | JSON
cqf-relativeDateTime relativeDateTime | XML | JSON
cqf-strengthOfRecommendation strengthOfRecommendation | XML | JSON
cqf-strengthOfRecommendation strengthOfRecommendation | XML | JSON
cqf-systemUserLanguage systemUserLanguage | XML | JSON
cqf-systemUserLanguage systemUserLanguage | XML | JSON
cqf-systemUserTaskContext systemUserTaskContext | XML | JSON
cqf-systemUserTaskContext systemUserTaskContext | XML | JSON
cqf-systemUserType systemUserType | XML | JSON
cqf-systemUserType systemUserType | XML | JSON
cqm-ValidityPeriod ValidityPeriod | XML | JSON
cqm-ValidityPeriod ValidityPeriod | XML | JSON
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designNote Design Note | XML | JSON
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device-implantStatus implantStatus | XML | JSON
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translation Translation | XML | JSON
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tz-code Timezone Code | XML | JSON
tz-offset Timezone Offset | XML | JSON
tz-offset Timezone Offset | XML | JSON
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valueset-usage usage | XML | JSON
valueset-usage usage | XML | JSON
valueset-warning warning | XML | JSON
valueset-warning warning | XML | JSON
valueset-workflowStatus workflowStatus | XML | JSON
valueset-workflowStatus workflowStatus | XML | JSON
variable Variable | XML | JSON
variable Variable | XML | JSON
workflow-episodeOfCare episodeOfCare | XML | JSON
workflow-episodeOfCare episodeOfCare | XML | JSON
workflow-instantiatesCanonical instantiatesCanonical | XML | JSON
workflow-instantiatesCanonical instantiatesCanonical | XML | JSON
workflow-instantiatesUri instantiatesUri | XML | JSON
workflow-instantiatesUri instantiatesUri | XML | JSON
workflow-reasonCode reasonCode | XML | JSON
workflow-reasonCode reasonCode | XML | JSON
workflow-reasonReference reasonReference | XML | JSON
workflow-reasonReference reasonReference | XML | JSON
workflow-relatedArtifact relatedArtifact | XML | JSON
workflow-relatedArtifact relatedArtifact | XML | JSON
workflow-researchStudy researchStudy | XML | JSON

workflow-researchStudy researchStudy | XML | JSON
workflow-supportingInfo supportingInfo | XML | JSON
workflow-supportingInfo supportingInfo | XML | JSON
openEHR Profiles
openEHR-administration administration | XML | JSON
openEHR-administration administration | XML | JSON
openEHR-careplan careplan | XML | JSON
openEHR-careplan careplan | XML | JSON
openEHR-exposureDate exposureDate | XML | JSON
openEHR-exposureDate exposureDate | XML | JSON
openEHR-exposureDescription exposureDescription | XML | JSON
openEHR-exposureDescription exposureDescription | XML | JSON
openEHR-exposureDuration exposureDuration | XML | JSON
openEHR-exposureDuration exposureDuration | XML | JSON
openEHR-location location | XML | JSON
openEHR-location location | XML | JSON
openEHR-management management | XML | JSON
openEHR-management management | XML | JSON
openEHR-test test | XML | JSON
openEHR-test test | XML | JSON

Example showing how to define Section Level Profiles | example-section-library | XML | JSON | Turtle
Example showing how to use Section Level Profiles | example-composition | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.3.8 Resource StructureDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/structuredefinition-definitions.html

5.3.8 Resource StructureDefinition - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the StructureDefinition resource.

StructureDefinition
Element Id | StructureDefinition
Definition | A definition of a FHIR structure. This resource is used to describe the underlying resources, data types defined in FHIR, and also for describing extensions and constraints on resources and data types.
Cardinality | 0..*
Type | DomainResource
Alternate Names | Template; Profile
Invariants | Defined on this element sdf-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}') sdf-1 Rule Element paths must be unique unless the structure is a constraint derivation = 'constraint' or snapshot.element.select(path).isDistinct() sdf-15a Rule If the first element in a differential has no "." in the path and it's not a logical model, it has no type (kind!='logical' and differential.element.first().path.contains('.').not()) implies differential.element.first().type.empty() sdf-4 Rule If the structure is not abstract, then there SHALL be a baseDefinition abstract = true or baseDefinition.exists() sdf-5 Rule If the structure defines an extension then the structure must have context information type != 'Extension' or derivation = 'specialization' or (context.exists()) sdf-6 Rule A structure must have either a differential, or a snapshot (or both) snapshot.exists() or differential.exists() sdf-9 Rule In any snapshot or differential, no label, code or requirements on an element without a "." in the path (e.g. the first element) children().element.where(path.contains('.').not()).label.empty() and children().element.where(path.contains('.').not()).code.empty() and children().element.where(path.contains('.').not()).requirements.empty() sdf-11 Rule If there's a type, its content must match the path name in the first element of a snapshot kind != 'logical' implies snapshot.empty() or snapshot.element.first().path = type sdf-14 Rule All element definitions must have an id snapshot.element.all(id.exists()) and differential.element.all(id.exists()) sdf-15 Rule The first element in a snapshot has no type unless model is a logical model. kind!='logical' implies snapshot.element.first().type.empty() sdf-16 Rule All element definitions must have unique ids (snapshot) snapshot.element.all(id.exists()) and snapshot.element.id.trace('ids').isDistinct() sdf-17 Rule All element definitions must have unique ids (diff) differential.element.all(id.exists()) and differential.element.id.trace('ids').isDistinct() sdf-18 Rule Context Invariants can only be used for extensions contextInvariant.exists() implies type = 'Extension' sdf-19 Rule FHIR Specification models only use FHIR defined types url.startsWith('http://hl7.org/fhir/StructureDefinition') implies (differential.element.type.code.all(matches('^[a-zA-Z0-9]+$') or matches('^http:\\/\\/hl7\\.org\\/fhirpath\\/System\\.[A-Z][A-Za-z]+$')) and snapshot.element.type.code.all(matches('^[a-zA-Z0-9\\.]+$') or matches('^http:\\/\\/hl7\\.org\\/fhirpath\\/System\\.[A-Z][A-Za-z]+$'))) sdf-21 Rule Default values can only be specified on specializations differential.element.defaultValue.exists() implies (derivation = 'specialization') sdf-22 Rule FHIR Specification models never have default values url.startsWith('http://hl7.org/fhir/StructureDefinition') implies (snapshot.element.defaultValue.empty() and differential.element.defaultValue.empty()) sdf-23 Rule No slice name on root (snapshot | differential).element.all(path.contains('.').not() implies sliceName.empty())
Defined on this element
sdf-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
sdf-1 | Rule | Element paths must be unique unless the structure is a constraint | derivation = 'constraint' or snapshot.element.select(path).isDistinct()

sdf-15a | Rule | If the first element in a differential has no "." in the path and it's not a logical model, it has no type | (kind!='logical' and differential.element.first().path.contains('.').not()) implies differential.element.first().type.empty()
sdf-4 | Rule | If the structure is not abstract, then there SHALL be a baseDefinition | abstract = true or baseDefinition.exists()
sdf-5 | Rule | If the structure defines an extension then the structure must have context information | type != 'Extension' or derivation = 'specialization' or (context.exists())
sdf-6 | Rule | A structure must have either a differential, or a snapshot (or both) | snapshot.exists() or differential.exists()
sdf-9 | Rule | In any snapshot or differential, no label, code or requirements on an element without a "." in the path (e.g. the first element) | children().element.where(path.contains('.').not()).label.empty() and children().element.where(path.contains('.').not()).code.empty() and children().element.where(path.contains('.').not()).requirements.empty()
sdf-11 | Rule | If there's a type, its content must match the path name in the first element of a snapshot | kind != 'logical' implies snapshot.empty() or snapshot.element.first().path = type
sdf-14 | Rule | All element definitions must have an id | snapshot.element.all(id.exists()) and differential.element.all(id.exists())
sdf-15 | Rule | The first element in a snapshot has no type unless model is a logical model. | kind!='logical' implies snapshot.element.first().type.empty()
sdf-16 | Rule | All element definitions must have unique ids (snapshot) | snapshot.element.all(id.exists()) and snapshot.element.id.trace('ids').isDistinct()
sdf-17 | Rule | All element definitions must have unique ids (diff) | differential.element.all(id.exists()) and differential.element.id.trace('ids').isDistinct()
sdf-18 | Rule | Context Invariants can only be used for extensions | contextInvariant.exists() implies type = 'Extension'
sdf-19 | Rule | FHIR Specification models only use FHIR defined types | url.startsWith('http://hl7.org/fhir/StructureDefinition') implies (differential.element.type.code.all(matches('^[a-zA-Z0-9]+$') or matches('^http:\\/\\/hl7\\.org\\/fhirpath\\/System\\.[A-Z][A-Za-z]+$')) and snapshot.element.type.code.all(matches('^[a-zA-Z0-9\\.]+$') or matches('^http:\\/\\/hl7\\.org\\/fhirpath\\/System\\.[A-Z][A-Za-z]+$')))
sdf-21 | Rule | Default values can only be specified on specializations | differential.element.defaultValue.exists() implies (derivation = 'specialization')
sdf-22 | Rule | FHIR Specification models never have default values | url.startsWith('http://hl7.org/fhir/StructureDefinition') implies (snapshot.element.defaultValue.empty() and differential.element.defaultValue.empty())
sdf-23 | Rule | No slice name on root | (snapshot | differential).element.all(path.contains('.').not() implies sliceName.empty())
StructureDefinition.url
Element Id | StructureDefinition.url
Definition | An absolute URI that is used to identify this structure definition when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this structure definition is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the structure definition is stored on different servers.
Cardinality | 1..1
Type | uri
Requirements | Allows the structure definition to be referenced by a single globally unique identifier.
Alternate Names | url; authoritative-url; destination; identity
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
StructureDefinition.identifier
Element Id | StructureDefinition.identifier
Definition | A formal identifier that is used to identify this structure definition when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this structure definition outside of FHIR, where it is not possible to use the logical URI.
StructureDefinition.version
Element Id | StructureDefinition.version
Definition | The identifier that is used to identify this version of the structure definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the structure definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different structure definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the structure definition with the format [url]|[version].
StructureDefinition.name
Element Id | StructureDefinition.name
Definition | A natural language name identifying the structure definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 1..1

Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.Some Examples: O2SatObservation PresentationReport Immunization2 AcmeAdmissionRecordOld.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
StructureDefinition.title
Element Id | StructureDefinition.title
Definition | A short, descriptive, user-friendly title for the structure definition.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.Applications don't have to use this name but can always fall back to it. The title also corresponds to the label for the root element.
StructureDefinition.status
Element Id | StructureDefinition.status
Definition | The status of this structure definition. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of structure definitions that are appropriate for use versus not.
StructureDefinition.experimental
Element Id | StructureDefinition.experimental
Definition | A Boolean value to indicate that this structure definition is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level structure definition.
Summary | true
Comments | Allows filtering of structure definitions that are appropriate for use versus not.
StructureDefinition.date
Element Id | StructureDefinition.date
Definition | The date (and optionally time) when the structure definition was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the structure definition changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the structure definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
StructureDefinition.publisher
Element Id | StructureDefinition.publisher
Definition | The name of the organization or individual that published the structure definition.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the structure definition. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the structure definition is the organization or individual primarily responsible for the maintenance and upkeep of the structure definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the structure definition. This item SHOULD be populated unless the information is available from context.
StructureDefinition.contact
Element Id | StructureDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
StructureDefinition.description
Element Id | StructureDefinition.description
Definition | A free text natural language description of the structure definition from a consumer's perspective.

Cardinality | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the structure definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the structure definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the structure definition is presumed to be the predominant language in the place the structure definition was created).
StructureDefinition.useContext
Element Id | StructureDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate structure definition instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
StructureDefinition.jurisdiction
Element Id | StructureDefinition.jurisdiction
Definition | A legal or geographic region in which the structure definition is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the structure definition to be used in jurisdictions other than those for which it was originally designed or intended.
StructureDefinition.purpose
Element Id | StructureDefinition.purpose
Definition | Explanation of why this structure definition is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the structure definition. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this structure definition.
StructureDefinition.copyright
Element Id | StructureDefinition.copyright
Definition | A copyright statement relating to the structure definition and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the structure definition.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the structure definition and/or its content.
Alternate Names | License; Restrictions
StructureDefinition.keyword
Element Id | StructureDefinition.keyword
Definition | A set of key words or terms from external terminologies that may be used to assist with indexing and searching of templates nby describing the use of this structure definition, or the content it describes.
Cardinality | 0..*
Terminology Binding | Structure Definition Use Codes / Keywords (Extensible)
Type | Coding
Requirements | Assist in searching for appropriate StructureDefinitions.
Summary | true
StructureDefinition.fhirVersion
Element Id | StructureDefinition.fhirVersion
Definition | The version of the FHIR specification on which this StructureDefinition is based - this is the formal version of the specification, without the revision number, e.g. [publication].[major].[minor], which is 4.0.1. for this version.
Cardinality | 0..1
Terminology Binding | FHIRVersion (Required)
Type | code
Summary | true
Comments | A StructureDefinition does not need to specify the target it applies to as StructureDefinitions will often be valid across multiple versions of FHIR. FHIR tooling can determine whether a StructureDefinition is consistent with a particular StructureDefinition if desired.
StructureDefinition.mapping
Element Id | StructureDefinition.mapping
Definition | An external specification that the content is mapped to.
Cardinality | 0..*
Invariants | Defined on this element sdf-2 Rule Must have at least a name or a uri (or both) name.exists() or uri.exists()
Defined on this element
sdf-2 | Rule | Must have at least a name or a uri (or both) | name.exists() or uri.exists()
StructureDefinition.mapping.identity
Element Id | StructureDefinition.mapping.identity
Definition | An Internal id that is used to identify this mapping set when specific mappings are made.
Cardinality | 1..1

Type | id
Comments | The specification is described once, with general comments, and then specific mappings are made that reference this declaration.
StructureDefinition.mapping.uri
Element Id | StructureDefinition.mapping.uri
Definition | An absolute URI that identifies the specification that this mapping is expressed to.
Cardinality | 0..1
Type | uri
Comments | A formal identity for the specification being mapped to helps with identifying maps consistently.
Invariants | Affect this element sdf-2 Rule Must have at least a name or a uri (or both) name.exists() or uri.exists()
Affect this element
sdf-2 | Rule | Must have at least a name or a uri (or both) | name.exists() or uri.exists()
StructureDefinition.mapping.name
Element Id | StructureDefinition.mapping.name
Definition | A name for the specification that is being mapped to.
Cardinality | 0..1
Type | string
Invariants | Affect this element sdf-2 Rule Must have at least a name or a uri (or both) name.exists() or uri.exists()
Affect this element
sdf-2 | Rule | Must have at least a name or a uri (or both) | name.exists() or uri.exists()
StructureDefinition.mapping.comment
Element Id | StructureDefinition.mapping.comment
Definition | Comments about this mapping, including version notes, issues, scope limitations, and other important notes for usage.
Cardinality | 0..1
Type | string
StructureDefinition.kind
Element Id | StructureDefinition.kind
Definition | Defines the kind of structure that this definition is describing.
Cardinality | 1..1
Terminology Binding | StructureDefinitionKind (Required)
Type | code
Summary | true
StructureDefinition.abstract
Element Id | StructureDefinition.abstract
Definition | Whether structure this definition describes is abstract or not - that is, whether the structure is not intended to be instantiated. For Resources and Data types, abstract types will never be exchanged between systems.
Cardinality | 1..1
Type | boolean
Summary | true
Comments | Abstract Resources cannot be instantiated - a concrete sub-type must be used. Abstract datatypes and extensions cannot be used in an instance. For logical models, the exact implication of "abstract" will rest with the author, depending how the logical model is used. Flagging a constraint structure as abstract conveys design intent but makes no difference to how the structure definition is handled. Note that inline declared elements that are given the type "Element" in the StructureDefinition, but have children described, are anonymous concrete types that specialize Element.
StructureDefinition.context
Element Id | StructureDefinition.context
Definition | Identifies the types of resource or data type elements to which the extension can be applied.
Cardinality | 0..*
Summary | true
Invariants | Affect this element sdf-5 Rule If the structure defines an extension then the structure must have context information type != 'Extension' or derivation = 'specialization' or (context.exists())
Affect this element
sdf-5 | Rule | If the structure defines an extension then the structure must have context information | type != 'Extension' or derivation = 'specialization' or (context.exists())
StructureDefinition.context.type
Element Id | StructureDefinition.context.type
Definition | Defines how to interpret the expression that defines what the context of the extension is.
Cardinality | 1..1
Terminology Binding | ExtensionContextType (Required)
Type | code
Summary | true
StructureDefinition.context.expression
Element Id | StructureDefinition.context.expression
Definition | An expression that defines where an extension can be used in resources.
Cardinality | 1..1
Type | string
Summary | true
StructureDefinition.contextInvariant
Element Id | StructureDefinition.contextInvariant
Definition | A set of rules as FHIRPath Invariants about when the extension can be used (e.g. co-occurrence variants for the extension). All the rules must be true.
Cardinality | 0..*
Type | string

Summary | true
Comments | The rules are only evaluated when the extension is present. When evaluating the invariant, the FHIRPath focus is the element that holds the extension, and %extension refers to the extension itself.
Invariants | Affect this element sdf-18 Rule Context Invariants can only be used for extensions contextInvariant.exists() implies type = 'Extension'
Affect this element
sdf-18 | Rule | Context Invariants can only be used for extensions | contextInvariant.exists() implies type = 'Extension'
StructureDefinition.type
Element Id | StructureDefinition.type
Definition | The type this structure describes. If the derivation kind is 'specialization' then this is the master definition for a type, and there is always one of these (a data type, an extension, a resource, including abstract ones). Otherwise the structure definition is a constraint on the stated type (and in this case, the type cannot be an abstract type). References are URLs that are relative to http://hl7.org/fhir/StructureDefinition e.g. "string" is a reference to http://hl7.org/fhir/StructureDefinition/string. Absolute URLs are only allowed in logical models.
Cardinality | 1..1
Terminology Binding | FHIRDefinedType (Extensible)
Type | uri
Summary | true
Comments | Note that in the case of constraints, the type could be determined by chasing through the baseDefinition references until a concrete structure (derivation = specialization) is reached, or by looking at the path of the first element in the snapshot - if present - but providing the type directly makes for simpler tooling and indexing. The type must match the elements defined in the differential and the snapshot. For all FHIR defined types, the path name of the element will start with the type name. For logical models, where the type is a URL, the type name SHOULD start with the tail of the type URL where required.
Invariants | Affect this element sdf-11 Rule If there's a type, its content must match the path name in the first element of a snapshot kind != 'logical' implies snapshot.empty() or snapshot.element.first().path = type
Affect this element
sdf-11 | Rule | If there's a type, its content must match the path name in the first element of a snapshot | kind != 'logical' implies snapshot.empty() or snapshot.element.first().path = type
StructureDefinition.baseDefinition
Element Id | StructureDefinition.baseDefinition
Definition | An absolute URI that is the base structure from which this type is derived, either by specialization or constraint.
Cardinality | 0..1
Type | canonical(StructureDefinition)
Hierarchy | This reference is part of a strict Hierarchy
Summary | true
Comments | If differential constraints are specified in this structure, they are applied to the base in a "differential" fashion. If there is no base, then the differential constraints cannot be provided (snapshot only). Differential structures are useful for the editing perspective, and snapshot structures are suitable for operational use. The FHIR Project provides a number of tools/services to populate snapshots from differential constraints. Logical Models have a base of "Element" or another logical model.
Invariants | Affect this element sdf-4 Rule If the structure is not abstract, then there SHALL be a baseDefinition abstract = true or baseDefinition.exists()
Affect this element
sdf-4 | Rule | If the structure is not abstract, then there SHALL be a baseDefinition | abstract = true or baseDefinition.exists()
StructureDefinition.derivation
Element Id | StructureDefinition.derivation
Definition | How the type relates to the baseDefinition.
Cardinality | 0..1
Terminology Binding | TypeDerivationRule (Required)
Type | code
Summary | true
Comments | If the definition is a specialization, then it adds new elements in the differential, and the snapshot includes the inherited elements. If the definition is a constraint, then it cannot define new elements, it can only make new rules about existing content (see Profiling Resources).
StructureDefinition.snapshot
Element Id | StructureDefinition.snapshot
Definition | A snapshot view is expressed in a standalone form that can be used and interpreted without considering the base StructureDefinition.
Cardinality | 0..1
Invariants | Defined on this element sdf-3 Rule Each element definition in a snapshot must have a formal definition and cardinalities element.all(definition.exists() and min.exists() and max.exists()) sdf-8 Rule All snapshot elements must start with the StructureDefinition's specified type for non-logical models, or with the same type name for logical models (%resource.kind = 'logical' or element.first().path = %resource.type) and element.tail().all(path.startsWith(%resource.snapshot.element.first().path&'.')) sdf-8b Rule All snapshot elements must have a base definition element.all(base.exists()) Affect this element sdf-6 Rule A structure must have either a differential, or a snapshot (or both) snapshot.exists() or differential.exists()
Defined on this element
sdf-3 | Rule | Each element definition in a snapshot must have a formal definition and cardinalities | element.all(definition.exists() and min.exists() and max.exists())
sdf-8 | Rule | All snapshot elements must start with the StructureDefinition's specified type for non-logical models, or with the same type name for logical models | (%resource.kind = 'logical' or element.first().path = %resource.type) and element.tail().all(path.startsWith(%resource.snapshot.element.first().path&'.'))
sdf-8b | Rule | All snapshot elements must have a base definition | element.all(base.exists())
Affect this element
sdf-6 | Rule | A structure must have either a differential, or a snapshot (or both) | snapshot.exists() or differential.exists()
StructureDefinition.snapshot.element

Element Id | StructureDefinition.snapshot.element
Definition | Captures constraints on each element within the resource.
Cardinality | 1..*
Type | ElementDefinition
Invariants | Defined on this element sdf-10 Rule provide either a binding reference or a description (or both) binding.empty() or binding.valueSet.exists() or binding.description.exists() Affect this element sdf-3 Rule Each element definition in a snapshot must have a formal definition and cardinalities element.all(definition.exists() and min.exists() and max.exists())
Defined on this element
sdf-10 | Rule | provide either a binding reference or a description (or both) | binding.empty() or binding.valueSet.exists() or binding.description.exists()
Affect this element
sdf-3 | Rule | Each element definition in a snapshot must have a formal definition and cardinalities | element.all(definition.exists() and min.exists() and max.exists())
StructureDefinition.differential
Element Id | StructureDefinition.differential
Definition | A differential view is expressed relative to the base StructureDefinition - a statement of differences that it applies.
Cardinality | 0..1
Invariants | Defined on this element sdf-20 Rule No slicing on the root element element.where(path.contains('.').not()).slicing.empty() sdf-8a Rule In any differential, all the elements must start with the StructureDefinition's specified type for non-logical models, or with the same type name for logical models (%resource.kind = 'logical' or element.first().path.startsWith(%resource.type)) and (element.tail().empty() or element.tail().all(path.startsWith(%resource.differential.element.first().path.replaceMatches('\\..*','')&'.'))) Affect this element sdf-6 Rule A structure must have either a differential, or a snapshot (or both) snapshot.exists() or differential.exists()
Defined on this element
sdf-20 | Rule | No slicing on the root element | element.where(path.contains('.').not()).slicing.empty()
sdf-8a | Rule | In any differential, all the elements must start with the StructureDefinition's specified type for non-logical models, or with the same type name for logical models | (%resource.kind = 'logical' or element.first().path.startsWith(%resource.type)) and (element.tail().empty() or element.tail().all(path.startsWith(%resource.differential.element.first().path.replaceMatches('\\..*','')&'.')))
Affect this element
sdf-6 | Rule | A structure must have either a differential, or a snapshot (or both) | snapshot.exists() or differential.exists()
StructureDefinition.differential.element
Element Id | StructureDefinition.differential.element
Definition | Captures constraints on each element within the resource.
Cardinality | 1..*
Type | ElementDefinition

5.3.9 Resource StructureDefinition - Mappings

URL: https://hl7.org/fhir/R4/structuredefinition-mappings.html

5.3.9 Resource StructureDefinition - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

ISO 11179

Mappings for the structuredefinition resource (see Mappings to Other Standards for further information & status).

5.3.9.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

StructureDefinition
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

5.3.9.2 Workflow Pattern (http://hl7.org/fhir/workflow)

StructureDefinition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status

experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright

5.3.9.3 RIM Mapping (http://hl7.org/v3)

StructureDefinition | n/a

5.3.9.4 ISO 11179 (http://metadata-standards.org/11179/)

These mappings are included to indicate where properties of the data elements defined by the FHIR specification relate to similar fields in the ISO 11179 specification.

StructureDefinition
url
identifier | Identifier part
version | Version
name
title
status | Registration status
experimental | Registration status
date
publisher | Registration Authority
contact
description
useContext
jurisdiction
purpose
copyright
keyword | keywords
fhirVersion
mapping
identity
uri
name
comment
kind
abstract
context
type
expression
contextInvariant
type
baseDefinition
derivation
snapshot
element | --
differential
element | --

5.3.10 Resource StructureDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/structuredefinition-profiles.html

5.3.10 Resource StructureDefinition - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the StructureDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

5.3.10.1 Profiles

No Profiles defined for this resource

5.3.10.2 Extensions

11179-permitted-value-conceptmap | StructureDefinition.snapshot.element.binding.valueSet, StructureDefinition.differential.element.binding.valueSet | permitted-value-conceptmap | for ISO 11179 Element Definition Profile

11179-permitted-value-valueset | StructureDefinition.snapshot.element.binding.valueSet, StructureDefinition.differential.element.binding.valueSet | permitted-value-valueset | for ISO 11179 Element Definition Profile
elementdefinition-namespace | StructureDefinition | namespace | for Common extensions for ElementDefinition data type
replaces | StructureDefinition | replaces | for General Extensions for use by FHIR Implementers
resource-approvalDate | StructureDefinition | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | StructureDefinition | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | StructureDefinition | lastReviewDate | for Resource HL7 Extensions
structuredefinition-ancestor | StructureDefinition | ancestor | for Standard Structure Definition Extensions
structuredefinition-applicable-version | StructureDefinition | applicable-version | for Standard Structure Definition Extensions
structuredefinition-category | StructureDefinition | category | for Standard Structure Definition Extensions
structuredefinition-codegen-super | StructureDefinition.baseDefinition | codegen-super | for Standard Structure Definition Extensions
structuredefinition-dependencies | StructureDefinition | dependencies | for Standard Structure Definition Extensions
structuredefinition-fmm-no-warnings | StructureDefinition | fmm-no-warnings | for Standard Structure Definition Extensions
structuredefinition-normative-version | StructureDefinition | normative-version | for Standard Structure Definition Extensions
structuredefinition-security-category | StructureDefinition | security-category | for Standard Structure Definition Extensions
structuredefinition-summary | StructureDefinition | summary | for Standard Structure Definition Extensions
structuredefinition-table-name | StructureDefinition.mapping | table-name | for Standard Structure Definition Extensions
structuredefinition-template-status | StructureDefinition | template-status | for Standard Structure Definition Extensions
structuredefinition-xml-no-order | StructureDefinition | xml-no-order | for Standard Structure Definition Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

questionnaire-referenceProfile | referenceProfile | for Core extensions for Questionnaire
structuredefinition-dependencies | dependencies | for Standard Structure Definition Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

5.3.10.3 Search Extensions

No Search Extensions defined for this resource

5.3.11 Resource StructureDefinition - Operations

URL: https://hl7.org/fhir/R4/structuredefinition-operations.html

5.3.11 Resource StructureDefinition - Operations

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This resource has 2 operations operations associated with it:

$questionnaire | Build Questionnaire | Trial Use
$snapshot | Generate Snapshot | Trial Use

For more information about operations, including how they are invoked, see Operations.

5.3.11.1 Operation $questionnaire on StructureDefinition

URL: https://hl7.org/fhir/R4/structuredefinition-operation-questionnaire.html

5.3.11.1 Operation $questionnaire on StructureDefinition

FHIR Infrastructure Work Group | Maturity Level: 5 | Trial Use | Compartments: N/A

Generates a Questionnaire instance based on a specified StructureDefinition, creating questions for each core element or extension element found in the StructureDefinition.

If the operation is not called at the instance level, one of the identifier, profile or url 'in' parameters must be provided. If more than one is specified, servers may raise an error or may resolve with the parameter of their choice. If called at the instance level, these parameters will be ignored. The response will contain a Questionnaire instance based on the specified StructureDefinition and/or an OperationOutcome resource with errors or warnings. Nested groups are used to handle complex structures and data types. If the 'supportedOnly' parameter is set to true, only those elements marked as "must support" will be included.

This operation is intended to enable auto-generation of simple interfaces for arbitrary profiles. The 'questionnaire' approach to data entry has limitations that will make it less optimal than custom-defined interfaces. However, this function may be useful for simple applications or for systems that wish to support

"non-core" resources with minimal development effort.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/StructureDefinition-questionnaire
```

Formal Definition (as a OperationDefinition).

URL: [base]/StructureDefinition/$questionnaire

URL: [base]/StructureDefinition/[id]/$questionnaire

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

identifier | 0..1 | canonical(StructureDefinition) | | | A logical identifier (i.e. 'StructureDefinition.identifier''). The server must know the StructureDefinition or be able to retrieve it from other known repositories.

profile | 0..1 | string (token) | | | The StructureDefinition is provided directly as part of the request. Servers may choose not to accept profiles in this fashion

url | 0..1 | canonical(StructureDefinition) | | | The StructureDefinition's official URL (i.e. 'StructureDefinition.url'). The server must know the StructureDefinition or be able to retrieve it from other known repositories.

supportedOnly | 0..1 | boolean | | | If true, the questionnaire will only include those elements marked as "mustSupport='true'" in the StructureDefinition.

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Questionnaire | | | The questionnaire form generated based on the StructureDefinition. Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

Open Issue: Ideally, extensions should be populated in the generated Questionnaire that will support taking QuestionnaireResponse resources generated from the Questionnaire and turning them back into the appropriate resources.

5.3.11.1.1 Examples

Request:

```
GET /fhir/StructureDefinition/daf-patient/$questionnaire
```

Response:

```
HTTP/1.1 200 OK
[other headers]

See DAF Patient Questionnaire
```

For more information about operations, including how they are invoked, see Operations.

5.3.11.2 Operation $snapshot on StructureDefinition

URL: https://hl7.org/fhir/R4/structuredefinition-operation-snapshot.html

5.3.11.2 Operation $snapshot on StructureDefinition

FHIR Infrastructure Work Group | Maturity Level: 5 | Trial Use | Compartments: N/A

Generates a StructureDefinition instance with a snapshot, based on a differential in a specified StructureDefinition.

If the operation is not called at the instance level, either definition or url 'in' parameters must be provided. If more than one is specified, servers may raise an error or may resolve with the parameter of their choice. If called at the instance level, these parameters will be ignored.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/StructureDefinition-snapshot
```

Formal Definition (as a OperationDefinition).

URL: [base]/StructureDefinition/$snapshot

URL: [base]/StructureDefinition/[id]/$snapshot

This is not an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

definition | 0..1 | StructureDefinition | | | The StructureDefinition is provided directly as part of the request. Servers may choose not to accept profiles in this fashion

url | 0..1 | string (token) | | | The StructureDefinition's canonical URL (i.e. 'StructureDefinition.url'). The server must know the structure definition, or be able to retrieve it from other known repositories.

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | StructureDefinition | | | The structure definition with a snapshot Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

5.3.11.2.1 Examples

Request:

```
GET /fhir/StructureDefinition/daf-patient/$snapshot
Accept: application/json+fhir
```

Response:

```
HTTP/1.1 200 OK
Content-Type: application/json+fhir
[other headers]

{
  "ResourceType" : "StructureDefinition",
  ... (including snapshot) ...
}
```

For more information about operations, including how they are invoked, see Operations.

5.3.12 Resource StructureDefinition - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/structuredefinition-version-maps.html

No validation errors - all conversions are clean

5.3.12 Resource StructureDefinition - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for StructureDefinition.

Functional status for this map: 206 tests that all execute ok. 4 fail round-trip testing and 206 r3 resources are invalid (0 errors). (see documentation)

5.3.12.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/StructureDefinition3to4" = "R3 to R4 Conversions for StructureDefinition"

uses "http://hl7.org/fhir/3.0/StructureDefinition/StructureDefinition" alias StructureDefinitionR3 as source
uses "http://hl7.org/fhir/StructureDefinition/StructureDefinition" alias StructureDefinition as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group StructureDefinition(source src : StructureDefinitionR3, target tgt : StructureDefinition) extends DomainResource
    <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.copyright -> tgt.copyright;
  src.keyword -> tgt.keyword;
  src.fhirVersion -> tgt.fhirVersion;
  src.mapping as vs0 -> tgt.mapping as vt0 then {
    vs0.identity -> vt0.identity;
    vs0.uri -> vt0.uri;
    vs0.name -> vt0.name;
    vs0.comment -> vt0.comment;
  };
  src.kind -> tgt.kind;
  src.abstract -> tgt.abstract;
  src.type -> tgt.type;
  src.baseDefinition -> tgt.baseDefinition;
  src.contextInvariant -> tgt.contextInvariant;
  src.derivation -> tgt.derivation;
  src.snapshot as vs0 -> tgt.snapshot as vt0 then {
    vs0.element -> vt0.element;
  };
  src.differential as vs0 -> tgt.differential as vt0 then {
    vs0.element -> vt0.element;
  };
}
```

5.3.12.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/StructureDefinition4to3" = "R4 to R3 Conversion for StructureDefinition"

uses "http://hl7.org/fhir/StructureDefinition/StructureDefinition" alias StructureDefinition as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/StructureDefinition" alias StructureDefinitionR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group StructureDefinition(source src : StructureDefinition, target tgt : StructureDefinitionR3) extends DomainResource
```

```
    <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.copyright -> tgt.copyright;
  src.keyword -> tgt.keyword;
  src.fhirVersion -> tgt.fhirVersion;
  src.mapping as vs0 -> tgt.mapping as vt0 then {
    vs0.identity -> vt0.identity;
    vs0.uri -> vt0.uri;
    vs0.name -> vt0.name;
    vs0.comment -> vt0.comment;
  };
  src.kind -> tgt.kind;
  src.abstract -> tgt.abstract;
  src.type -> tgt.type;
  src.baseDefinition -> tgt.baseDefinition;
  src.contextInvariant -> tgt.contextInvariant;
  src.derivation -> tgt.derivation;
  src.snapshot as vs0 -> tgt.snapshot as vt0 then {
    vs0.element -> vt0.element;
  };
  src.differential as vs0 -> tgt.differential as vt0 then {
    vs0.element -> vt0.element;
  };
}
```

5.3.12.3 R4 Validation Errors

5.3.13 Standard Structure Definition Extensions

URL: https://hl7.org/fhir/R4/structuredefinition-extensions.html

This profile describes common extensions that are used with Structure Definitions

5.3.13 Standard Structure Definition Extensions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

5.3.13.1 Content

Extensions:

structuredefinition-display-hint | display-hint : Hinting information for the narrative generator - a series of name: value; pairs.

structuredefinition-template-status | template-status : Status code taken from HL7 template specification - allows for alignment with the template DSTU, and has more authoring status codes.

structuredefinition-fmm | fmm : The FMM level assigned to the artifact.

structuredefinition-fmm-no-warnings | fmm-no-warnings : The FMM level that would be assigned to the artifact if it had no warnings.

structuredefinition-wg | wg : The work group that owns and maintains this resource.

structuredefinition-explicit-type-name | explicit-type-name : A name to use for the type, in implementations. This is a suggestion; it's not a normative part of the FHIR specification, but it does appear in the UML diagrams, and is used in generated code, schemas, etc.to identify the type.

structuredefinition-fhir-type | fhir-type : The formal FHIR type of a property - used for the value property of a primitive type (for which there is no type in the FHIR typing system), and Element.id and Extension.url.

structuredefinition-summary | summary : Additional text for the summary presentation.

structuredefinition-ancestor | ancestor : A canonical reference to a StructureDefinition that this is derived from. This is a de-normalization of a chain of StructureDefinition.baseDefinition.

structuredefinition-category | category : The category under which the resource type is presented on the official resource list.

structuredefinition-table-name | table-name : A name to use to show mappings of this type in the generated summary tables.

structuredefinition-standards-status | standards-status : The Current HL7 ballot/Standards status of this artifact.

structuredefinition-hierarchy | hierarchy : For circular references (references back to the same type of resource): whether they are allowed to transitively point back to the same instance (false), or whether the relationship must be a strict one-way hierarchy (true).

structuredefinition-xml-no-order | xml-no-order : Whether elements can come in any order in XML.

structuredefinition-codegen-super | codegen-super : A specific instruction to use an intermediate parent class when generating code for the classes.

structuredefinition-dependencies | dependencies : Instances can only be valid against this StructureDefinition, if they also sucessfully validate against the dependent profile identified in this extension.

structuredefinition-applicable-version | applicable-version : An additional version that this profile apples to, other than the version explicitly stated in StructureDefinition.fhirVersion.

structuredefinition-normative-version | normative-version : If this StructureDefinition is normative, which was the first normative version.

structuredefinition-security-category | security-category : Provides general guidance around the kind of access Control to Read, Search, Create, Update, or Delete the resource.

5.3.14 JSON Schema for StructureDefinition

URL: https://hl7.org/fhir/R4/structuredefinition.schema.json.html

5.3.14 JSON Schema for StructureDefinition

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for StructureDefinition, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/StructureDefinition",
  "$ref": "#/definitions/StructureDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "StructureDefinition": {
      "description": "A definition of a FHIR structure. This resource is used to describe the underlying resources, data
     types defined in FHIR, and also for describing extensions and constraints on resources and data types.",
      "properties": {
        "resourceType": {
          "description": "This is a StructureDefinition resource",
          "const": "StructureDefinition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
```

```
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this structure definition when it is referenced in a
specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which at which an authoritative instance of this structure definition is (or will
 be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the structure
definition is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this structure definition when it is represented
in other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the structure definition when it is
referenced in a specification, model, design or instance. This is an arbitrary value managed by the structure
definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if
 a managed version is not available. There is also no expectation that versions can be placed in a lexicographical
sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the structure definition. This name should be usable as an
 identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the structure definition.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this structure definition. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this structure definition is authored for testing purposes
(or education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the structure definition was published. The date must
change when the business version changes and it must change if the status code changes. In addition, it should
change when the substantive content of the structure definition changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
    "publisher": {
      "description": "The name of the organization or individual that published the structure definition.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the structure definition from a consumer\u0027s
perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate structure definition
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the structure definition is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this structure definition is needed and why it has been designed as it
has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the structure definition and/or its contents. Copyright
statements are generally legal restrictions on the use and publishing of the structure definition.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "keyword": {
      "description": "A set of key words or terms from external terminologies that may be used to assist with
indexing and searching of templates nby describing the use of this structure definition, or the content it
describes.",
      "items": {
        "$ref": "Coding.schema.json#/definitions/Coding"
      },
      "type": "array"
    },
    "fhirVersion": {
      "description": "The version of the FHIR specification on which this StructureDefinition is based - this is the
 formal version of the specification, without the revision number, e.g. [publication].[major].[minor], which is
4.0.1. for this version.",
      "enum": [
        "0.01",
        "0.05",
        "0.06",
        "0.11",
        "0.0.80",
        "0.0.81",
        "0.0.82",
        "0.4.0",
        "0.5.0",
        "1.0.0",
        "1.0.1",
        "1.0.2",
        "1.1.0",
        "1.4.0",
        "1.6.0",
        "1.8.0",
```

```
        "3.0.0",
        "3.0.1",
        "3.3.0",
        "3.5.0",
        "4.0.0",
        "4.0.1"
      ]
    },
    "_fhirVersion": {
      "description": "Extensions for fhirVersion",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "mapping": {
      "description": "An external specification that the content is mapped to.",
      "items": {
        "$ref": "#/definitions/StructureDefinition_Mapping"
      },
      "type": "array"
    },
    "kind": {
      "description": "Defines the kind of structure that this definition is describing.",
      "enum": [
        "primitive-type",
        "complex-type",
        "resource",
        "logical"
      ]
    },
    "_kind": {
      "description": "Extensions for kind",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "abstract": {
      "description": "Whether structure this definition describes is abstract or not  - that is, whether the
structure is not intended to be instantiated. For Resources and Data types, abstract types will never be exchanged
between systems.",
      "$ref": "#/definitions/boolean"
    },
    "_abstract": {
      "description": "Extensions for abstract",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "context": {
      "description": "Identifies the types of resource or data type elements to which the extension can be
applied.",
      "items": {
        "$ref": "#/definitions/StructureDefinition_Context"
      },
      "type": "array"
    },
    "contextInvariant": {
      "description": "A set of rules as FHIRPath Invariants about when the extension can be used (e.g. co-occurrence
 variants for the extension). All the rules must be true.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_contextInvariant": {
      "description": "Extensions for contextInvariant",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "type": {
      "description": "The type this structure describes. If the derivation kind is \u0027specialization\u0027 then
this is the master definition for a type, and there is always one of these (a data type, an extension, a resource,
including abstract ones). Otherwise the structure definition is a constraint on the stated type (and in this case,
the type cannot be an abstract type).  References are URLs that are relative to
http://hl7.org/fhir/StructureDefinition e.g. \"string\" is a reference to
http://hl7.org/fhir/StructureDefinition/string. Absolute URLs are only allowed in logical models.",
      "$ref": "#/definitions/uri"
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "baseDefinition": {
      "description": "An absolute URI that is the base structure from which this type is derived, either by
specialization or constraint.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "derivation": {
      "description": "How the type relates to the baseDefinition.",
      "enum": [
        "specialization",
        "constraint"
      ]
```

```
    },
    "_derivation": {
      "description": "Extensions for derivation",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "snapshot": {
      "description": "A snapshot view is expressed in a standalone form that can be used and interpreted without
considering the base StructureDefinition.",
      "$ref": "#/definitions/StructureDefinition_Snapshot"
    },
    "differential": {
      "description": "A differential view is expressed relative to the base StructureDefinition - a statement of
differences that it applies.",
      "$ref": "#/definitions/StructureDefinition_Differential"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"StructureDefinition_Mapping": {
  "description": "A definition of a FHIR structure. This resource is used to describe the underlying resources, data
 types defined in FHIR, and also for describing extensions and constraints on resources and data types.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identity": {
      "description": "An Internal id that is used to identify this mapping set when specific mappings are made.",
      "$ref": "#/definitions/id"
    },
    "_identity": {
      "description": "Extensions for identity",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "uri": {
      "description": "An absolute URI that identifies the specification that this mapping is expressed to.",
      "$ref": "#/definitions/uri"
    },
    "_uri": {
      "description": "Extensions for uri",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A name for the specification that is being mapped to.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "comment": {
      "description": "Comments about this mapping, including version notes, issues, scope limitations, and other
important notes for usage.",
      "$ref": "#/definitions/string"
    },
    "_comment": {
      "description": "Extensions for comment",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
```

```
},
"StructureDefinition_Context": {
  "description": "A definition of a FHIR structure. This resource is used to describe the underlying resources, data
 types defined in FHIR, and also for describing extensions and constraints on resources and data types.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Defines how to interpret the expression that defines what the context of the extension is.",
      "enum": [
        "fhirpath",
        "element",
        "extension"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "expression": {
      "description": "An expression that defines where an extension can be used in resources.",
      "$ref": "#/definitions/string"
    },
    "_expression": {
      "description": "Extensions for expression",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"StructureDefinition_Snapshot": {
  "description": "A definition of a FHIR structure. This resource is used to describe the underlying resources, data
 types defined in FHIR, and also for describing extensions and constraints on resources and data types.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
```

```
        },
        "element": {
          "description": "Captures constraints on each element within the resource.",
          "items": {
            "$ref": "ElementDefinition.schema.json#/definitions/ElementDefinition"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "element"
      ]
    },
    "StructureDefinition_Differential": {
      "description": "A definition of a FHIR structure. This resource is used to describe the underlying resources, data
     types defined in FHIR, and also for describing extensions and constraints on resources and data types.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "element": {
          "description": "Captures constraints on each element within the resource.",
          "items": {
            "$ref": "ElementDefinition.schema.json#/definitions/ElementDefinition"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "element"
      ]
    }
  }
}
```


5.3.15 Resource StructureDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/structuredefinition-dependencies.html

5.3.15 Resource StructureDefinition - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| Some of the content on this page (marked clearly) has been approved as part of an ANSI standard. See the Conformance Package for further details.

Dependency Graph for StructureDefinition FMM level 5

Name | Card. | Type | Dependency Analysis
StructureDefinition | | DomainResource
url | 1..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 1..1 | string
title | 0..1 | string
status | 1..1 | code

experimental | 0..1 | boolean
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext TU | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
copyright | 0..1 | markdown
keyword | 0..* | Coding
fhirVersion | 0..1 | code
mapping | 0..* | BackboneElement
identity | 1..1 | id
uri | 0..1 | uri
name | 0..1 | string
comment | 0..1 | string
kind | 1..1 | code
abstract | 1..1 | boolean
context | 0..* | BackboneElement
type | 1..1 | code
expression | 1..1 | string
contextInvariant | 0..* | string
type | 1..1 | uri
baseDefinition | 0..1 | canonical(StructureDefinition)
derivation | 0..1 | code
snapshot | 0..1 | BackboneElement
element | 1..* | ElementDefinition | Type Error: (ElementDefinition = FMM2-Normative vs. Element = FMM5-Normative)
differential | 0..1 | BackboneElement
element | 1..* | ElementDefinition | Type Error: (ElementDefinition = FMM2-Normative vs. Element = FMM5-Normative)

5.4 Resource OperationDefinition - Content

URL: https://hl7.org/fhir/R4/operationdefinition.html
FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Conformance Package for further details.

A formal computable definition of an operation (on the RESTful interface) or a named query (using the search interaction).

5.4.1 Scope and Usage

The OperationDefinition resource provides a formal computable definition of an operation or a named query. The OperationDefinition serves two principal purposes:

• To allow for automatic determination of system compatibility

• To allow for dynamic generation of forms to drive the operations

See below for further information about these, and about how operations and named queries are executed.

5.4.2 Boundaries and Relationships

OperationDefinitions are published to define operations that servers can implement in a common fashion. The FHIR specification itself describes some (see below), and other organizations (including IHE, national programs, jurisdictions and vendors) are able to publish additional OperationDefinitions.

OperationDefinition resources are referred to from two different places:

• From a CapabilityStatement, to declare what operations a system does or should implement

• From another OperationDefinition resource. This allows for a server to describe a limited implementation of a standard operation, or to allow traceability if the server has to rename the operation due to a name clash

This resource is referenced by CapabilityStatement, CarePlan, Communication, FamilyMemberHistory, itself and Procedure

5.4.3 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff

- All

Structure

Name | Flags | Card. | Type | Description & Constraints
OperationDefinition | IN | | DomainResource | Definition of an operation or a named query + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this operation definition, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the operation definition
name | ΣI | 1..1 | string | Name for this operation definition (computer friendly)
title | Σ | 0..1 | string | Name for this operation definition (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
kind | Σ | 1..1 | code | operation | query OperationKind (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the operation definition
useContext | ΣTU | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for operation definition (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this operation definition is defined
affectsState | Σ | 0..1 | boolean | Whether content is changed by the operation
code | Σ | 1..1 | code | Name used to invoke the operation
comment | | 0..1 | markdown | Additional information about use
base | Σ | 0..1 | canonical(OperationDefinition) | Marks this as a profile of the base
resource | Σ | 0..* | code | Types this operation applies to ResourceType (Required)
system | Σ | 1..1 | boolean | Invoke at the system level?
type | Σ | 1..1 | boolean | Invoke at the type level?
instance | Σ | 1..1 | boolean | Invoke on an instance?
inputProfile | | 0..1 | canonical(StructureDefinition) | Validation information for in parameters
outputProfile | | 0..1 | canonical(StructureDefinition) | Validation information for out parameters
parameter | I | 0..* | BackboneElement | Parameters for the operation/query + Rule: Either a type must be provided, or parts + Rule: A search type can only be specified for parameters of type string + Rule: A targetProfile can only be specified for parameters of type Reference or Canonical
name | | 1..1 | code | Name in Parameters.parameter.name or in URL
use | | 1..1 | code | in | out OperationParameterUse (Required)
min | | 1..1 | integer | Minimum Cardinality
max | | 1..1 | string | Maximum Cardinality (a number or *)
documentation | | 0..1 | string | Description of meaning/use
type | I | 0..1 | code | What type this parameter has FHIRAllTypes (Required)
targetProfile | | 0..* | canonical(StructureDefinition) | If type is Reference | canonical, allowed targets
searchType | I | 0..1 | code | number | date | string | token | reference | composite | quantity | uri | special SearchParamType (Required)
binding | | 0..1 | BackboneElement | ValueSet details if this is coded
strength | | 1..1 | code | required | extensible | preferred | example BindingStrength (Required)
valueSet | | 1..1 | canonical(ValueSet) | Source of value set
referencedFrom | TU | 0..* | BackboneElement | References to this parameter
source | | 1..1 | string | Referencing parameter
sourceId | | 0..1 | string | Element id of reference
part | I | 0..* | see parameter | Parts of a nested Parameter
overload | | 0..* | BackboneElement | Define overloaded variants for when generating code
parameterName | | 0..* | string | Name of parameter to include in overload
comment | | 0..1 | string | Comments to go on overload
Documentation for this format

UML Diagram (Legend)

XML Template

```
<OperationDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this operation definition, represented as a URI (globally
    unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the operation definition -->
 <name value="[string]"/><!--  1..1 Name for this operation definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this operation definition (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <kind value="[code]"/><!-- 1..1 operation | query -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the operation definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for operation definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this operation definition is defined -->
 <affectsState value="[boolean]"/><!-- 0..1 Whether content is changed by the operation -->
```

```
 <code value="[code]"/><!-- 1..1 Name used to invoke the operation -->
 <comment value="[markdown]"/><!-- 0..1 Additional information about use -->
 <base><!-- 0..1 canonical(OperationDefinition) Marks this as a profile of the base --></base>
 <resource value="[code]"/><!-- 0..* Types this operation applies to -->
 <system value="[boolean]"/><!-- 1..1 Invoke at the system level? -->
 <type value="[boolean]"/><!-- 1..1 Invoke at the type level? -->
 <instance value="[boolean]"/><!-- 1..1 Invoke on an instance? -->
 <inputProfile><!-- 0..1 canonical(StructureDefinition) Validation information for in parameters --></inputProfile>
 <outputProfile><!-- 0..1 canonical(StructureDefinition) Validation information for out parameters --></outputProfile>
 <parameter>  <!-- 0..* Parameters for the operation/query -->
  <name value="[code]"/><!-- 1..1 Name in Parameters.parameter.name or in URL -->
  <use value="[code]"/><!-- 1..1 in | out -->
  <min value="[integer]"/><!-- 1..1 Minimum Cardinality -->
  <max value="[string]"/><!-- 1..1 Maximum Cardinality (a number or *) -->
  <documentation value="[string]"/><!-- 0..1 Description of meaning/use -->
  <type value="[code]"/><!--  0..1 What type this parameter has -->
  <targetProfile><!-- 0..* canonical(StructureDefinition) If type is Reference | canonical, allowed targets
    --></targetProfile>
  <searchType value="[code]"/><!--  0..1 number | date | string | token | reference | composite | quantity | uri |
    special -->
  <binding>  <!-- 0..1 ValueSet details if this is coded -->
   <strength value="[code]"/><!-- 1..1 required | extensible | preferred | example -->
   <valueSet><!-- 1..1 canonical(ValueSet) Source of value set --></valueSet>
  </binding>
  <referencedFrom>  <!-- 0..* References to this parameter -->
   <source value="[string]"/><!-- 1..1 Referencing parameter -->
   <sourceId value="[string]"/><!-- 0..1 Element id of reference -->
  </referencedFrom>
  <part><!--  0..* Content as for OperationDefinition.parameter Parts of a nested Parameter --></part>
 </parameter>
 <overload>  <!-- 0..* Define overloaded variants for when  generating code -->
  <parameterName value="[string]"/><!-- 0..* Name of parameter to include in overload -->
  <comment value="[string]"/><!-- 0..1 Comments to go on overload -->
 </overload>
</OperationDefinition>
```

JSON Template

```
{
  "resourceType" : "OperationDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this operation definition, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the operation definition
  "name" : "<string>", // C? R!  Name for this operation definition (computer friendly)
  "title" : "<string>", // Name for this operation definition (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "kind" : "<code>", // R!  operation | query
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the operation definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for operation definition (if applicable)
  "purpose" : "<markdown>", // Why this operation definition is defined
  "affectsState" : <boolean>, // Whether content is changed by the operation
  "code" : "<code>", // R!  Name used to invoke the operation
  "comment" : "<markdown>", // Additional information about use
  "base" : { canonical(OperationDefinition) }, // Marks this as a profile of the base
  "resource" : ["<code>"], // Types this operation applies to
  "system" : <boolean>, // R!  Invoke at the system level?
  "type" : <boolean>, // R!  Invoke at the type level?
  "instance" : <boolean>, // R!  Invoke on an instance?
  "inputProfile" : { canonical(StructureDefinition) }, // Validation information for in parameters
  "outputProfile" : { canonical(StructureDefinition) }, // Validation information for out parameters
  "parameter" : [{ // Parameters for the operation/query
    "name" : "<code>", // R!  Name in Parameters.parameter.name or in URL
    "use" : "<code>", // R!  in | out
    "min" : <integer>, // R!  Minimum Cardinality
    "max" : "<string>", // R!  Maximum Cardinality (a number or *)
    "documentation" : "<string>", // Description of meaning/use
    "type" : "<code>", // C? What type this parameter has
    "targetProfile" : [{ canonical(StructureDefinition) }], // If type is Reference | canonical, allowed targets
    "searchType" : "<code>", // C? number | date | string | token | reference | composite | quantity | uri | special
    "binding" : { // ValueSet details if this is coded
      "strength" : "<code>", // R!  required | extensible | preferred | example
      "valueSet" : { canonical(ValueSet) } // R!  Source of value set
    },
    "referencedFrom" : [{ // References to this parameter
      "source" : "<string>", // R!  Referencing parameter
      "sourceId" : "<string>" // Element id of reference
    }],
    "part" : [{ Content as for OperationDefinition.parameter }] // C? Parts of a nested Parameter
  }],
  "overload" : [{ // Define overloaded variants for when  generating code
    "parameterName" : ["<string>"], // Name of parameter to include in overload
    "comment" : "<string>" // Comments to go on overload
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:OperationDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:OperationDefinition.url [ uri ]; # 0..1 Canonical identifier for this operation definition, represented as a URI
    (globally unique)
  fhir:OperationDefinition.version [ string ]; # 0..1 Business version of the operation definition
  fhir:OperationDefinition.name [ string ]; # 1..1 Name for this operation definition (computer friendly)
  fhir:OperationDefinition.title [ string ]; # 0..1 Name for this operation definition (human friendly)
  fhir:OperationDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:OperationDefinition.kind [ code ]; # 1..1 operation | query
  fhir:OperationDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:OperationDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:OperationDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:OperationDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:OperationDefinition.description [ markdown ]; # 0..1 Natural language description of the operation definition
  fhir:OperationDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:OperationDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for operation definition
     (if applicable)
  fhir:OperationDefinition.purpose [ markdown ]; # 0..1 Why this operation definition is defined
  fhir:OperationDefinition.affectsState [ boolean ]; # 0..1 Whether content is changed by the operation
  fhir:OperationDefinition.code [ code ]; # 1..1 Name used to invoke the operation
  fhir:OperationDefinition.comment [ markdown ]; # 0..1 Additional information about use
  fhir:OperationDefinition.base [ canonical(OperationDefinition) ]; # 0..1 Marks this as a profile of the base
  fhir:OperationDefinition.resource [ code ], ... ; # 0..* Types this operation applies to
  fhir:OperationDefinition.system [ boolean ]; # 1..1 Invoke at the system level?
  fhir:OperationDefinition.type [ boolean ]; # 1..1 Invoke at the type level?
  fhir:OperationDefinition.instance [ boolean ]; # 1..1 Invoke on an instance?
  fhir:OperationDefinition.inputProfile [ canonical(StructureDefinition) ]; # 0..1 Validation information for in
    parameters
  fhir:OperationDefinition.outputProfile [ canonical(StructureDefinition) ]; # 0..1 Validation information for out
    parameters
  fhir:OperationDefinition.parameter [ # 0..* Parameters for the operation/query
    fhir:OperationDefinition.parameter.name [ code ]; # 1..1 Name in Parameters.parameter.name or in URL
    fhir:OperationDefinition.parameter.use [ code ]; # 1..1 in | out
    fhir:OperationDefinition.parameter.min [ integer ]; # 1..1 Minimum Cardinality
    fhir:OperationDefinition.parameter.max [ string ]; # 1..1 Maximum Cardinality (a number or *)
    fhir:OperationDefinition.parameter.documentation [ string ]; # 0..1 Description of meaning/use
    fhir:OperationDefinition.parameter.type [ code ]; # 0..1 What type this parameter has
    fhir:OperationDefinition.parameter.targetProfile [ canonical(StructureDefinition) ], ... ; # 0..* If type is
    Reference | canonical, allowed targets
    fhir:OperationDefinition.parameter.searchType [ code ]; # 0..1 number | date | string | token | reference |
    composite | quantity | uri | special
    fhir:OperationDefinition.parameter.binding [ # 0..1 ValueSet details if this is coded
      fhir:OperationDefinition.parameter.binding.strength [ code ]; # 1..1 required | extensible | preferred | example
      fhir:OperationDefinition.parameter.binding.valueSet [ canonical(ValueSet) ]; # 1..1 Source of value set
    ];
    fhir:OperationDefinition.parameter.referencedFrom [ # 0..* References to this parameter
      fhir:OperationDefinition.parameter.referencedFrom.source [ string ]; # 1..1 Referencing parameter
      fhir:OperationDefinition.parameter.referencedFrom.sourceId [ string ]; # 0..1 Element id of reference
    ], ...;
    fhir:OperationDefinition.parameter.part [ See OperationDefinition.parameter ], ... ; # 0..* Parts of a nested
    Parameter
  ], ...;
  fhir:OperationDefinition.overload [ # 0..* Define overloaded variants for when  generating code
    fhir:OperationDefinition.overload.parameterName [ string ], ... ; # 0..* Name of parameter to include in overload
    fhir:OperationDefinition.overload.comment [ string ]; # 0..1 Comments to go on overload
  ], ...;
]
```

Changes since R3

OperationDefinition
OperationDefinition.title | Added Element
OperationDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
OperationDefinition.kind | Change value set from http://hl7.org/fhir/ValueSet/operation-kind to http://hl7.org/fhir/ValueSet/operation-kind|4.0.1
OperationDefinition.experimental | No longer marked as Modifier
OperationDefinition.affectsState | Renamed from idempotent to affectsState
OperationDefinition.comment | Type changed from string to markdown
OperationDefinition.base | Type changed from Reference(OperationDefinition) to canonical(OperationDefinition)
OperationDefinition.resource | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
OperationDefinition.inputProfile | Added Element
OperationDefinition.outputProfile | Added Element
OperationDefinition.parameter.use | Change value set from http://hl7.org/fhir/ValueSet/operation-parameter-use to http://hl7.org/fhir/ValueSet/operation-parameter-use|4.0.1
OperationDefinition.parameter.type | Change value set from http://hl7.org/fhir/ValueSet/all-types to http://hl7.org/fhir/ValueSet/all-types|4.0.1

OperationDefinition.parameter.targetProfile | Renamed from profile to targetProfile Max Cardinality changed from 1 to * Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
OperationDefinition.parameter.searchType | Change value set from http://hl7.org/fhir/ValueSet/search-param-type to http://hl7.org/fhir/ValueSet/search-param-type|4.0.1
OperationDefinition.parameter.binding.strength | Change value set from http://hl7.org/fhir/ValueSet/binding-strength to http://hl7.org/fhir/ValueSet/binding-strength|4.0.1
OperationDefinition.parameter.binding.valueSet | Renamed from valueSet[x] to valueSet Add Type canonical(ValueSet) Remove Types uri, Reference(ValueSet)
OperationDefinition.parameter.referencedFrom | Added Element
OperationDefinition.parameter.referencedFrom.source | Added Mandatory Element
OperationDefinition.parameter.referencedFrom.sourceId | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 37 tests that all execute ok. All tests pass round-trip testing and 34 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
OperationDefinition | IN | | DomainResource | Definition of an operation or a named query + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this operation definition, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the operation definition
name | ΣI | 1..1 | string | Name for this operation definition (computer friendly)
title | Σ | 0..1 | string | Name for this operation definition (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
kind | Σ | 1..1 | code | operation | query OperationKind (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the operation definition
useContext | ΣTU | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for operation definition (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this operation definition is defined
affectsState | Σ | 0..1 | boolean | Whether content is changed by the operation
code | Σ | 1..1 | code | Name used to invoke the operation
comment | | 0..1 | markdown | Additional information about use
base | Σ | 0..1 | canonical(OperationDefinition) | Marks this as a profile of the base
resource | Σ | 0..* | code | Types this operation applies to ResourceType (Required)
system | Σ | 1..1 | boolean | Invoke at the system level?
type | Σ | 1..1 | boolean | Invoke at the type level?
instance | Σ | 1..1 | boolean | Invoke on an instance?
inputProfile | | 0..1 | canonical(StructureDefinition) | Validation information for in parameters
outputProfile | | 0..1 | canonical(StructureDefinition) | Validation information for out parameters
parameter | I | 0..* | BackboneElement | Parameters for the operation/query + Rule: Either a type must be provided, or parts + Rule: A search type can only be specified for parameters of type string + Rule: A targetProfile can only be specified for parameters of type Reference or Canonical
name | | 1..1 | code | Name in Parameters.parameter.name or in URL
use | | 1..1 | code | in | out OperationParameterUse (Required)
min | | 1..1 | integer | Minimum Cardinality
max | | 1..1 | string | Maximum Cardinality (a number or *)
documentation | | 0..1 | string | Description of meaning/use
type | I | 0..1 | code | What type this parameter has FHIRAllTypes (Required)
targetProfile | | 0..* | canonical(StructureDefinition) | If type is Reference | canonical, allowed targets
searchType | I | 0..1 | code | number | date | string | token | reference | composite | quantity | uri | special SearchParamType (Required)
binding | | 0..1 | BackboneElement | ValueSet details if this is coded
strength | | 1..1 | code | required | extensible | preferred | example BindingStrength (Required)
valueSet | | 1..1 | canonical(ValueSet) | Source of value set
referencedFrom | TU | 0..* | BackboneElement | References to this parameter
source | | 1..1 | string | Referencing parameter
sourceId | | 0..1 | string | Element id of reference
part | I | 0..* | see parameter | Parts of a nested Parameter
overload | | 0..* | BackboneElement | Define overloaded variants for when generating code
parameterName | | 0..* | string | Name of parameter to include in overload
comment | | 0..1 | string | Comments to go on overload
Documentation for this format

UML Diagram (Legend)

XML Template

```
<OperationDefinition xmlns="http://hl7.org/fhir">
```

```
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this operation definition, represented as a URI (globally
    unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the operation definition -->
 <name value="[string]"/><!--  1..1 Name for this operation definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this operation definition (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <kind value="[code]"/><!-- 1..1 operation | query -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the operation definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for operation definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this operation definition is defined -->
 <affectsState value="[boolean]"/><!-- 0..1 Whether content is changed by the operation -->
 <code value="[code]"/><!-- 1..1 Name used to invoke the operation -->
 <comment value="[markdown]"/><!-- 0..1 Additional information about use -->
 <base><!-- 0..1 canonical(OperationDefinition) Marks this as a profile of the base --></base>
 <resource value="[code]"/><!-- 0..* Types this operation applies to -->
 <system value="[boolean]"/><!-- 1..1 Invoke at the system level? -->
 <type value="[boolean]"/><!-- 1..1 Invoke at the type level? -->
 <instance value="[boolean]"/><!-- 1..1 Invoke on an instance? -->
 <inputProfile><!-- 0..1 canonical(StructureDefinition) Validation information for in parameters --></inputProfile>
 <outputProfile><!-- 0..1 canonical(StructureDefinition) Validation information for out parameters --></outputProfile>
 <parameter>  <!-- 0..* Parameters for the operation/query -->
  <name value="[code]"/><!-- 1..1 Name in Parameters.parameter.name or in URL -->
  <use value="[code]"/><!-- 1..1 in | out -->
  <min value="[integer]"/><!-- 1..1 Minimum Cardinality -->
  <max value="[string]"/><!-- 1..1 Maximum Cardinality (a number or *) -->
  <documentation value="[string]"/><!-- 0..1 Description of meaning/use -->
  <type value="[code]"/><!--  0..1 What type this parameter has -->
  <targetProfile><!-- 0..* canonical(StructureDefinition) If type is Reference | canonical, allowed targets
     --></targetProfile>
  <searchType value="[code]"/><!--  0..1 number | date | string | token | reference | composite | quantity | uri |
     special -->
  <binding>  <!-- 0..1 ValueSet details if this is coded -->
   <strength value="[code]"/><!-- 1..1 required | extensible | preferred | example -->
   <valueSet><!-- 1..1 canonical(ValueSet) Source of value set --></valueSet>
  </binding>
  <referencedFrom>  <!-- 0..* References to this parameter -->
   <source value="[string]"/><!-- 1..1 Referencing parameter -->
   <sourceId value="[string]"/><!-- 0..1 Element id of reference -->
  </referencedFrom>
  <part><!--  0..* Content as for OperationDefinition.parameter Parts of a nested Parameter --></part>
 </parameter>
 <overload>  <!-- 0..* Define overloaded variants for when  generating code -->
  <parameterName value="[string]"/><!-- 0..* Name of parameter to include in overload -->
  <comment value="[string]"/><!-- 0..1 Comments to go on overload -->
 </overload>
</OperationDefinition>
```

JSON Template

```
{
  "resourceType" : "OperationDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this operation definition, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the operation definition
  "name" : "<string>", // C? R!  Name for this operation definition (computer friendly)
  "title" : "<string>", // Name for this operation definition (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "kind" : "<code>", // R!  operation | query
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the operation definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for operation definition (if applicable)
  "purpose" : "<markdown>", // Why this operation definition is defined
  "affectsState" : <boolean>, // Whether content is changed by the operation
  "code" : "<code>", // R!  Name used to invoke the operation
  "comment" : "<markdown>", // Additional information about use
  "base" : { canonical(OperationDefinition) }, // Marks this as a profile of the base
  "resource" : ["<code>"], // Types this operation applies to
  "system" : <boolean>, // R!  Invoke at the system level?
  "type" : <boolean>, // R!  Invoke at the type level?
  "instance" : <boolean>, // R!  Invoke on an instance?
  "inputProfile" : { canonical(StructureDefinition) }, // Validation information for in parameters
  "outputProfile" : { canonical(StructureDefinition) }, // Validation information for out parameters
  "parameter" : [{ // Parameters for the operation/query
    "name" : "<code>", // R!  Name in Parameters.parameter.name or in URL
    "use" : "<code>", // R!  in | out
    "min" : <integer>, // R!  Minimum Cardinality
    "max" : "<string>", // R!  Maximum Cardinality (a number or *)
```

```
    "documentation" : "<string>", // Description of meaning/use
    "type" : "<code>", // C? What type this parameter has
    "targetProfile" : [{ canonical(StructureDefinition) }], // If type is Reference | canonical, allowed targets
    "searchType" : "<code>", // C? number | date | string | token | reference | composite | quantity | uri | special
    "binding" : { // ValueSet details if this is coded
      "strength" : "<code>", // R!  required | extensible | preferred | example
      "valueSet" : { canonical(ValueSet) } // R!  Source of value set
    },
    "referencedFrom" : [{ // References to this parameter
      "source" : "<string>", // R!  Referencing parameter
      "sourceId" : "<string>" // Element id of reference
    }],
    "part" : [{ Content as for OperationDefinition.parameter }] // C? Parts of a nested Parameter
  }],
  "overload" : [{ // Define overloaded variants for when  generating code
    "parameterName" : ["<string>"], // Name of parameter to include in overload
    "comment" : "<string>" // Comments to go on overload
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:OperationDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:OperationDefinition.url [ uri ]; # 0..1 Canonical identifier for this operation definition, represented as a URI
    (globally unique)
  fhir:OperationDefinition.version [ string ]; # 0..1 Business version of the operation definition
  fhir:OperationDefinition.name [ string ]; # 1..1 Name for this operation definition (computer friendly)
  fhir:OperationDefinition.title [ string ]; # 0..1 Name for this operation definition (human friendly)
  fhir:OperationDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:OperationDefinition.kind [ code ]; # 1..1 operation | query
  fhir:OperationDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:OperationDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:OperationDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:OperationDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:OperationDefinition.description [ markdown ]; # 0..1 Natural language description of the operation definition
  fhir:OperationDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:OperationDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for operation definition
     (if applicable)
  fhir:OperationDefinition.purpose [ markdown ]; # 0..1 Why this operation definition is defined
  fhir:OperationDefinition.affectsState [ boolean ]; # 0..1 Whether content is changed by the operation
  fhir:OperationDefinition.code [ code ]; # 1..1 Name used to invoke the operation
  fhir:OperationDefinition.comment [ markdown ]; # 0..1 Additional information about use
  fhir:OperationDefinition.base [ canonical(OperationDefinition) ]; # 0..1 Marks this as a profile of the base
  fhir:OperationDefinition.resource [ code ], ... ; # 0..* Types this operation applies to
  fhir:OperationDefinition.system [ boolean ]; # 1..1 Invoke at the system level?
  fhir:OperationDefinition.type [ boolean ]; # 1..1 Invoke at the type level?
  fhir:OperationDefinition.instance [ boolean ]; # 1..1 Invoke on an instance?
  fhir:OperationDefinition.inputProfile [ canonical(StructureDefinition) ]; # 0..1 Validation information for in
    parameters
  fhir:OperationDefinition.outputProfile [ canonical(StructureDefinition) ]; # 0..1 Validation information for out
    parameters
  fhir:OperationDefinition.parameter [ # 0..* Parameters for the operation/query
    fhir:OperationDefinition.parameter.name [ code ]; # 1..1 Name in Parameters.parameter.name or in URL
    fhir:OperationDefinition.parameter.use [ code ]; # 1..1 in | out
    fhir:OperationDefinition.parameter.min [ integer ]; # 1..1 Minimum Cardinality
    fhir:OperationDefinition.parameter.max [ string ]; # 1..1 Maximum Cardinality (a number or *)
    fhir:OperationDefinition.parameter.documentation [ string ]; # 0..1 Description of meaning/use
    fhir:OperationDefinition.parameter.type [ code ]; # 0..1 What type this parameter has
    fhir:OperationDefinition.parameter.targetProfile [ canonical(StructureDefinition) ], ... ; # 0..* If type is
    Reference | canonical, allowed targets
    fhir:OperationDefinition.parameter.searchType [ code ]; # 0..1 number | date | string | token | reference |
    composite | quantity | uri | special
    fhir:OperationDefinition.parameter.binding [ # 0..1 ValueSet details if this is coded
      fhir:OperationDefinition.parameter.binding.strength [ code ]; # 1..1 required | extensible | preferred | example
      fhir:OperationDefinition.parameter.binding.valueSet [ canonical(ValueSet) ]; # 1..1 Source of value set
    ];
    fhir:OperationDefinition.parameter.referencedFrom [ # 0..* References to this parameter
      fhir:OperationDefinition.parameter.referencedFrom.source [ string ]; # 1..1 Referencing parameter
      fhir:OperationDefinition.parameter.referencedFrom.sourceId [ string ]; # 0..1 Element id of reference
    ], ...;
    fhir:OperationDefinition.parameter.part [ See OperationDefinition.parameter ], ... ; # 0..* Parts of a nested
    Parameter
  ], ...;
  fhir:OperationDefinition.overload [ # 0..* Define overloaded variants for when  generating code
    fhir:OperationDefinition.overload.parameterName [ string ], ... ; # 0..* Name of parameter to include in overload
    fhir:OperationDefinition.overload.comment [ string ]; # 0..1 Comments to go on overload
  ], ...;
]
```

Changes since Release 3

OperationDefinition

OperationDefinition.title | Added Element

OperationDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
OperationDefinition.kind | Change value set from http://hl7.org/fhir/ValueSet/operation-kind to http://hl7.org/fhir/ValueSet/operation-kind|4.0.1
OperationDefinition.experimental | No longer marked as Modifier
OperationDefinition.affectsState | Renamed from idempotent to affectsState
OperationDefinition.comment | Type changed from string to markdown
OperationDefinition.base | Type changed from Reference(OperationDefinition) to canonical(OperationDefinition)
OperationDefinition.resource | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
OperationDefinition.inputProfile | Added Element
OperationDefinition.outputProfile | Added Element
OperationDefinition.parameter.use | Change value set from http://hl7.org/fhir/ValueSet/operation-parameter-use to http://hl7.org/fhir/ValueSet/operation-parameter-use|4.0.1
OperationDefinition.parameter.type | Change value set from http://hl7.org/fhir/ValueSet/all-types to http://hl7.org/fhir/ValueSet/all-types|4.0.1
OperationDefinition.parameter.targetProfile | Renamed from profile to targetProfile Max Cardinality changed from 1 to * Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)
OperationDefinition.parameter.searchType | Change value set from http://hl7.org/fhir/ValueSet/search-param-type to http://hl7.org/fhir/ValueSet/search-param-type|4.0.1
OperationDefinition.parameter.binding.strength | Change value set from http://hl7.org/fhir/ValueSet/binding-strength to http://hl7.org/fhir/ValueSet/binding-strength|4.0.1
OperationDefinition.parameter.binding.valueSet | Renamed from valueSet[x] to valueSet Add Type canonical(ValueSet) Remove Types uri, Reference(ValueSet)
OperationDefinition.parameter.referencedFrom | Added Element
OperationDefinition.parameter.referencedFrom.source | Added Mandatory Element
OperationDefinition.parameter.referencedFrom.sourceId | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 37 tests that all execute ok. All tests pass round-trip testing and 34 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

5.4.3.1 Terminology Bindings

Path | Definition | Type | Reference
OperationDefinition.status | The lifecycle status of an artifact. | Required | PublicationStatus
OperationDefinition.kind | Whether an operation is a normal operation or a query. | Required | OperationKind
OperationDefinition.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
OperationDefinition.resource | One of the resource types defined as part of this version of FHIR. | Required | Resource Types
OperationDefinition.parameter.use | Whether an operation parameter is an input or an output parameter. | Required | OperationParameterUse
OperationDefinition.parameter.type | A list of all the concrete types defined in this version of the FHIR specification - Abstract Types, Data Types and Resource Types. | Required | All Types
OperationDefinition.parameter.searchType | Data types allowed to be used for search parameters. | Required | SearchParamType
OperationDefinition.parameter.binding.strength | Indication of the degree of conformance expectations associated with a binding. | Required | BindingStrength

5.4.3.2 Constraints

id | Level | Location | Description | Expression
opd-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
opd-1 | Rule | OperationDefinition.parameter | Either a type must be provided, or parts | type.exists() or part.exists()
opd-2 | Rule | OperationDefinition.parameter | A search type can only be specified for parameters of type string | searchType.exists() implies type = 'string'
opd-3 | Rule | OperationDefinition.parameter | A targetProfile can only be specified for parameters of type Reference or Canonical | targetProfile.exists() implies (type = 'Reference' or type = 'canonical')

5.4.3.3 Operations defined as part of this Specification

Base Operations (All resource types)
Validate a resource | [base]/[Resource]/$validate | [base]/[Resource]/[id]/$validate
Access a list of profiles, tags, and security labels | [base]/$meta | [base]/[Resource]/$meta | [base]/[Resource]/[id]/$meta
Add profiles, tags, and security labels to a resource | [base]/[Resource]/[id]/$meta-add
Delete profiles, tags, and security labels for a resource | [base]/[Resource]/[id]/$meta-delete
Convert from one form to another | [base]/$convert
Execute a graphql statement | [base]/$graphql | [base]/[Resource]/[id]/$graphql
Return a graph of resources | [base]/[Resource]/[id]/$graph
Operations Defined by Resource Types
Apply | [base]/ActivityDefinition/$apply | [base]/ActivityDefinition/[id]/$apply
Data Requirements | [base]/ActivityDefinition/[id]/$data-requirements

Fetch a subset of the CapabilityStatement resource | [base]/CapabilityStatement/$subset | [base]/CapabilityStatement/[id]/$subset
Test if a server implements a client's required operations | [base]/CapabilityStatement/$implements | [base]/CapabilityStatement/[id]/$implements
Test if a server implements a client's required operations | [base]/CapabilityStatement/$conforms
Discover what versions a server supports | [base]/$versions
Apply | [base]/ChargeItemDefinition/[id]/$apply
Submit a Claim resource for adjudication | [base]/Claim/$submit
Concept Look Up & Decomposition | [base]/CodeSystem/$lookup
Code System based Validation | [base]/CodeSystem/$validate-code | [base]/CodeSystem/[id]/$validate-code
Subsumption Testing | [base]/CodeSystem/$subsumes | [base]/CodeSystem/[id]/$subsumes
Finding codes based on supplied properties | [base]/CodeSystem/$find-matches | [base]/CodeSystem/[id]/$find-matches
Generate a Document | [base]/Composition/$document | [base]/Composition/[id]/$document
Concept Translation | [base]/ConceptMap/$translate | [base]/ConceptMap/[id]/$translate
Closure Table Maintenance | [base]/$closure
Submit an EligibilityRequest resource for assessment | [base]/CoverageEligibilityRequest/$submit
Fetch Encounter Record | [base]/Encounter/[id]/$everything
Fetch a group of Patient Records | [base]/Group/[id]/$everything
Data Requirements | [base]/$data-requirements | [base]/Library/[id]/$data-requirements
Find a functional list | [base]/List/$find
Evaluate Measure | [base]/Measure/$evaluate-measure | [base]/Measure/[id]/$evaluate-measure
Data Requirements | [base]/Measure/[id]/$data-requirements
Submit Data | [base]/Measure/$submit-data | [base]/Measure/[id]/$submit-data
Collect Data | [base]/Measure/$collect-data | [base]/Measure/[id]/$collect-data
Care Gaps | [base]/Measure/$care-gaps
Fetch Product Record | [base]/MedicinalProduct/$everything | [base]/MedicinalProduct/[id]/$everything
Process Message | [base]/$process-message
Fetch Preferred it | [base]/NamingSystem/$preferred-id
Observation Statistics | [base]/Observation/$stats
Last N Observations Query | [base]/Observation/$lastn
Find patient matches using MPI based logic | [base]/Patient/$match
Fetch Patient Record | [base]/Patient/$everything | [base]/Patient/[id]/$everything
Apply | [base]/PlanDefinition/$apply | [base]/PlanDefinition/[id]/$apply
Data Requirements | [base]/PlanDefinition/[id]/$data-requirements
Build Questionnaire | [base]/StructureDefinition/$questionnaire | [base]/StructureDefinition/[id]/$questionnaire
Generate Snapshot | [base]/StructureDefinition/$snapshot | [base]/StructureDefinition/[id]/$snapshot
Model Instance Transformation | [base]/StructureMap/$transform | [base]/StructureMap/[id]/$transform
Value Set Expansion | [base]/ValueSet/$expand | [base]/ValueSet/[id]/$expand
Value Set based Validation | [base]/ValueSet/$validate-code | [base]/ValueSet/[id]/$validate-code

5.4.3.4 Describing Operation Parameters

There are 2 ways to describe the input and output parameters for an operation:

• Using OperationDefinition.parameter to describe the parameters

• Using OperationDefinition.inputProfile and OperationDefinition.outputProfile

The parameters is a simple list of possible parameters, along with cardinalities and types. The profiles allow a rich set of validation rules etc. to be provided. OperationDefinitions SHALL always define the parameters using OperationDefinition.parameter in the resource, and MAY also provide profiles for the parameters. If present, the profiles SHALL NOT disagree with the parameters defined for the operation.

As an example, consider an operation that defines 3 parameters, 2 in and 1 out:

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
a | 0..1 | integer | | | An integer parameter
b | 0..1 | Patient | | | A patient parameter
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
c | 1..1 | decimal | | | A result parameter

In addition to specifying the operation parameters directly, an operation definition can also provide a profile:

```
"inputProfile" : "http://example.org/StructureDefinition/op.x.in.profile"
```

This profile would describe a parameters resource with 2 parameters (using slicing), with the same details as the table above. The operation definition would still list the parameters directly to save applications consuming the definition (e.g. to produce an OpenAPI document) from parsing and interpreting the profile.

5.4.3.5 Executing Operations

OperationDefinitions with kind = operation are executed as defined in the Operations Framework.

5.4.3.6 Executing Named Queries

Named queries (OperationDefinitions with kind = query) are executed by performing a search with the value of the search parameter "_query" set to the name provided in the definition.

If the named query is to be performed over the RESTful API, all the parameters must be simple search parameters, so that they can be represented directly in the URL without tricky encoding issues. Named queries always return a bundle containing a set of resources, so all the out parameters must be resources, not data types.

For named queries, all the standard search parameters are automatically in scope (though servers do not need to support them unless explicitly documented).

5.4.3.7 Passing Resources to Operations

There are two ways to pass resources to an operation: directly or by reference. The definition of an operation distinguishes between these two, since they have very different behaviors and consequences.

As an example, take the ValueSet.$expand operation. This operation takes a valueset as a direct parameter. The type of the parameter is defined as 'ValueSet'. In a Parameters resource, it would be represented like this:

```
  <parameter>
    <name value="valueset"/>
    <resource>
      <ValueSet>
        <!-- Valueset contents -->
      </ValueSet>
    </resource>
  </parameter>
```

or, in JSON:

```
 "parameter": [
    {
      "name": "valueset",
      "resource": {
        "resourceType": "ValueSet",
        // Valueset contents
      }
    }
  ]
```

Other parameters are passed by reference. For example, the ChargeItemDefinition.$apply operation takes two parameters of type Reference - one to the chargeItem and the other to the account. The type of the parameters is Reference(Charge) and Reference(Account), respectively. The expectation is that the server performing the operation will resolve those references as part of the operation execution. In a parameters resource, the chargeItem parameter would be represented like this:

```
  <parameter>
    <name value="chargeItem"/>
    <valueReference>
      <reference value="ChargeItem/123"/>
    </valueReference>
  </parameter>
```

or, in JSON:

```
 "parameter": [
    {
      "name": "chargeItem",
      "valueReference" : {
        "reference" : "ChargeItem/123"
      }
    }
  ]
```

Some operations can take either form; in that case, two distinct parameters must be defined: one for a resource as a direct parameter, and one for a reference.

5.4.3.8 Renaming OperationDefinition.name

It is possible for two different organizations to create different operation definitions with the same name or, perhaps more likely, to define equivalent operations that have the same name but incompatible approaches in their parameter lists.

It is also possible, though unlikely, that a server will be required to support both of these operations. If this is the case, the server is able to do this by giving one of them a new name and referring to it by definition in the capability statement. To illustrate this, assume that two different organizations, "orgA" and "orgB", both define an operation called "dothis", and the definitions are incompatible. OrgA publishes its operation definition at http://orga.com/fhir/dothis.xml, and OrgB publishes its operation at http://fhir.orgb.com/meta/OperationDefinition/dothis. The server is able to implement both. Its capability statement will say:

```
<CapabilityStatement xmlns="http://hl7.org/fhir">
  <!-- snip -->
  <rest>
    <!-- snip -->
    <operation>
      <name value="dothis"/>
      <definition>
        <reference value="http://orga.com/fhir/dothis.xml"/>
      </definition>
```

```
    </operation>
    <operation>
      <name value="dothis2"/>
      <definition>
        <reference value="http://fhir.orgb.com/meta/OperationDefinition/dothis"/>
      </definition>
    </operation>
    <!-- snip -->
  </rest>
  <!-- snip -->
</CapabilityStatement>
```

If a general purpose cross server client is looking for the implementation of the http://fhir.orgb.com/meta/OperationDefinition/dothis operation and wants to be robust against this name clash problem, instead of simply executing the $dothis operation, it can look at the server's CapabilityStatement for the underlying definition URI and then execute the name given in the capability statement.

5.4.3.9 Determining System Compatibility

A client can determine the compatibility of the server by processing its capability statement and ensuring that the server implements the specific operation definitions and parameters required by the client. The client can then report a useful error to the user rather than allowing mystifying operational errors to occur.

However, there are fundamental limitations to this approach because there are many aspects of these operations that are not (or cannot be) defined in a formal fashion using OperationDefinition, for example, co-occurrence constraints among parameters.

In the same sense, a 3rd party tool can examine a server's CapabilityStatement and a client's definition of an acceptable server to determine whether those two system are interoperable or not.

5.4.3.10 Dynamically Generating Forms

Finally, it is possible to generate user interface forms automatically from the OperationDefinition. The documentation in the OperationDefinition.description and OperationDefinition.parameter.documentation should be sufficient to allow moderately technical users to guess at the correct content of the form. For this reason, highly technical documentation should go in OperationDefinition.comment.

It is anticipated that this would be used to automate development processes, rather than to generate end-user forms: such users will usually need more support than can be offered in a generated form.

5.4.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
base TU | reference | Marks this as a profile of the base | OperationDefinition.base (OperationDefinition)
code TU | token | Name used to invoke the operation | OperationDefinition.code
context TU | token | A use context assigned to the operation definition | (OperationDefinition.useContext.value as CodeableConcept)
context-quantity TU | quantity | A quantity- or range-valued use context assigned to the operation definition | (OperationDefinition.useContext.value as Quantity) | (OperationDefinition.useContext.value as Range)
context-type TU | token | A type of use context assigned to the operation definition | OperationDefinition.useContext.code
context-type-quantity TU | composite | A use context type and quantity- or range-based value assigned to the operation definition | On OperationDefinition.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value TU | composite | A use context type and value assigned to the operation definition | On OperationDefinition.useContext: context-type: code context: value.as(CodeableConcept)
date TU | date | The operation definition publication date | OperationDefinition.date
description TU | string | The description of the operation definition | OperationDefinition.description
input-profile TU | reference | Validation information for in parameters | OperationDefinition.inputProfile (StructureDefinition)
instance TU | token | Invoke on an instance? | OperationDefinition.instance
jurisdiction TU | token | Intended jurisdiction for the operation definition | OperationDefinition.jurisdiction
kind TU | token | operation | query | OperationDefinition.kind
name TU | string | Computationally friendly name of the operation definition | OperationDefinition.name
output-profile TU | reference | Validation information for out parameters | OperationDefinition.outputProfile (StructureDefinition)
publisher TU | string | Name of the publisher of the operation definition | OperationDefinition.publisher
status TU | token | The current status of the operation definition | OperationDefinition.status
system TU | token | Invoke at the system level? | OperationDefinition.system
title TU | string | The human-friendly name of the operation definition | OperationDefinition.title
type TU | token | Invoke at the type level? | OperationDefinition.type
url TU | uri | The uri that identifies the operation definition | OperationDefinition.url
version TU | token | The business version of the operation definition | OperationDefinition.version

5.4.5 Resource OperationDefinition - Examples

URL: https://hl7.org/fhir/R4/operationdefinition-examples.html

5.4.5 Resource OperationDefinition - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
OperationDefinition example | example | XML | JSON | Turtle
ValueSet Expansion | ValueSet-expand | XML | JSON | Turtle
ValueSet Validation | ValueSet-validate-code | XML | JSON | Turtle
ValueSet Lookup | CodeSystem-lookup | XML | JSON | Turtle
ConceptMap Translate | ConceptMap-translate | XML | JSON | Turtle
ConceptMap Closure | ConceptMap-closure | XML | JSON | Turtle
Library data requirements | Library-data-requirements | XML | JSON | Turtle
Measure data requirements | Measure-data-requirements | XML | JSON | Turtle
Measure evaluate measure | Measure-evaluate-measure | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.4.6 Resource OperationDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/operationdefinition-definitions.html

5.4.6 Resource OperationDefinition - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the OperationDefinition resource.

OperationDefinition
Element Id | OperationDefinition
Definition | A formal computable definition of an operation (on the RESTful interface) or a named query (using the search interaction).
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element opd-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
opd-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
OperationDefinition.url
Element Id | OperationDefinition.url
Definition | An absolute URI that is used to identify this operation definition when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this operation definition is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the operation definition is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the operation definition to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
OperationDefinition.version
Element Id | OperationDefinition.version
Definition | The identifier that is used to identify this version of the operation definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the operation definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different operation definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the operation definition with the format [url]|[version].
OperationDefinition.name
Element Id | OperationDefinition.name
Definition | A natural language name identifying the operation definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 1..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true

Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
OperationDefinition.title
Element Id | OperationDefinition.title
Definition | A short, descriptive, user-friendly title for the operation definition.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
OperationDefinition.status
Element Id | OperationDefinition.status
Definition | The status of this operation definition. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of operation definitions that are appropriate for use versus not.
OperationDefinition.kind
Element Id | OperationDefinition.kind
Definition | Whether this is an operation or a named query.
Cardinality | 1..1
Terminology Binding | OperationKind (Required)
Type | code
Summary | true
Comments | Named queries are invoked differently, and have different capabilities.
OperationDefinition.experimental
Element Id | OperationDefinition.experimental
Definition | A Boolean value to indicate that this operation definition is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level operation definition.
Summary | true
Comments | Allows filtering of operation definitions that are appropriate for use versus not.
OperationDefinition.date
Element Id | OperationDefinition.date
Definition | The date (and optionally time) when the operation definition was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the operation definition changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the operation definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
OperationDefinition.publisher
Element Id | OperationDefinition.publisher
Definition | The name of the organization or individual that published the operation definition.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the operation definition. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the operation definition is the organization or individual primarily responsible for the maintenance and upkeep of the operation definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the operation definition. This item SHOULD be populated unless the information is available from context.
OperationDefinition.contact
Element Id | OperationDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.

OperationDefinition.description
Element Id | OperationDefinition.description
Definition | A free text natural language description of the operation definition from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the operation definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the operation definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the operation definition is presumed to be the predominant language in the place the operation definition was created).
OperationDefinition.useContext
Element Id | OperationDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate operation definition instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
OperationDefinition.jurisdiction
Element Id | OperationDefinition.jurisdiction
Definition | A legal or geographic region in which the operation definition is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the operation definition to be used in jurisdictions other than those for which it was originally designed or intended.
OperationDefinition.purpose
Element Id | OperationDefinition.purpose
Definition | Explanation of why this operation definition is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the operation definition. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this operation definition.
OperationDefinition.affectsState
Element Id | OperationDefinition.affectsState
Definition | Whether the operation affects state. Side effects such as producing audit trail entries do not count as 'affecting state'.
Cardinality | 0..1
Type | boolean
Summary | true
Comments | What http methods can be used for the operation depends on the .affectsState value and whether the input parameters are primitive or complex: Servers SHALL support POST method for all operations. Servers SHALL support GET method if all the parameters for the operation are primitive or there are no parameters and the operation has affectsState = false.
OperationDefinition.code
Element Id | OperationDefinition.code
Definition | The name used to invoke the operation.
Cardinality | 1..1
Type | code
Summary | true
OperationDefinition.comment
Element Id | OperationDefinition.comment
Definition | Additional information about how to use this operation or named query.
Cardinality | 0..1
Type | markdown
OperationDefinition.base
Element Id | OperationDefinition.base
Definition | Indicates that this operation definition is a constraining profile on the base.
Cardinality | 0..1
Type | canonical(OperationDefinition)
Summary | true
Comments | A constrained profile can make optional parameters required or not used and clarify documentation.
OperationDefinition.resource
Element Id | OperationDefinition.resource
Definition | The types on which this operation can be executed.
Cardinality | 0..*
Terminology Binding | Any defined Resource Type name
Type | code
Summary | true
Comments | If the type is an abstract resource ("Resource" or "DomainResource") then the operation can be invoked on any concrete specialization.

OperationDefinition.system
Element Id | OperationDefinition.system
Definition | Indicates whether this operation or named query can be invoked at the system level (e.g. without needing to choose a resource type for the context).
Cardinality | 1..1
Type | boolean
Summary | true
OperationDefinition.type
Element Id | OperationDefinition.type
Definition | Indicates whether this operation or named query can be invoked at the resource type level for any given resource type level (e.g. without needing to choose a specific resource id for the context).
Cardinality | 1..1
Type | boolean
Summary | true
OperationDefinition.instance
Element Id | OperationDefinition.instance
Definition | Indicates whether this operation can be invoked on a particular instance of one of the given types.
Cardinality | 1..1
Type | boolean
Summary | true
OperationDefinition.inputProfile
Element Id | OperationDefinition.inputProfile
Definition | Additional validation information for the in parameters - a single profile that covers all the parameters. The profile is a constraint on the parameters resource as a whole.
Cardinality | 0..1
Type | canonical(StructureDefinition)
Comments | If present the profile shall not conflict with what is specified in the parameters in the operation definition (max/min etc.), though it may provide additional constraints. The constraints expressed in the profile apply whether the operation is invoked by a POST wih parameters or not.
OperationDefinition.outputProfile
Element Id | OperationDefinition.outputProfile
Definition | Additional validation information for the out parameters - a single profile that covers all the parameters. The profile is a constraint on the parameters resource.
Cardinality | 0..1
Type | canonical(StructureDefinition)
Comments | If present the profile shall not conflict with what is specified in the parameters in the operation definition (max/min etc.), though it may provide additional constraints. The constraints expressed in the profile apply whether the operation is invoked by a POST wih parameters or not.
OperationDefinition.parameter
Element Id | OperationDefinition.parameter
Definition | The parameters for the operation/query.
Cardinality | 0..*
Comments | Query Definitions only have one output parameter, named "result". This might not be described, but can be to allow a profile to be defined.
Invariants | Defined on this element opd-1 Rule Either a type must be provided, or parts type.exists() or part.exists() opd-2 Rule A search type can only be specified for parameters of type string searchType.exists() implies type = 'string' opd-3 Rule A targetProfile can only be specified for parameters of type Reference or Canonical targetProfile.exists() implies (type = 'Reference' or type = 'canonical')
Defined on this element
opd-1 | Rule | Either a type must be provided, or parts | type.exists() or part.exists()
opd-2 | Rule | A search type can only be specified for parameters of type string | searchType.exists() implies type = 'string'
opd-3 | Rule | A targetProfile can only be specified for parameters of type Reference or Canonical | targetProfile.exists() implies (type = 'Reference' or type = 'canonical')
OperationDefinition.parameter.name
Element Id | OperationDefinition.parameter.name
Definition | The name of used to identify the parameter.
Cardinality | 1..1
Type | code
Comments | This name must be a token (start with a letter in a..z, and only contain letters, numerals, and underscore. Note that for search parameters (type = string, with a search type), the name may be altered by the search modifiers.
OperationDefinition.parameter.use
Element Id | OperationDefinition.parameter.use
Definition | Whether this is an input or an output parameter.
Cardinality | 1..1
Terminology Binding | OperationParameterUse (Required)
Type | code
Comments | If a parameter name is used for both an input and an output parameter, the parameter should be defined twice.
OperationDefinition.parameter.min
Element Id | OperationDefinition.parameter.min
Definition | The minimum number of times this parameter SHALL appear in the request or response.
Cardinality | 1..1
Type | integer
OperationDefinition.parameter.max

Element Id | OperationDefinition.parameter.max
Definition | The maximum number of times this element is permitted to appear in the request or response.
Cardinality | 1..1
Type | string
OperationDefinition.parameter.documentation
Element Id | OperationDefinition.parameter.documentation
Definition | Describes the meaning or use of this parameter.
Cardinality | 0..1
Type | string
OperationDefinition.parameter.type
Element Id | OperationDefinition.parameter.type
Definition | The type for this parameter.
Cardinality | 0..1
Terminology Binding | Any defined Resource or Data Type name (including "Any" and "Type")
Type | code
Comments | if there is no stated parameter, then the parameter is a multi-part parameter; type and must have at least one part defined.
Invariants | Affect this element opd-1 Rule Either a type must be provided, or parts type.exists() or part.exists()
Affect this element
opd-1 | Rule | Either a type must be provided, or parts | type.exists() or part.exists()
OperationDefinition.parameter.targetProfile
Element Id | OperationDefinition.parameter.targetProfile
Definition | Used when the type is "Reference" or "canonical", and identifies a profile structure or implementation Guide that applies to the target of the reference this parameter refers to. If any profiles are specified, then the content must conform to at least one of them. The URL can be a local reference - to a contained StructureDefinition, or a reference to another StructureDefinition or Implementation Guide by a canonical URL. When an implementation guide is specified, the target resource SHALL conform to at least one profile defined in the implementation guide.
Cardinality | 0..*
Type | canonical(StructureDefinition)
Comments | Often, these profiles are the base definitions from the spec (e.g. http://hl7.org/fhir/StructureDefinition/Patient).
OperationDefinition.parameter.searchType
Element Id | OperationDefinition.parameter.searchType
Definition | How the parameter is understood as a search parameter. This is only used if the parameter type is 'string'.
Cardinality | 0..1
Terminology Binding | SearchParamType (Required)
Type | code
Invariants | Affect this element opd-2 Rule A search type can only be specified for parameters of type string searchType.exists() implies type = 'string'
Affect this element
opd-2 | Rule | A search type can only be specified for parameters of type string | searchType.exists() implies type = 'string'
OperationDefinition.parameter.binding
Element Id | OperationDefinition.parameter.binding
Definition | Binds to a value set if this parameter is coded (code, Coding, CodeableConcept).
Cardinality | 0..1
OperationDefinition.parameter.binding.strength
Element Id | OperationDefinition.parameter.binding.strength
Definition | Indicates the degree of conformance expectations associated with this binding - that is, the degree to which the provided value set must be adhered to in the instances.
Cardinality | 1..1
Terminology Binding | BindingStrength (Required)
Type | code
Alternate Names | conformance; extensibility
Comments | For further discussion, see Using Terminologies.
OperationDefinition.parameter.binding.valueSet
Element Id | OperationDefinition.parameter.binding.valueSet
Definition | Points to the value set or external definition (e.g. implicit value set) that identifies the set of codes to be used.
Cardinality | 1..1
Type | canonical(ValueSet)
Comments | For value sets with a referenceResource, the display can contain the value set description. The reference may be version-specific or not.
OperationDefinition.parameter.referencedFrom
Element Id | OperationDefinition.parameter.referencedFrom
Definition | Identifies other resource parameters within the operation invocation that are expected to resolve to this resource.
Cardinality | 0..*
Comments | Resolution applies if the referenced parameter exists.
OperationDefinition.parameter.referencedFrom.source
Element Id | OperationDefinition.parameter.referencedFrom.source
Definition | The name of the parameter or dot-separated path of parameter names pointing to the resource parameter that is expected to contain a reference to this resource.
Cardinality | 1..1
Type | string
OperationDefinition.parameter.referencedFrom.sourceId
Element Id | OperationDefinition.parameter.referencedFrom.sourceId

Definition | The id of the element in the referencing resource that is expected to resolve to this resource.
Cardinality | 0..1
Type | string
OperationDefinition.parameter.part
Element Id | OperationDefinition.parameter.part
Definition | The parts of a nested Parameter.
Cardinality | 0..*
Type | See OperationDefinition.parameter
Comments | Query Definitions only have one output parameter, named "result". This might not be described, but can be to allow a profile to be defined.
Invariants | Affect this element opd-1 Rule Either a type must be provided, or parts type.exists() or part.exists()
Affect this element
opd-1 | Rule | Either a type must be provided, or parts | type.exists() or part.exists()
OperationDefinition.overload
Element Id | OperationDefinition.overload
Definition | Defines an appropriate combination of parameters to use when invoking this operation, to help code generators when generating overloaded parameter sets for this operation.
Cardinality | 0..*
Comments | The combinations are suggestions as to which sets of parameters to use together, but the combinations are not intended to be authoritative.
OperationDefinition.overload.parameterName
Element Id | OperationDefinition.overload.parameterName
Definition | Name of parameter to include in overload.
Cardinality | 0..*
Type | string
OperationDefinition.overload.comment
Element Id | OperationDefinition.overload.comment
Definition | Comments to go on overload.
Cardinality | 0..1
Type | string

5.4.7 Resource OperationDefinition - Mappings

URL: https://hl7.org/fhir/R4/operationdefinition-mappings.html

5.4.7 Resource OperationDefinition - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the operationdefinition resource (see Mappings to Other Standards for further information & status).

5.4.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

OperationDefinition
url | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
kind | FiveWs.class
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

5.4.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

OperationDefinition | Definition
url | Definition.url
version | Definition.version
name | Definition.title
title | Definition.title
status | Definition.status {different ValueSet}
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext

jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
base | Definition.derivedFrom
resource | Definition.subject[x]

5.4.7.3 RIM Mapping (http://hl7.org/v3)

OperationDefinition | n/a

5.4.8 Resource OperationDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/operationdefinition-profiles.html

5.4.8 Resource OperationDefinition - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the OperationDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

5.4.8.1 Profiles

No Profiles defined for this resource

5.4.8.2 Extensions

operationdefinition-allowed-type | OperationDefinition.parameter | allowed-type | for Standard Operation Definition Extensions
operationdefinition-profile | OperationDefinition.parameter | profile | for Standard Operation Definition Extensions
replaces | OperationDefinition | replaces | for General Extensions for use by FHIR Implementers
resource-approvalDate | OperationDefinition | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | OperationDefinition | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | OperationDefinition | lastReviewDate | for Resource HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

workflow-instantiatesCanonical | instantiatesCanonical | for Workflow Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

5.4.8.3 Search Extensions

No Search Extensions defined for this resource

5.4.9 Resource OperationDefinition - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/operationdefinition-version-maps.html

No validation errors - all conversions are clean

5.4.9 Resource OperationDefinition - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for OperationDefinition.

Functional status for this map: 37 tests that all execute ok. All tests pass round-trip testing and 34 r3 resources are invalid (0 errors). (see documentation)

5.4.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/OperationDefinition3to4" = "R3 to R4 Conversions for OperationDefinition"

uses "http://hl7.org/fhir/3.0/StructureDefinition/OperationDefinition" alias OperationDefinitionR3 as source
uses "http://hl7.org/fhir/StructureDefinition/OperationDefinition" alias OperationDefinition as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group OperationDefinition(source src : OperationDefinitionR3, target tgt : OperationDefinition) extends DomainResource
    <<type+>> {
  src.url -> tgt.url;
  src.version -> tgt.version;
```

```
  src.name -> tgt.name;
  src.status -> tgt.status;
  src.kind -> tgt.kind;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.idempotent -> tgt.affectsState "affectsState";
  src.code -> tgt.code;
  src.comment as vs -> tgt.comment as vt then stringToMarkdown(vs, vt);
  src.base -> tgt.base;
  src.resource -> tgt.resource;
  src.system -> tgt.system;
  src.type -> tgt.type;
  src.instance -> tgt.instance;
  src.parameter as vs0 -> tgt.parameter as vt0 then parameter(vs0, vt0);
  src.overload as vs0 -> tgt.overload as vt0 then {
    vs0.parameterName -> vt0.parameterName;
    vs0.comment -> vt0.comment;
  };
}

group parameter(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.use -> tgt.use;
  src.min -> tgt.min;
  src.max -> tgt.max;
  src.documentation -> tgt.documentation;
  src.type -> tgt.type;
  src.searchType -> tgt.searchType;
  src.profile -> tgt.targetProfile;
  src.binding as vs0 -> tgt.binding as vt0 then binding(vs0, vt0);
  src.part as vs0 -> tgt.part as vt0 then parameter(vs0, vt0);
}

group binding(source src, target tgt) extends BackboneElement {
  src.strength -> tgt.strength;
  src.valueSet : uri as vs0 -> tgt.valueSet as vt0 then uri2Canonical(vs0, vt0) "valueSetU";
  src.valueSet : Reference as vs0 -> tgt.valueSet as vt0 then Reference2Canonical(vs0, vt0) "valueSetR";
}
```

5.4.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/OperationDefinition4to3" = "R4 to R3 Conversion for OperationDefinition"

uses "http://hl7.org/fhir/StructureDefinition/OperationDefinition" alias OperationDefinition as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/OperationDefinition" alias OperationDefinitionR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group OperationDefinition(source src : OperationDefinition, target tgt : OperationDefinitionR3) extends DomainResource
    <<type+>> {
  src.url -> tgt.url;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.status -> tgt.status;
  src.kind -> tgt.kind;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.affectsState -> tgt.idempotent;
  src.code -> tgt.code;
  src.comment as vs -> tgt.comment as vt then markdownToString(vs, vt);
  src.base -> tgt.base;
  src.resource -> tgt.resource;
  src.system -> tgt.system;
  src.type -> tgt.type;
  src.instance -> tgt.instance;
  src.parameter as vs0 -> tgt.parameter as vt0 then parameter(vs0, vt0);
  src.overload as vs0 -> tgt.overload as vt0 then {
    vs0.parameterName -> vt0.parameterName;
    vs0.comment -> vt0.comment;
  };
}

group parameter(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.use -> tgt.use;
  src.min -> tgt.min;
  src.max -> tgt.max;
  src.documentation -> tgt.documentation;
```

```
  src.type -> tgt.type;
  src.searchType -> tgt.searchType;
  src.targetProfile -> tgt.profile "profile";
  src.binding as vs0 -> tgt.binding as vt0 then binding(vs0, vt0);
  src.part as vs0 -> tgt.part as vt0 then parameter(vs0, vt0);
}

group binding(source src, target tgt) extends BackboneElement {
  src.strength -> tgt.strength;
  src.valueSet as vs0 where value.startsWith('http://hl7.org').not() -> tgt.valueSet = create('uri') as vt0 then
    canonical2Uri(vs0, vt0) "valueSetU";
  src.valueSet as vs0 where value.startsWith('http://hl7.org') -> tgt.valueSet = create('Reference') as vt0 then
    canonical2Reference(vs0, vt0) "valueSetR";
}
```

5.4.9.3 R4 Validation Errors

5.4.10 Standard Operation Definition Extensions

URL: https://hl7.org/fhir/R4/operationdefinition-extensions.html

This profile describes common extensions that are used with Operation Definitions

5.4.10 Standard Operation Definition Extensions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

5.4.10.1 Content

Extensions:

operationdefinition-profile | profile : Identifies a profile structure or implementation Guide that applies to the datatype this element refers to. If any profiles are specified, then the content must conform to at least one of them. The URL can be a local reference - to a contained StructureDefinition, or a reference to another StructureDefinition or Implementation Guide by a canonical URL. When an implementation guide is specified, the type SHALL conform to at least one profile defined in the implementation guide.

operationdefinition-allowed-type | allowed-type : When the base type is an abstract type (e.g. Resource or Element) then this extension defines which concrete types are allowed to be used for a parameter. In the absence of this extension, any type is allowed.

5.4.11 JSON Schema for OperationDefinition

URL: https://hl7.org/fhir/R4/operationdefinition.schema.json.html

5.4.11 JSON Schema for OperationDefinition

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for OperationDefinition, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/OperationDefinition",
  "$ref": "#/definitions/OperationDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "OperationDefinition": {
      "description": "A formal computable definition of an operation (on the RESTful interface) or a named query (using
    the search interaction).",
      "properties": {
        "resourceType": {
          "description": "This is a OperationDefinition resource",
          "const": "OperationDefinition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
```

```
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this operation definition when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this operation definition is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the operation definition is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the operation definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the operation definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the operation definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the operation definition.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this operation definition. Enables tracking the life-cycle of the content.",
      "enum": [
```

```
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "kind": {
      "description": "Whether this is an operation or a named query.",
      "enum": [
        "operation",
        "query"
      ]
    },
    "_kind": {
      "description": "Extensions for kind",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this operation definition is authored for testing purposes
(or education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the operation definition was published. The date must
change when the business version changes and it must change if the status code changes. In addition, it should
change when the substantive content of the operation definition changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the operation definition.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the operation definition from a consumer\u0027s
perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate operation definition
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the operation definition is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this operation definition is needed and why it has been designed as it
has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
```

```
    },
    "affectsState": {
      "description": "Whether the operation affects state. Side effects such as producing audit trail entries do not
 count as \u0027affecting  state\u0027.",
      "$ref": "#/definitions/boolean"
    },
    "_affectsState": {
      "description": "Extensions for affectsState",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "The name used to invoke the operation.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "comment": {
      "description": "Additional information about how to use this operation or named query.",
      "$ref": "#/definitions/markdown"
    },
    "_comment": {
      "description": "Extensions for comment",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "base": {
      "description": "Indicates that this operation definition is a constraining profile on the base.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "resource": {
      "description": "The types on which this operation can be executed.",
      "items": {
        "$ref": "#/definitions/code"
      },
      "type": "array"
    },
    "_resource": {
      "description": "Extensions for resource",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "system": {
      "description": "Indicates whether this operation or named query can be invoked at the system level (e.g.
without needing to choose a resource type for the context).",
      "$ref": "#/definitions/boolean"
    },
    "_system": {
      "description": "Extensions for system",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Indicates whether this operation or named query can be invoked at the resource type level for
any given resource type level (e.g. without needing to choose a specific resource id for the context).",
      "$ref": "#/definitions/boolean"
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "instance": {
      "description": "Indicates whether this operation can be invoked on a particular instance of one of the given
types.",
      "$ref": "#/definitions/boolean"
    },
    "_instance": {
      "description": "Extensions for instance",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "inputProfile": {
      "description": "Additional validation information for the in parameters - a single profile that covers all the
 parameters. The profile is a constraint on the parameters resource as a whole.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "outputProfile": {
      "description": "Additional validation information for the out parameters - a single profile that covers all
the parameters. The profile is a constraint on the parameters resource.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "parameter": {
      "description": "The parameters for the operation/query.",
      "items": {
        "$ref": "#/definitions/OperationDefinition_Parameter"
      },
      "type": "array"
    },
    "overload": {
```

```
      "description": "Defines an appropriate combination of parameters to use when invoking this operation, to help code generators when generating overloaded parameter sets for this operation.",
      "items": {
        "$ref": "#/definitions/OperationDefinition_Overload"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"OperationDefinition_Parameter": {
  "description": "A formal computable definition of an operation (on the RESTful interface) or a named query (using the search interaction).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part  of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "The name of used to identify the parameter.",
      "$ref": "#/definitions/code"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "use": {
      "description": "Whether this is an input or an output parameter.",
      "enum": [
        "in",
        "out"
      ]
    },
    "_use": {
      "description": "Extensions for use",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "min": {
      "description": "The minimum number of times this parameter SHALL appear in the request or response.",
      "$ref": "#/definitions/integer"
    },
    "_min": {
      "description": "Extensions for min",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "max": {
      "description": "The maximum number of times this element is permitted to appear in the request or response.",
      "$ref": "#/definitions/string"
    },
    "_max": {
      "description": "Extensions for max",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "documentation": {
      "description": "Describes the meaning or use of this parameter.",
      "$ref": "#/definitions/string"
    },
    "_documentation": {
      "description": "Extensions for documentation",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
    "type": {
      "description": "The type for this parameter.",
      "$ref": "#/definitions/code"
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "targetProfile": {
      "description": "Used when the type is \"Reference\" or \"canonical\", and identifies a profile structure or
implementation Guide that applies to the target of the reference this parameter refers to. If any profiles are
specified, then the content must conform to at least one of them. The URL can be a local reference - to a contained
StructureDefinition, or a reference to another StructureDefinition or Implementation Guide by a canonical URL. When
an implementation guide is specified, the target resource SHALL conform to at least one profile defined in the
implementation guide.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "searchType": {
      "description": "How the parameter is understood as a search parameter. This is only used if the parameter type
 is \u0027string\u0027.",
      "enum": [
        "number",
        "date",
        "string",
        "token",
        "reference",
        "composite",
        "quantity",
        "uri",
        "special"
      ]
    },
    "_searchType": {
      "description": "Extensions for searchType",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "binding": {
      "description": "Binds to a value set if this parameter is coded (code, Coding, CodeableConcept).",
      "$ref": "#/definitions/OperationDefinition_Binding"
    },
    "referencedFrom": {
      "description": "Identifies other resource parameters within the operation invocation that are expected to
resolve to this resource.",
      "items": {
        "$ref": "#/definitions/OperationDefinition_ReferencedFrom"
      },
      "type": "array"
    },
    "part": {
      "description": "The parts of a nested Parameter.",
      "items": {
        "$ref": "#/definitions/OperationDefinition_Parameter"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"OperationDefinition_Binding": {
  "description": "A formal computable definition of an operation (on the RESTful interface) or a named query (using
the search interaction).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
```

```
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "strength": {
      "description": "Indicates the degree of conformance expectations associated with this binding - that is, the
degree to which the provided value set must be adhered to in the instances.",
      "enum": [
        "required",
        "extensible",
        "preferred",
        "example"
      ]
    },
    "_strength": {
      "description": "Extensions for strength",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueSet": {
      "description": "Points to the value set or external definition (e.g. implicit value set) that identifies the
set of codes to be used.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    }
  },
  "additionalProperties": false,
  "required": [
    "valueSet"
  ]
},
"OperationDefinition_ReferencedFrom": {
  "description": "A formal computable definition of an operation (on the RESTful interface) or a named query (using
the search interaction).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "source": {
      "description": "The name of the parameter or dot-separated path of parameter names pointing to the resource
parameter that is expected to contain a reference to this resource.",
      "$ref": "#/definitions/string"
    },
    "_source": {
      "description": "Extensions for source",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "sourceId": {
      "description": "The id of the element in the referencing resource that is expected to resolve to this
resource.",
      "$ref": "#/definitions/string"
    },
    "_sourceId": {
      "description": "Extensions for sourceId",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"OperationDefinition_Overload": {
  "description": "A formal computable definition of an operation (on the RESTful interface) or a named query (using
the search interaction).",
  "properties": {
    "id": {
```

```
        "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
        "$ref": "string.schema.json#/definitions/string"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "parameterName": {
        "description": "Name of parameter to include in overload.",
        "items": {
          "$ref": "#/definitions/string"
        },
        "type": "array"
      },
      "_parameterName": {
        "description": "Extensions for parameterName",
        "items": {
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "type": "array"
      },
      "comment": {
        "description": "Comments to go on overload.",
        "$ref": "#/definitions/string"
      },
      "_comment": {
        "description": "Extensions for comment",
        "$ref": "Element.schema.json#/definitions/Element"
      }
    },
    "additionalProperties": false
  }
 }
}
```


5.4.12 Resource OperationDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/operationdefinition-dependencies.html

5.4.12 Resource OperationDefinition - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Conformance Package for further details.

Dependency Graph for OperationDefinition FMM level 5

Name | Card. | Type | Dependency Analysis
OperationDefinition | | | DomainResource
url | 0..1 | uri
version | 0..1 | string
name | 1..1 | string
title | 0..1 | string
status | 1..1 | code
kind | 1..1 | code
experimental | 0..1 | boolean
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext TU | 0..* | UsageContext

jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
affectsState | 0..1 | boolean
code | 1..1 | code
comment | 0..1 | markdown
base | 0..1 | canonical(OperationDefinition)
resource | 0..* | code
system | 1..1 | boolean
type | 1..1 | boolean
instance | 1..1 | boolean
inputProfile | 0..1 | canonical(StructureDefinition)
outputProfile | 0..1 | canonical(StructureDefinition)
parameter | 0..* | BackboneElement
name | 1..1 | code
use | 1..1 | code
min | 1..1 | integer
max | 1..1 | string
documentation | 0..1 | string
type | 0..1 | code
targetProfile | 0..* | canonical(StructureDefinition)
searchType | 0..1 | code
binding | 0..1 | BackboneElement
strength | 1..1 | code
valueSet | 1..1 | canonical(ValueSet)
referencedFrom TU | 0..* | BackboneElement
source | 1..1 | string
sourceId | 0..1 | string
part | 0..* | see parameter
overload | 0..* | BackboneElement
parameterName | 0..* | string
comment | 0..1 | string

5.5 Resource SearchParameter - Content

URL: https://hl7.org/fhir/R4/searchparameter.html
FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

A search parameter that defines a named search item that can be used to search/filter on a resource.

5.5.1 Scope and Usage

A SearchParameter resource specifies a search parameter that may be used on the RESTful API to search or filter on a resource. The SearchParameter resource declares:

• how to refer to the search parameter from a client
• how the search parameter is to be understood by the server
• where in the source resource the parameter matches

5.5.2 Boundaries and Relationships

• Search Parameters are referred to by CapabilityStatement resources via the canonical URL for a search parameter (CapabilityStatement.rest.resource.searchParam.definition)
• Search Parameters can appear in an Implementation Guide to specify how resources can be found

5.5.3 Background and Context

Implementers should be familiar with the background and concepts described in Search on the RESTful API before working with this resource.

There is a registry of all Search Parameters.

This resource is referenced by CapabilityStatement and itself

5.5.4 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
SearchParameter | ITU | | DomainResource | Search parameter for a resource + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: If an xpath is present, there SHALL be an xpathUsage + Rule: Search parameters can only have chain names when the search parameter type is 'reference' Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 1..1 | uri | Canonical identifier for this search parameter, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the search parameter
name | ΣI | 1..1 | string | Name for this search parameter (computer friendly)
derivedFrom | | 0..1 | canonical(SearchParameter) | Original definition for the search parameter
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 1..1 | markdown | Natural language description of the search parameter
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for search parameter (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this search parameter is defined
code | Σ | 1..1 | code | Code used in URL
base | Σ | 1..* | code | The resource type(s) this search parameter applies to ResourceType (Required)
type | Σ | 1..1 | code | number | date | string | token | reference | composite | quantity | uri | special SearchParamType (Required)
expression | I | 0..1 | string | FHIRPath expression that extracts the values
xpath | I | 0..1 | string | XPath that extracts the values
xpathUsage | I | 0..1 | code | normal | phonetic | nearby | distance | other XPathUsageType (Required)
target | | 0..* | code | Types of resource (if a resource reference) ResourceType (Required)
multipleOr | | 0..1 | boolean | Allow multiple values per parameter (or)
multipleAnd | | 0..1 | boolean | Allow multiple parameters (and)
comparator | | 0..* | code | eq | ne | gt | lt | ge | le | sa | eb | ap SearchComparator (Required)
modifier | | 0..* | code | missing | exact | contains | not | text | in | not-in | below | above | type | identifier | ofType SearchModifierCode (Required)
chain | | 0..* | string | Chained names supported
component | | 0..* | BackboneElement | For Composite resources to define the parts
definition | | 1..1 | canonical(SearchParameter) | Defines how the part works
expression | | 1..1 | string | Subexpression relative to main expression
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SearchParameter xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Canonical identifier for this search parameter, represented as a URI (globally unique)
    -->
 <version value="[string]"/><!-- 0..1 Business version of the search parameter -->
 <name value="[string]"/><!--  1..1 Name for this search parameter (computer friendly) -->
 <derivedFrom><!-- 0..1 canonical(SearchParameter) Original definition for the search parameter --></derivedFrom>
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 1..1 Natural language description of the search parameter -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for search parameter (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this search parameter is defined -->
 <code value="[code]"/><!-- 1..1 Code used in URL -->
 <base value="[code]"/><!-- 1..* The resource type(s) this search parameter applies to -->
 <type value="[code]"/><!-- 1..1 number | date | string | token | reference | composite | quantity | uri | special -->
 <expression value="[string]"/><!--  0..1 FHIRPath expression that extracts the values -->
 <xpath value="[string]"/><!--  0..1 XPath that extracts the values -->
 <xpathUsage value="[code]"/><!--  0..1 normal | phonetic | nearby | distance | other -->
 <target value="[code]"/><!-- 0..* Types of resource (if a resource reference) -->
 <multipleOr value="[boolean]"/><!-- 0..1 Allow multiple values per parameter (or) -->
 <multipleAnd value="[boolean]"/><!-- 0..1 Allow multiple parameters (and) -->
 <comparator value="[code]"/><!-- 0..* eq | ne | gt | lt | ge | le | sa | eb | ap -->
 <modifier value="[code]"/><!-- 0..* missing | exact | contains | not | text | in | not-in | below | above | type |
    identifier | ofType -->
 <chain value="[string]"/><!-- 0..* Chained names supported -->
 <component>  <!-- 0..* For Composite resources to define the parts -->
  <definition><!-- 1..1 canonical(SearchParameter) Defines how the part works --></definition>
  <expression value="[string]"/><!-- 1..1 Subexpression relative to main expression -->
 </component>
</SearchParameter>
```

JSON Template

```
{
  "resourceType" : "SearchParameter",
```

```
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Canonical identifier for this search parameter, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the search parameter
  "name" : "<string>", // C? R!  Name for this search parameter (computer friendly)
  "derivedFrom" : { canonical(SearchParameter) }, // Original definition for the search parameter
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // R!  Natural language description of the search parameter
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for search parameter (if applicable)
  "purpose" : "<markdown>", // Why this search parameter is defined
  "code" : "<code>", // R!  Code used in URL
  "base" : ["<code>"], // R!  The resource type(s) this search parameter applies to
  "type" : "<code>", // R!  number | date | string | token | reference | composite | quantity | uri | special
  "expression" : "<string>", // C? FHIRPath expression that extracts the values
  "xpath" : "<string>", // C? XPath that extracts the values
  "xpathUsage" : "<code>", // C? normal | phonetic | nearby | distance | other
  "target" : ["<code>"], // Types of resource (if a resource reference)
  "multipleOr" : <boolean>, // Allow multiple values per parameter (or)
  "multipleAnd" : <boolean>, // Allow multiple parameters (and)
  "comparator" : ["<code>"], // eq | ne | gt | lt | ge | le | sa | eb | ap
  "modifier" : ["<code>"], // missing | exact | contains | not | text | in | not-in | below | above | type | identifier
    | ofType
  "chain" : ["<string>"], // Chained names supported
  "component" : [{ // For Composite resources to define the parts
    "definition" : { canonical(SearchParameter) }, // R!  Defines how the part works
    "expression" : "<string>" // R!  Subexpression relative to main expression
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SearchParameter;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SearchParameter.url [ uri ]; # 1..1 Canonical identifier for this search parameter, represented as a URI
    (globally unique)
  fhir:SearchParameter.version [ string ]; # 0..1 Business version of the search parameter
  fhir:SearchParameter.name [ string ]; # 1..1 Name for this search parameter (computer friendly)
  fhir:SearchParameter.derivedFrom [ canonical(SearchParameter) ]; # 0..1 Original definition for the search parameter
  fhir:SearchParameter.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:SearchParameter.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:SearchParameter.date [ dateTime ]; # 0..1 Date last changed
  fhir:SearchParameter.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:SearchParameter.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:SearchParameter.description [ markdown ]; # 1..1 Natural language description of the search parameter
  fhir:SearchParameter.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:SearchParameter.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for search parameter (if
    applicable)
  fhir:SearchParameter.purpose [ markdown ]; # 0..1 Why this search parameter is defined
  fhir:SearchParameter.code [ code ]; # 1..1 Code used in URL
  fhir:SearchParameter.base [ code ], ... ; # 1..* The resource type(s) this search parameter applies to
  fhir:SearchParameter.type [ code ]; # 1..1 number | date | string | token | reference | composite | quantity | uri |
    special
  fhir:SearchParameter.expression [ string ]; # 0..1 FHIRPath expression that extracts the values
  fhir:SearchParameter.xpath [ string ]; # 0..1 XPath that extracts the values
  fhir:SearchParameter.xpathUsage [ code ]; # 0..1 normal | phonetic | nearby | distance | other
  fhir:SearchParameter.target [ code ], ... ; # 0..* Types of resource (if a resource reference)
  fhir:SearchParameter.multipleOr [ boolean ]; # 0..1 Allow multiple values per parameter (or)
  fhir:SearchParameter.multipleAnd [ boolean ]; # 0..1 Allow multiple parameters (and)
  fhir:SearchParameter.comparator [ code ], ... ; # 0..* eq | ne | gt | lt | ge | le | sa | eb | ap
  fhir:SearchParameter.modifier [ code ], ... ; # 0..* missing | exact | contains | not | text | in | not-in | below |
    above | type | identifier | ofType
  fhir:SearchParameter.chain [ string ], ... ; # 0..* Chained names supported
  fhir:SearchParameter.component [ # 0..* For Composite resources to define the parts
    fhir:SearchParameter.component.definition [ canonical(SearchParameter) ]; # 1..1 Defines how the part works
    fhir:SearchParameter.component.expression [ string ]; # 1..1 Subexpression relative to main expression
  ], ...;
]
```

Changes since R3

SearchParameter

SearchParameter | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *

SearchParameter.derivedFrom | Type changed from uri to canonical(SearchParameter)

SearchParameter.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

SearchParameter.experimental | No longer marked as Modifier

SearchParameter.base | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1

SearchParameter.type | Change value set from http://hl7.org/fhir/ValueSet/search-param-type to http://hl7.org/fhir/ValueSet/search-param-type|4.0.1
SearchParameter.xpathUsage | Change value set from http://hl7.org/fhir/ValueSet/search-xpath-usage to http://hl7.org/fhir/ValueSet/search-xpath-usage|4.0.1
SearchParameter.target | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
SearchParameter.multipleOr | Added Element
SearchParameter.multipleAnd | Added Element
SearchParameter.comparator | Change value set from http://hl7.org/fhir/ValueSet/search-comparator to http://hl7.org/fhir/ValueSet/search-comparator|4.0.1
SearchParameter.modifier | Change value set from http://hl7.org/fhir/ValueSet/search-modifier-code to http://hl7.org/fhir/ValueSet/search-modifier-code|4.0.1
SearchParameter.component.definition | Type changed from Reference(SearchParameter) to canonical(SearchParameter)

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
SearchParameter | ITU | | DomainResource | Search parameter for a resource + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: If an xpath is present, there SHALL be an xpathUsage + Rule: Search parameters can only have chain names when the search parameter type is 'reference' Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 1..1 | uri | Canonical identifier for this search parameter, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the search parameter
name | ΣI | 1..1 | string | Name for this search parameter (computer friendly)
derivedFrom | | 0..1 | canonical(SearchParameter) | Original definition for the search parameter
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 1..1 | markdown | Natural language description of the search parameter
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for search parameter (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this search parameter is defined
code | Σ | 1..1 | code | Code used in URL
base | Σ | 1..* | code | The resource type(s) this search parameter applies to ResourceType (Required)
type | Σ | 1..1 | code | number | date | string | token | reference | composite | quantity | uri | special SearchParamType (Required)
expression | I | 0..1 | string | FHIRPath expression that extracts the values
xpath | I | 0..1 | string | XPath that extracts the values
xpathUsage | I | 0..1 | code | normal | phonetic | nearby | distance | other XPathUsageType (Required)
target | | 0..* | code | Types of resource (if a resource reference) ResourceType (Required)
multipleOr | | 0..1 | boolean | Allow multiple values per parameter (or)
multipleAnd | | 0..1 | boolean | Allow multiple parameters (and)
comparator | | 0..* | code | eq | ne | gt | lt | ge | le | sa | eb | ap SearchComparator (Required)
modifier | | 0..* | code | missing | exact | contains | not | text | in | not-in | below | above | type | identifier | ofType SearchModifierCode (Required)
chain | | 0..* | string | Chained names supported
component | | 0..* | BackboneElement | For Composite resources to define the parts
definition | | 1..1 | canonical(SearchParameter) | Defines how the part works
expression | | 1..1 | string | Subexpression relative to main expression
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SearchParameter xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Canonical identifier for this search parameter, represented as a URI (globally unique)
    -->
 <version value="[string]"/><!-- 0..1 Business version of the search parameter -->
 <name value="[string]"/><!--  1..1 Name for this search parameter (computer friendly) -->
 <derivedFrom><!-- 0..1 canonical(SearchParameter) Original definition for the search parameter --></derivedFrom>
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 1..1 Natural language description of the search parameter -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for search parameter (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this search parameter is defined -->
 <code value="[code]"/><!-- 1..1 Code used in URL -->
 <base value="[code]"/><!-- 1..* The resource type(s) this search parameter applies to -->
```

```
 <type value="[code]"/><!-- 1..1 number | date | string | token | reference | composite | quantity | uri | special -->
 <expression value="[string]"/><!--  0..1 FHIRPath expression that extracts the values -->
 <xpath value="[string]"/><!--  0..1 XPath that extracts the values -->
 <xpathUsage value="[code]"/><!--  0..1 normal | phonetic | nearby | distance | other -->
 <target value="[code]"/><!-- 0..* Types of resource (if a resource reference) -->
 <multipleOr value="[boolean]"/><!-- 0..1 Allow multiple values per parameter (or) -->
 <multipleAnd value="[boolean]"/><!-- 0..1 Allow multiple parameters (and) -->
 <comparator value="[code]"/><!-- 0..* eq | ne | gt | lt | ge | le | sa | eb | ap -->
 <modifier value="[code]"/><!-- 0..* missing | exact | contains | not | text | in | not-in | below | above | type |
    identifier | ofType -->
 <chain value="[string]"/><!-- 0..* Chained names supported -->
 <component>  <!-- 0..* For Composite resources to define the parts -->
  <definition><!-- 1..1 canonical(SearchParameter) Defines how the part works --></definition>
  <expression value="[string]"/><!-- 1..1 Subexpression relative to main expression -->
 </component>
</SearchParameter>
```

JSON Template

```
{
  "resourceType" : "SearchParameter",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Canonical identifier for this search parameter, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the search parameter
  "name" : "<string>", // C? R!  Name for this search parameter (computer friendly)
  "derivedFrom" : { canonical(SearchParameter) }, // Original definition for the search parameter
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // R!  Natural language description of the search parameter
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for search parameter (if applicable)
  "purpose" : "<markdown>", // Why this search parameter is defined
  "code" : "<code>", // R!  Code used in URL
  "base" : ["<code>"], // R!  The resource type(s) this search parameter applies to
  "type" : "<code>", // R!  number | date | string | token | reference | composite | quantity | uri | special
  "expression" : "<string>", // C? FHIRPath expression that extracts the values
  "xpath" : "<string>", // C? XPath that extracts the values
  "xpathUsage" : "<code>", // C? normal | phonetic | nearby | distance | other
  "target" : ["<code>"], // Types of resource (if a resource reference)
  "multipleOr" : <boolean>, // Allow multiple values per parameter (or)
  "multipleAnd" : <boolean>, // Allow multiple parameters (and)
  "comparator" : ["<code>"], // eq | ne | gt | lt | ge | le | sa | eb | ap
  "modifier" : ["<code>"], // missing | exact | contains | not | text | in | not-in | below | above | type | identifier
    | ofType
  "chain" : ["<string>"], // Chained names supported
  "component" : [{ // For Composite resources to define the parts
    "definition" : { canonical(SearchParameter) }, // R!  Defines how the part works
    "expression" : "<string>" // R!  Subexpression relative to main expression
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SearchParameter;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SearchParameter.url [ uri ]; # 1..1 Canonical identifier for this search parameter, represented as a URI
    (globally unique)
  fhir:SearchParameter.version [ string ]; # 0..1 Business version of the search parameter
  fhir:SearchParameter.name [ string ]; # 1..1 Name for this search parameter (computer friendly)
  fhir:SearchParameter.derivedFrom [ canonical(SearchParameter) ]; # 0..1 Original definition for the search parameter
  fhir:SearchParameter.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:SearchParameter.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:SearchParameter.date [ dateTime ]; # 0..1 Date last changed
  fhir:SearchParameter.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:SearchParameter.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:SearchParameter.description [ markdown ]; # 1..1 Natural language description of the search parameter
  fhir:SearchParameter.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:SearchParameter.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for search parameter (if
    applicable)
  fhir:SearchParameter.purpose [ markdown ]; # 0..1 Why this search parameter is defined
  fhir:SearchParameter.code [ code ]; # 1..1 Code used in URL
  fhir:SearchParameter.base [ code ], ... ; # 1..* The resource type(s) this search parameter applies to
  fhir:SearchParameter.type [ code ]; # 1..1 number | date | string | token | reference | composite | quantity | uri |
    special
  fhir:SearchParameter.expression [ string ]; # 0..1 FHIRPath expression that extracts the values
  fhir:SearchParameter.xpath [ string ]; # 0..1 XPath that extracts the values
  fhir:SearchParameter.xpathUsage [ code ]; # 0..1 normal | phonetic | nearby | distance | other
  fhir:SearchParameter.target [ code ], ... ; # 0..* Types of resource (if a resource reference)
  fhir:SearchParameter.multipleOr [ boolean ]; # 0..1 Allow multiple values per parameter (or)
  fhir:SearchParameter.multipleAnd [ boolean ]; # 0..1 Allow multiple parameters (and)
  fhir:SearchParameter.comparator [ code ], ... ; # 0..* eq | ne | gt | lt | ge | le | sa | eb | ap
```

```
  fhir:SearchParameter.modifier [ code ], ... ; # 0..* missing | exact | contains | not | text | in | not-in | below |
    above | type | identifier | ofType
  fhir:SearchParameter.chain [ string ], ... ; # 0..* Chained names supported
  fhir:SearchParameter.component [ # 0..* For Composite resources to define the parts
    fhir:SearchParameter.component.definition [ canonical(SearchParameter) ]; # 1..1 Defines how the part works
    fhir:SearchParameter.component.expression [ string ]; # 1..1 Subexpression relative to main expression
  ], ...;
]
```

Changes since Release 3

SearchParameter
SearchParameter | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *
SearchParameter.derivedFrom | Type changed from uri to canonical(SearchParameter)
SearchParameter.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
SearchParameter.experimental | No longer marked as Modifier
SearchParameter.base | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
SearchParameter.type | Change value set from http://hl7.org/fhir/ValueSet/search-param-type to http://hl7.org/fhir/ValueSet/search-param-type|4.0.1
SearchParameter.xpathUsage | Change value set from http://hl7.org/fhir/ValueSet/search-xpath-usage to http://hl7.org/fhir/ValueSet/search-xpath-usage|4.0.1
SearchParameter.target | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
SearchParameter.multipleOr | Added Element
SearchParameter.multipleAnd | Added Element
SearchParameter.comparator | Change value set from http://hl7.org/fhir/ValueSet/search-comparator to http://hl7.org/fhir/ValueSet/search-comparator|4.0.1
SearchParameter.modifier | Change value set from http://hl7.org/fhir/ValueSet/search-modifier-code to http://hl7.org/fhir/ValueSet/search-modifier-code|4.0.1
SearchParameter.component.definition | Type changed from Reference(SearchParameter) to canonical(SearchParameter)

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

5.5.4.1 Terminology Bindings

Path | Definition | Type | Reference
SearchParameter.status | The lifecycle status of an artifact. | Required | PublicationStatus
SearchParameter.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
SearchParameter.base SearchParameter.target | One of the resource types defined as part of this version of FHIR. | Required | Resource Types
SearchParameter.type | Data types allowed to be used for search parameters. | Required | SearchParamType
SearchParameter.xpathUsage | How a search parameter relates to the set of elements returned by evaluating its xpath query. | Required | XPathUsageType
SearchParameter.comparator | What Search Comparator Codes are supported in search. | Required | SearchComparator
SearchParameter.modifier | A supported modifier for a search parameter. | Required | SearchModifierCode

5.5.4.2 Constraints

id | Level | Location | Description | Expression
spd-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
spd-1 | Rule | (base) | If an xpath is present, there SHALL be an xpathUsage | xpath.empty() or xpathUsage.exists()
spd-2 | Rule | (base) | Search parameters can only have chain names when the search parameter type is 'reference' | chain.empty() or type = 'reference'

5.5.5 SearchParameters on Extensions

The SearchParameter resource may be used to define searches on extensions. Depending on the context of the extension (as defined in the StructureDefinition), this may require multiple instances of the SearchParameter.xpath element like so:

• If the extension's context is scoped by a single resource, SearchParameter.base must be set to that resource, and there may be one or more xpath elements corresponding to the elements specified in the extension's context, e.g. "f:Patient/f:name/f:extension[@url='extension url']"

• If the extension's context specifies multiple resources, SearchParameter.base must be set to "DomainResource".

• You may use a single xpath element with the full URL of the Extension to indicate that the SearchParameter would search in all elements of the given SearchParameter.base resource for the given extension.

5.5.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
base | token | The resource type(s) this search parameter applies to | SearchParameter.base
code | token | Code used in URL | SearchParameter.code
component | reference | Defines how the part works | SearchParameter.component.definition (SearchParameter)
context | token | A use context assigned to the search parameter | (SearchParameter.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the search parameter | (SearchParameter.useContext.value as Quantity) | (SearchParameter.useContext.value as Range)
context-type | token | A type of use context assigned to the search parameter | SearchParameter.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the search parameter | On SearchParameter.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the search parameter | On SearchParameter.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The search parameter publication date | SearchParameter.date
derived-from | reference | Original definition for the search parameter | SearchParameter.derivedFrom (SearchParameter)
description | string | The description of the search parameter | SearchParameter.description
jurisdiction | token | Intended jurisdiction for the search parameter | SearchParameter.jurisdiction
name | string | Computationally friendly name of the search parameter | SearchParameter.name
publisher | string | Name of the publisher of the search parameter | SearchParameter.publisher
status | token | The current status of the search parameter | SearchParameter.status
target | token | Types of resource (if a resource reference) | SearchParameter.target
type | token | number | date | string | token | reference | composite | quantity | uri | special | SearchParameter.type
url | uri | The uri that identifies the search parameter | SearchParameter.url
version | token | The business version of the search parameter | SearchParameter.version

5.5.7 Resource SearchParameter - Examples

URL: https://hl7.org/fhir/R4/searchparameter-examples.html

5.5.7 Resource SearchParameter - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
General example | example | XML | JSON | Turtle
Search parameter for an extension example | example-extension | XML | JSON | Turtle
Search parameter for a reference example | example-reference | XML | JSON | Turtle
Formal Definition for _filter parameter | filter | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.5.8 Resource SearchParameter - Detailed Descriptions

URL: https://hl7.org/fhir/R4/searchparameter-definitions.html

5.5.8 Resource SearchParameter - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the SearchParameter resource.

SearchParameter
Element Id | SearchParameter
Definition | A search parameter that defines a named search item that can be used to search/filter on a resource.
Cardinality | 0..*
Type | DomainResource
Comments | In FHIR, search is not performed directly on a resource (by XML or JSON path), but on a named parameter that maps into the resource content.
Invariants | Defined on this element spd-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}') spd-1 Rule If an xpath is present, there SHALL be an xpathUsage xpath.empty() or xpathUsage.exists() spd-2 Rule Search parameters can only have chain names when the search parameter type is 'reference' chain.empty() or type = 'reference'
Defined on this element
spd-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
spd-1 | Rule | If an xpath is present, there SHALL be an xpathUsage | xpath.empty() or xpathUsage.exists()
spd-2 | Rule | Search parameters can only have chain names when the search parameter type is 'reference' | chain.empty() or type = 'reference'
SearchParameter.url
Element Id | SearchParameter.url

Definition | An absolute URI that is used to identify this search parameter when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which an authoritative instance of this search parameter is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the search parameter is stored on different servers.
Cardinality | 1..1
Type | uri
Requirements | Allows the search parameter to be referenced by a single globally unique identifier. To allow referencing and reusing search parameter definitions under other names on operational systems (in case of name clash).
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
SearchParameter.version
Element Id | SearchParameter.version
Definition | The identifier that is used to identify this version of the search parameter when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the search parameter author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different search parameter instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the search parameter with the format [url]|[version].
SearchParameter.name
Element Id | SearchParameter.name
Definition | A natural language name identifying the search parameter. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 1..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
SearchParameter.derivedFrom
Element Id | SearchParameter.derivedFrom
Definition | Where this search parameter is originally defined. If a derivedFrom is provided, then the details in the search parameter must be consistent with the definition from which it is defined. i.e. the parameter should have the same meaning, and (usually) the functionality should be a proper subset of the underlying search parameter.
Cardinality | 0..1
Type | canonical(SearchParameter)
Comments | The intent of this is that a server can designate that it provides support for a search parameter defined in the specification itself (e.g. value-quantity , but detail how it is supported by the server.
SearchParameter.status
Element Id | SearchParameter.status
Definition | The status of this search parameter. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of search parameters that are appropriate for use versus not.
SearchParameter.experimental
Element Id | SearchParameter.experimental
Definition | A Boolean value to indicate that this search parameter is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level search parameter.
Summary | true
Comments | Allows filtering of search parameters that are appropriate for use versus not.
SearchParameter.date
Element Id | SearchParameter.date

Definition | The date (and optionally time) when the search parameter was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the search parameter changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the search parameter. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
SearchParameter.publisher
Element Id | SearchParameter.publisher
Definition | The name of the organization or individual that published the search parameter.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the search parameter. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the search parameter is the organization or individual primarily responsible for the maintenance and upkeep of the search parameter. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the search parameter. This item SHOULD be populated unless the information is available from context.
SearchParameter.contact
Element Id | SearchParameter.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
SearchParameter.description
Element Id | SearchParameter.description
Definition | And how it used.
Cardinality | 1..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the search parameter was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the search parameter as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the search parameter is presumed to be the predominant language in the place the search parameter was created).
SearchParameter.useContext
Element Id | SearchParameter.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate search parameter instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
SearchParameter.jurisdiction
Element Id | SearchParameter.jurisdiction
Definition | A legal or geographic region in which the search parameter is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the search parameter to be used in jurisdictions other than those for which it was originally designed or intended.
SearchParameter.purpose
Element Id | SearchParameter.purpose
Definition | Explanation of why this search parameter is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the search parameter. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this search parameter.
SearchParameter.code
Element Id | SearchParameter.code
Definition | The code used in the URL or the parameter name in a parameters resource for this search parameter.
Cardinality | 1..1
Type | code
Summary | true
Comments | For maximum compatibility, use only lowercase ASCII characters.

SearchParameter.base
Element Id | SearchParameter.base
Definition | The base resource type(s) that this search parameter can be used against.
Cardinality | 1..*
Terminology Binding | Any defined Resource Type name
Type | code
Summary | true
Comments | A search parameter must always apply to at least one resource type. When search parameters apply to more than one resource type, they can be used against any of the listed resource types, or in a cross-type search (see Cross Resource Search).
SearchParameter.type
Element Id | SearchParameter.type
Definition | The type of value that a search parameter may contain, and how the content is interpreted.
Cardinality | 1..1
Terminology Binding | SearchParamType (Required)
Type | code
Summary | true
SearchParameter.expression
Element Id | SearchParameter.expression
Definition | A FHIRPath expression that returns a set of elements for the search parameter.
Cardinality | 0..1
Type | string
Comments | Note that the elements returned by the expression are sometimes complex elements where logic is required to determine quite how to handle them; e.g. CodeableConcepts may contain text and/or multiple codings, where the codings themselves contain a code and a system. For composite search parameters, the outcome of the expression must a collection of base elements from which the composites are derived.
Invariants | Affect this element spd-1 Rule If an xpath is present, there SHALL be an xpathUsage xpath.empty() or xpathUsage.exists()
Affect this element
spd-1 | Rule | If an xpath is present, there SHALL be an xpathUsage | xpath.empty() or xpathUsage.exists()
SearchParameter.xpath
Element Id | SearchParameter.xpath
Definition | An XPath expression that returns a set of elements for the search parameter.
Cardinality | 0..1
Type | string
Comments | Note that the elements returned by the XPath are sometimes complex elements where logic is required to determine quite how to handle them; e.g. CodeableConcepts may contain text and/or multiple codings, where the codings themselves contain a code and a system.
Invariants | Affect this element spd-1 Rule If an xpath is present, there SHALL be an xpathUsage xpath.empty() or xpathUsage.exists()
Affect this element
spd-1 | Rule | If an xpath is present, there SHALL be an xpathUsage | xpath.empty() or xpathUsage.exists()
SearchParameter.xpathUsage
Element Id | SearchParameter.xpathUsage
Definition | How the search parameter relates to the set of elements returned by evaluating the xpath query.
Cardinality | 0..1
Terminology Binding | XPathUsageType (Required)
Type | code
Invariants | Affect this element spd-1 Rule If an xpath is present, there SHALL be an xpathUsage xpath.empty() or xpathUsage.exists()
Affect this element
spd-1 | Rule | If an xpath is present, there SHALL be an xpathUsage | xpath.empty() or xpathUsage.exists()
SearchParameter.target
Element Id | SearchParameter.target
Definition | Types of resource (if a resource is referenced).
Cardinality | 0..*
Terminology Binding | Any defined Resource Type name
Type | code
SearchParameter.multipleOr
Element Id | SearchParameter.multipleOr
Definition | Whether multiple values are allowed for each time the parameter exists. Values are separated by commas, and the parameter matches if any of the values match.
Cardinality | 0..1
Type | boolean
SearchParameter.multipleAnd
Element Id | SearchParameter.multipleAnd
Definition | Whether multiple parameters are allowed - e.g. more than one parameter with the same name. The search matches if all the parameters match.
Cardinality | 0..1
Type | boolean
SearchParameter.comparator
Element Id | SearchParameter.comparator
Definition | Comparators supported for the search parameter.

Cardinality | 0..*
Terminology Binding | SearchComparator (Required)
Type | code
Comments | If no comparators are listed, clients should not expect servers to support any comparators.
SearchParameter.modifier
Element Id | SearchParameter.modifier
Definition | A modifier supported for the search parameter.
Cardinality | 0..*
Terminology Binding | SearchModifierCode (Required)
Type | code
SearchParameter.chain
Element Id | SearchParameter.chain
Definition | Contains the names of any search parameters which may be chained to the containing search parameter. Chained parameters may be added to search parameters of type reference and specify that resources will only be returned if they contain a reference to a resource which matches the chained parameter value. Values for this field should be drawn from SearchParameter.code for a parameter on the target resource type.
Cardinality | 0..*
Type | string
Comments | Systems are not required to list all the chain names they support, but if they don't list them, clients might not know to use them.
SearchParameter.component
Element Id | SearchParameter.component
Definition | Used to define the parts of a composite search parameter.
Cardinality | 0..*
SearchParameter.component.definition
Element Id | SearchParameter.component.definition
Definition | The definition of the search parameter that describes this part.
Cardinality | 1..1
Type | canonical(SearchParameter)
SearchParameter.component.expression
Element Id | SearchParameter.component.expression
Definition | A sub-expression that defines how to extract values for this component from the output of the main SearchParameter.expression.
Cardinality | 1..1
Type | string
Comments | This expression overrides the expression in the definition and extracts the index values from the outcome of the composite expression.

5.5.9 Resource SearchParameter - Mappings

URL: https://hl7.org/fhir/R4/searchparameter-mappings.html

5.5.9 Resource SearchParameter - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the searchparameter resource (see Mappings to Other Standards for further information & status).

5.5.9.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

SearchParameter
url | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

5.5.9.2 Workflow Pattern (http://hl7.org/fhir/workflow)

SearchParameter
url | Definition.url
version | Definition.version
status | Definition.status
experimental | Definition.experimental

date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose

5.5.9.3 HL7 v2 Mapping (http://hl7.org/v2)

SearchParameter | n/a

5.5.9.4 RIM Mapping (http://hl7.org/v3)

SearchParameter | n/a

5.5.10 Resource SearchParameter - Extensions & Profiles

URL: https://hl7.org/fhir/R4/searchparameter-profiles.html

5.5.10 Resource SearchParameter - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the SearchParameter resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

5.5.10.1 Profiles

No Profiles defined for this resource

5.5.10.2 Extensions

replaces | SearchParameter | replaces | for General Extensions for use by FHIR Implementers
resource-approvalDate | SearchParameter | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | SearchParameter | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | SearchParameter | lastReviewDate | for Resource HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

5.5.10.3 Search Extensions

No Search Extensions defined for this resource

5.5.11 Resource SearchParameter - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/searchparameter-version-maps.html

No validation errors - all conversions are clean

5.5.11 Resource SearchParameter - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for SearchParameter.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors). (see documentation)

5.5.11.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/SearchParameter3to4" = "R3 to R4 Conversions for SearchParameter"

uses "http://hl7.org/fhir/3.0/StructureDefinition/SearchParameter" alias SearchParameterR3 as source
uses "http://hl7.org/fhir/StructureDefinition/SearchParameter" alias SearchParameter as target

imports "http://hl7.org/fhir/StructureMap/*3to4"
```

```
group SearchParameter(source src : SearchParameterR3, target tgt : SearchParameter) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.code -> tgt.code;
  src.base -> tgt.base;
  src.type -> tgt.type;
  src.derivedFrom -> tgt.derivedFrom;
  src.description -> tgt.description;
  src.expression -> tgt.expression;
  src.xpath -> tgt.xpath;
  src.xpathUsage -> tgt.xpathUsage;
  src.target -> tgt.target;
  src.comparator -> tgt.comparator;
  src.modifier -> tgt.modifier;
  src.chain -> tgt.chain;
  src.component as vs0 -> tgt.component as vt0 then {
    vs0.definition -> vt0.definition;
    vs0.expression -> vt0.expression;
  };
}
```

5.5.11.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/SearchParameter4to3" = "R4 to R3 Conversion for SearchParameter"

uses "http://hl7.org/fhir/StructureDefinition/SearchParameter" alias SearchParameter as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/SearchParameter" alias SearchParameterR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group SearchParameter(source src : SearchParameter, target tgt : SearchParameterR3) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.code -> tgt.code;
  src.base -> tgt.base;
  src.type -> tgt.type;
  src.derivedFrom -> tgt.derivedFrom;
  src.description -> tgt.description;
  src.expression -> tgt.expression;
  src.xpath -> tgt.xpath;
  src.xpathUsage -> tgt.xpathUsage;
  src.target -> tgt.target;
  src.comparator -> tgt.comparator;
  src.modifier -> tgt.modifier;
  src.chain -> tgt.chain;
  src.component as vs0 -> tgt.component as vt0 then {
    vs0.definition -> vt0.definition;
    vs0.expression -> vt0.expression;
  };
}
```

5.5.11.3 R4 Validation Errors

5.5.12 Standard Search Parameter Extensions

URL: https://hl7.org/fhir/R4/searchparameter-extensions.html

This profile describes common extensions that are used with Search Parameters

5.5.12 Standard Search Parameter Extensions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

5.5.12.1 Content

5.5.13 JSON Schema for SearchParameter

URL: https://hl7.org/fhir/R4/searchparameter.schema.json.html

5.5.13 JSON Schema for SearchParameter

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for SearchParameter, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/SearchParameter",
  "$ref": "#/definitions/SearchParameter",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "SearchParameter": {
      "description": "A search parameter that defines a named search item that can be used to search/filter on a
    resource.",
      "properties": {
        "resourceType": {
          "description": "This is a SearchParameter resource",
          "const": "SearchParameter"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "url": {
          "description": "An absolute URI that is used to identify this search parameter when it is referenced in a
```

```
specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this search parameter is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the search parameter is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the search parameter when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the search parameter author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the search parameter. This name should be usable as an identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "derivedFrom": {
      "description": "Where this search parameter is originally defined. If a derivedFrom is provided, then the details in the search parameter must be consistent with the definition from which it is defined. i.e. the parameter should have the same meaning, and (usually) the functionality should be a proper subset of the underlying search parameter.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "status": {
      "description": "The status of this search parameter. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this search parameter is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the search parameter was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the search parameter changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the search parameter.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "And how it used.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
```

```
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate search parameter instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the search parameter is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this search parameter is needed and why it has been designed as it has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "The code used in the URL or the parameter name in a parameters resource for this search parameter.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "base": {
      "description": "The base resource type(s) that this search parameter can be used against.",
      "items": {
        "$ref": "#/definitions/code"
      },
      "type": "array"
    },
    "_base": {
      "description": "Extensions for base",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of value that a search parameter may contain, and how the content is interpreted.",
      "enum": [
        "number",
        "date",
        "string",
        "token",
        "reference",
        "composite",
        "quantity",
        "uri",
        "special"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "expression": {
      "description": "A FHIRPath expression that returns a set of elements for the search parameter.",
      "$ref": "#/definitions/string"
    },
    "_expression": {
      "description": "Extensions for expression",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "xpath": {
      "description": "An XPath expression that returns a set of elements for the search parameter.",
      "$ref": "#/definitions/string"
    },
    "_xpath": {
      "description": "Extensions for xpath",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "xpathUsage": {
      "description": "How the search parameter relates to the set of elements returned by evaluating the xpath query.",
      "enum": [
```

```
        "normal",
        "phonetic",
        "nearby",
        "distance",
        "other"
      ]
    },
    "_xpathUsage": {
      "description": "Extensions for xpathUsage",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "target": {
      "description": "Types of resource (if a resource is referenced).",
      "items": {
        "$ref": "#/definitions/code"
      },
      "type": "array"
    },
    "_target": {
      "description": "Extensions for target",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "multipleOr": {
      "description": "Whether multiple values are allowed for each time the parameter exists. Values are separated
by commas, and the parameter matches if any of the values match.",
      "$ref": "#/definitions/boolean"
    },
    "_multipleOr": {
      "description": "Extensions for multipleOr",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "multipleAnd": {
      "description": "Whether multiple parameters are allowed - e.g. more than one parameter with the same name. The
 search matches if all the parameters match.",
      "$ref": "#/definitions/boolean"
    },
    "_multipleAnd": {
      "description": "Extensions for multipleAnd",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "comparator": {
      "description": "Comparators supported for the search parameter.",
      "items": {
        "enum": [
          "eq",
          "ne",
          "gt",
          "lt",
          "ge",
          "le",
          "sa",
          "eb",
          "ap"
        ]
      },
      "type": "array"
    },
    "_comparator": {
      "description": "Extensions for comparator",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "modifier": {
      "description": "A modifier supported for the search parameter.",
      "items": {
        "enum": [
          "missing",
          "exact",
          "contains",
          "not",
          "text",
          "in",
          "not-in",
          "below",
          "above",
          "type",
          "identifier",
          "ofType"
        ]
      },
      "type": "array"
    },
    "_modifier": {
      "description": "Extensions for modifier",
```

```
        "items": {
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "type": "array"
      },
      "chain": {
        "description": "Contains the names of any search parameters which may be chained to the containing search
    parameter. Chained parameters may be added to search parameters of type reference and specify that resources will
    only be returned if they contain a reference to a resource which matches the chained parameter value. Values for
    this field should be drawn from SearchParameter.code for a parameter on the target resource type.",
        "items": {
          "$ref": "#/definitions/string"
        },
        "type": "array"
      },
      "_chain": {
        "description": "Extensions for chain",
        "items": {
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "type": "array"
      },
      "component": {
        "description": "Used to define the parts of a composite search parameter.",
        "items": {
          "$ref": "#/definitions/SearchParameter_Component"
        },
        "type": "array"
      }
    },
    "additionalProperties": false,
    "required": [
      "resourceType"
    ]
  },
  "SearchParameter_Component": {
    "description": "A search parameter that defines a named search item that can be used to search/filter on a
    resource.",
    "properties": {
      "id": {
        "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
        "$ref": "string.schema.json#/definitions/string"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "definition": {
        "description": "The definition of the search parameter that describes this part.",
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "expression": {
        "description": "A sub-expression that defines how to extract values for this component from the output of the
    main SearchParameter.expression.",
        "$ref": "#/definitions/string"
      },
      "_expression": {
        "description": "Extensions for expression",
        "$ref": "Element.schema.json#/definitions/Element"
      }
    },
    "additionalProperties": false,
    "required": [
      "definition"
    ]
  }
 }
}
```

5.5.14 Resource SearchParameter - Maturity Dependency View

URL: https://hl7.org/fhir/R4/searchparameter-dependencies.html

5.5.14 Resource SearchParameter - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for SearchParameter FMM level 3

Name | Card. | Type | Dependency Analysis
SearchParameter | | DomainResource
url | 1..1 | uri
version | 0..1 | string
name | 1..1 | string
derivedFrom | 0..1 | canonical(SearchParameter)
status | 1..1 | code
experimental | 0..1 | boolean
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 1..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
code | 1..1 | code
base | 1..* | code
type | 1..1 | code
expression | 0..1 | string
xpath | 0..1 | string
xpathUsage | 0..1 | code
target | 0..* | code
multipleOr | 0..1 | boolean
multipleAnd | 0..1 | boolean
comparator | 0..* | code
modifier | 0..* | code
chain | 0..* | string
component | 0..* | BackboneElement
definition | 1..1 | canonical(SearchParameter)
expression | 1..1 | string

5.6 Resource CompartmentDefinition - Content

URL: https://hl7.org/fhir/R4/compartmentdefinition.html

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

A compartment definition that defines how resources are accessed on a server.

5.6.1 Scope and Usage

Each resource may belong to one or more logical compartments. A compartment is a logical grouping of resources which share a common property. Compartments have two principal roles:

• Function as an access mechanism for finding a set of related resources quickly

• Provide a definitional basis for applying access control to resources quickly

Note:

At present, compartment definitions can only be defined by HL7 International. This is because their existence creates significant impact on the behavior of servers.

5.6.2 Boundaries and Relationships

Compartment definitions describe how particular compartment instances are named and identified, and how systems know which resources are in the compartment. The following compartments are defined by this specification:

Title | Description | Identity | Membership
Patient | The set of resources associated with a particular patient | There is an instance of the patient compartment for each patient resource, and the identity of the compartment is the same as the patient. When a patient is linked to another patient, all the records associated with the linked patient are in the compartment associated with the target of the link. | The patient compartment includes any resources where the subject of the resource is the patient, and some other resources that are directly linked to resources in the patient compartment
Encounter | The set of resources associated with a particular encounter | There is an instance of the encounter compartment for each encounter resource, and the identity of the compartment is the same as the encounter | The encounter compartment includes any resources where the resource has an explicitly nominated encounter, and some other resources that themselves link to resources in the encounter compartment. Note that for many resources, the exact nature of the link to encounter can be

ambiguous (e.g. for a DiagnosticReport, is it the encounter when it was initiated, or when it was reported?)
RelatedPerson | The set of resources associated with a particular 'related person' | There is an instance of the relatedPerson compartment for each relatedPerson resource, and the identity of the compartment is the same as the relatedPerson | The relatedPerson compartment includes any resources where the resource is explicitly linked to relatedPerson (usually as author)
Practitioner | The set of resources associated with a particular practitioner | There is an instance of the practitioner compartment for each Practitioner resource, and the identity of the compartment is the same as the Practitioner | The practitioner compartment includes any resources where the resource is explicitly linked to a Practitioner (usually as author, but other kinds of linkage exist)
Device | The set of resources associated with a particular device | There is an instance of the device compartment for each Device resource, and the identity of the compartment is the same as the Device | The device compartment includes any resources where the resource is explicitly linked to a Device (mostly subject or performer)

The full definitions of these compartments are published as CompartmentDefinition resources. Servers typically do not support the full definition of a compartment, and are not required to. Systems may publish CompartmentDefinition resources so that other systems may make use of compartments properly.

• CompartmentDefinitions are used by CapabilityStatement instances for specifying how resources are accessed

This resource is referenced by CapabilityStatement

5.6.3 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
CompartmentDefinition | ITU | | DomainResource | Compartment Definition for a resource + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 1..1 | uri | Canonical identifier for this compartment definition, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the compartment definition
name | ΣI | 1..1 | string | Name for this compartment definition (computer friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the compartment definition
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
purpose | | 0..1 | markdown | Why this compartment definition is defined
code | Σ | 1..1 | code | Patient | Encounter | RelatedPerson | Practitioner | Device CompartmentType (Required)
search | Σ | 1..1 | boolean | Whether the search syntax is supported
resource | Σ | 0..* | BackboneElement | How a resource is related to the compartment
code | Σ | 1..1 | code | Name of resource type ResourceType (Required)
param | Σ | 0..* | string | Search Parameter Name, or chained parameters
documentation | | 0..1 | string | Additional documentation about the resource and compartment
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CompartmentDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Canonical identifier for this compartment definition, represented as a URI (globally
    unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the compartment definition -->
 <name value="[string]"/><!--  1..1 Name for this compartment definition (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the compartment definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <purpose value="[markdown]"/><!-- 0..1 Why this compartment definition is defined -->
 <code value="[code]"/><!-- 1..1 Patient | Encounter | RelatedPerson | Practitioner | Device -->
 <search value="[boolean]"/><!-- 1..1 Whether the search syntax is supported -->
 <resource>  <!-- 0..* How a resource is related to the compartment -->
  <code value="[code]"/><!-- 1..1 Name of resource type -->
  <param value="[string]"/><!-- 0..* Search Parameter Name, or chained parameters -->
  <documentation value="[string]"/><!-- 0..1 Additional documentation about the resource and compartment -->
```

```
 </resource>
</CompartmentDefinition>
```

JSON Template

```
{
  "resourceType" : "CompartmentDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Canonical identifier for this compartment definition, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the compartment definition
  "name" : "<string>", // C? R!  Name for this compartment definition (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the compartment definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "purpose" : "<markdown>", // Why this compartment definition is defined
  "code" : "<code>", // R!  Patient | Encounter | RelatedPerson | Practitioner | Device
  "search" : <boolean>, // R!  Whether the search syntax is supported
  "resource" : [{ // How a resource is related to the compartment
    "code" : "<code>", // R!  Name of resource type
    "param" : ["<string>"], // Search Parameter Name, or chained parameters
    "documentation" : "<string>" // Additional documentation about the resource and compartment
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CompartmentDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CompartmentDefinition.url [ uri ]; # 1..1 Canonical identifier for this compartment definition, represented as a
    URI (globally unique)
  fhir:CompartmentDefinition.version [ string ]; # 0..1 Business version of the compartment definition
  fhir:CompartmentDefinition.name [ string ]; # 1..1 Name for this compartment definition (computer friendly)
  fhir:CompartmentDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:CompartmentDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:CompartmentDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:CompartmentDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:CompartmentDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:CompartmentDefinition.description [ markdown ]; # 0..1 Natural language description of the compartment definition
  fhir:CompartmentDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to
    support
  fhir:CompartmentDefinition.purpose [ markdown ]; # 0..1 Why this compartment definition is defined
  fhir:CompartmentDefinition.code [ code ]; # 1..1 Patient | Encounter | RelatedPerson | Practitioner | Device
  fhir:CompartmentDefinition.search [ boolean ]; # 1..1 Whether the search syntax is supported
  fhir:CompartmentDefinition.resource [ # 0..* How a resource is related to the compartment
    fhir:CompartmentDefinition.resource.code [ code ]; # 1..1 Name of resource type
    fhir:CompartmentDefinition.resource.param [ string ], ... ; # 0..* Search Parameter Name, or chained parameters
    fhir:CompartmentDefinition.resource.documentation [ string ]; # 0..1 Additional documentation about the resource and
     compartment
  ], ...;
]
```

Changes since R3

CompartmentDefinition

CompartmentDefinition | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *

CompartmentDefinition.version | Added Element

CompartmentDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

CompartmentDefinition.experimental | No longer marked as Modifier

CompartmentDefinition.code | Change value set from http://hl7.org/fhir/ValueSet/compartment-type to http://hl7.org/fhir/ValueSet/compartment-type|4.0.1

CompartmentDefinition.resource.code | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1

CompartmentDefinition.title | deleted

CompartmentDefinition.jurisdiction | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. 1 fail round-trip testing and 5 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

CompartmentDefinition | ITU | | DomainResource | Compartment Definition for a resource + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | Σ | 1..1 | uri | Canonical identifier for this compartment definition, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the compartment definition
name | ΣI | 1..1 | string | Name for this compartment definition (computer friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the compartment definition
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
purpose | | 0..1 | markdown | Why this compartment definition is defined
code | Σ | 1..1 | code | Patient | Encounter | RelatedPerson | Practitioner | Device CompartmentType (Required)
search | Σ | 1..1 | boolean | Whether the search syntax is supported
resource | Σ | 0..* | BackboneElement | How a resource is related to the compartment
code | Σ | 1..1 | code | Name of resource type ResourceType (Required)
param | Σ | 0..* | string | Search Parameter Name, or chained parameters
documentation | | 0..1 | string | Additional documentation about the resource and compartment
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CompartmentDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Canonical identifier for this compartment definition, represented as a URI (globally
    unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the compartment definition -->
 <name value="[string]"/><!--  1..1 Name for this compartment definition (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the compartment definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <purpose value="[markdown]"/><!-- 0..1 Why this compartment definition is defined -->
 <code value="[code]"/><!-- 1..1 Patient | Encounter | RelatedPerson | Practitioner | Device -->
 <search value="[boolean]"/><!-- 1..1 Whether the search syntax is supported -->
 <resource>  <!-- 0..* How a resource is related to the compartment -->
  <code value="[code]"/><!-- 1..1 Name of resource type -->
  <param value="[string]"/><!-- 0..* Search Parameter Name, or chained parameters -->
  <documentation value="[string]"/><!-- 0..1 Additional documentation about the resource and compartment -->
 </resource>
</CompartmentDefinition>
```

JSON Template

```
{
  "resourceType" : "CompartmentDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Canonical identifier for this compartment definition, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the compartment definition
  "name" : "<string>", // C? R!  Name for this compartment definition (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the compartment definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "purpose" : "<markdown>", // Why this compartment definition is defined
  "code" : "<code>", // R!  Patient | Encounter | RelatedPerson | Practitioner | Device
  "search" : <boolean>, // R!  Whether the search syntax is supported
  "resource" : [{ // How a resource is related to the compartment
    "code" : "<code>", // R!  Name of resource type
    "param" : ["<string>"], // Search Parameter Name, or chained parameters
    "documentation" : "<string>" // Additional documentation about the resource and compartment
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CompartmentDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CompartmentDefinition.url [ uri ]; # 1..1 Canonical identifier for this compartment definition, represented as a
    URI (globally unique)
  fhir:CompartmentDefinition.version [ string ]; # 0..1 Business version of the compartment definition
  fhir:CompartmentDefinition.name [ string ]; # 1..1 Name for this compartment definition (computer friendly)
  fhir:CompartmentDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
```

```
  fhir:CompartmentDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:CompartmentDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:CompartmentDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:CompartmentDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:CompartmentDefinition.description [ markdown ]; # 0..1 Natural language description of the compartment definition
  fhir:CompartmentDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to
    support
  fhir:CompartmentDefinition.purpose [ markdown ]; # 0..1 Why this compartment definition is defined
  fhir:CompartmentDefinition.code [ code ]; # 1..1 Patient | Encounter | RelatedPerson | Practitioner | Device
  fhir:CompartmentDefinition.search [ boolean ]; # 1..1 Whether the search syntax is supported
  fhir:CompartmentDefinition.resource [ # 0..* How a resource is related to the compartment
    fhir:CompartmentDefinition.resource.code [ code ]; # 1..1 Name of resource type
    fhir:CompartmentDefinition.resource.param [ string ], ... ; # 0..* Search Parameter Name, or chained parameters
    fhir:CompartmentDefinition.resource.documentation [ string ]; # 0..1 Additional documentation about the resource and
     compartment
  ], ...;
]
```

Changes since Release 3

CompartmentDefinition

CompartmentDefinition | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *

CompartmentDefinition.version | Added Element

CompartmentDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

CompartmentDefinition.experimental | No longer marked as Modifier

CompartmentDefinition.code | Change value set from http://hl7.org/fhir/ValueSet/compartment-type to http://hl7.org/fhir/ValueSet/compartment-type|4.0.1

CompartmentDefinition.resource.code | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1

CompartmentDefinition.title | deleted

CompartmentDefinition.jurisdiction | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. 1 fail round-trip testing and 5 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

5.6.3.1 Terminology Bindings

Path | Definition | Type | Reference

CompartmentDefinition.status | The lifecycle status of an artifact. | Required | PublicationStatus

CompartmentDefinition.code | Which type a compartment definition describes. | Required | CompartmentType

CompartmentDefinition.resource.code | One of the resource types defined as part of this version of FHIR. | Required | Resource Types

5.6.3.2 Constraints

id | Level | Location | Description | Expression

cpd-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

5.6.3.3 Using Compartments

As an example of compartment usage, to retrieve a list of a patient's conditions, use the URL:

```
GET [base]/Patient/[id]/Condition
```

Additional search parameters can be defined, such as this hypothetical search for acute conditions:

```
GET [base]/Patient/[id]/Condition?code:in=http://hspc.org/ValueSet/acute-concerns
```

Note that as searches, these are syntactic variations on these two search URLs respectively:

```
GET [base]/Condition?patient=[id]
GET [base]/Condition?patient=[id]&code:in=http://hspc.org/ValueSet/acute-concerns
```

The outcome of a compartment search is the same as the equivalent normal search. For example, both these searches return the same outcome if there is no patient 333:

```
GET [base]/Patient/333/Condition
GET [base]/Condition?patient=333
```

Whether the patient doesn't exist, or the user has no access to the patient, both these searches return an empty bundle with no matches. Some systems will include an operation outcome warning that there is no matching patient.

However, there is a key difference in functionality between compartment based searches and direct searches with parameters. Consider this search:

```
GET [base]/Patient/[id]/Communication
```

Because the definition of the patient compartment for Communication says that a Communication resource is in the patient compartment if the subject, sender, or recipient is the patient, the compartment search is

actually the same as the union of these 3 searches:

```
GET [base]/Communication?subject=[id]
GET [base]/Communication?sender=[id]
GET [base]/Communication?recipient=[id]
```

There is no way to do this as a single search, except by using the _filter:

```
GET [base]/Communication?_filter=subject re [id] or sender re [id] or recipient re [id]
```

Further details of searching by compartment are described under the search operation. As a search related operation, the assignment of resources to compartments is only based on the current version of any of the resources involved. Note that contained patient resources cannot create a patient compartment of their own.

Note that while this specification describes how to use the compartment syntax to find resources that are logically associated with the compartment, the compartment is not part of the identity of the resource. E.g. the response to the following is not defined by this specification:

```
GET [base]/Patient/[patient-id]/Condition/[resource-id]
```

The response for write operations (PUT/DELETE/PATCH) are also not defined by this specification. Nor is the response to a POST defined:

```
POST [base]/Patient/[patient-id]/Condition
```

There is no expectation for servers to support either read or write to such URLs.

5.6.3.4 Logical Meaning of Compartments

Compartments may be used explicitly, like this, but can also be used implicitly. For instance, if a FHIR server is providing a patient view of a record, the authorized user associated with use of the FHIR RESTful API may be limited to accessing records from the compartment instance(s) logically associated with their identity. Irrespective of whether compartments are being used explicitly or implicitly, servers will need to make arrangements to make some resources with no direct link to a patient available to the client (medications, substances, etc.).

Note that resources may cross between compartments, or interlink them. Examples of this would be where a Diagnostic Report identifies a subject, but an Observation it references identifies a different subject, or where a List resource references items that identify different subjects. Such cross-linking may arise for many valid reasons, including:

• Cases where subject records are inter-linked - Transplants, Perinatal care, family therapy etc.

• Workflow management where action lists link multiple patients and/or practitioners

Given the wide variety of use cases and contexts in which FHIR is used, compartments do not define how cross-linking is handled. Systems may reject resources, remove them from both compartments, or place them in both, or act in some other fashion.

The graph definition resource provides a method by which rules about cross-linking may be made and enforced.

It is at the discretion of the server whether to include resources in a compartment when the reference to the resource that establishes the compartment is in an extension.

Some resources are not in any compartment, e.g. Medication, Substance, Location. These resources are not linked directly to a patient or authored record, and are sometimes called 'master files'. Servers will need to make arrangements to make these resources available to the clients that are limited to particular compartments. For example, a Medication resource describes a medication itself and does not link to a patient; however, a resource such as MedicationAdministration connects the Medication (details of what was administered) to the patient (for whom was it administered), and so is required to interpret the administration.

5.6.4 Defining New Compartments

Compartments are defined and added to the list above when implementer communities identify them as common access points for data. As described below, compartments have both syntactical and logical consequences, and both these aspects of their functionality are evaluated when deciding whether to define compartments.

5.6.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code | token | Patient | Encounter | RelatedPerson | Practitioner | Device | CompartmentDefinition.code
context | token | A use context assigned to the compartment definition | (CompartmentDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the compartment definition | (CompartmentDefinition.useContext.value as Quantity) | (CompartmentDefinition.useContext.value as Range)
context-type | token | A type of use context assigned to the compartment definition | CompartmentDefinition.useContext.code

context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the compartment definition | On CompartmentDefinition.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the compartment definition | On CompartmentDefinition.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The compartment definition publication date | CompartmentDefinition.date
description | string | The description of the compartment definition | CompartmentDefinition.description
name | string | Computationally friendly name of the compartment definition | CompartmentDefinition.name
publisher | string | Name of the publisher of the compartment definition | CompartmentDefinition.publisher
resource | token | Name of resource type | CompartmentDefinition.resource.code
status | token | The current status of the compartment definition | CompartmentDefinition.status
url | uri | The uri that identifies the compartment definition | CompartmentDefinition.url
version | token | The business version of the compartment definition | CompartmentDefinition.version

5.6.6 Resource CompartmentDefinition - Examples

URL: https://hl7.org/fhir/R4/compartmentdefinition-examples.html

5.6.6 Resource CompartmentDefinition - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Full Patient Compartment | patient | XML | JSON | Turtle
Full Practitioner Compartment | practitioner | XML | JSON | Turtle
Full RelatedPerson Compartment | relatedperson | XML | JSON | Turtle
Full Device Compartment | device | XML | JSON | Turtle
Full Encounter Compartment | encounter | XML | JSON | Turtle
Example Compartment | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.6.7 Resource CompartmentDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/compartmentdefinition-definitions.html

5.6.7 Resource CompartmentDefinition - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the CompartmentDefinition resource.

CompartmentDefinition
Element Id | CompartmentDefinition
Definition | A compartment definition that defines how resources are accessed on a server.
Cardinality | 0..*
Type | DomainResource
Comments | In FHIR, search is not performed directly on a resource (by XML or JSON path), but on a named parameter that maps into the resource content.
Invariants | Defined on this element cpd-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
cpd-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
CompartmentDefinition.url
Element Id | CompartmentDefinition.url
Definition | An absolute URI that is used to identify this compartment definition when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which an authoritative instance of this compartment definition is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the compartment definition is stored on different servers.
Cardinality | 1..1
Type | uri
Requirements | Allows the compartment definition to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
CompartmentDefinition.version
Element Id | CompartmentDefinition.version
Definition | The identifier that is used to identify this version of the compartment definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the compartment definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.

Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different compartment definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the compartment definition with the format [url]|[version].
CompartmentDefinition.name
Element Id | CompartmentDefinition.name
Definition | A natural language name identifying the compartment definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 1..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.This is often the same as the code for the parameter, but does not need to be.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
CompartmentDefinition.status
Element Id | CompartmentDefinition.status
Definition | The status of this compartment definition. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of compartment definitions that are appropriate for use versus not.
CompartmentDefinition.experimental
Element Id | CompartmentDefinition.experimental
Definition | A Boolean value to indicate that this compartment definition is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level compartment definition.
Summary | true
Comments | Allows filtering of compartment definitions that are appropriate for use versus not.
CompartmentDefinition.date
Element Id | CompartmentDefinition.date
Definition | The date (and optionally time) when the compartment definition was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the compartment definition changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the compartment definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
CompartmentDefinition.publisher
Element Id | CompartmentDefinition.publisher
Definition | The name of the organization or individual that published the compartment definition.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the compartment definition. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the compartment definition is the organization or individual primarily responsible for the maintenance and upkeep of the compartment definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the compartment definition. This item SHOULD be populated unless the information is available from context.
CompartmentDefinition.contact
Element Id | CompartmentDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
CompartmentDefinition.description

Element Id | CompartmentDefinition.description
Definition | A free text natural language description of the compartment definition from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the compartment definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the compartment definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the compartment definition is presumed to be the predominant language in the place the compartment definition was created).
CompartmentDefinition.useContext
Element Id | CompartmentDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate compartment definition instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
CompartmentDefinition.purpose
Element Id | CompartmentDefinition.purpose
Definition | Explanation of why this compartment definition is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the compartment definition. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this compartment definition.
CompartmentDefinition.code
Element Id | CompartmentDefinition.code
Definition | Which compartment this definition describes.
Cardinality | 1..1
Terminology Binding | CompartmentType (Required)
Type | code
Summary | true
Comments | Only the specification can define the compartments that can exist. Servers can choose to support them.
CompartmentDefinition.search
Element Id | CompartmentDefinition.search
Definition | Whether the search syntax is supported,.
Cardinality | 1..1
Type | boolean
Summary | true
Comments | Servers may define and use compartments to manage logical access without implementing the compartment related syntax.
CompartmentDefinition.resource
Element Id | CompartmentDefinition.resource
Definition | Information about how a resource is related to the compartment.
Cardinality | 0..*
Summary | true
CompartmentDefinition.resource.code
Element Id | CompartmentDefinition.resource.code
Definition | The name of a resource supported by the server.
Cardinality | 1..1
Terminology Binding | Any defined Resource Type name
Type | code
Summary | true
CompartmentDefinition.resource.param
Element Id | CompartmentDefinition.resource.param
Definition | The name of a search parameter that represents the link to the compartment. More than one may be listed because a resource may be linked to a compartment in more than one way,.
Cardinality | 0..*
Type | string
Summary | true
Comments | If no search parameters are listed, then the resource is not linked to the compartment.
CompartmentDefinition.resource.documentation
Element Id | CompartmentDefinition.resource.documentation
Definition | Additional documentation about the resource and compartment.
Cardinality | 0..1
Type | string

5.6.8 Resource CompartmentDefinition - Mappings

URL: https://hl7.org/fhir/R4/compartmentdefinition-mappings.html

5.6.8 Resource CompartmentDefinition - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the compartmentdefinition resource (see Mappings to Other Standards for further information & status).

5.6.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

CompartmentDefinition
url | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

5.6.8.2 Workflow Pattern (http://hl7.org/fhir/workflow)

CompartmentDefinition
url | Definition.url
version | Definition.version
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
purpose | Definition.purpose

5.6.8.3 HL7 v2 Mapping (http://hl7.org/v2)

CompartmentDefinition | n/a

5.6.8.4 RIM Mapping (http://hl7.org/v3)

CompartmentDefinition | n/a

5.6.9 Resource CompartmentDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/compartmentdefinition-profiles.html

5.6.9 Resource CompartmentDefinition - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the CompartmentDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

5.6.9.1 Profiles

No Profiles defined for this resource

5.6.9.2 Extensions

replaces | CompartmentDefinition | replaces | for General Extensions for use by FHIR Implementers
resource-approvalDate | CompartmentDefinition | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | CompartmentDefinition | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | CompartmentDefinition | lastReviewDate | for Resource HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions

event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

5.6.9.3 Search Extensions

No Search Extensions defined for this resource

5.6.10 Resource CompartmentDefinition - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/compartmentdefinition-version-maps.html

No validation errors - all conversions are clean

5.6.10 Resource CompartmentDefinition - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for CompartmentDefinition.

Functional status for this map: 6 tests that all execute ok. 1 fail round-trip testing and 5 r3 resources are invalid (0 errors). (see documentation)

5.6.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/CompartmentDefinition3to4" = "R3 to R4 Conversions for CompartmentDefinition"

uses "http://hl7.org/fhir/3.0/StructureDefinition/CompartmentDefinition" alias CompartmentDefinitionR3 as source
uses "http://hl7.org/fhir/StructureDefinition/CompartmentDefinition" alias CompartmentDefinition as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group CompartmentDefinition(source src : CompartmentDefinitionR3, target tgt : CompartmentDefinition) extends
    DomainResource <<type+>> {
  src.url -> tgt.url;
  src.name -> tgt.name;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.purpose -> tgt.purpose;
  src.useContext -> tgt.useContext;
  src.code -> tgt.code;
  src.search -> tgt.search;
  src.resource as vs0 -> tgt.resource as vt0 then {
    vs0.code -> vt0.code;
    vs0.param -> vt0.param;
    vs0.documentation -> vt0.documentation;
  };
}
```

5.6.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/CompartmentDefinition4to3" = "R4 to R3 Conversion for CompartmentDefinition"

uses "http://hl7.org/fhir/StructureDefinition/CompartmentDefinition" alias CompartmentDefinition as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/CompartmentDefinition" alias CompartmentDefinitionR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group CompartmentDefinition(source src : CompartmentDefinition, target tgt : CompartmentDefinitionR3) extends
    DomainResource <<type+>> {
  src.url -> tgt.url;
  src.name -> tgt.name;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.purpose -> tgt.purpose;
  src.useContext -> tgt.useContext;
  src.code -> tgt.code;
  src.search -> tgt.search;
  src.resource as vs0 -> tgt.resource as vt0 then {
    vs0.code -> vt0.code;
    vs0.param -> vt0.param;
    vs0.documentation -> vt0.documentation;
  };
}
```

5.6.10.3 R4 Validation Errors

5.6.11 JSON Schema for CompartmentDefinition

URL: https://hl7.org/fhir/R4/compartmentdefinition.schema.json.html

5.6.11 JSON Schema for CompartmentDefinition

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for CompartmentDefinition, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/CompartmentDefinition",
  "$ref": "#/definitions/CompartmentDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "CompartmentDefinition": {
      "description": "A compartment definition that defines how resources are accessed on a server.",
      "properties": {
        "resourceType": {
          "description": "This is a CompartmentDefinition resource",
          "const": "CompartmentDefinition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
```

```
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this compartment definition when it is referenced in
a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which at which an authoritative instance of this compartment definition is (or
will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the
compartment definition is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the compartment definition when it is
referenced in a specification, model, design or instance. This is an arbitrary value managed by the compartment
definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if
 a managed version is not available. There is also no expectation that versions can be placed in a lexicographical
sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the compartment definition. This name should be usable as
an identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this compartment definition. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"

      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this compartment definition is authored for testing purposes
(or education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the compartment definition was published. The date must
change when the business version changes and it must change if the status code changes. In addition, it should
change when the substantive content of the compartment definition changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the compartment definition.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the compartment definition from a consumer\u0027s
perspective.",
```

```
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate compartment definition instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this compartment definition is needed and why it has been designed as it has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "Which compartment this definition describes.",
      "enum": [
        "Patient",
        "Encounter",
        "RelatedPerson",
        "Practitioner",
        "Device"
      ]
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "search": {
      "description": "Whether the search syntax is supported,.",
      "$ref": "#/definitions/boolean"
    },
    "_search": {
      "description": "Extensions for search",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "resource": {
      "description": "Information about how a resource is related to the compartment.",
      "items": {
        "$ref": "#/definitions/CompartmentDefinition_Resource"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"CompartmentDefinition_Resource": {
  "description": "A compartment definition that defines how resources are accessed on a server.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
```

```
          },
          "type": "array"
        },
        "code": {
          "description": "The name of a resource supported by the server.",
          "$ref": "#/definitions/code"
        },
        "_code": {
          "description": "Extensions for code",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "param": {
          "description": "The name of a search parameter that represents the link to the compartment. More than one may
    be listed because a resource may be linked to a compartment in more than one way,.",
          "items": {
            "$ref": "#/definitions/string"
          },
          "type": "array"
        },
        "_param": {
          "description": "Extensions for param",
          "items": {
            "$ref": "Element.schema.json#/definitions/Element"
          },
          "type": "array"
        },
        "documentation": {
          "description": "Additional documentation about the resource and compartment.",
          "$ref": "#/definitions/string"
        },
        "_documentation": {
          "description": "Extensions for documentation",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```


5.6.12 Resource CompartmentDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/compartmentdefinition-dependencies.html

5.6.12 Resource CompartmentDefinition - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for CompartmentDefinition FMM level 1

Name | Card. | Type | Dependency Analysis
CompartmentDefinition | | DomainResource
url | 1..1 | uri
version | 0..1 | string
name | 1..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext | 0..* | UsageContext
purpose | 0..1 | markdown
code | 1..1 | code
search | 1..1 | boolean
resource | 0..* | BackboneElement
code | 1..1 | code
param | 0..* | string
documentation | 0..1 | string

5.8 Resource ImplementationGuide - Content

URL: https://hl7.org/fhir/R4/implementationguide.html
FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

A set of rules of how a particular interoperability or standards problem is solved - typically through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a logical whole and to publish a computable definition of all the parts.

5.8.1 Scope and Usage

An implementation guide (IG) is a set of rules about how FHIR resources are used (or should be used) to solve a particular problem, with associated documentation to support and clarify the usage. Classically, FHIR implementation guides are published on the web after they are generated using the FHIR Implementation Guide Publisher .

The ImplementationGuide resource is a single resource that defines the logical content of the IG, along with the important entry pages into the publication, so that the logical package that the IG represents, so that the contents are computable.

In particular, validators are able to use the ImplementationGuide resource to validate content against the implementation guide as a whole. The significant conformance expectation introduced by the ImplementationGuide resource is the idea of Default Profiles. Implementations may conform to multiple implementation guides at once, but this requires that the implementation guides are compatible (see below).

5.8.2 Boundaries and Relationships

Implementation Guides contain two different types of resource references:

• Contents: A set of logical statements which implementations must conform to. These are almost always conformance resources

• Examples: Examples that illustrate the intent of the profiles defined in the implementation guide. These can be any kind of resource

An application's Capability Statement may identify one or more implementation guides that an application conforms to.

This resource is referenced by ElementDefinition, CapabilityStatement and itself

5.8.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
ImplementationGuide | ITU | | DomainResource | A set of rules about how FHIR is used + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: If a resource has a fhirVersion, it must be oe of the versions defined for the Implementation Guide Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 1..1 | uri | Canonical identifier for this implementation guide, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the implementation guide
name | ΣI | 1..1 | string | Name for this implementation guide (computer friendly)
title | Σ | 0..1 | string | Name for this implementation guide (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the implementation guide
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for implementation guide (if applicable) Jurisdiction (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
packageId | Σ | 1..1 | id | NPM Package name for IG
license | Σ | 0..1 | code | SPDX license code for this IG (or not-open-source) SPDXLicense (Required)
fhirVersion | Σ | 1..* | code | FHIR Version(s) this Implementation Guide targets FHIRVersion (Required)
dependsOn | Σ | 0..* | BackboneElement | Another Implementation guide this depends on
uri | Σ | 1..1 | canonical(ImplementationGuide) | Identity of the IG that this depends on
packageId | Σ | 0..1 | id | NPM Package name for IG this depends on
version | Σ | 0..1 | string | Version of the IG
global | Σ | 0..* | BackboneElement | Profiles that apply globally
type | Σ | 1..1 | code | Type this profile applies to ResourceType (Required)
profile | Σ | 1..1 | canonical(StructureDefinition) | Profile that all resources must conform to
definition | I | 0..1 | BackboneElement | Information needed to build the IG + Rule: If a resource has a groupingId, it must refer to a grouping defined in the Implementation Guide
grouping | | 0..* | BackboneElement | Grouping used to present related resources in the IG
name | | 1..1 | string | Descriptive name for the package

description | | 0..1 | string | Human readable text describing the package
resource | | 1..* | BackboneElement | Resource in the implementation guide
reference | | 1..1 | Reference(Any) | Location of the resource
fhirVersion | | 0..* | code | Versions this applies to (if different to IG) FHIRVersion (Required)
name | | 0..1 | string | Human Name for the resource
description | | 0..1 | string | Reason why included in guide
example[x] | | 0..1 | | Is an example/What is this an example of?
exampleBoolean | | | boolean
exampleCanonical | | | canonical(StructureDefinition)
groupingId | | 0..1 | id | Grouping this is part of
page | | 0..1 | BackboneElement | Page/Section in the Guide
name[x] | | 1..1 | | Where to find that page
nameUrl | | | url
nameReference | | | Reference(Binary)
title | | 1..1 | string | Short title shown for navigational assistance
generation | | 1..1 | code | html | markdown | xml | generated GuidePageGeneration (Required)
page | | 0..* | see page | Nested Pages / Sections
parameter | | 0..* | BackboneElement | Defines how IG is built by tools
code | | 1..1 | code | apply | path-resource | path-pages | path-tx-cache | expansion-parameter | rule-broken-links | generate-xml | generate-json | generate-turtle | html-template GuideParameterCode (Required)
value | | 1..1 | string | Value for named type
template | | 0..* | BackboneElement | A template for building resources
code | | 1..1 | code | Type of template specified
source | | 1..1 | string | The source location for the template
scope | | 0..1 | string | The scope in which the template applies
manifest | | 0..1 | BackboneElement | Information about an assembled IG
rendering | Σ | 0..1 | url | Location of rendered implementation guide
resource | Σ | 1..* | BackboneElement | Resource in the implementation guide
reference | Σ | 1..1 | Reference(Any) | Location of the resource
example[x] | | 0..1 | | Is an example/What is this an example of?
exampleBoolean | | | boolean
exampleCanonical | | | canonical(StructureDefinition)
relativePath | | 0..1 | url | Relative path for page in IG
page | | 0..* | BackboneElement | HTML page within the parent IG
name | | 1..1 | string | HTML page name
title | | 0..1 | string | Title of the page, for references
anchor | | 0..* | string | Anchor available on the page
image | | 0..* | string | Image within the IG
other | | 0..* | string | Additional linkable file in IG
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImplementationGuide xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Canonical identifier for this implementation guide, represented as a URI (globally
    unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the implementation guide -->
 <name value="[string]"/><!--  1..1 Name for this implementation guide (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this implementation guide (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the implementation guide -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for implementation guide (if applicable)
    --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <packageId value="[id]"/><!-- 1..1 NPM Package name for IG -->
 <license value="[code]"/><!-- 0..1 SPDX license code for this IG (or not-open-source) -->
 <fhirVersion value="[code]"/><!-- 1..* FHIR Version(s) this Implementation Guide targets -->
 <dependsOn>  <!-- 0..* Another Implementation guide this depends on -->
  <uri><!-- 1..1 canonical(ImplementationGuide) Identity of the IG that this depends on --></uri>
  <packageId value="[id]"/><!-- 0..1 NPM Package name for IG this depends on -->
  <version value="[string]"/><!-- 0..1 Version of the IG -->
 </dependsOn>
 <global>  <!-- 0..* Profiles that apply globally -->
  <type value="[code]"/><!-- 1..1 Type this profile applies to -->
  <profile><!-- 1..1 canonical(StructureDefinition) Profile that all resources must conform to --></profile>
 </global>
 <definition>  <!-- 0..1 Information needed to build the IG -->
  <grouping>  <!-- 0..* Grouping used to present related resources in the IG -->
   <name value="[string]"/><!-- 1..1 Descriptive name for the package -->
   <description value="[string]"/><!-- 0..1 Human readable text describing the package -->
  </grouping>
  <resource>  <!-- 1..* Resource in the implementation guide -->
```

```
   <reference><!-- 1..1 Reference(Any) Location of the resource --></reference>
   <fhirVersion value="[code]"/><!-- 0..* Versions this applies to (if different to IG) -->
   <name value="[string]"/><!-- 0..1 Human Name for the resource -->
   <description value="[string]"/><!-- 0..1 Reason why included in guide -->
   <example[x]><!-- 0..1 boolean|canonical(StructureDefinition) Is an example/What is this an example of?
    --></example[x]>
   <groupingId value="[id]"/><!-- 0..1 Grouping this is part of -->
  </resource>
  <page>  <!-- 0..1 Page/Section in the Guide -->
   <name[x]><!-- 1..1 url|Reference(Binary) Where to find that page --></name[x]>
   <title value="[string]"/><!-- 1..1 Short title shown for navigational assistance -->
   <generation value="[code]"/><!-- 1..1 html | markdown | xml | generated -->
   <page><!-- 0..* Content as for ImplementationGuide.definition.page Nested Pages / Sections --></page>
  </page>
  <parameter>  <!-- 0..* Defines how IG is built by tools -->
   <code value="[code]"/><!-- 1..1 apply | path-resource | path-pages | path-tx-cache | expansion-parameter | rule-
    broken-links | generate-xml | generate-json | generate-turtle | html-template -->
   <value value="[string]"/><!-- 1..1 Value for named type -->
  </parameter>
  <template>  <!-- 0..* A template for building resources -->
   <code value="[code]"/><!-- 1..1 Type of template specified -->
   <source value="[string]"/><!-- 1..1 The source location for the template -->
   <scope value="[string]"/><!-- 0..1 The scope in which the template applies -->
  </template>
 </definition>
 <manifest>  <!-- 0..1 Information about an assembled IG -->
  <rendering value="[url]"/><!-- 0..1 Location of rendered implementation guide -->
  <resource>  <!-- 1..* Resource in the implementation guide -->
   <reference><!-- 1..1 Reference(Any) Location of the resource --></reference>
   <example[x]><!-- 0..1 boolean|canonical(StructureDefinition) Is an example/What is this an example of?
    --></example[x]>
   <relativePath value="[url]"/><!-- 0..1 Relative path for page in IG -->
  </resource>
  <page>  <!-- 0..* HTML page within the parent IG -->
   <name value="[string]"/><!-- 1..1 HTML page name -->
   <title value="[string]"/><!-- 0..1 Title of the page, for references -->
   <anchor value="[string]"/><!-- 0..* Anchor available on the page -->
  </page>
  <image value="[string]"/><!-- 0..* Image within the IG -->
  <other value="[string]"/><!-- 0..* Additional linkable file in IG -->
 </manifest>
</ImplementationGuide>
```

JSON Template

```
{
  "resourceType" : "ImplementationGuide",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Canonical identifier for this implementation guide, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the implementation guide
  "name" : "<string>", // C? R!  Name for this implementation guide (computer friendly)
  "title" : "<string>", // Name for this implementation guide (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the implementation guide
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for implementation guide (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "packageId" : "<id>", // R!  NPM Package name for IG
  "license" : "<code>", // SPDX license code for this IG (or not-open-source)
  "fhirVersion" : ["<code>"], // R!  FHIR Version(s) this Implementation Guide targets
  "dependsOn" : [{ // Another Implementation guide this depends on
    "uri" : { canonical(ImplementationGuide) }, // R!  Identity of the IG that this depends on
    "packageId" : "<id>", // NPM Package name for IG this depends on
    "version" : "<string>" // Version of the IG
  }],
  "global" : [{ // Profiles that apply globally
    "type" : "<code>", // R!  Type this profile applies to
    "profile" : { canonical(StructureDefinition) } // R!  Profile that all resources must conform to
  }],
  "definition" : { // Information needed to build the IG
    "grouping" : [{ // Grouping used to present related resources in the IG
      "name" : "<string>", // R!  Descriptive name for the package
      "description" : "<string>" // Human readable text describing the package
    }],
    "resource" : [{ // R!  Resource in the implementation guide
      "reference" : { Reference(Any) }, // R!  Location of the resource
      "fhirVersion" : ["<code>"], // Versions this applies to (if different to IG)
      "name" : "<string>", // Human Name for the resource
      "description" : "<string>", // Reason why included in guide
      // example[x]: Is an example/What is this an example of?. One of these 2:
      "exampleBoolean" : <boolean>,
      "exampleCanonical" : { canonical(StructureDefinition) },
      "groupingId" : "<id>" // Grouping this is part of
    }],
    "page" : { // Page/Section in the Guide
```

```
      // name[x]: Where to find that page. One of these 2:
      "nameUrl" : "<url>",
      "nameReference" : { Reference(Binary) },
      "title" : "<string>", // R!  Short title shown for navigational assistance
      "generation" : "<code>", // R!  html | markdown | xml | generated
      "page" : [{ Content as for ImplementationGuide.definition.page }] // Nested Pages / Sections
    },
    "parameter" : [{ // Defines how IG is built by tools
      "code" : "<code>", // R!  apply | path-resource | path-pages | path-tx-cache | expansion-parameter | rule-broken-
    links | generate-xml | generate-json | generate-turtle | html-template
      "value" : "<string>" // R!  Value for named type
    }],
    "template" : [{ // A template for building resources
      "code" : "<code>", // R!  Type of template specified
      "source" : "<string>", // R!  The source location for the template
      "scope" : "<string>" // The scope in which the template applies
    }]
  },
  "manifest" : { // Information about an assembled IG
    "rendering" : "<url>", // Location of rendered implementation guide
    "resource" : [{ // R!  Resource in the implementation guide
      "reference" : { Reference(Any) }, // R!  Location of the resource
      // example[x]: Is an example/What is this an example of?. One of these 2:
      "exampleBoolean" : <boolean>,
      "exampleCanonical" : { canonical(StructureDefinition) },
      "relativePath" : "<url>" // Relative path for page in IG
    }],
    "page" : [{ // HTML page within the parent IG
      "name" : "<string>", // R!  HTML page name
      "title" : "<string>", // Title of the page, for references
      "anchor" : ["<string>"] // Anchor available on the page
    }],
    "image" : ["<string>"], // Image within the IG
    "other" : ["<string>"] // Additional linkable file in IG
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImplementationGuide;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImplementationGuide.url [ uri ]; # 1..1 Canonical identifier for this implementation guide, represented as a URI
    (globally unique)
  fhir:ImplementationGuide.version [ string ]; # 0..1 Business version of the implementation guide
  fhir:ImplementationGuide.name [ string ]; # 1..1 Name for this implementation guide (computer friendly)
  fhir:ImplementationGuide.title [ string ]; # 0..1 Name for this implementation guide (human friendly)
  fhir:ImplementationGuide.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ImplementationGuide.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ImplementationGuide.date [ dateTime ]; # 0..1 Date last changed
  fhir:ImplementationGuide.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ImplementationGuide.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ImplementationGuide.description [ markdown ]; # 0..1 Natural language description of the implementation guide
  fhir:ImplementationGuide.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:ImplementationGuide.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for implementation guide
     (if applicable)
  fhir:ImplementationGuide.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ImplementationGuide.packageId [ id ]; # 1..1 NPM Package name for IG
  fhir:ImplementationGuide.license [ code ]; # 0..1 SPDX license code for this IG (or not-open-source)
  fhir:ImplementationGuide.fhirVersion [ code ], ... ; # 1..* FHIR Version(s) this Implementation Guide targets
  fhir:ImplementationGuide.dependsOn [ # 0..* Another Implementation guide this depends on
    fhir:ImplementationGuide.dependsOn.uri [ canonical(ImplementationGuide) ]; # 1..1 Identity of the IG that this
    depends on
    fhir:ImplementationGuide.dependsOn.packageId [ id ]; # 0..1 NPM Package name for IG this depends on
    fhir:ImplementationGuide.dependsOn.version [ string ]; # 0..1 Version of the IG
  ], ...;
  fhir:ImplementationGuide.global [ # 0..* Profiles that apply globally
    fhir:ImplementationGuide.global.type [ code ]; # 1..1 Type this profile applies to
    fhir:ImplementationGuide.global.profile [ canonical(StructureDefinition) ]; # 1..1 Profile that all resources must
    conform to
  ], ...;
  fhir:ImplementationGuide.definition [ # 0..1 Information needed to build the IG
    fhir:ImplementationGuide.definition.grouping [ # 0..* Grouping used to present related resources in the IG
      fhir:ImplementationGuide.definition.grouping.name [ string ]; # 1..1 Descriptive name for the package
      fhir:ImplementationGuide.definition.grouping.description [ string ]; # 0..1 Human readable text describing the
    package
    ], ...;
    fhir:ImplementationGuide.definition.resource [ # 1..* Resource in the implementation guide
      fhir:ImplementationGuide.definition.resource.reference [ Reference(Any) ]; # 1..1 Location of the resource
      fhir:ImplementationGuide.definition.resource.fhirVersion [ code ], ... ; # 0..* Versions this applies to (if
    different to IG)
      fhir:ImplementationGuide.definition.resource.name [ string ]; # 0..1 Human Name for the resource
      fhir:ImplementationGuide.definition.resource.description [ string ]; # 0..1 Reason why included in guide
      # ImplementationGuide.definition.resource.example[x] : 0..1 Is an example/What is this an example of?. One of
    these 2
```

```
        fhir:ImplementationGuide.definition.resource.exampleBoolean [ boolean ]
        fhir:ImplementationGuide.definition.resource.exampleCanonical [ canonical(StructureDefinition) ]
      fhir:ImplementationGuide.definition.resource.groupingId [ id ]; # 0..1 Grouping this is part of
    ], ...;
    fhir:ImplementationGuide.definition.page [ # 0..1 Page/Section in the Guide
      # ImplementationGuide.definition.page.name[x] : 1..1 Where to find that page. One of these 2
        fhir:ImplementationGuide.definition.page.nameUrl [ url ]
        fhir:ImplementationGuide.definition.page.nameReference [ Reference(Binary) ]
      fhir:ImplementationGuide.definition.page.title [ string ]; # 1..1 Short title shown for navigational assistance
      fhir:ImplementationGuide.definition.page.generation [ code ]; # 1..1 html | markdown | xml | generated
      fhir:ImplementationGuide.definition.page.page [ See ImplementationGuide.definition.page ], ... ; # 0..* Nested
    Pages / Sections
    ];
    fhir:ImplementationGuide.definition.parameter [ # 0..* Defines how IG is built by tools
      fhir:ImplementationGuide.definition.parameter.code [ code ]; # 1..1 apply | path-resource | path-pages | path-tx-
    cache | expansion-parameter | rule-broken-links | generate-xml | generate-json | generate-turtle | html-template
      fhir:ImplementationGuide.definition.parameter.value [ string ]; # 1..1 Value for named type
    ], ...;
    fhir:ImplementationGuide.definition.template [ # 0..* A template for building resources
      fhir:ImplementationGuide.definition.template.code [ code ]; # 1..1 Type of template specified
      fhir:ImplementationGuide.definition.template.source [ string ]; # 1..1 The source location for the template
      fhir:ImplementationGuide.definition.template.scope [ string ]; # 0..1 The scope in which the template applies
    ], ...;
  ];
  fhir:ImplementationGuide.manifest [ # 0..1 Information about an assembled IG
    fhir:ImplementationGuide.manifest.rendering [ url ]; # 0..1 Location of rendered implementation guide
    fhir:ImplementationGuide.manifest.resource [ # 1..* Resource in the implementation guide
      fhir:ImplementationGuide.manifest.resource.reference [ Reference(Any) ]; # 1..1 Location of the resource
      # ImplementationGuide.manifest.resource.example[x] : 0..1 Is an example/What is this an example of?. One of these
    2
        fhir:ImplementationGuide.manifest.resource.exampleBoolean [ boolean ]
        fhir:ImplementationGuide.manifest.resource.exampleCanonical [ canonical(StructureDefinition) ]
      fhir:ImplementationGuide.manifest.resource.relativePath [ url ]; # 0..1 Relative path for page in IG
    ], ...;
    fhir:ImplementationGuide.manifest.page [ # 0..* HTML page within the parent IG
      fhir:ImplementationGuide.manifest.page.name [ string ]; # 1..1 HTML page name
      fhir:ImplementationGuide.manifest.page.title [ string ]; # 0..1 Title of the page, for references
      fhir:ImplementationGuide.manifest.page.anchor [ string ], ... ; # 0..* Anchor available on the page
    ], ...;
    fhir:ImplementationGuide.manifest.image [ string ], ... ; # 0..* Image within the IG
    fhir:ImplementationGuide.manifest.other [ string ], ... ; # 0..* Additional linkable file in IG
  ];
]
```

Changes since R3

ImplementationGuide

ImplementationGuide | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *

ImplementationGuide.title | Added Element

ImplementationGuide.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

ImplementationGuide.experimental | No longer marked as Modifier

ImplementationGuide.packageId | Added Mandatory Element

ImplementationGuide.license | Added Element

ImplementationGuide.fhirVersion | Min Cardinality changed from 0 to 1 Max Cardinality changed from 1 to * Type changed from id to code Add Binding http://hl7.org/fhir/ValueSet/FHIR-version|4.0.1 (required)

ImplementationGuide.dependsOn | Renamed from dependency to dependsOn

ImplementationGuide.dependsOn.uri | Moved from ImplementationGuide.dependency to ImplementationGuide.dependsOn Type changed from uri to canonical(ImplementationGuide)

ImplementationGuide.dependsOn.packageId | Added Element

ImplementationGuide.dependsOn.version | Added Element

ImplementationGuide.global.type | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1

ImplementationGuide.global.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)

ImplementationGuide.definition | Added Element

ImplementationGuide.definition.grouping | Added Element

ImplementationGuide.definition.grouping.name | Added Mandatory Element

ImplementationGuide.definition.grouping.description | Added Element

ImplementationGuide.definition.resource | Moved from ImplementationGuide.package to ImplementationGuide.definition

ImplementationGuide.definition.resource.reference | Added Mandatory Element

ImplementationGuide.definition.resource.fhirVersion | Added Element

ImplementationGuide.definition.resource.example[x] | Moved from ImplementationGuide.package.resource.example to example[x] Min Cardinality changed from 1 to 0 Add Type canonical(StructureDefinition)

ImplementationGuide.definition.resource.groupingId | Added Element

ImplementationGuide.definition.page | Moved from ImplementationGuide to ImplementationGuide.definition

ImplementationGuide.definition.page.name[x] | Added Mandatory Element

ImplementationGuide.definition.page.generation | Added Mandatory Element

ImplementationGuide.definition.parameter | Added Element

ImplementationGuide.definition.parameter.code | Added Mandatory Element

ImplementationGuide.definition.parameter.value | Added Mandatory Element

ImplementationGuide.definition.template | Added Element

ImplementationGuide.definition.template.code | Added Mandatory Element

ImplementationGuide.definition.template.source | Added Mandatory Element
ImplementationGuide.definition.template.scope | Added Element
ImplementationGuide.manifest | Added Element
ImplementationGuide.manifest.rendering | Added Element
ImplementationGuide.manifest.resource | Added Mandatory Element
ImplementationGuide.manifest.resource.reference | Added Mandatory Element
ImplementationGuide.manifest.resource.example[x] | Added Element
ImplementationGuide.manifest.resource.relativePath | Added Element
ImplementationGuide.manifest.page | Added Element
ImplementationGuide.manifest.page.name | Added Mandatory Element
ImplementationGuide.manifest.page.title | Added Element
ImplementationGuide.manifest.page.anchor | Added Element
ImplementationGuide.manifest.image | Added Element
ImplementationGuide.manifest.other | Added Element
ImplementationGuide.dependency.type | deleted
ImplementationGuide.package | deleted
ImplementationGuide.binary | deleted
ImplementationGuide.page.source | deleted
ImplementationGuide.page.kind | deleted
ImplementationGuide.page.type | deleted
ImplementationGuide.page.package | deleted
ImplementationGuide.page.format | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
ImplementationGuide | ITU | | DomainResource | A set of rules about how FHIR is used + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: If a resource has a fhirVersion, it must be oe of the versions defined for the Implementation Guide Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 1..1 | uri | Canonical identifier for this implementation guide, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the implementation guide
name | ΣI | 1..1 | string | Name for this implementation guide (computer friendly)
title | Σ | 0..1 | string | Name for this implementation guide (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the implementation guide
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for implementation guide (if applicable) Jurisdiction (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
packageId | Σ | 1..1 | id | NPM Package name for IG
license | Σ | 0..1 | code | SPDX license code for this IG (or not-open-source) SPDXLicense (Required)
fhirVersion | Σ | 1..* | code | FHIR Version(s) this Implementation Guide targets FHIRVersion (Required)
dependsOn | Σ | 0..* | BackboneElement | Another Implementation guide this depends on
uri | Σ | 1..1 | canonical(ImplementationGuide) | Identity of the IG that this depends on
packageId | Σ | 0..1 | id | NPM Package name for IG this depends on
version | Σ | 0..1 | string | Version of the IG
global | Σ | 0..* | BackboneElement | Profiles that apply globally
type | Σ | 1..1 | code | Type this profile applies to ResourceType (Required)
profile | Σ | 1..1 | canonical(StructureDefinition) | Profile that all resources must conform to
definition | I | 0..1 | BackboneElement | Information needed to build the IG + Rule: If a resource has a groupingId, it must refer to a grouping defined in the Implementation Guide
grouping | | 0..* | BackboneElement | Grouping used to present related resources in the IG
name | | 1..1 | string | Descriptive name for the package
description | | 0..1 | string | Human readable text describing the package
resource | | 1..* | BackboneElement | Resource in the implementation guide
reference | | 1..1 | Reference(Any) | Location of the resource
fhirVersion | | 0..* | code | Versions this applies to (if different to IG) FHIRVersion (Required)
name | | 0..1 | string | Human Name for the resource
description | | 0..1 | string | Reason why included in guide
example[x] | | 0..1 | | Is an example/What is this an example of?
exampleBoolean | | | boolean
exampleCanonical | | | canonical(StructureDefinition)
groupingId | | 0..1 | id | Grouping this is part of
page | | 0..1 | BackboneElement | Page/Section in the Guide
name[x] | | 1..1 | | Where to find that page

nameUrl | | | url
nameReference | | | Reference(Binary)
title | | 1..1 | string | Short title shown for navigational assistance
generation | | 1..1 | code | html | markdown | xml | generated GuidePageGeneration (Required)
page | | 0..* | see page | Nested Pages / Sections
parameter | | 0..* | BackboneElement | Defines how IG is built by tools
code | | 1..1 | code | apply | path-resource | path-pages | path-tx-cache | expansion-parameter | rule-broken-links | generate-xml | generate-json | generate-turtle | html-template GuideParameterCode (Required)
value | | 1..1 | string | Value for named type
template | | 0..* | BackboneElement | A template for building resources
code | | 1..1 | code | Type of template specified
source | | 1..1 | string | The source location for the template
scope | | 0..1 | string | The scope in which the template applies
manifest | | 0..1 | BackboneElement | Information about an assembled IG
rendering | Σ | 0..1 | url | Location of rendered implementation guide
resource | Σ | 1..* | BackboneElement | Resource in the implementation guide
reference | Σ | 1..1 | Reference(Any) | Location of the resource
example[x] | | 0..1 | | Is an example/What is this an example of?
exampleBoolean | | | boolean
exampleCanonical | | | canonical(StructureDefinition)
relativePath | | 0..1 | url | Relative path for page in IG
page | | 0..* | BackboneElement | HTML page within the parent IG
name | | 1..1 | string | HTML page name
title | | 0..1 | string | Title of the page, for references
anchor | | 0..* | string | Anchor available on the page
image | | 0..* | string | Image within the IG
other | | 0..* | string | Additional linkable file in IG
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImplementationGuide xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Canonical identifier for this implementation guide, represented as a URI (globally
    unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the implementation guide -->
 <name value="[string]"/><!--  1..1 Name for this implementation guide (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this implementation guide (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the implementation guide -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for implementation guide (if applicable)
    --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <packageId value="[id]"/><!-- 1..1 NPM Package name for IG -->
 <license value="[code]"/><!-- 0..1 SPDX license code for this IG (or not-open-source) -->
 <fhirVersion value="[code]"/><!-- 1..* FHIR Version(s) this Implementation Guide targets -->
 <dependsOn>  <!-- 0..* Another Implementation guide this depends on -->
  <uri><!-- 1..1 canonical(ImplementationGuide) Identity of the IG that this depends on --></uri>
  <packageId value="[id]"/><!-- 0..1 NPM Package name for IG this depends on -->
  <version value="[string]"/><!-- 0..1 Version of the IG -->
 </dependsOn>
 <global>  <!-- 0..* Profiles that apply globally -->
  <type value="[code]"/><!-- 1..1 Type this profile applies to -->
  <profile><!-- 1..1 canonical(StructureDefinition) Profile that all resources must conform to --></profile>
 </global>
 <definition>  <!-- 0..1 Information needed to build the IG -->
  <grouping>  <!-- 0..* Grouping used to present related resources in the IG -->
   <name value="[string]"/><!-- 1..1 Descriptive name for the package -->
   <description value="[string]"/><!-- 0..1 Human readable text describing the package -->
  </grouping>
  <resource>  <!-- 1..* Resource in the implementation guide -->
   <reference><!-- 1..1 Reference(Any) Location of the resource --></reference>
   <fhirVersion value="[code]"/><!-- 0..* Versions this applies to (if different to IG) -->
   <name value="[string]"/><!-- 0..1 Human Name for the resource -->
   <description value="[string]"/><!-- 0..1 Reason why included in guide -->
   <example[x]><!-- 0..1 boolean|canonical(StructureDefinition) Is an example/What is this an example of?
    --></example[x]>
   <groupingId value="[id]"/><!-- 0..1 Grouping this is part of -->
  </resource>
  <page>  <!-- 0..1 Page/Section in the Guide -->
   <name[x]><!-- 1..1 url|Reference(Binary) Where to find that page --></name[x]>
   <title value="[string]"/><!-- 1..1 Short title shown for navigational assistance -->
   <generation value="[code]"/><!-- 1..1 html | markdown | xml | generated -->
   <page><!-- 0..* Content as for ImplementationGuide.definition.page Nested Pages / Sections --></page>
  </page>
  <parameter>  <!-- 0..* Defines how IG is built by tools -->
```

```
   <code value="[code]"/><!-- 1..1 apply | path-resource | path-pages | path-tx-cache | expansion-parameter | rule-
    broken-links | generate-xml | generate-json | generate-turtle | html-template -->
   <value value="[string]"/><!-- 1..1 Value for named type -->
  </parameter>
  <template>  <!-- 0..* A template for building resources -->
   <code value="[code]"/><!-- 1..1 Type of template specified -->
   <source value="[string]"/><!-- 1..1 The source location for the template -->
   <scope value="[string]"/><!-- 0..1 The scope in which the template applies -->
  </template>
 </definition>
 <manifest>  <!-- 0..1 Information about an assembled IG -->
  <rendering value="[url]"/><!-- 0..1 Location of rendered implementation guide -->
  <resource>  <!-- 1..* Resource in the implementation guide -->
   <reference><!-- 1..1 Reference(Any) Location of the resource --></reference>
   <example[x]><!-- 0..1 boolean|canonical(StructureDefinition) Is an example/What is this an example of?
    --></example[x]>
   <relativePath value="[url]"/><!-- 0..1 Relative path for page in IG -->
  </resource>
  <page>  <!-- 0..* HTML page within the parent IG -->
   <name value="[string]"/><!-- 1..1 HTML page name -->
   <title value="[string]"/><!-- 0..1 Title of the page, for references -->
   <anchor value="[string]"/><!-- 0..* Anchor available on the page -->
  </page>
  <image value="[string]"/><!-- 0..* Image within the IG -->
  <other value="[string]"/><!-- 0..* Additional linkable file in IG -->
 </manifest>
</ImplementationGuide>
```

JSON Template

```
{
  "resourceType" : "ImplementationGuide",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Canonical identifier for this implementation guide, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the implementation guide
  "name" : "<string>", // C? R!  Name for this implementation guide (computer friendly)
  "title" : "<string>", // Name for this implementation guide (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the implementation guide
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for implementation guide (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "packageId" : "<id>", // R!  NPM Package name for IG
  "license" : "<code>", // SPDX license code for this IG (or not-open-source)
  "fhirVersion" : ["<code>"], // R!  FHIR Version(s) this Implementation Guide targets
  "dependsOn" : [{ // Another Implementation guide this depends on
    "uri" : { canonical(ImplementationGuide) }, // R!  Identity of the IG that this depends on
    "packageId" : "<id>", // NPM Package name for IG this depends on
    "version" : "<string>" // Version of the IG
  }],
  "global" : [{ // Profiles that apply globally
    "type" : "<code>", // R!  Type this profile applies to
    "profile" : { canonical(StructureDefinition) } // R!  Profile that all resources must conform to
  }],
  "definition" : { // Information needed to build the IG
    "grouping" : [{ // Grouping used to present related resources in the IG
      "name" : "<string>", // R!  Descriptive name for the package
      "description" : "<string>" // Human readable text describing the package
    }],
    "resource" : [{ // R!  Resource in the implementation guide
      "reference" : { Reference(Any) }, // R!  Location of the resource
      "fhirVersion" : ["<code>"], // Versions this applies to (if different to IG)
      "name" : "<string>", // Human Name for the resource
      "description" : "<string>", // Reason why included in guide
      // example[x]: Is an example/What is this an example of?. One of these 2:
      "exampleBoolean" : <boolean>,
      "exampleCanonical" : { canonical(StructureDefinition) },
      "groupingId" : "<id>" // Grouping this is part of
    }],
    "page" : { // Page/Section in the Guide
      // name[x]: Where to find that page. One of these 2:
      "nameUrl" : "<url>",
      "nameReference" : { Reference(Binary) },
      "title" : "<string>", // R!  Short title shown for navigational assistance
      "generation" : "<code>", // R!  html | markdown | xml | generated
      "page" : [{ Content as for ImplementationGuide.definition.page }] // Nested Pages / Sections
    },
    "parameter" : [{ // Defines how IG is built by tools
      "code" : "<code>", // R!  apply | path-resource | path-pages | path-tx-cache | expansion-parameter | rule-broken-
    links | generate-xml | generate-json | generate-turtle | html-template
      "value" : "<string>" // R!  Value for named type
    }],
    "template" : [{ // A template for building resources
      "code" : "<code>", // R!  Type of template specified
      "source" : "<string>", // R!  The source location for the template
```

```
      "scope" : "<string>" // The scope in which the template applies
    }]
  },
  "manifest" : { // Information about an assembled IG
    "rendering" : "<url>", // Location of rendered implementation guide
    "resource" : [{ // R!  Resource in the implementation guide
      "reference" : { Reference(Any) }, // R!  Location of the resource
      // example[x]: Is an example/What is this an example of?. One of these 2:
      "exampleBoolean" : <boolean>,
      "exampleCanonical" : { canonical(StructureDefinition) },
      "relativePath" : "<url>" // Relative path for page in IG
    }],
    "page" : [{ // HTML page within the parent IG
      "name" : "<string>", // R!  HTML page name
      "title" : "<string>", // Title of the page, for references
      "anchor" : ["<string>"] // Anchor available on the page
    }],
    "image" : ["<string>"], // Image within the IG
    "other" : ["<string>"] // Additional linkable file in IG
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImplementationGuide;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImplementationGuide.url [ uri ]; # 1..1 Canonical identifier for this implementation guide, represented as a URI
    (globally unique)
  fhir:ImplementationGuide.version [ string ]; # 0..1 Business version of the implementation guide
  fhir:ImplementationGuide.name [ string ]; # 1..1 Name for this implementation guide (computer friendly)
  fhir:ImplementationGuide.title [ string ]; # 0..1 Name for this implementation guide (human friendly)
  fhir:ImplementationGuide.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ImplementationGuide.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ImplementationGuide.date [ dateTime ]; # 0..1 Date last changed
  fhir:ImplementationGuide.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ImplementationGuide.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ImplementationGuide.description [ markdown ]; # 0..1 Natural language description of the implementation guide
  fhir:ImplementationGuide.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:ImplementationGuide.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for implementation guide
     (if applicable)
  fhir:ImplementationGuide.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ImplementationGuide.packageId [ id ]; # 1..1 NPM Package name for IG
  fhir:ImplementationGuide.license [ code ]; # 0..1 SPDX license code for this IG (or not-open-source)
  fhir:ImplementationGuide.fhirVersion [ code ], ... ; # 1..* FHIR Version(s) this Implementation Guide targets
  fhir:ImplementationGuide.dependsOn [ # 0..* Another Implementation guide this depends on
    fhir:ImplementationGuide.dependsOn.uri [ canonical(ImplementationGuide) ]; # 1..1 Identity of the IG that this
    depends on
    fhir:ImplementationGuide.dependsOn.packageId [ id ]; # 0..1 NPM Package name for IG this depends on
    fhir:ImplementationGuide.dependsOn.version [ string ]; # 0..1 Version of the IG
  ], ...;
  fhir:ImplementationGuide.global [ # 0..* Profiles that apply globally
    fhir:ImplementationGuide.global.type [ code ]; # 1..1 Type this profile applies to
    fhir:ImplementationGuide.global.profile [ canonical(StructureDefinition) ]; # 1..1 Profile that all resources must
    conform to
  ], ...;
  fhir:ImplementationGuide.definition [ # 0..1 Information needed to build the IG
    fhir:ImplementationGuide.definition.grouping [ # 0..* Grouping used to present related resources in the IG
      fhir:ImplementationGuide.definition.grouping.name [ string ]; # 1..1 Descriptive name for the package
      fhir:ImplementationGuide.definition.grouping.description [ string ]; # 0..1 Human readable text describing the
    package
    ], ...;
    fhir:ImplementationGuide.definition.resource [ # 1..* Resource in the implementation guide
      fhir:ImplementationGuide.definition.resource.reference [ Reference(Any) ]; # 1..1 Location of the resource
      fhir:ImplementationGuide.definition.resource.fhirVersion [ code ], ... ; # 0..* Versions this applies to (if
    different to IG)
      fhir:ImplementationGuide.definition.resource.name [ string ]; # 0..1 Human Name for the resource
      fhir:ImplementationGuide.definition.resource.description [ string ]; # 0..1 Reason why included in guide
      # ImplementationGuide.definition.resource.example[x] : 0..1 Is an example/What is this an example of?. One of
    these 2
        fhir:ImplementationGuide.definition.resource.exampleBoolean [ boolean ]
        fhir:ImplementationGuide.definition.resource.exampleCanonical [ canonical(StructureDefinition) ]
      fhir:ImplementationGuide.definition.resource.groupingId [ id ]; # 0..1 Grouping this is part of
    ], ...;
    fhir:ImplementationGuide.definition.page [ # 0..1 Page/Section in the Guide
      # ImplementationGuide.definition.page.name[x] : 1..1 Where to find that page. One of these 2
        fhir:ImplementationGuide.definition.page.nameUrl [ url ]
        fhir:ImplementationGuide.definition.page.nameReference [ Reference(Binary) ]
      fhir:ImplementationGuide.definition.page.title [ string ]; # 1..1 Short title shown for navigational assistance
      fhir:ImplementationGuide.definition.page.generation [ code ]; # 1..1 html | markdown | xml | generated
      fhir:ImplementationGuide.definition.page.page [ See ImplementationGuide.definition.page ], ... ; # 0..* Nested
    Pages / Sections
    ];
    fhir:ImplementationGuide.definition.parameter [ # 0..* Defines how IG is built by tools
      fhir:ImplementationGuide.definition.parameter.code [ code ]; # 1..1 apply | path-resource | path-pages | path-tx-
```

```
    cache | expansion-parameter | rule-broken-links | generate-xml | generate-json | generate-turtle | html-template
      fhir:ImplementationGuide.definition.parameter.value [ string ]; # 1..1 Value for named type
    ], ...;
    fhir:ImplementationGuide.definition.template [ # 0..* A template for building resources
      fhir:ImplementationGuide.definition.template.code [ code ]; # 1..1 Type of template specified
      fhir:ImplementationGuide.definition.template.source [ string ]; # 1..1 The source location for the template
      fhir:ImplementationGuide.definition.template.scope [ string ]; # 0..1 The scope in which the template applies
    ], ...;
  ];
  fhir:ImplementationGuide.manifest [ # 0..1 Information about an assembled IG
    fhir:ImplementationGuide.manifest.rendering [ url ]; # 0..1 Location of rendered implementation guide
    fhir:ImplementationGuide.manifest.resource [ # 1..* Resource in the implementation guide
      fhir:ImplementationGuide.manifest.resource.reference [ Reference(Any) ]; # 1..1 Location of the resource
      # ImplementationGuide.manifest.resource.example[x] : 0..1 Is an example/What is this an example of?. One of these
    2
        fhir:ImplementationGuide.manifest.resource.exampleBoolean [ boolean ]
        fhir:ImplementationGuide.manifest.resource.exampleCanonical [ canonical(StructureDefinition) ]
      fhir:ImplementationGuide.manifest.resource.relativePath [ url ]; # 0..1 Relative path for page in IG
    ], ...;
    fhir:ImplementationGuide.manifest.page [ # 0..* HTML page within the parent IG
      fhir:ImplementationGuide.manifest.page.name [ string ]; # 1..1 HTML page name
      fhir:ImplementationGuide.manifest.page.title [ string ]; # 0..1 Title of the page, for references
      fhir:ImplementationGuide.manifest.page.anchor [ string ], ... ; # 0..* Anchor available on the page
    ], ...;
    fhir:ImplementationGuide.manifest.image [ string ], ... ; # 0..* Image within the IG
    fhir:ImplementationGuide.manifest.other [ string ], ... ; # 0..* Additional linkable file in IG
  ];
]
```

Changes since Release 3

ImplementationGuide

ImplementationGuide | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *

ImplementationGuide.title | Added Element

ImplementationGuide.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

ImplementationGuide.experimental | No longer marked as Modifier

ImplementationGuide.packageId | Added Mandatory Element

ImplementationGuide.license | Added Element

ImplementationGuide.fhirVersion | Min Cardinality changed from 0 to 1 Max Cardinality changed from 1 to * Type changed from id to code Add Binding http://hl7.org/fhir/ValueSet/FHIR-version|4.0.1 (required)

ImplementationGuide.dependsOn | Renamed from dependency to dependsOn

ImplementationGuide.dependsOn.uri | Moved from ImplementationGuide.dependency to ImplementationGuide.dependsOn Type changed from uri to canonical(ImplementationGuide)

ImplementationGuide.dependsOn.packageId | Added Element

ImplementationGuide.dependsOn.version | Added Element

ImplementationGuide.global.type | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1

ImplementationGuide.global.profile | Type changed from Reference(StructureDefinition) to canonical(StructureDefinition)

ImplementationGuide.definition | Added Element

ImplementationGuide.definition.grouping | Added Element

ImplementationGuide.definition.grouping.name | Added Mandatory Element

ImplementationGuide.definition.grouping.description | Added Element

ImplementationGuide.definition.resource | Moved from ImplementationGuide.package to ImplementationGuide.definition

ImplementationGuide.definition.resource.reference | Added Mandatory Element

ImplementationGuide.definition.resource.fhirVersion | Added Element

ImplementationGuide.definition.resource.example[x] | Moved from ImplementationGuide.package.resource.example to example[x] Min Cardinality changed from 1 to 0 Add Type canonical(StructureDefinition)

ImplementationGuide.definition.resource.groupingId | Added Element

ImplementationGuide.definition.page | Moved from ImplementationGuide to ImplementationGuide.definition

ImplementationGuide.definition.page.name[x] | Added Mandatory Element

ImplementationGuide.definition.page.generation | Added Mandatory Element

ImplementationGuide.definition.parameter | Added Element

ImplementationGuide.definition.parameter.code | Added Mandatory Element

ImplementationGuide.definition.parameter.value | Added Mandatory Element

ImplementationGuide.definition.template | Added Element

ImplementationGuide.definition.template.code | Added Mandatory Element

ImplementationGuide.definition.template.source | Added Mandatory Element

ImplementationGuide.definition.template.scope | Added Element

ImplementationGuide.manifest | Added Element

ImplementationGuide.manifest.rendering | Added Element

ImplementationGuide.manifest.resource | Added Mandatory Element

ImplementationGuide.manifest.resource.reference | Added Mandatory Element

ImplementationGuide.manifest.resource.example[x] | Added Element

ImplementationGuide.manifest.resource.relativePath | Added Element

ImplementationGuide.manifest.page | Added Element

ImplementationGuide.manifest.page.name | Added Mandatory Element

ImplementationGuide.manifest.page.title | Added Element

ImplementationGuide.manifest.page.anchor | Added Element

ImplementationGuide.manifest.image | Added Element
ImplementationGuide.manifest.other | Added Element
ImplementationGuide.dependency.type | deleted
ImplementationGuide.package | deleted
ImplementationGuide.binary | deleted
ImplementationGuide.page.source | deleted
ImplementationGuide.page.kind | deleted
ImplementationGuide.page.type | deleted
ImplementationGuide.page.package | deleted
ImplementationGuide.page.format | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

5.8.3.1 Terminology Bindings

Path | Definition | Type | Reference
ImplementationGuide.status | The lifecycle status of an artifact. | Required | PublicationStatus
ImplementationGuide.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
ImplementationGuide.license | The license that applies to an Implementation Guide (using an SPDX license Identifiers, or 'not-open-source'). The binding is required but new SPDX license Identifiers are allowed to be used (https://spdx.org/licenses/). | Required | SPDXLicense
ImplementationGuide.fhirVersion ImplementationGuide.definition.resource.fhirVersion | All published FHIR Versions. | Required | FHIRVersion
ImplementationGuide.global.type | One of the resource types defined as part of this version of FHIR. | Required | Resource Types
ImplementationGuide.definition.page.generation | A code that indicates how the page is generated. | Required | GuidePageGeneration
ImplementationGuide.definition.parameter.code | Code of parameter that is input to the guide. | Required | GuideParameterCode

5.8.3.2 Constraints

id | Level | Location | Description | Expression
ig-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ig-1 | Rule | ImplementationGuide.definition | If a resource has a groupingId, it must refer to a grouping defined in the Implementation Guide | resource.groupingId.all(%context.grouping.id contains $this)
ig-2 | Rule | (base) | If a resource has a fhirVersion, it must be oe of the versions defined for the Implementation Guide | definition.resource.fhirVersion.all(%context.fhirVersion contains $this)

5.8.3.3 Distributing Implementation Guides

Implementation Guides are published through the FHIR Package distribution system. For further details, see the FHIR Confluence site . This content may be moved into the specification in a future version.

5.8.3.4 Multi-Version Implementation Guides

Most implementation guides target a single version - that is, they describe how to use a particular version, and all the profiles, value sets and examples they contain etc are valid for that particular version.

In other cases, however, implementation of an implementation guide is not confined to a single version. Typically, the requirement to support multiple versions arises as implementation matures and different implementation communities are stuck at different versions by regulation or market dynamics. Applications may be stuck at different versions of the specification. See Managing Multiple Versions for further information about cross-version support.

For this reason, implementation guides might describe how to use multiple different versions of FHIR for the same purpose. The different versions might have different profiles, extensions, and examples, while sharing common value set definitions, for example. For some reasons, profiles and examples could be common across all versions. And an implementation guide will generally have a lot of common narrative describing the problem, security approaches, and other deployment information irrespective of specific FHIR versions.

An implementation guide specifies which versions of FHIR it describes in the ImplementationGuide.fhirVersion property:

```
  "fhirVersion" : ["3.0", "4.0"],
```

This specifies that the implementation guide applies to both release 3 and Release 4 . Note that the patch version (".1" for Release 3) is omitted, since the patch releases never make changes that make any difference to Implementation Guides.

Note that it is possible to have an Implementation Guide that declares support for one version and imports an Implementation Guide that declares support for a different version. It is up to the importing IG ad/or the tooling that supports Implenentation Guides to determine whether it's safe and/or appropriate to reference

the different resources from the new IG, and what to make of this.

5.8.3.5 Default Profiles

An implementation guide can define default profiles using ImplementationGuide.global- these are profiles that apply to any resource that does not otherwise have an explicit profile assigned by the implementation guide. Default profiles are always references to profiles (StructureDefinition resources) that are also contained in the resources. By defining default profiles, an implementation guide can save itself from exhaustively defining profiles on every resource type just to profile every reference to a particular resource type.

Note that a resource can conform to the default profile by conforming to any profile derived from it.

5.8.3.6 Compatibility list

This table declares the compatibility between the various resources as determined by the Implementation Guide comparison tool:

Yet to be done

5.8.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
context | token | A use context assigned to the implementation guide | (ImplementationGuide.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the implementation guide | (ImplementationGuide.useContext.value as Quantity) | (ImplementationGuide.useContext.value as Range)
context-type | token | A type of use context assigned to the implementation guide | ImplementationGuide.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the implementation guide | On ImplementationGuide.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the implementation guide | On ImplementationGuide.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The implementation guide publication date | ImplementationGuide.date
depends-on | reference | Identity of the IG that this depends on | ImplementationGuide.dependsOn.uri (ImplementationGuide)
description | string | The description of the implementation guide | ImplementationGuide.description
experimental | token | For testing purposes, not real usage | ImplementationGuide.experimental
global | reference | Profile that all resources must conform to | ImplementationGuide.global.profile (StructureDefinition)
jurisdiction | token | Intended jurisdiction for the implementation guide | ImplementationGuide.jurisdiction
name | string | Computationally friendly name of the implementation guide | ImplementationGuide.name
publisher | string | Name of the publisher of the implementation guide | ImplementationGuide.publisher
resource | reference | Location of the resource | ImplementationGuide.definition.resource.reference (Any)
status | token | The current status of the implementation guide | ImplementationGuide.status
title | string | The human-friendly name of the implementation guide | ImplementationGuide.title
url | uri | The uri that identifies the implementation guide | ImplementationGuide.url
version | token | The business version of the implementation guide | ImplementationGuide.version

5.8.5 Resource ImplementationGuide - Examples

URL: https://hl7.org/fhir/R4/implementationguide-examples.html

5.8.5 Resource ImplementationGuide - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of implementationguide | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.8.6 Resource ImplementationGuide - Detailed Descriptions

URL: https://hl7.org/fhir/R4/implementationguide-definitions.html

5.8.6 Resource ImplementationGuide - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the ImplementationGuide resource.

ImplementationGuide
Element Id | ImplementationGuide
Definition | A set of rules of how a particular interoperability or standards problem is solved - typically through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a logical whole and to publish a computable definition of all the parts.
Cardinality | 0..*

Type | DomainResource
Requirements | An implementation guide is able to define default profiles that must apply to any use of a resource, so validation services may need to take one or more implementation guide resources when validating.
Invariants | Defined on this element ig-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}') ig-2 Rule If a resource has a fhirVersion, it must be oe of the versions defined for the Implementation Guide definition.resource.fhirVersion.all(%context.fhirVersion contains $this)
Defined on this element
ig-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ig-2 | Rule | If a resource has a fhirVersion, it must be oe of the versions defined for the Implementation Guide | definition.resource.fhirVersion.all(%context.fhirVersion contains $this)
ImplementationGuide.url
Element Id | ImplementationGuide.url
Definition | An absolute URI that is used to identify this implementation guide when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this implementation guide is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the implementation guide is stored on different servers.
Cardinality | 1..1
Type | uri
Requirements | Allows the implementation guide to be referenced by a single globally unique identifier. This is required to allow hosting Implementation Guides on multiple different servers, and to allow for the editorial process.
Alternate Names | url; authoritative-url; destination; identity
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
ImplementationGuide.version
Element Id | ImplementationGuide.version
Definition | The identifier that is used to identify this version of the implementation guide when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the implementation guide author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different implementation guide instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the implementation guide with the format [url]|[version].
ImplementationGuide.name
Element Id | ImplementationGuide.name
Definition | A natural language name identifying the implementation guide. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 1..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ImplementationGuide.title
Element Id | ImplementationGuide.title
Definition | A short, descriptive, user-friendly title for the implementation guide.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
ImplementationGuide.status
Element Id | ImplementationGuide.status
Definition | The status of this implementation guide. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of implementation guides that are appropriate for use versus not.
ImplementationGuide.experimental

Element Id | ImplementationGuide.experimental
Definition | A Boolean value to indicate that this implementation guide is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level implementation guide.
Summary | true
Comments | Allows filtering of implementation guides that are appropriate for use versus not.
ImplementationGuide.date
Element Id | ImplementationGuide.date
Definition | The date (and optionally time) when the implementation guide was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the implementation guide changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the implementation guide. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
ImplementationGuide.publisher
Element Id | ImplementationGuide.publisher
Definition | The name of the organization or individual that published the implementation guide.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the implementation guide. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the implementation guide is the organization or individual primarily responsible for the maintenance and upkeep of the implementation guide. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the implementation guide. This item SHOULD be populated unless the information is available from context.
ImplementationGuide.contact
Element Id | ImplementationGuide.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
ImplementationGuide.description
Element Id | ImplementationGuide.description
Definition | A free text natural language description of the implementation guide from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the implementation guide was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the implementation guide as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the implementation guide is presumed to be the predominant language in the place the implementation guide was created).
ImplementationGuide.useContext
Element Id | ImplementationGuide.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate implementation guide instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
ImplementationGuide.jurisdiction
Element Id | ImplementationGuide.jurisdiction
Definition | A legal or geographic region in which the implementation guide is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the implementation guide to be used in jurisdictions other than those for which it was originally designed or intended.
ImplementationGuide.copyright
Element Id | ImplementationGuide.copyright
Definition | A copyright statement relating to the implementation guide and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the implementation guide.
Cardinality | 0..1

Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the implementation guide and/or its content.
Alternate Names | License; Restrictions
ImplementationGuide.packageId
Element Id | ImplementationGuide.packageId
Definition | The NPM package name for this Implementation Guide, used in the NPM package distribution, which is the primary mechanism by which FHIR based tooling manages IG dependencies. This value must be globally unique, and should be assigned with care.
Cardinality | 1..1
Type | id
Summary | true
Comments | Many (if not all) IG publishing tools will require that this element be present. For implementation guides published through HL7 or the FHIR foundation, the FHIR product director assigns package IDs.
ImplementationGuide.license
Element Id | ImplementationGuide.license
Definition | The license that applies to this Implementation Guide, using an SPDX license code, or 'not-open-source'.
Cardinality | 0..1
Terminology Binding | SPDXLicense (Required)
Type | code
Summary | true
ImplementationGuide.fhirVersion
Element Id | ImplementationGuide.fhirVersion
Definition | The version(s) of the FHIR specification that this ImplementationGuide targets - e.g. describes how to use. The value of this element is the formal version of the specification, without the revision number, e.g. [publication].[major].[minor], which is 4.0.1. for this version.
Cardinality | 1..*
Terminology Binding | FHIRVersion (Required)
Type | code
Summary | true
Comments | Most implementation guides target a single version - e.g. they describe how to use a particular version, and the profiles and examples etc are valid for that version. But some implementation guides describe how to use multiple different versions of FHIR to solve the same problem, or in concert with each other. Typically, the requirement to support multiple versions arises as implementation matures and different implementation communities are stuck at different versions by regulation or market dynamics.
ImplementationGuide.dependsOn
Element Id | ImplementationGuide.dependsOn
Definition | Another implementation guide that this implementation depends on. Typically, an implementation guide uses value sets, profiles etc.defined in other implementation guides.
Cardinality | 0..*
Summary | true
ImplementationGuide.dependsOn.uri
Element Id | ImplementationGuide.dependsOn.uri
Definition | A canonical reference to the Implementation guide for the dependency.
Cardinality | 1..1
Type | canonical(ImplementationGuide)
Summary | true
Comments | Usually, A canonical reference to the implementation guide is the same as the master location at which the implementation guide is published.
ImplementationGuide.dependsOn.packageId
Element Id | ImplementationGuide.dependsOn.packageId
Definition | The NPM package name for the Implementation Guide that this IG depends on.
Cardinality | 0..1
Type | id
Summary | true
ImplementationGuide.dependsOn.version
Element Id | ImplementationGuide.dependsOn.version
Definition | The version of the IG that is depended on, when the correct version is required to understand the IG correctly.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | This follows the syntax of the NPM packaging version field - see [[reference]].
ImplementationGuide.global
Element Id | ImplementationGuide.global
Definition | A set of profiles that all resources covered by this implementation guide must conform to.
Cardinality | 0..*
Summary | true
Comments | See Default Profiles for a discussion of which resources are 'covered' by an implementation guide.
ImplementationGuide.global.type
Element Id | ImplementationGuide.global.type
Definition | The type of resource that all instances must conform to.
Cardinality | 1..1

Terminology Binding | Any defined Resource Type name
Type | code
Summary | true
Comments | The type must match that of the profile that is referred to but is made explicit here as a denormalization so that a system processing the implementation guide resource knows which resources the profile applies to even if the profile itself is not available.
ImplementationGuide.global.profile
Element Id | ImplementationGuide.global.profile
Definition | A reference to the profile that all instances must conform to.
Cardinality | 1..1
Type | canonical(StructureDefinition)
Summary | true
ImplementationGuide.definition
Element Id | ImplementationGuide.definition
Definition | The information needed by an IG publisher tool to publish the whole implementation guide.
Cardinality | 0..1
Comments | Principally, this consists of information abuot source resource and file locations, and build parameters and templates.
Invariants | Defined on this element ig-1 Rule If a resource has a groupingId, it must refer to a grouping defined in the Implementation Guide resource.groupingId.all(%context.grouping.id contains $this)
Defined on this element
ig-1 | Rule | If a resource has a groupingId, it must refer to a grouping defined in the Implementation Guide | resource.groupingId.all(%context.grouping.id contains $this)
ImplementationGuide.definition.grouping
Element Id | ImplementationGuide.definition.grouping
Definition | A logical group of resources. Logical groups can be used when building pages.
Cardinality | 0..*
Comments | Groupings are arbitrary sub-divisions of content. Typically, they are used to help build Table of Contents automatically.
ImplementationGuide.definition.grouping.name
Element Id | ImplementationGuide.definition.grouping.name
Definition | The human-readable title to display for the package of resources when rendering the implementation guide.
Cardinality | 1..1
Type | string
ImplementationGuide.definition.grouping.description
Element Id | ImplementationGuide.definition.grouping.description
Definition | Human readable text describing the package.
Cardinality | 0..1
Type | string
ImplementationGuide.definition.resource
Element Id | ImplementationGuide.definition.resource
Definition | A resource that is part of the implementation guide. Conformance resources (value set, structure definition, capability statements etc.) are obvious candidates for inclusion, but any kind of resource can be included as an example resource.
Cardinality | 1..*
ImplementationGuide.definition.resource.reference
Element Id | ImplementationGuide.definition.resource.reference
Definition | Where this resource is found.
Cardinality | 1..1
Type | Reference(Any)
Comments | Usually this is a relative URL that locates the resource within the implementation guide. If you authoring an implementation guide, and will publish it using the FHIR publication tooling, use a URI that may point to a resource, or to one of various alternative representations (e.g. spreadsheet). The tooling will convert this when it publishes it.
ImplementationGuide.definition.resource.fhirVersion
Element Id | ImplementationGuide.definition.resource.fhirVersion
Definition | Indicates the FHIR Version(s) this artifact is intended to apply to. If no versions are specified, the resource is assumed to apply to all the versions stated in ImplementationGuide.fhirVersion.
Cardinality | 0..*
Terminology Binding | FHIRVersion (Required)
Type | code
Comments | The resource SHALL be valid against all the versions it is specified to apply to. If the resource referred to is a StructureDefinition, the fhirVersion stated in the StructureDefinition cannot disagree with the version specified here; the specified versions SHALL include the version specified by the StructureDefinition, and may include additional versions using the applicable-version extension.
ImplementationGuide.definition.resource.name
Element Id | ImplementationGuide.definition.resource.name
Definition | A human assigned name for the resource. All resources SHOULD have a name, but the name may be extracted from the resource (e.g. ValueSet.name).
Cardinality | 0..1
Type | string
ImplementationGuide.definition.resource.description
Element Id | ImplementationGuide.definition.resource.description
Definition | A description of the reason that a resource has been included in the implementation guide.
Cardinality | 0..1
Type | string

Comments | This is mostly used with examples to explain why it is present (though they can have extensive comments in the examples).
ImplementationGuide.definition.resource.example[x]
Element Id | ImplementationGuide.definition.resource.example[x]
Definition | If true or a reference, indicates the resource is an example instance. If a reference is present, indicates that the example is an example of the specified profile.
Cardinality | 0..1
Type | boolean|canonical(StructureDefinition)
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | Examples: StructureDefinition -> Any ValueSet -> expansion OperationDefinition -> Parameters Questionnaire -> QuestionnaireResponse.
ImplementationGuide.definition.resource.groupingId
Element Id | ImplementationGuide.definition.resource.groupingId
Definition | Reference to the id of the grouping this resource appears in.
Cardinality | 0..1
Type | id
Comments | This must correspond to a package.id element within this implementation guide.
ImplementationGuide.definition.page
Element Id | ImplementationGuide.definition.page
Definition | A page / section in the implementation guide. The root page is the implementation guide home page.
Cardinality | 0..1
Comments | Pages automatically become sections if they have sub-pages. By convention, the home page is called index.html.
ImplementationGuide.definition.page.name[x]
Element Id | ImplementationGuide.definition.page.name[x]
Definition | The source address for the page.
Cardinality | 1..1
Type | url|Reference(Binary)
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | The publishing tool will autogenerate source for list (source = n/a) and inject included implementations for include (source = uri of guide to include).
ImplementationGuide.definition.page.title
Element Id | ImplementationGuide.definition.page.title
Definition | A short title used to represent this page in navigational structures such as table of contents, bread crumbs, etc.
Cardinality | 1..1
Type | string
ImplementationGuide.definition.page.generation
Element Id | ImplementationGuide.definition.page.generation
Definition | A code that indicates how the page is generated.
Cardinality | 1..1
Terminology Binding | GuidePageGeneration (Required)
Type | code
ImplementationGuide.definition.page.page
Element Id | ImplementationGuide.definition.page.page
Definition | Nested Pages/Sections under this page.
Cardinality | 0..*
Type | See ImplementationGuide.definition.page
Comments | The implementation guide breadcrumbs are generated from this structure.
ImplementationGuide.definition.parameter
Element Id | ImplementationGuide.definition.parameter
Definition | Defines how IG is built by tools.
Cardinality | 0..*
ImplementationGuide.definition.parameter.code
Element Id | ImplementationGuide.definition.parameter.code
Definition | apply | path-resource | path-pages | path-tx-cache | expansion-parameter | rule-broken-links | generate-xml | generate-json | generate-turtle | html-template.
Cardinality | 1..1
Terminology Binding | GuideParameterCode (Required)
Type | code
ImplementationGuide.definition.parameter.value
Element Id | ImplementationGuide.definition.parameter.value
Definition | Value for named type.
Cardinality | 1..1
Type | string
ImplementationGuide.definition.template
Element Id | ImplementationGuide.definition.template
Definition | A template for building resources.
Cardinality | 0..*
ImplementationGuide.definition.template.code
Element Id | ImplementationGuide.definition.template.code
Definition | Type of template specified.
Cardinality | 1..1
Type | code
ImplementationGuide.definition.template.source

Element Id | ImplementationGuide.definition.template.source
Definition | The source location for the template.
Cardinality | 1..1
Type | string
ImplementationGuide.definition.template.scope
Element Id | ImplementationGuide.definition.template.scope
Definition | The scope in which the template applies.
Cardinality | 0..1
Type | string
ImplementationGuide.manifest
Element Id | ImplementationGuide.manifest
Definition | Information about an assembled implementation guide, created by the publication tooling.
Cardinality | 0..1
ImplementationGuide.manifest.rendering
Element Id | ImplementationGuide.manifest.rendering
Definition | A pointer to official web page, PDF or other rendering of the implementation guide.
Cardinality | 0..1
Type | url
Summary | true
ImplementationGuide.manifest.resource
Element Id | ImplementationGuide.manifest.resource
Definition | A resource that is part of the implementation guide. Conformance resources (value set, structure definition, capability statements etc.) are obvious candidates for inclusion, but any kind of resource can be included as an example resource.
Cardinality | 1..*
Summary | true
ImplementationGuide.manifest.resource.reference
Element Id | ImplementationGuide.manifest.resource.reference
Definition | Where this resource is found.
Cardinality | 1..1
Type | Reference(Any)
Summary | true
Comments | Usually this is a relative URL that locates the resource within the implementation guide. If you authoring an implementation guide, and will publish it using the FHIR publication tooling, use a URI that may point to a resource, or to one of various alternative representations (e.g. spreadsheet). The tooling will convert this when it publishes it.
ImplementationGuide.manifest.resource.example[x]
Element Id | ImplementationGuide.manifest.resource.example[x]
Definition | If true or a reference, indicates the resource is an example instance. If a reference is present, indicates that the example is an example of the specified profile.
Cardinality | 0..1
Type | boolean|canonical(StructureDefinition)
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | Typically, conformance resources and knowledge resources are directly part of the implementation guide, with their normal meaning, and patient linked resources are usually examples. However this is not always true.
ImplementationGuide.manifest.resource.relativePath
Element Id | ImplementationGuide.manifest.resource.relativePath
Definition | The relative path for primary page for this resource within the IG.
Cardinality | 0..1
Type | url
Comments | Appending 'rendering' + "/" + this should resolve to the resource page.
ImplementationGuide.manifest.page
Element Id | ImplementationGuide.manifest.page
Definition | Information about a page within the IG.
Cardinality | 0..*
Requirements | Allows validation of hyperlinks from a derived IG to this IG without a local copy of the IG.
ImplementationGuide.manifest.page.name
Element Id | ImplementationGuide.manifest.page.name
Definition | Relative path to the page.
Cardinality | 1..1
Type | string
Comments | Appending 'rendering' + "/" + this should resolve to the page.
ImplementationGuide.manifest.page.title
Element Id | ImplementationGuide.manifest.page.title
Definition | Label for the page intended for human display.
Cardinality | 0..1
Type | string
Requirements | Allows generation of labels for markdown-generated hyperlinks.
ImplementationGuide.manifest.page.anchor
Element Id | ImplementationGuide.manifest.page.anchor
Definition | The name of an anchor available on the page.
Cardinality | 0..*
Type | string
Requirements | Allows validation of hyperlinks from a derived IG to this IG without a local copy of the IG.
Comments | Appending 'rendering' + "/" + page.name + "#" + page.anchor should resolve to the anchor.

ImplementationGuide.manifest.image
Element Id | ImplementationGuide.manifest.image
Definition | Indicates a relative path to an image that exists within the IG.
Cardinality | 0..*
Type | string
Requirements | Allows validation of image links from a derived IG to this IG without a local copy of the IG.
ImplementationGuide.manifest.other
Element Id | ImplementationGuide.manifest.other
Definition | Indicates the relative path of an additional non-page, non-image file that is part of the IG - e.g. zip, jar and similar files that could be the target of a hyperlink in a derived IG.
Cardinality | 0..*
Type | string
Requirements | Allows validation of links from a derived IG to this IG without a local copy of the IG.

5.8.7 Resource ImplementationGuide - Mappings

URL: https://hl7.org/fhir/R4/implementationguide-mappings.html

5.8.7 Resource ImplementationGuide - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the implementationguide resource (see Mappings to Other Standards for further information & status).

5.8.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ImplementationGuide
url | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness

5.8.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ImplementationGuide
url | Definition.url
version | Definition.version
title | Definition.title
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
copyright | Definition.copyright

5.8.7.3 RIM Mapping (http://hl7.org/v3)

ImplementationGuide | n/a

5.8.8 Resource ImplementationGuide - Extensions & Profiles

URL: https://hl7.org/fhir/R4/implementationguide-profiles.html

5.8.8 Resource ImplementationGuide - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the ImplementationGuide resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

5.8.8.1 Profiles

No Profiles defined for this resource

5.8.8.2 Extensions

replaces | ImplementationGuide | replaces | for General Extensions for use by FHIR Implementers
resource-approvalDate | ImplementationGuide | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | ImplementationGuide | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | ImplementationGuide | lastReviewDate | for Resource HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

5.8.8.3 Search Extensions

No Search Extensions defined for this resource

5.8.9 Resource ImplementationGuide - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/implementationguide-version-maps.html

No validation errors - all conversions are clean

5.8.9 Resource ImplementationGuide - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for ImplementationGuide.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

5.8.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/ImplementationGuide3to4" = "R3 to R4 Conversions for ImplementationGuide"

uses "http://hl7.org/fhir/3.0/StructureDefinition/ImplementationGuide" alias ImplementationGuideR3 as source
uses "http://hl7.org/fhir/StructureDefinition/ImplementationGuide" alias ImplementationGuide as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group ImplementationGuide(source src : ImplementationGuideR3, target tgt : ImplementationGuide) extends DomainResource
    <<type+>> {
  src.url -> tgt.url;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.copyright -> tgt.copyright;
  src.fhirVersion -> tgt.fhirVersion;
  src.dependency as vs0 check type = 'reference' -> tgt.dependsOn as vt0 then dependency(vs0, vt0);
  src.global as vs0 -> tgt.global as vt0 then global(vs0, vt0);
  src -> tgt.definition as dfn then definition(src, dfn) "definition";
  src where binary.exists() -> tgt.manifest as mft then manifest(src, mft) "manifest";
}

group dependency(source src, target tgt) extends BackboneElement {
  src.uri -> tgt.uri;
}

group global(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.profile -> tgt.profile;
}

group definition(source src, target tgt) extends BackboneElement {
  src.package as pck -> tgt.grouping as grp then grouping(pck, tgt, grp);
  src.page as vs0 -> tgt.page as vt0 then page(vs0, vt0);
```

```
}

group grouping(source src, target dfn, target grp) extends BackboneElement {
  src.name -> grp.name;
  src.description -> grp.description;
  src.resource as rs -> dfn.resource as rt then resource(rs, rt, src);
}

group resource(source src, target tgt, source pck) extends BackboneElement {
  pck.id -> tgt.groupingId "packageId";
  src.example where src.exampleFor.empty() as vs0 -> tgt.example = create('boolean') as vt0 then boolean(vs0, vt0);
  src.name -> tgt.name;
  src.description -> tgt.description;
  src.acronym as vs ->  tgt.extension as ext,  ext.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    ImplementationGuide.package.resource.acronym',  ext.value = vs;
  src.source : uri as vs ->  tgt.reference as ref,  ref.reference = vs,  ref.extension as ext,  ext.url =
    'http://hl7.org/fhir/3.0/StructureDefinition/BaseType',  ext.value = 'uri' "sourceUri";
  src.source : Reference as vs0 -> tgt.reference as vt0 then reference(vs0, vt0) "sourceRef";
  src.exampleFor as vs0 -> tgt.example = create('canonical') as vt0 then Reference2Canonical(vs0, vt0);
}

group page(source src, target tgt) extends BackboneElement {
  src.source as vs0 -> tgt.name = create('uri') as vt0 then uri(vs0, vt0);
  src.title -> tgt.title;
  src.kind as vs ->  tgt.extension as ext,  ext.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    ImplementationGuide.page.kind',  ext.value = vs;
  src.type as vs ->  tgt.extension as ext,  ext.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    ImplementationGuide.page.type',  ext.value = vs;
  src.package as vs ->  tgt.extension as ext,  ext.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    ImplementationGuide.page.package',  ext.value = vs;
  src.format as vs ->  tgt.extension as ext,  ext.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    ImplementationGuide.page.format',  ext.value = vs;
  src.page as vs0 -> tgt.page as vt0 then page(vs0, vt0);
}

group manifest(source src, target tgt) extends BackboneElement {
  src.binary -> tgt.other;
}
```

5.8.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/ImplementationGuide4to3" = "R4 to R3 Conversion for ImplementationGuide"

uses "http://hl7.org/fhir/StructureDefinition/ImplementationGuide" alias ImplementationGuide as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/ImplementationGuide" alias ImplementationGuideR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group ImplementationGuide(source src : ImplementationGuideR3, target tgt : ImplementationGuide) extends DomainResource
    <<type+>> {
  src.url -> tgt.url;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.copyright -> tgt.copyright;
  src.fhirVersion -> tgt.fhirVersion;
  src.dependsOn as vs0 -> tgt.dependency as vt0 then dependency(vs0, vt0) "dependency";
  src.global as vs0 -> tgt.global as vt0 then global(vs0, vt0);
  src.definition as vs0 then definition(vs0, tgt);
  src.manifest as mft then manifest(mft, tgt);
}

group dependency(source src, target tgt) extends BackboneElement {
  src -> tgt.type = 'reference' "type";
  src.uri -> tgt.uri;
}

group global(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.profile -> tgt.profile;
}

group definition(source src, target tgt) extends BackboneElement {
  src.grouping as grp -> tgt.package as pck then grouping(grp, pck, src) "package";
  src.page as vs0 -> tgt.page as vt0 then page(vs0, vt0);
}

group grouping(source src, target tgt, source dfn) extends BackboneElement {
  src.name -> tgt.name;
  src.description -> tgt.description;
  dfn.resource as rs where groupingId = src.id -> tgt.resource as rt then resource(rs, rt);
}
```

```
group resource(source src, target tgt) extends BackboneElement {
  src.example : boolean as vs0 -> tgt.example as vt0 then boolean(vs0, vt0);
  src.name -> tgt.name;
  src.description -> tgt.description;
  src.extension as ext where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    ImplementationGuide.package.resource.acronym' then {
    ext.value as vs0 -> tgt.acronym = vs0 "acronym2";
  } "acronym";
  src.reference as vs0 where extension('http://hl7.org/fhir/3.0/StructureDefinition/BaseType').empty() -> tgt.source =
    create('Reference') as vt0 then Reference(vs0, vt0) "source1";
  src.reference as vs0 where extension('http://hl7.org/fhir/3.0/StructureDefinition/BaseType').exists() -> tgt.source =
    create('uri') as vt0 then {
    vs0.reference as vs1 then uri(vs1, vt0) "source2a";
  } "source2";
  src.example : canonical as vs0 ->  tgt.example = true,  tgt.exampleFor as vt0 then canonical2Reference(vs0, vt0)
    "exampleFor";
}

group page(source src, target tgt) extends BackboneElement {
  src.name : uri as vs0 -> tgt.source as vt0 then uri(vs0, vt0) "source";
  src.title -> tgt.title;
  src.extension as ext where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-ImplementationGuide.page.kind'
     then {
    ext.value as vs0 -> tgt.kind = vs0 "acronym2";
  } "acronym";
  src.extension as ext where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-ImplementationGuide.page.type'
     then {
    ext.value as vs0 -> tgt.type = vs0 "acronym2";
  } "acronym";
  src.extension as ext where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    ImplementationGuide.page.package' then {
    ext.value as vs0 -> tgt.package = vs0 "acronym2";
  } "acronym";
  src.extension as ext where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    ImplementationGuide.page.format' then {
    ext.value as vs0 -> tgt.format = vs0 "acronym2";
  } "acronym";
  src.page as vs0 -> tgt.page as vt0 then page(vs0, vt0);
}

group manifest(source src, target tgt) extends BackboneElement {
  src.other -> tgt.binary "binary";
}
```

5.8.9.3 R4 Validation Errors

5.8.10 FHIR Core ImplementationGuide Profile

URL: https://hl7.org/fhir/R4/implementationguide-extensions.html

This profile holds core ImplementationGuide extensions

5.8.10 FHIR Core ImplementationGuide Profile

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

5.8.10.1 Content

5.8.11 JSON Schema for ImplementationGuide

URL: https://hl7.org/fhir/R4/implementationguide.schema.json.html

5.8.11 JSON Schema for ImplementationGuide

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for ImplementationGuide, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ImplementationGuide",
  "$ref": "#/definitions/ImplementationGuide",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ImplementationGuide": {
      "description": "A set of rules of how a particular interoperability or standards problem is solved - typically
    through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a
    logical whole and to publish a computable definition of all the parts.",
      "properties": {
        "resourceType": {
          "description": "This is a ImplementationGuide resource",
          "const": "ImplementationGuide"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
```

```
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this implementation guide when it is referenced in a
specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which an authoritative instance of this implementation guide is (or will
 be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the
implementation guide is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the implementation guide when it is
referenced in a specification, model, design or instance. This is an arbitrary value managed by the implementation
guide author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a
managed version is not available. There is also no expectation that versions can be placed in a lexicographical
sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
    "name": {
      "description": "A natural language name identifying the implementation guide. This name should be usable as an
 identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the implementation guide.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this implementation guide. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this implementation guide is authored for testing purposes
(or education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the implementation guide was published. The date must
change when the business version changes and it must change if the status code changes. In addition, it should
change when the substantive content of the implementation guide changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the implementation guide.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the implementation guide from a consumer\u0027s
perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate implementation guide
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the implementation guide is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
```

```
    },
    "copyright": {
      "description": "A copyright statement relating to the implementation guide and/or its contents. Copyright
statements are generally legal restrictions on the use and publishing of the implementation guide.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "packageId": {
      "description": "The NPM package name for this Implementation Guide, used in the NPM package distribution,
which is the primary mechanism by which FHIR based tooling manages IG dependencies. This value must be globally
unique, and should be assigned with care.",
      "$ref": "#/definitions/id"
    },
    "_packageId": {
      "description": "Extensions for packageId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "license": {
      "description": "The license that applies to this Implementation Guide, using an SPDX license code, or
\u0027not-open-source\u0027.",
      "enum": [
        "not-open-source",
        "0BSD",
        "AAL",
        "Abstyles",
        "Adobe-2006",
        "Adobe-Glyph",
        "ADSL",
        "AFL-1.1",
        "AFL-1.2",
        "AFL-2.0",
        "AFL-2.1",
        "AFL-3.0",
        "Afmparse",
        "AGPL-1.0-only",
        "AGPL-1.0-or-later",
        "AGPL-3.0-only",
        "AGPL-3.0-or-later",
        "Aladdin",
        "AMDPLPA",
        "AML",
        "AMPAS",
        "ANTLR-PD",
        "Apache-1.0",
        "Apache-1.1",
        "Apache-2.0",
        "APAFML",
        "APL-1.0",
        "APSL-1.0",
        "APSL-1.1",
        "APSL-1.2",
        "APSL-2.0",
        "Artistic-1.0-cl8",
        "Artistic-1.0-Perl",
        "Artistic-1.0",
        "Artistic-2.0",
        "Bahyph",
        "Barr",
        "Beerware",
        "BitTorrent-1.0",
        "BitTorrent-1.1",
        "Borceux",
        "BSD-1-Clause",
        "BSD-2-Clause-FreeBSD",
        "BSD-2-Clause-NetBSD",
        "BSD-2-Clause-Patent",
        "BSD-2-Clause",
        "BSD-3-Clause-Attribution",
        "BSD-3-Clause-Clear",
        "BSD-3-Clause-LBNL",
        "BSD-3-Clause-No-Nuclear-License-2014",
        "BSD-3-Clause-No-Nuclear-License",
        "BSD-3-Clause-No-Nuclear-Warranty",
        "BSD-3-Clause",
        "BSD-4-Clause-UC",
        "BSD-4-Clause",
        "BSD-Protection",
        "BSD-Source-Code",
        "BSL-1.0",
        "bzip2-1.0.5",
        "bzip2-1.0.6",
        "Caldera",
        "CATOSL-1.1",
        "CC-BY-1.0",
        "CC-BY-2.0",
        "CC-BY-2.5",
```

```
"CC-BY-3.0",
"CC-BY-4.0",
"CC-BY-NC-1.0",
"CC-BY-NC-2.0",
"CC-BY-NC-2.5",
"CC-BY-NC-3.0",
"CC-BY-NC-4.0",
"CC-BY-NC-ND-1.0",
"CC-BY-NC-ND-2.0",
"CC-BY-NC-ND-2.5",
"CC-BY-NC-ND-3.0",
"CC-BY-NC-ND-4.0",
"CC-BY-NC-SA-1.0",
"CC-BY-NC-SA-2.0",
"CC-BY-NC-SA-2.5",
"CC-BY-NC-SA-3.0",
"CC-BY-NC-SA-4.0",
"CC-BY-ND-1.0",
"CC-BY-ND-2.0",
"CC-BY-ND-2.5",
"CC-BY-ND-3.0",
"CC-BY-ND-4.0",
"CC-BY-SA-1.0",
"CC-BY-SA-2.0",
"CC-BY-SA-2.5",
"CC-BY-SA-3.0",
"CC-BY-SA-4.0",
"CC0-1.0",
"CDDL-1.0",
"CDDL-1.1",
"CDLA-Permissive-1.0",
"CDLA-Sharing-1.0",
"CECILL-1.0",
"CECILL-1.1",
"CECILL-2.0",
"CECILL-2.1",
"CECILL-B",
"CECILL-C",
"ClArtistic",
"CNRI-Jython",
"CNRI-Python-GPL-Compatible",
"CNRI-Python",
"Condor-1.1",
"CPAL-1.0",
"CPL-1.0",
"CPOL-1.02",
"Crossword",
"CrystalStacker",
"CUA-OPL-1.0",
"Cube",
"curl",
"D-FSL-1.0",
"diffmark",
"DOC",
"Dotseqn",
"DSDP",
"dvipdfm",
"ECL-1.0",
"ECL-2.0",
"EFL-1.0",
"EFL-2.0",
"eGenix",
"Entessa",
"EPL-1.0",
"EPL-2.0",
"ErlPL-1.1",
"EUDatagrid",
"EUPL-1.0",
"EUPL-1.1",
"EUPL-1.2",
"Eurosym",
"Fair",
"Frameworx-1.0",
"FreeImage",
"FSFAP",
"FSFUL",
"FSFULLR",
"FTL",
"GFDL-1.1-only",
"GFDL-1.1-or-later",
"GFDL-1.2-only",
"GFDL-1.2-or-later",
"GFDL-1.3-only",
"GFDL-1.3-or-later",
"Giftware",
"GL2PS",
"Glide",
"Glulxe",
"gnuplot",
```

```
"GPL-1.0-only",
"GPL-1.0-or-later",
"GPL-2.0-only",
"GPL-2.0-or-later",
"GPL-3.0-only",
"GPL-3.0-or-later",
"gSOAP-1.3b",
"HaskellReport",
"HPND",
"IBM-pibs",
"ICU",
"IJG",
"ImageMagick",
"iMatix",
"Imlib2",
"Info-ZIP",
"Intel-ACPI",
"Intel",
"Interbase-1.0",
"IPA",
"IPL-1.0",
"ISC",
"JasPer-2.0",
"JSON",
"LAL-1.2",
"LAL-1.3",
"Latex2e",
"Leptonica",
"LGPL-2.0-only",
"LGPL-2.0-or-later",
"LGPL-2.1-only",
"LGPL-2.1-or-later",
"LGPL-3.0-only",
"LGPL-3.0-or-later",
"LGPLLR",
"Libpng",
"libtiff",
"LiLiQ-P-1.1",
"LiLiQ-R-1.1",
"LiLiQ-Rplus-1.1",
"Linux-OpenIB",
"LPL-1.0",
"LPL-1.02",
"LPPL-1.0",
"LPPL-1.1",
"LPPL-1.2",
"LPPL-1.3a",
"LPPL-1.3c",
"MakeIndex",
"MirOS",
"MIT-0",
"MIT-advertising",
"MIT-CMU",
"MIT-enna",
"MIT-feh",
"MIT",
"MITNFA",
"Motosoto",
"mpich2",
"MPL-1.0",
"MPL-1.1",
"MPL-2.0-no-copyleft-exception",
"MPL-2.0",
"MS-PL",
"MS-RL",
"MTLL",
"Multics",
"Mup",
"NASA-1.3",
"Naumen",
"NBPL-1.0",
"NCSA",
"Net-SNMP",
"NetCDF",
"Newsletr",
"NGPL",
"NLOD-1.0",
"NLPL",
"Nokia",
"NOSL",
"Noweb",
"NPL-1.0",
"NPL-1.1",
"NPOSL-3.0",
"NRL",
"NTP",
"OCCT-PL",
"OCLC-2.0",
"ODbL-1.0",
```

```
"OFL-1.0",
"OFL-1.1",
"OGTSL",
"OLDAP-1.1",
"OLDAP-1.2",
"OLDAP-1.3",
"OLDAP-1.4",
"OLDAP-2.0.1",
"OLDAP-2.0",
"OLDAP-2.1",
"OLDAP-2.2.1",
"OLDAP-2.2.2",
"OLDAP-2.2",
"OLDAP-2.3",
"OLDAP-2.4",
"OLDAP-2.5",
"OLDAP-2.6",
"OLDAP-2.7",
"OLDAP-2.8",
"OML",
"OpenSSL",
"OPL-1.0",
"OSET-PL-2.1",
"OSL-1.0",
"OSL-1.1",
"OSL-2.0",
"OSL-2.1",
"OSL-3.0",
"PDDL-1.0",
"PHP-3.0",
"PHP-3.01",
"Plexus",
"PostgreSQL",
"psfrag",
"psutils",
"Python-2.0",
"Qhull",
"QPL-1.0",
"Rdisc",
"RHeCos-1.1",
"RPL-1.1",
"RPL-1.5",
"RPSL-1.0",
"RSA-MD",
"RSCPL",
"Ruby",
"SAX-PD",
"Saxpath",
"SCEA",
"Sendmail",
"SGI-B-1.0",
"SGI-B-1.1",
"SGI-B-2.0",
"SimPL-2.0",
"SISSL-1.2",
"SISSL",
"Sleepycat",
"SMLNJ",
"SMPPL",
"SNIA",
"Spencer-86",
"Spencer-94",
"Spencer-99",
"SPL-1.0",
"SugarCRM-1.1.3",
"SWL",
"TCL",
"TCP-wrappers",
"TMate",
"TORQUE-1.1",
"TOSL",
"Unicode-DFS-2015",
"Unicode-DFS-2016",
"Unicode-TOU",
"Unlicense",
"UPL-1.0",
"Vim",
"VOSTROM",
"VSL-1.0",
"W3C-19980720",
"W3C-20150513",
"W3C",
"Watcom-1.0",
"Wsuipa",
"WTFPL",
"X11",
"Xerox",
"XFree86-1.1",
"xinetd",
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        "Xnet",
        "xpp",
        "XSkat",
        "YPL-1.0",
        "YPL-1.1",
        "Zed",
        "Zend-2.0",
        "Zimbra-1.3",
        "Zimbra-1.4",
        "zlib-acknowledgement",
        "Zlib",
        "ZPL-1.1",
        "ZPL-2.0",
        "ZPL-2.1"
      ]
    },
    "_license": {
      "description": "Extensions for license",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "fhirVersion": {
      "description": "The version(s) of the FHIR specification that this ImplementationGuide targets - e.g.
describes how to use. The value of this element is the formal version of the specification, without the revision
number, e.g. [publication].[major].[minor], which is 4.0.1. for this version.",
      "items": {
        "enum": [
          "0.01",
          "0.05",
          "0.06",
          "0.11",
          "0.0.80",
          "0.0.81",
          "0.0.82",
          "0.4.0",
          "0.5.0",
          "1.0.0",
          "1.0.1",
          "1.0.2",
          "1.1.0",
          "1.4.0",
          "1.6.0",
          "1.8.0",
          "3.0.0",
          "3.0.1",
          "3.3.0",
          "3.5.0",
          "4.0.0",
          "4.0.1"
        ]
      },
      "type": "array"
    },
    "_fhirVersion": {
      "description": "Extensions for fhirVersion",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "dependsOn": {
      "description": "Another implementation guide that this implementation depends on. Typically, an implementation
 guide uses value sets, profiles etc.defined in other implementation guides.",
      "items": {
        "$ref": "#/definitions/ImplementationGuide_DependsOn"
      },
      "type": "array"
    },
    "global": {
      "description": "A set of profiles that all resources covered by this implementation guide must conform to.",
      "items": {
        "$ref": "#/definitions/ImplementationGuide_Global"
      },
      "type": "array"
    },
    "definition": {
      "description": "The information needed by an IG publisher tool to publish the whole implementation guide.",
      "$ref": "#/definitions/ImplementationGuide_Definition"
    },
    "manifest": {
      "description": "Information about an assembled implementation guide, created by the publication tooling.",
      "$ref": "#/definitions/ImplementationGuide_Manifest"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"ImplementationGuide_DependsOn": {
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  "description": "A set of rules of how a particular interoperability or standards problem is solved - typically through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a logical whole and to publish a computable definition of all the parts.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "uri": {
      "description": "A canonical reference to the Implementation guide for the dependency.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "packageId": {
      "description": "The NPM package name for the Implementation Guide that this IG depends on.",
      "$ref": "#/definitions/id"
    },
    "_packageId": {
      "description": "Extensions for packageId",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "The version of the IG that is depended on, when the correct version is required to understand the IG correctly.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false,
  "required": [
    "uri"
  ]
},
"ImplementationGuide_Global": {
  "description": "A set of rules of how a particular interoperability or standards problem is solved - typically through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a logical whole and to publish a computable definition of all the parts.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
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      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of resource that all instances must conform to.",
      "$ref": "#/definitions/code"
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "profile": {
      "description": "A reference to the profile that all instances must conform to.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    }
  },
  "additionalProperties": false,
  "required": [
    "profile"
  ]
},
"ImplementationGuide_Definition": {
  "description": "A set of rules of how a particular interoperability or standards problem is solved - typically
through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a
logical whole and to publish a computable definition of all the parts.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "grouping": {
      "description": "A logical group of resources. Logical groups can be used when building pages.",
      "items": {
        "$ref": "#/definitions/ImplementationGuide_Grouping"
      },
      "type": "array"
    },
    "resource": {
      "description": "A resource that is part of the implementation guide. Conformance resources (value set,
structure definition, capability statements etc.) are obvious candidates for inclusion, but any kind of resource can
 be included as an example resource.",
      "items": {
        "$ref": "#/definitions/ImplementationGuide_Resource"
      },
      "type": "array"
    },
    "page": {
      "description": "A page / section in the implementation guide. The root page is the implementation guide home
page.",
      "$ref": "#/definitions/ImplementationGuide_Page"
    },
    "parameter": {
      "description": "Defines how IG is built by tools.",
      "items": {
        "$ref": "#/definitions/ImplementationGuide_Parameter"
      },
      "type": "array"
    },
    "template": {
      "description": "A template for building resources.",
      "items": {
        "$ref": "#/definitions/ImplementationGuide_Template"
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      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resource"
  ]
},
"ImplementationGuide_Grouping": {
  "description": "A set of rules of how a particular interoperability or standards problem is solved - typically through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a logical whole and to publish a computable definition of all the parts.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "The human-readable title to display for the package of resources when rendering the implementation guide.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "Human readable text describing the package.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"ImplementationGuide_Resource": {
  "description": "A set of rules of how a particular interoperability or standards problem is solved - typically through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a logical whole and to publish a computable definition of all the parts.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
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 of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "reference": {
      "description": "Where this resource is found.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "fhirVersion": {
      "description": "Indicates the FHIR Version(s) this artifact is intended to apply to. If no versions are specified, the resource is assumed to apply to all the versions stated in ImplementationGuide.fhirVersion.",
      "items": {
        "enum": [
          "0.01",
          "0.05",
          "0.06",
          "0.11",
          "0.0.80",
          "0.0.81",
          "0.0.82",
          "0.4.0",
          "0.5.0",
          "1.0.0",
          "1.0.1",
          "1.0.2",
          "1.1.0",
          "1.4.0",
          "1.6.0",
          "1.8.0",
          "3.0.0",
          "3.0.1",
          "3.3.0",
          "3.5.0",
          "4.0.0",
          "4.0.1"
        ]
      },
      "type": "array"
    },
    "_fhirVersion": {
      "description": "Extensions for fhirVersion",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "name": {
      "description": "A human assigned name for the resource. All resources SHOULD have a name, but the name may be extracted from the resource (e.g. ValueSet.name).",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "A description of the reason that a resource has been included in the implementation guide.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "exampleBoolean": {
      "description": "If true or a reference, indicates the resource is an example instance.  If a reference is present, indicates that the example is an example of the specified profile.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_exampleBoolean": {
      "description": "Extensions for exampleBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "exampleCanonical": {
      "description": "If true or a reference, indicates the resource is an example instance.  If a reference is present, indicates that the example is an example of the specified profile.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_exampleCanonical": {
      "description": "Extensions for exampleCanonical",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "groupingId": {
      "description": "Reference to the id of the grouping this resource appears in.",
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      "$ref": "#/definitions/id"
    },
    "_groupingId": {
      "description": "Extensions for groupingId",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false,
  "required": [
    "reference"
  ]
},
"ImplementationGuide_Page": {
  "description": "A set of rules of how a particular interoperability or standards problem is solved - typically
through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a
logical whole and to publish a computable definition of all the parts.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "nameUrl": {
      "description": "The source address for the page.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_nameUrl": {
      "description": "Extensions for nameUrl",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "nameReference": {
      "description": "The source address for the page.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "title": {
      "description": "A short title used to represent this page in navigational structures such as table of
contents, bread crumbs, etc.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "generation": {
      "description": "A code that indicates how the page is generated.",
      "enum": [
        "html",
        "markdown",
        "xml",
        "generated"
      ]
    },
    "_generation": {
      "description": "Extensions for generation",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "page": {
      "description": "Nested Pages/Sections under this page.",
      "items": {
        "$ref": "#/definitions/ImplementationGuide_Page"
      },
      "type": "array"
    }
  },
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  "additionalProperties": false
},
"ImplementationGuide_Parameter": {
  "description": "A set of rules of how a particular interoperability or standards problem is solved - typically
through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a
logical whole and to publish a computable definition of all the parts.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "apply | path-resource | path-pages | path-tx-cache | expansion-parameter | rule-broken-links |
 generate-xml | generate-json | generate-turtle | html-template.",
      "enum": [
        "apply",
        "path-resource",
        "path-pages",
        "path-tx-cache",
        "expansion-parameter",
        "rule-broken-links",
        "generate-xml",
        "generate-json",
        "generate-turtle",
        "html-template"
      ]
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "value": {
      "description": "Value for named type.",
      "$ref": "#/definitions/string"
    },
    "_value": {
      "description": "Extensions for value",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"ImplementationGuide_Template": {
  "description": "A set of rules of how a particular interoperability or standards problem is solved - typically
through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a
logical whole and to publish a computable definition of all the parts.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
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the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "Type of template specified.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "source": {
      "description": "The source location for the template.",
      "$ref": "#/definitions/string"
    },
    "_source": {
      "description": "Extensions for source",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "scope": {
      "description": "The scope in which the template applies.",
      "$ref": "#/definitions/string"
    },
    "_scope": {
      "description": "Extensions for scope",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"ImplementationGuide_Manifest": {
  "description": "A set of rules of how a particular interoperability or standards problem is solved - typically
through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a
logical whole and to publish a computable definition of all the parts.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "rendering": {
      "description": "A pointer to official web page, PDF or other rendering of the implementation guide.",
      "$ref": "#/definitions/url"
    },
    "_rendering": {
      "description": "Extensions for rendering",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "resource": {
      "description": "A resource that is part of the implementation guide. Conformance resources (value set,
structure definition, capability statements etc.) are obvious candidates for inclusion, but any kind of resource can
 be included as an example resource.",
      "items": {
        "$ref": "#/definitions/ImplementationGuide_Resource1"
      },
      "type": "array"
```

```
    },
    "page": {
      "description": "Information about a page within the IG.",
      "items": {
        "$ref": "#/definitions/ImplementationGuide_Page1"
      },
      "type": "array"
    },
    "image": {
      "description": "Indicates a relative path to an image that exists within the IG.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_image": {
      "description": "Extensions for image",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "other": {
      "description": "Indicates the relative path of an additional non-page, non-image file that is part of the IG -
 e.g. zip, jar and similar files that could be the target of a hyperlink in a derived IG.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_other": {
      "description": "Extensions for other",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resource"
  ]
},
"ImplementationGuide_Resource1": {
  "description": "A set of rules of how a particular interoperability or standards problem is solved - typically
through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a
logical whole and to publish a computable definition of all the parts.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "reference": {
      "description": "Where this resource is found.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "exampleBoolean": {
      "description": "If true or a reference, indicates the resource is an example instance.  If a reference is
present, indicates that the example is an example of the specified profile.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_exampleBoolean": {
      "description": "Extensions for exampleBoolean",
```

```
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "exampleCanonical": {
      "description": "If true or a reference, indicates the resource is an example instance.  If a reference is
present, indicates that the example is an example of the specified profile.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_exampleCanonical": {
      "description": "Extensions for exampleCanonical",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "relativePath": {
      "description": "The relative path for primary page for this resource within the IG.",
      "$ref": "#/definitions/url"
    },
    "_relativePath": {
      "description": "Extensions for relativePath",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false,
  "required": [
    "reference"
  ]
},
"ImplementationGuide_Page1": {
  "description": "A set of rules of how a particular interoperability or standards problem is solved - typically
through the use of FHIR resources. This resource is used to gather all the parts of an implementation guide into a
logical whole and to publish a computable definition of all the parts.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "Relative path to the page.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "Label for the page intended for human display.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "anchor": {
      "description": "The name of an anchor available on the page.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_anchor": {
      "description": "Extensions for anchor",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
```

```
        }
      },
      "additionalProperties": false
    }
  }
}
```

5.8.12 Resource ImplementationGuide - Maturity Dependency View

URL: https://hl7.org/fhir/R4/implementationguide-dependencies.html

5.8.12 Resource ImplementationGuide - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for ImplementationGuide FMM level 1

Name | Card. | Type | Dependency Analysis
ImplementationGuide | | DomainResource
url | 1..1 | uri
version | 0..1 | string
name | 1..1 | string
title | 0..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
copyright | 0..1 | markdown
packageId | 1..1 | id
license | 0..1 | code
fhirVersion | 1..* | code
dependsOn | 0..* | BackboneElement
uri | 1..1 | canonical(ImplementationGuide)
packageId | 0..1 | id
version | 0..1 | string
global | 0..* | BackboneElement
type | 1..1 | code
profile | 1..1 | canonical(StructureDefinition)
definition | 0..1 | BackboneElement
grouping | 0..* | BackboneElement
name | 1..1 | string
description | 0..1 | string
resource | 1..* | BackboneElement
reference | 1..1 | Reference(Any)
fhirVersion | 0..* | code
name | 0..1 | string
description | 0..1 | string
example[x] | 0..1
exampleBoolean | | boolean
exampleCanonical | | canonical
groupingId | 0..1 | id
page | 0..1 | BackboneElement
name[x] | 1..1
nameUrl | | url
nameReference | | Reference(Binary
title | 1..1 | string
generation | 1..1 | code
page | 0..* | see page
parameter | 0..* | BackboneElement
code | 1..1 | code
value | 1..1 | string
template | 0..* | BackboneElement
code | 1..1 | code
source | 1..1 | string
scope | 0..1 | string
manifest | 0..1 | BackboneElement
rendering | 0..1 | url
resource | 1..* | BackboneElement
reference | 1..1 | Reference(Any)

example[x] | 0..1
exampleBoolean | | boolean
exampleCanonical | | canonical
relativePath | 0..1 | url
page | 0..* | BackboneElement
name | 1..1 | string
title | 0..1 | string
anchor | 0..* | string
image | 0..* | string
other | 0..* | string

5.9 Resource GraphDefinition - Content

URL: https://hl7.org/fhir/R4/graphdefinition.html
FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

A formal computable definition of a graph of resources - that is, a coherent set of resources that form a graph by following references. The Graph Definition resource defines a set and makes rules about the set.

5.9.1 Scope and Usage

The GraphDefinition resource provides a formal computable definition of a graph of resources - that is, a coherent set of resources that form a graph by following references. The Graph Definition resource defines a set and makes rules about the set. The GraphDefinition resource can be used to:

• Summarize a set of profiles on resources

• Define a graph of resources to return in a query

• Define a graph of resources to include in a document

• Document rules about the relationship between a set of resources e.g. must all resources concern the same patient?

5.9.2 Boundaries and Relationships

There is a close relationship between Profiles and GraphDefinitions:

• A StructureDefinition defines a profile, and profiles can make rules about the relationships between resources. A carefully defined set of profiles implies part of what is in a GraphDefinition

• A GraphDefinition defines rules about the relationships between resources, and in so doing, implies some constraints that would need to be represented in their profiles

Profiles and Graph Definitions can be used together, or separately. When used together, they should be consistent. Note, though, that a graph definition may contain a subset or a superset of the relationships explicitly described in the profiles it refers to.

It is possible that in some circumstances, a graph definition makes incompatible rules with the Profiles it refers to - in this case, no graph if resources will meet the constraints expressed. Applications should - but are not required - detect when such incompatibilities arise.

This resource is referenced by MessageDefinition

5.9.3 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
GraphDefinition | ITU | | DomainResource | Definition of a graph of resources + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this graph definition, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the graph definition
name | ΣI | 1..1 | string | Name for this graph definition (computer friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)

contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the graph definition
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for graph definition (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this graph definition is defined
start | Σ | 1..1 | code | Type of resource at which the graph starts ResourceType (Required)
profile | | 0..1 | canonical(StructureDefinition) | Profile on base resource
link | | 0..* | BackboneElement | Links this graph makes rules about
path | | 0..1 | string | Path in the resource that contains the link
sliceName | | 0..1 | string | Which slice (if profiled)
min | | 0..1 | integer | Minimum occurrences for this link
max | | 0..1 | string | Maximum occurrences for this link
description | | 0..1 | string | Why this link is specified
target | | 0..* | BackboneElement | Potential target for the link
type | | 1..1 | code | Type of resource this link refers to ResourceType (Required)
params | | 0..1 | string | Criteria for reverse lookup
profile | | 0..1 | canonical(StructureDefinition) | Profile for the target resource
compartment | | 0..* | BackboneElement | Compartment Consistency Rules
use | | 1..1 | code | condition | requirement GraphCompartmentUse (Required)
code | | 1..1 | code | Patient | Encounter | RelatedPerson | Practitioner | Device CompartmentType (Required)
rule | | 1..1 | code | identical | matching | different | custom GraphCompartmentRule (Required)
expression | | 0..1 | string | Custom rule, as a FHIRPath expression
description | | 0..1 | string | Documentation for FHIRPath expression
link | | 0..* | see link | Additional links from target resource
Documentation for this format

UML Diagram (Legend)

XML Template

```
<GraphDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this graph definition, represented as a URI (globally unique)
    -->
 <version value="[string]"/><!-- 0..1 Business version of the graph definition -->
 <name value="[string]"/><!--  1..1 Name for this graph definition (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the graph definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for graph definition (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this graph definition is defined -->
 <start value="[code]"/><!-- 1..1 Type of resource at which the graph starts -->
 <profile><!-- 0..1 canonical(StructureDefinition) Profile on base resource --></profile>
 <link>  <!-- 0..* Links this graph makes rules about -->
  <path value="[string]"/><!-- 0..1 Path in the resource that contains the link -->
  <sliceName value="[string]"/><!-- 0..1 Which slice (if profiled) -->
  <min value="[integer]"/><!-- 0..1 Minimum occurrences for this link -->
  <max value="[string]"/><!-- 0..1 Maximum occurrences for this link -->
  <description value="[string]"/><!-- 0..1 Why this link is specified -->
  <target>  <!-- 0..* Potential target for the link -->
   <type value="[code]"/><!-- 1..1 Type of resource this link refers to -->
   <params value="[string]"/><!-- 0..1 Criteria for reverse lookup -->
   <profile><!-- 0..1 canonical(StructureDefinition) Profile for the target resource --></profile>
   <compartment>  <!-- 0..* Compartment Consistency Rules -->
    <use value="[code]"/><!-- 1..1 condition | requirement -->
    <code value="[code]"/><!-- 1..1 Patient | Encounter | RelatedPerson | Practitioner | Device -->
    <rule value="[code]"/><!-- 1..1 identical | matching | different | custom -->
    <expression value="[string]"/><!-- 0..1 Custom rule, as a FHIRPath expression -->
    <description value="[string]"/><!-- 0..1 Documentation for FHIRPath expression -->
   </compartment>
   <link><!-- 0..* Content as for GraphDefinition.link Additional links from target resource --></link>
  </target>
 </link>
</GraphDefinition>
```

JSON Template

```
{
  "resourceType" : "GraphDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this graph definition, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the graph definition
  "name" : "<string>", // C? R!  Name for this graph definition (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the graph definition
```

```
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for graph definition (if applicable)
  "purpose" : "<markdown>", // Why this graph definition is defined
  "start" : "<code>", // R!  Type of resource at which the graph starts
  "profile" : { canonical(StructureDefinition) }, // Profile on base resource
  "link" : [{ // Links this graph makes rules about
    "path" : "<string>", // Path in the resource that contains the link
    "sliceName" : "<string>", // Which slice (if profiled)
    "min" : <integer>, // Minimum occurrences for this link
    "max" : "<string>", // Maximum occurrences for this link
    "description" : "<string>", // Why this link is specified
    "target" : [{ // Potential target for the link
      "type" : "<code>", // R!  Type of resource this link refers to
      "params" : "<string>", // Criteria for reverse lookup
      "profile" : { canonical(StructureDefinition) }, // Profile for the target resource
      "compartment" : [{ // Compartment Consistency Rules
        "use" : "<code>", // R!  condition | requirement
        "code" : "<code>", // R!  Patient | Encounter | RelatedPerson | Practitioner | Device
        "rule" : "<code>", // R!  identical | matching | different | custom
        "expression" : "<string>", // Custom rule, as a FHIRPath expression
        "description" : "<string>" // Documentation for FHIRPath expression
      }],
      "link" : [{ Content as for GraphDefinition.link }] // Additional links from target resource
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:GraphDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:GraphDefinition.url [ uri ]; # 0..1 Canonical identifier for this graph definition, represented as a URI
    (globally unique)
  fhir:GraphDefinition.version [ string ]; # 0..1 Business version of the graph definition
  fhir:GraphDefinition.name [ string ]; # 1..1 Name for this graph definition (computer friendly)
  fhir:GraphDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:GraphDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:GraphDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:GraphDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:GraphDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:GraphDefinition.description [ markdown ]; # 0..1 Natural language description of the graph definition
  fhir:GraphDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:GraphDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for graph definition (if
    applicable)
  fhir:GraphDefinition.purpose [ markdown ]; # 0..1 Why this graph definition is defined
  fhir:GraphDefinition.start [ code ]; # 1..1 Type of resource at which the graph starts
  fhir:GraphDefinition.profile [ canonical(StructureDefinition) ]; # 0..1 Profile on base resource
  fhir:GraphDefinition.link [ # 0..* Links this graph makes rules about
    fhir:GraphDefinition.link.path [ string ]; # 0..1 Path in the resource that contains the link
    fhir:GraphDefinition.link.sliceName [ string ]; # 0..1 Which slice (if profiled)
    fhir:GraphDefinition.link.min [ integer ]; # 0..1 Minimum occurrences for this link
    fhir:GraphDefinition.link.max [ string ]; # 0..1 Maximum occurrences for this link
    fhir:GraphDefinition.link.description [ string ]; # 0..1 Why this link is specified
    fhir:GraphDefinition.link.target [ # 0..* Potential target for the link
      fhir:GraphDefinition.link.target.type [ code ]; # 1..1 Type of resource this link refers to
      fhir:GraphDefinition.link.target.params [ string ]; # 0..1 Criteria for reverse lookup
      fhir:GraphDefinition.link.target.profile [ canonical(StructureDefinition) ]; # 0..1 Profile for the target
    resource
      fhir:GraphDefinition.link.target.compartment [ # 0..* Compartment Consistency Rules
        fhir:GraphDefinition.link.target.compartment.use [ code ]; # 1..1 condition | requirement
        fhir:GraphDefinition.link.target.compartment.code [ code ]; # 1..1 Patient | Encounter | RelatedPerson |
    Practitioner | Device
        fhir:GraphDefinition.link.target.compartment.rule [ code ]; # 1..1 identical | matching | different | custom
        fhir:GraphDefinition.link.target.compartment.expression [ string ]; # 0..1 Custom rule, as a FHIRPath expression
        fhir:GraphDefinition.link.target.compartment.description [ string ]; # 0..1 Documentation for FHIRPath
    expression
      ], ...;
      fhir:GraphDefinition.link.target.link [ See GraphDefinition.link ], ... ; # 0..* Additional links from target
    resource
    ], ...;
  ], ...;
]
```

Changes since R3

GraphDefinition

GraphDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

GraphDefinition.experimental | No longer marked as Modifier

GraphDefinition.start | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1

GraphDefinition.profile | Type changed from uri to canonical(StructureDefinition)

GraphDefinition.link.path | Min Cardinality changed from 1 to 0

GraphDefinition.link.target | Min Cardinality changed from 1 to 0
GraphDefinition.link.target.type | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
GraphDefinition.link.target.params | Added Element
GraphDefinition.link.target.profile | Type changed from uri to canonical(StructureDefinition)
GraphDefinition.link.target.compartment.use | Added Mandatory Element
GraphDefinition.link.target.compartment.code | Change value set from http://hl7.org/fhir/ValueSet/compartment-type to http://hl7.org/fhir/ValueSet/compartment-type|4.0.1
GraphDefinition.link.target.compartment.rule | Change value set from http://hl7.org/fhir/ValueSet/graph-compartment-rule to http://hl7.org/fhir/ValueSet/graph-compartment-rule|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
GraphDefinition | ITU | | DomainResource | Definition of a graph of resources + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this graph definition, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the graph definition
name | ΣI | 1..1 | string | Name for this graph definition (computer friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the graph definition
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for graph definition (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this graph definition is defined
start | Σ | 1..1 | code | Type of resource at which the graph starts ResourceType (Required)
profile | | 0..1 | canonical(StructureDefinition) | Profile on base resource
link | | 0..* | BackboneElement | Links this graph makes rules about
path | | 0..1 | string | Path in the resource that contains the link
sliceName | | 0..1 | string | Which slice (if profiled)
min | | 0..1 | integer | Minimum occurrences for this link
max | | 0..1 | string | Maximum occurrences for this link
description | | 0..1 | string | Why this link is specified
target | | 0..* | BackboneElement | Potential target for the link
type | | 1..1 | code | Type of resource this link refers to ResourceType (Required)
params | | 0..1 | string | Criteria for reverse lookup
profile | | 0..1 | canonical(StructureDefinition) | Profile for the target resource
compartment | | 0..* | BackboneElement | Compartment Consistency Rules
use | | 1..1 | code | condition | requirement GraphCompartmentUse (Required)
code | | 1..1 | code | Patient | Encounter | RelatedPerson | Practitioner | Device CompartmentType (Required)
rule | | 1..1 | code | identical | matching | different | custom GraphCompartmentRule (Required)
expression | | 0..1 | string | Custom rule, as a FHIRPath expression
description | | 0..1 | string | Documentation for FHIRPath expression
link | | 0..* | see link | Additional links from target resource
Documentation for this format

UML Diagram (Legend)

XML Template

```
<GraphDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this graph definition, represented as a URI (globally unique)
    -->
 <version value="[string]"/><!-- 0..1 Business version of the graph definition -->
 <name value="[string]"/><!--  1..1 Name for this graph definition (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the graph definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for graph definition (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this graph definition is defined -->
 <start value="[code]"/><!-- 1..1 Type of resource at which the graph starts -->
 <profile><!-- 0..1 canonical(StructureDefinition) Profile on base resource --></profile>
 <link>  <!-- 0..* Links this graph makes rules about -->
  <path value="[string]"/><!-- 0..1 Path in the resource that contains the link -->
  <sliceName value="[string]"/><!-- 0..1 Which slice (if profiled) -->
```

```
  <min value="[integer]"/><!-- 0..1 Minimum occurrences for this link -->
  <max value="[string]"/><!-- 0..1 Maximum occurrences for this link -->
  <description value="[string]"/><!-- 0..1 Why this link is specified -->
  <target>  <!-- 0..* Potential target for the link -->
   <type value="[code]"/><!-- 1..1 Type of resource this link refers to -->
   <params value="[string]"/><!-- 0..1 Criteria for reverse lookup -->
   <profile><!-- 0..1 canonical(StructureDefinition) Profile for the target resource --></profile>
   <compartment>  <!-- 0..* Compartment Consistency Rules -->
    <use value="[code]"/><!-- 1..1 condition | requirement -->
    <code value="[code]"/><!-- 1..1 Patient | Encounter | RelatedPerson | Practitioner | Device -->
    <rule value="[code]"/><!-- 1..1 identical | matching | different | custom -->
    <expression value="[string]"/><!-- 0..1 Custom rule, as a FHIRPath expression -->
    <description value="[string]"/><!-- 0..1 Documentation for FHIRPath expression -->
   </compartment>
   <link><!-- 0..* Content as for GraphDefinition.link Additional links from target resource --></link>
  </target>
 </link>
</GraphDefinition>
```

JSON Template

```
{
  "resourceType" : "GraphDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this graph definition, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the graph definition
  "name" : "<string>", // C? R!  Name for this graph definition (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the graph definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for graph definition (if applicable)
  "purpose" : "<markdown>", // Why this graph definition is defined
  "start" : "<code>", // R!  Type of resource at which the graph starts
  "profile" : { canonical(StructureDefinition) }, // Profile on base resource
  "link" : [{ // Links this graph makes rules about
    "path" : "<string>", // Path in the resource that contains the link
    "sliceName" : "<string>", // Which slice (if profiled)
    "min" : <integer>, // Minimum occurrences for this link
    "max" : "<string>", // Maximum occurrences for this link
    "description" : "<string>", // Why this link is specified
    "target" : [{ // Potential target for the link
      "type" : "<code>", // R!  Type of resource this link refers to
      "params" : "<string>", // Criteria for reverse lookup
      "profile" : { canonical(StructureDefinition) }, // Profile for the target resource
      "compartment" : [{ // Compartment Consistency Rules
        "use" : "<code>", // R!  condition | requirement
        "code" : "<code>", // R!  Patient | Encounter | RelatedPerson | Practitioner | Device
        "rule" : "<code>", // R!  identical | matching | different | custom
        "expression" : "<string>", // Custom rule, as a FHIRPath expression
        "description" : "<string>" // Documentation for FHIRPath expression
      }],
      "link" : [{ Content as for GraphDefinition.link }] // Additional links from target resource
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:GraphDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:GraphDefinition.url [ uri ]; # 0..1 Canonical identifier for this graph definition, represented as a URI
    (globally unique)
  fhir:GraphDefinition.version [ string ]; # 0..1 Business version of the graph definition
  fhir:GraphDefinition.name [ string ]; # 1..1 Name for this graph definition (computer friendly)
  fhir:GraphDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:GraphDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:GraphDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:GraphDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:GraphDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:GraphDefinition.description [ markdown ]; # 0..1 Natural language description of the graph definition
  fhir:GraphDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:GraphDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for graph definition (if
    applicable)
  fhir:GraphDefinition.purpose [ markdown ]; # 0..1 Why this graph definition is defined
  fhir:GraphDefinition.start [ code ]; # 1..1 Type of resource at which the graph starts
  fhir:GraphDefinition.profile [ canonical(StructureDefinition) ]; # 0..1 Profile on base resource
  fhir:GraphDefinition.link [ # 0..* Links this graph makes rules about
    fhir:GraphDefinition.link.path [ string ]; # 0..1 Path in the resource that contains the link
    fhir:GraphDefinition.link.sliceName [ string ]; # 0..1 Which slice (if profiled)
    fhir:GraphDefinition.link.min [ integer ]; # 0..1 Minimum occurrences for this link
```

```
    fhir:GraphDefinition.link.max [ string ]; # 0..1 Maximum occurrences for this link
    fhir:GraphDefinition.link.description [ string ]; # 0..1 Why this link is specified
    fhir:GraphDefinition.link.target [ # 0..* Potential target for the link
      fhir:GraphDefinition.link.target.type [ code ]; # 1..1 Type of resource this link refers to
      fhir:GraphDefinition.link.target.params [ string ]; # 0..1 Criteria for reverse lookup
      fhir:GraphDefinition.link.target.profile [ canonical(StructureDefinition) ]; # 0..1 Profile for the target
    resource
      fhir:GraphDefinition.link.target.compartment [ # 0..* Compartment Consistency Rules
        fhir:GraphDefinition.link.target.compartment.use [ code ]; # 1..1 condition | requirement
        fhir:GraphDefinition.link.target.compartment.code [ code ]; # 1..1 Patient | Encounter | RelatedPerson |
    Practitioner | Device
        fhir:GraphDefinition.link.target.compartment.rule [ code ]; # 1..1 identical | matching | different | custom
        fhir:GraphDefinition.link.target.compartment.expression [ string ]; # 0..1 Custom rule, as a FHIRPath expression
        fhir:GraphDefinition.link.target.compartment.description [ string ]; # 0..1 Documentation for FHIRPath
    expression
      ], ...;
      fhir:GraphDefinition.link.target.link [ See GraphDefinition.link ], ... ; # 0..* Additional links from target
    resource
    ], ...;
  ], ...;
]
```

Changes since Release 3

GraphDefinition
GraphDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
GraphDefinition.experimental | No longer marked as Modifier
GraphDefinition.start | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
GraphDefinition.profile | Type changed from uri to canonical(StructureDefinition)
GraphDefinition.link.path | Min Cardinality changed from 1 to 0
GraphDefinition.link.target | Min Cardinality changed from 1 to 0
GraphDefinition.link.target.type | Change value set from http://hl7.org/fhir/ValueSet/resource-types to http://hl7.org/fhir/ValueSet/resource-types|4.0.1
GraphDefinition.link.target.params | Added Element
GraphDefinition.link.target.profile | Type changed from uri to canonical(StructureDefinition)
GraphDefinition.link.target.compartment.use | Added Mandatory Element
GraphDefinition.link.target.compartment.code | Change value set from http://hl7.org/fhir/ValueSet/compartment-type to http://hl7.org/fhir/ValueSet/compartment-type|4.0.1
GraphDefinition.link.target.compartment.rule | Change value set from http://hl7.org/fhir/ValueSet/graph-compartment-rule to http://hl7.org/fhir/ValueSet/graph-compartment-rule|4.0.1

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

5.9.3.1 Terminology Bindings

Path | Definition | Type | Reference
GraphDefinition.status | The lifecycle status of an artifact. | Required | PublicationStatus
GraphDefinition.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
GraphDefinition.start GraphDefinition.link.target.type | One of the resource types defined as part of this version of FHIR. | Required | Resource Types
GraphDefinition.link.target.compartment.use | Defines how a compartment rule is used. | Required | GraphCompartmentUse
GraphDefinition.link.target.compartment.code | Identifies a compartment. | Required | CompartmentType
GraphDefinition.link.target.compartment.rule | How a compartment must be linked. | Required | GraphCompartmentRule

5.9.3.2 Constraints

id | Level | Location | Description | Expression
gdf-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

5.9.3.3 Using GraphDefinitions

The GraphDefinition resource can be used to:

• Summarize a set of profiles on resources

• Define a graph of resources to return in a query

• Define a graph of resources to include in a document

• Document rules about the relationship between a set of resources e.g. must all resources concern the same patient?

5.9.3.3.1 Summarize a set of profiles on resources

FHIR resources are relatively granular. In many/most cases, many resources are needed to handle any particular task. A typical example of this is a complex diagnostic report: it will start with a DiagnosticReport, which will link to a set of panels (Observation resources), each of which link to a set of Observation resources for atomic data items.

One way to represent this is to profile each of the resources, creating hundreds of profiles, and then leave it to the user to infer the overall pattern of the report from the detailed profiles for each observation in the report. But it's not easy to see the forest for the trees. A GraphDefinition can summarize the overall picture and present a summary to the user.

Here's an example of the kind of summary this represents. (Todo: make this an actual graph definition, and clone into the main spec)

5.9.3.3.2 Fetching a graph of resources

As another example of using many resources, to completely represent a medication dispense, an application needs not only the MedicationDispense resource, but also resources to represent the patient, provider, organizations, and the associated prescription.

A client can retrieve a single resource:

```
GET [base]/MedicationDispense/example
```

Then, when it reads the returned resource, it can fetch the referenced resources:

```
GET [base]/Patient/example
GET [base]/Practitioner/example
GET [base]/MedicationRequest/example
... etc.
```

This is a very inefficient way to retrieve all the required resources. An alternative approach is to do a search, and _include the required resources:

```
GET [base]/MedicationDispense?_id=example
  &_include=MedicationDispense:authorizingPrescription
  &_include=MedicationDispense:subject
```

But scaling this approach to fetch a full package with its dependencies becomes increasingly difficult as the package gets deeper. A graph definition can be used instead to inform the server what to return with the resource using the $graph operation:

```
GET [base]/MedicationDispense/example/$graph?graph=med-package
```

This is a request to return a graph of resource, using the graph definition 'med-package'. In this case, the graph definition would look approximately like this:

```
MedicationDispense
  .subject
  .context
  .performer.actor
  .authorizingPrescription
     .requester.agent
  .substitution.responsibleParty
```

Systems may either provide a pre-defined list of graph definitions that clients may choose from, or allow clients to define their own GraphDefinition resources and then refer to them.

Server may also allow clients also pass in their own graph definition using a text representation:

```
GET [base]/MedicationDispense/example/$graph?definition=Patient{managingOrganization:Organization{endpoint:Endpoint}}
```

See below for further details.

5.9.3.3.3 Building a document

A very similar issue applies when building a document using the $document operation. A document must include all the resources linked directly from the composition, but whether to include additional linked resources is at the discretion of the document author. How does the user inform the $document operation which linked resources to include? One option is a boolean flag for including all linked data, but this may be extensive - up to an entire patient record - and may include resources that are not desired.

An operation can use a graph definition as a parameter to the $document operation:

```
GET [base]/Composition/example/$document?graph=example
```

This tells the server to include the graph of resources defined in the example GraphDefinition - in this case, any resources referred to from lists, when the section content is a list. Alternatively, servers may allow a client to pass in a definition directly (as shown above) using the parameter definition.

5.9.3.3.4 Document Relationship rules

One important question about the use of resources is cross-resource consistency. For example, if an Observation refers to both a Patient and Encounter, does the Encounter have to refer to the same patient?

In general, the answer to this is that it usually should - the record needs to be consistent. However there are edge cases where the references may differ. For example, with regard to patient references, they may differ for:

• Health Records concerning mother and baby

• Organ Transplants
• Counselling, particularly family counselling

Other reasons for the references to differ - mixing records about the same patient from different servers, or specific records about patients mixed with records about groups of patients (particularly common in veterinarian care).

The GraphDefinition resource allows for compartment consistency rules to be made regarding the links between resources. For each link in the graph, the graph definition can make a rule about the compartment consistency. The rule can specify one of the following consistencies:

Code | Meaning
identical | The compartment must be identical (the same literal reference)
matching | The compartment must be the same - the record must be about the same patient, but the reference may be different
different | The compartment must be different
custom | The compartment rule is defined in the accompanying FHIRPath expression

Todo: how would this be validated? - where is the graph referred to?

5.9.3.4 Simple Examples

GraphDefinition can walk a graph forward through the links, which is the natural order. As an example, consider a profile on composition that wants to specify that the graph follows the section entry into a list:

```
  <!--
    starting from a Composition...
  -->
<link>
  <!--
    follow and Composition.section.entry that refers to a list
  -->
  <path value="Composition.section.entry"/>
  <target>
    <type value="List"/>
    <!-- and inside this list, follow any items -->
    <link>
      <path value="List.entry.item"/>
      <target>
        <type value="Resource"/>
      </target>
    </link>
  </target>
</link>
```

The pattern can be extended to require that the none of the resources in this little graph reference a different patient - that is, it is a requirement when traversing the links that the patient compartment remain identical:

```
<link>
  <path value="Composition.section.entry"/>
  <target>
    <type value="List"/>
      <compartment>
        <use value="requirement"/>
        <code value="Patient"/>
        <rule value="identical"/>
      </compartment>
    </target>
    <link>
      <path value="List.entry.item"/>
      <description value="Include any list entries"/>
      <target>
        <type value="Resource"/>
        <compartment>
          <use value="requirement"/>
          <code value="Patient"/>
          <rule value="identical"/>
        </compartment>
      </target>
    </link>
  </target>
</link>
```

Another useful approach is to walk any link * (e.g. wildcard support):

```
<link>
  <!--
    follow all links that refers to a list
  -->
  <path value="*"/>
  <target>
    <type value="List"/>
    <!-- and inside this list, follow any references -->
    <link>
      <path value="*"/>
      <target>
        <type value="Resource"/>
      </target>
    </link>
  </target>
```

```
  </link>
```

or you can follow any links in any resource:

```
  <link>
    <!--
      follow all links
    -->
    <path value="*"/>
    <target>
      <type value="Resource"/>
      <!-- and inside the target, follow any links -->
      <link>
        <path value="*"/>
        <target>
          <type value="Resource"/>
        </target>
      </link>
    </target>
  </link>
```

It's also possible to build a graph by walking links in reverse. This technique is useful, for example, when including provenance resources in a document:

```
  <link>
    <!-- if the path is missing <path value=""/> -->
    <target>
      <!--
        any provenance
      -->
      <type value="Provenance"/>
      <!--
        and specify the search parameters to include
        all the provenances that refer to this resource
      -->
      <params value="target={ref}"/>
    </target>
  </link>
```

5.9.3.4.1 Text Representation

For convenience, a graph definition may be represented using a text short hand form, loosely based on the graphQL language. A graph definition is introduced by the name of the Resource (GraphDefinition.start, followed by an optional profile in parentheses (GraphDefinition.profile, then a group of links, surrounded by { and }:

```
Patient(http://hl7.org/fhir/us/core/Patient) {
}
```

Between the brackets are one or more links (GraphDefinition.link), separated by commas. Links statements can either be path-based look ups, or a search to perform a reverse lookup:

```
Patient {
  managingOrganization : Organization,
  generalPractitioner : Practitioner,
  search Observation?patient={ref}
}
```

The format for a path-based look up is:

```
  [path] cardinality [min]..[max] '[description]' : [ResoureType rules]
```

where:

• path = GraphDefinition.link.path

• cardinality [min]..[max] is optional, and min = GraphDefinition.link.min and max = GraphDefinition.link.max

• description is optional, and = GraphDefinition.link.description

The resource type rules statement consists of one or more of the following, separated by ; characters:

```
  [ResourceType] (profile)
```

where:

• ResourceType = GraphDefinition.link.target.type

• profile is optional, and = GraphDefinition.link.target.profile

• description is optional, and = GraphDefinition.link.description

In addition, the resource type rule may be followed by a links section, surrounded by {}, with the content as described above.

Search statements for reverse look ups have a slightly different format:

```
Patient{
  search [ResourceType]?[params] cardinality [min]..[max] '[description]'
}
```

where:

• ResourceType = GraphDefinition.link.target.type

• params = GraphDefinition.link.target.params

• cardinality [min]..[max] is optional, and min = GraphDefinition.link.min and max = GraphDefinition.link.max
• description is optional, and = GraphDefinition.link.description

In this format, the amount of whitespace, and its form, is irrelevant. For the purposes of clarity, the examples here are laid out on different lines, but this is not required. When a GraphDefinition is used as a parameter in a url, no whitespace is used.

Here is a full example that uses all the features of the syntax:

```
Patient(http://hl7.org/fhir/us/core/Patient) {
  managingOrganization cardinality 0..1 'description of item' :
    Organization(http://hl7.org/fhir/us/core/Organization) {
      endpoint : Endpoint
    };
    Basic;
    Group {
      item : Patient
    },
  generalPractitioner : Organization,
  search Observation?patient={ref} cardinality 0..10 'Observations for the patient' {
    performer : Practitioner,
    related.where(type='has-member').target : Observation require matching Patient,
    related.where(type='derived-from').target : Observation where identical Patient,
    related.where(type='sequel-to').target : Observation where different Patient,
    related.where(type='qualified-by').target : Observation where custom Patient = path
  }
}
```

5.9.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
context | token | A use context assigned to the graph definition | (GraphDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the graph definition | (GraphDefinition.useContext.value as Quantity) | (GraphDefinition.useContext.value as Range)
context-type | token | A type of use context assigned to the graph definition | GraphDefinition.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the graph definition | On GraphDefinition.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the graph definition | On GraphDefinition.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The graph definition publication date | GraphDefinition.date
description | string | The description of the graph definition | GraphDefinition.description
jurisdiction | token | Intended jurisdiction for the graph definition | GraphDefinition.jurisdiction
name | string | Computationally friendly name of the graph definition | GraphDefinition.name
publisher | string | Name of the publisher of the graph definition | GraphDefinition.publisher
start | token | Type of resource at which the graph starts | GraphDefinition.start
status | token | The current status of the graph definition | GraphDefinition.status
url | uri | The uri that identifies the graph definition | GraphDefinition.url
version | token | The business version of the graph definition | GraphDefinition.version

5.9.5 Resource GraphDefinition - Examples

URL: https://hl7.org/fhir/R4/graphdefinition-examples.html

5.9.5 Resource GraphDefinition - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
GraphDefinition example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.9.6 Resource GraphDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/graphdefinition-definitions.html

5.9.6 Resource GraphDefinition - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the GraphDefinition resource.

GraphDefinition
Element Id | GraphDefinition
Definition | A formal computable definition of a graph of resources - that is, a coherent set of resources that form a graph by following references. The Graph Definition resource defines a set and makes rules about the set.

Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element gdf-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
gdf-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
GraphDefinition.url
Element Id | GraphDefinition.url
Definition | An absolute URI that is used to identify this graph definition when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this graph definition is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the graph definition is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the graph definition to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
GraphDefinition.version
Element Id | GraphDefinition.version
Definition | The identifier that is used to identify this version of the graph definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the graph definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different graph definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the graph definition with the format [url]|[version].
GraphDefinition.name
Element Id | GraphDefinition.name
Definition | A natural language name identifying the graph definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 1..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
GraphDefinition.status
Element Id | GraphDefinition.status
Definition | The status of this graph definition. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of graph definitions that are appropriate for use versus not.
GraphDefinition.experimental
Element Id | GraphDefinition.experimental
Definition | A Boolean value to indicate that this graph definition is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level graph definition.
Summary | true
Comments | Allows filtering of graph definitions that are appropriate for use versus not.
GraphDefinition.date
Element Id | GraphDefinition.date
Definition | The date (and optionally time) when the graph definition was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the

graph definition changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the graph definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
GraphDefinition.publisher
Element Id | GraphDefinition.publisher
Definition | The name of the organization or individual that published the graph definition.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the graph definition. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the graph definition is the organization or individual primarily responsible for the maintenance and upkeep of the graph definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the graph definition. This item SHOULD be populated unless the information is available from context.
GraphDefinition.contact
Element Id | GraphDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
GraphDefinition.description
Element Id | GraphDefinition.description
Definition | A free text natural language description of the graph definition from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the graph definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the graph definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the graph definition is presumed to be the predominant language in the place the graph definition was created).
GraphDefinition.useContext
Element Id | GraphDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate graph definition instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
GraphDefinition.jurisdiction
Element Id | GraphDefinition.jurisdiction
Definition | A legal or geographic region in which the graph definition is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the graph definition to be used in jurisdictions other than those for which it was originally designed or intended.
GraphDefinition.purpose
Element Id | GraphDefinition.purpose
Definition | Explanation of why this graph definition is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the graph definition. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this graph definition.
GraphDefinition.start
Element Id | GraphDefinition.start
Definition | The type of FHIR resource at which instances of this graph start.
Cardinality | 1..1
Terminology Binding | Any defined Resource Type name
Type | code
Summary | true
GraphDefinition.profile
Element Id | GraphDefinition.profile
Definition | The profile that describes the use of the base resource.

Cardinality | 0..1
Type | canonical(StructureDefinition)
Comments | The code does not include the '$' prefix that is always included in the URL when the operation is invoked.
GraphDefinition.link
Element Id | GraphDefinition.link
Definition | Links this graph makes rules about.
Cardinality | 0..*
GraphDefinition.link.path
Element Id | GraphDefinition.link.path
Definition | A FHIR expression that identifies one of FHIR References to other resources.
Cardinality | 0..1
Type | string
Comments | The path expression cannot contain a resolve() function. If there is no path, the link is a reverse lookup, using target.params. If the path is "*" then this means all references in the resource.
GraphDefinition.link.sliceName
Element Id | GraphDefinition.link.sliceName
Definition | Which slice (if profiled).
Cardinality | 0..1
Type | string
GraphDefinition.link.min
Element Id | GraphDefinition.link.min
Definition | Minimum occurrences for this link.
Cardinality | 0..1
Type | integer
GraphDefinition.link.max
Element Id | GraphDefinition.link.max
Definition | Maximum occurrences for this link.
Cardinality | 0..1
Type | string
GraphDefinition.link.description
Element Id | GraphDefinition.link.description
Definition | Information about why this link is of interest in this graph definition.
Cardinality | 0..1
Type | string
GraphDefinition.link.target
Element Id | GraphDefinition.link.target
Definition | Potential target for the link.
Cardinality | 0..*
GraphDefinition.link.target.type
Element Id | GraphDefinition.link.target.type
Definition | Type of resource this link refers to.
Cardinality | 1..1
Terminology Binding | Any defined Resource Type name
Type | code
GraphDefinition.link.target.params
Element Id | GraphDefinition.link.target.params
Definition | A set of parameters to look up.
Cardinality | 0..1
Type | string
Comments | At least one of the parameters must have the value {ref} which identifies the focus resource.
GraphDefinition.link.target.profile
Element Id | GraphDefinition.link.target.profile
Definition | Profile for the target resource.
Cardinality | 0..1
Type | canonical(StructureDefinition)
GraphDefinition.link.target.compartment
Element Id | GraphDefinition.link.target.compartment
Definition | Compartment Consistency Rules.
Cardinality | 0..*
GraphDefinition.link.target.compartment.use
Element Id | GraphDefinition.link.target.compartment.use
Definition | Defines how the compartment rule is used - whether it it is used to test whether resources are subject to the rule, or whether it is a rule that must be followed.
Cardinality | 1..1
Terminology Binding | GraphCompartmentUse (Required)
Type | code
Comments | All conditional rules are evaluated; if they are true, then the rules are evaluated.
GraphDefinition.link.target.compartment.code
Element Id | GraphDefinition.link.target.compartment.code
Definition | Identifies the compartment.
Cardinality | 1..1
Terminology Binding | CompartmentType (Required)

Type | code
GraphDefinition.link.target.compartment.rule
Element Id | GraphDefinition.link.target.compartment.rule
Definition | identical | matching | different | no-rule | custom.
Cardinality | 1..1
Terminology Binding | GraphCompartmentRule (Required)
Type | code
GraphDefinition.link.target.compartment.expression
Element Id | GraphDefinition.link.target.compartment.expression
Definition | Custom rule, as a FHIRPath expression.
Cardinality | 0..1
Type | string
GraphDefinition.link.target.compartment.description
Element Id | GraphDefinition.link.target.compartment.description
Definition | Documentation for FHIRPath expression.
Cardinality | 0..1
Type | string
GraphDefinition.link.target.link
Element Id | GraphDefinition.link.target.link
Definition | Additional links from target resource.
Cardinality | 0..*
Type | See GraphDefinition.link

5.9.7 Resource GraphDefinition - Mappings

URL: https://hl7.org/fhir/R4/graphdefinition-mappings.html

5.9.7 Resource GraphDefinition - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the graphdefinition resource (see Mappings to Other Standards for further information & status).

5.9.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

GraphDefinition
url | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

5.9.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

GraphDefinition
url | Definition.url
version | Definition.version
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose

5.9.7.3 RIM Mapping (http://hl7.org/v3)

GraphDefinition | n/a

5.9.8 Resource GraphDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/graphdefinition-profiles.html

5.9.8 Resource GraphDefinition - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the GraphDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

5.9.8.1 Profiles

No Profiles defined for this resource

5.9.8.2 Extensions

replaces | GraphDefinition | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

5.9.8.3 Search Extensions

No Search Extensions defined for this resource

5.9.9 Resource GraphDefinition - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/graphdefinition-version-maps.html

No validation errors - all conversions are clean

5.9.9 Resource GraphDefinition - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for GraphDefinition.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (0 errors). (see documentation)

5.9.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/GraphDefinition3to4" = "R3 to R4 Conversions for GraphDefinition"

uses "http://hl7.org/fhir/3.0/StructureDefinition/GraphDefinition" alias GraphDefinitionR3 as source
uses "http://hl7.org/fhir/StructureDefinition/GraphDefinition" alias GraphDefinition as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group GraphDefinition(source src : GraphDefinitionR3, target tgt : GraphDefinition) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.start -> tgt.start;
  src.profile -> tgt.profile;
  src.link as vs0 -> tgt.link as vt0 then link(vs0, vt0);
}

group link(source src, target tgt) extends BackboneElement {
  src.path -> tgt.path;
  src.sliceName -> tgt.sliceName;
  src.min -> tgt.min;
  src.max -> tgt.max;
  src.description -> tgt.description;
  src.target as vs0 -> tgt.target as vt0 then target(vs0, vt0);
}

group target(source src, target tgt) extends BackboneElement {
```

```
  src.type -> tgt.type;
  src.profile -> tgt.profile;
  src.compartment as vs0 -> tgt.compartment as vt0 then compartment(vs0, vt0);
  src.link as vs0 -> tgt.link as vt0 then link(vs0, vt0);
}

group compartment(source src, target tgt) extends BackboneElement {
  src -> tgt.use = 'requirement' "use";
  src.code -> tgt.code;
  src.rule -> tgt.rule;
  src.expression -> tgt.expression;
  src.description -> tgt.description;
}
```

5.9.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/GraphDefinition4to3" = "R4 to R3 Conversion for GraphDefinition"

uses "http://hl7.org/fhir/StructureDefinition/GraphDefinition" alias GraphDefinition as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/GraphDefinition" alias GraphDefinitionR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group GraphDefinition(source src : GraphDefinition, target tgt : GraphDefinitionR3) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.start -> tgt.start;
  src.profile -> tgt.profile;
  src.link as vs0 -> tgt.link as vt0 then link(vs0, vt0);
}

group link(source src, target tgt) extends BackboneElement {
  src.path -> tgt.path;
  src.sliceName -> tgt.sliceName;
  src.min -> tgt.min;
  src.max -> tgt.max;
  src.description -> tgt.description;
  src.target as vs0 -> tgt.target as vt0 then target(vs0, vt0);
}

group target(source src, target tgt) extends BackboneElement {
  src.type -> tgt.type;
  src.profile -> tgt.profile;
  src.compartment as vs0 where use = 'requirement' -> tgt.compartment as vt0 then compartment(vs0, vt0);
  src.link as vs0 -> tgt.link as vt0 then link(vs0, vt0);
}

group compartment(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.rule -> tgt.rule;
  src.expression -> tgt.expression;
  src.description -> tgt.description;
}
```

5.9.9.3 R4 Validation Errors

5.9.10 JSON Schema for GraphDefinition

URL: https://hl7.org/fhir/R4/graphdefinition.schema.json.html

5.9.10 JSON Schema for GraphDefinition

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for GraphDefinition, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/GraphDefinition",
  "$ref": "#/definitions/GraphDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "GraphDefinition": {
      "description": "A formal computable definition of a graph of resources - that is, a coherent set of resources that
     form a graph by following references. The Graph Definition resource defines a set and makes rules about the set.",
      "properties": {
        "resourceType": {
          "description": "This is a GraphDefinition resource",
```

```
      "const": "GraphDefinition"
    },
    "id": {
      "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this graph definition when it is referenced in a
specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which at which an authoritative instance of this graph definition is (or will be)
 published. This URL can be the target of a canonical reference. It SHALL remain the same when the graph definition
is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the graph definition when it is
referenced in a specification, model, design or instance. This is an arbitrary value managed by the graph definition
 author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed
 version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
```

```
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the graph definition. This name should be usable as an
identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this graph definition. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this graph definition is authored for testing purposes (or
education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the graph definition was published. The date must change
when the business version changes and it must change if the status code changes. In addition, it should change when
the substantive content of the graph definition changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the graph definition.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the graph definition from a consumer\u0027s
perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate graph definition
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the graph definition is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this graph definition is needed and why it has been designed as it has.",
      "$ref": "#/definitions/markdown"
    },
```

```
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "start": {
      "description": "The type of FHIR resource at which instances of this graph start.",
      "$ref": "#/definitions/code"
    },
    "_start": {
      "description": "Extensions for start",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "profile": {
      "description": "The profile that describes the use of the base resource.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "link": {
      "description": "Links this graph makes rules about.",
      "items": {
        "$ref": "#/definitions/GraphDefinition_Link"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"GraphDefinition_Link": {
  "description": "A formal computable definition of a graph of resources - that is, a coherent set of resources that
 form a graph by following references. The Graph Definition resource defines a set and makes rules about the set.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "path": {
      "description": "A FHIR expression that identifies one of FHIR References to other resources.",
      "$ref": "#/definitions/string"
    },
    "_path": {
      "description": "Extensions for path",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "sliceName": {
      "description": "Which slice (if profiled).",
      "$ref": "#/definitions/string"
    },
    "_sliceName": {
      "description": "Extensions for sliceName",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "min": {
      "description": "Minimum occurrences for this link.",
      "$ref": "#/definitions/integer"
    },
    "_min": {
      "description": "Extensions for min",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "max": {
      "description": "Maximum occurrences for this link.",
      "$ref": "#/definitions/string"
```

```
    },
    "_max": {
      "description": "Extensions for max",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "Information about why this link is of interest in this graph definition.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "target": {
      "description": "Potential target for the link.",
      "items": {
        "$ref": "#/definitions/GraphDefinition_Target"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"GraphDefinition_Target": {
  "description": "A formal computable definition of a graph of resources - that is, a coherent set of resources that
 form a graph by following references. The Graph Definition resource defines a set and makes rules about the set.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Type of resource this link refers to.",
      "$ref": "#/definitions/code"
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "params": {
      "description": "A set of parameters to look up.",
      "$ref": "#/definitions/string"
    },
    "_params": {
      "description": "Extensions for params",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "profile": {
      "description": "Profile for the target resource.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "compartment": {
      "description": "Compartment Consistency Rules.",
      "items": {
        "$ref": "#/definitions/GraphDefinition_Compartment"
      },
      "type": "array"
    },
    "link": {
      "description": "Additional links from target resource.",
      "items": {
        "$ref": "#/definitions/GraphDefinition_Link"
      },
      "type": "array"
```

```
    }
  },
  "additionalProperties": false
},
"GraphDefinition_Compartment": {
  "description": "A formal computable definition of a graph of resources - that is, a coherent set of resources that
 form a graph by following references. The Graph Definition resource defines a set and makes rules about the set.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "use": {
      "description": "Defines how the compartment rule is used - whether it it is used to test whether resources are
 subject to the rule, or whether it is a rule that must be followed.",
      "enum": [
        "condition",
        "requirement"
      ]
    },
    "_use": {
      "description": "Extensions for use",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "Identifies the compartment.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "rule": {
      "description": "identical | matching | different | no-rule | custom.",
      "enum": [
        "identical",
        "matching",
        "different",
        "custom"
      ]
    },
    "_rule": {
      "description": "Extensions for rule",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "expression": {
      "description": "Custom rule, as a FHIRPath expression.",
      "$ref": "#/definitions/string"
    },
    "_expression": {
      "description": "Extensions for expression",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "Documentation for FHIRPath expression.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
}
```

```
  }
}
```

5.9.11 Resource GraphDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/graphdefinition-dependencies.html

5.9.11 Resource GraphDefinition - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for GraphDefinition FMM level 1

Name | Card. | Type | Dependency Analysis
GraphDefinition | | DomainResource
url | 0..1 | uri
version | 0..1 | string
name | 1..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
start | 1..1 | code
profile | 0..1 | canonical(StructureDefinition)
link | 0..* | BackboneElement
path | 0..1 | string
sliceName | 0..1 | string
min | 0..1 | integer
max | 0..1 | string
description | 0..1 | string
target | 0..* | BackboneElement
type | 1..1 | code
params | 0..1 | string
profile | 0..1 | canonical(StructureDefinition)
compartment | 0..* | BackboneElement
use | 1..1 | code
code | 1..1 | code
rule | 1..1 | code
expression | 0..1 | string
description | 0..1 | string
link | 0..* | see link